

Vapor residence time required for the droplets to settle to liquid surface:

$$h_v / u_s = 0.5D_v / 0.066 = 7.58D_v$$

Actual residence time = vessel length/vapor velocity:

$$\frac{L_v}{u_v} = \frac{4D_v}{0.374D_v^{-2}} = 10.70D_v^3$$

For satisfactory separation, required residence time = actual:

$$\begin{aligned} \text{So, } 7.58D_v &= 10.70D_v^3 \\ D_v &= 0.84 \text{ m, say } 0.92 \text{ m (3 ft, standard pipe size)} \end{aligned}$$

Liquid hold-up time:

$$\text{liquid volumetric flow - rate} = \frac{10,000}{3600 \times 962.0} = 0.00289 \text{ m}^3/\text{s}$$

$$\text{liquid cross-sectional area} = \frac{\pi \times 0.92^2}{4} \times 0.5 = 0.332 \text{ m}^2$$

$$\text{Length, } L_v = 4 \times 0.92 = 3.7 \text{ m}$$

$$\text{Hold-up volume} = 0.332 \times 3.7 = 1.23 \text{ m}^3$$

$$\text{Hold-up time} = \text{liquid volume/liquid flow rate}$$

$$= 1.23/0.00289 = 426 \text{ s} = 7 \text{ minutes}$$

This is unsatisfactory, 10 minutes minimum required.

Need to increase the liquid volume. This is best done by increasing the vessel diameter. If the liquid height is kept at half the vessel diameter, the diameter must be increased by a factor of roughly $(10/7)0.5 = 1.2$.

$$\text{New } D_v = 0.92 \times 1.2 = 1.1 \text{ m}$$

Check liquid residence time:

$$\text{new liquid volume} = \frac{\pi \times 1.1^2}{4} \times 0.5 \times (4 \times 1.1) = 2.09 \text{ m}^3$$

$$\text{new residence time} = 2.09/0.00289 = 723 \text{ s} = 12 \text{ minutes, satisfactory}$$

Increasing the vessel diameter will have also changed the vapor velocity and the height above the liquid surface. The liquid separation will still be satisfactory, as the velocity, and hence the residence time, is inversely proportional to the diameter squared, whereas the distance the droplets have to fall is directly proportional to the diameter.

In practice, the distance traveled by the vapor will be less than the vessel length, L_v , as the vapor inlet and outlet nozzles will be set in from the ends. This could be allowed for in the design but will make little difference.

16.4 Liquid-liquid separation

Separation of two liquid phases, immiscible or partially miscible liquids, is a common requirement in the process industries. For example, in single-stage liquid-liquid extraction, the liquid-contacting step must be followed by a settling stage. It is also frequently necessary to separate small quantities of entrained water from organic-phase process streams. The simplest form of equipment used to separate liquid phases is the gravity settling tank, the decanter. Various types of proprietary equipment are also used to promote coalescence and improve separation in difficult systems or where emulsions are likely to form. Centrifuges and hydrocyclones are also used.

Liquid-liquid extraction can also be carried out as a multistage countercurrent process. This is described with other multistage separation processes in [Chapter 17](#), Section 17.16.

16.4.1 Decanters (settlers)

Decanters are used to separate liquids where there is a sufficient difference in density between the liquids for the droplets to settle readily. Decanters are essentially tanks that give sufficient residence time for the droplets of the dispersed phase to rise (or settle) to the interface between the phases and coalesce. In an operating decanter there will be three distinct zones or bands: clear heavy liquid; separating dispersed liquid (the dispersion zone); and clear light liquid.

Decanters are normally designed for continuous operation, but the same design principles apply to batch-operated units. Many vessel shapes are used for decanters, but for most applications a cylindrical vessel will be suitable and will be the cheapest shape. Typical designs are shown in Figs. 16.13 and 16.14. The position of the interface can be controlled, with or without the use of instruments, by use of a siphon take-off for the heavy liquid (see Fig. 16.13).

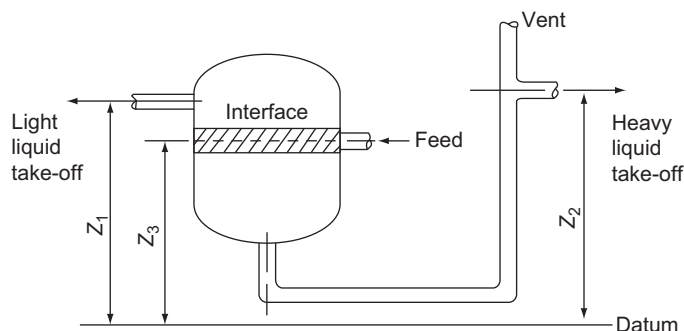


FIG. 16.13 Vertical decanter.

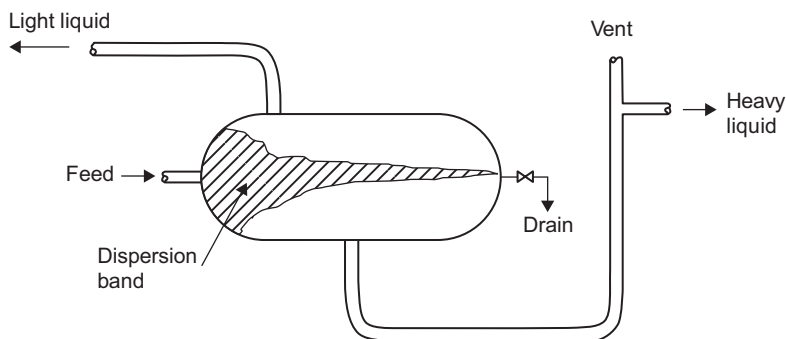


FIG. 16.14 Horizontal decanter.

The height of the take-off can be determined by making a pressure balance. Neglecting friction loss in the pipes, the pressure exerted by the combined height of the heavy and light liquid in the vessel must be balanced by the height of the heavy liquid in the take-off leg (see Fig. 16.13).

$$(z_1 - z_3)\rho_1 g + z_3 \rho_2 g = z_2 \rho_2 g$$

hence

$$z_2 = \frac{(z_1 - z_3)\rho_1}{\rho_2} + z_3 \quad (16.8)$$

where ρ_1 = density of the light liquid, kg/m³

ρ_2 = density of the heavy liquid, kg/m³

z_1 = height from datum to light liquid overflow, m

z_2 = height from datum to heavy liquid overflow, m

z_3 = height from datum to the interface, m

The height of the liquid interface should be measured accurately when the liquid densities are close, when one component is present only in small quantities, or when the throughput is very small. A typical scheme for the automatic control of the interface, using a level instrument that can detect the position of the interface, is shown

in Fig. 16.15. Where one phase is present only in small amounts, it is often recycled to the decanter feed to give more stable operation.

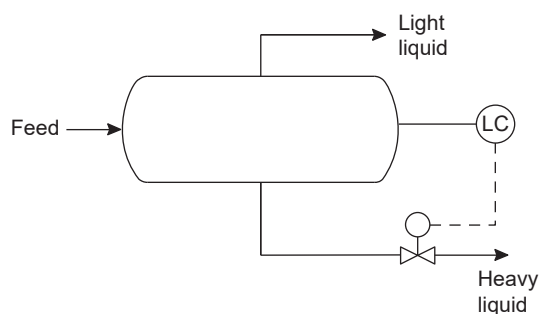


FIG. 16.15 Automatic control, level controller detecting interface.

Decanter design

A rough estimate of the decanter volume required can be made by taking a hold-up time of 5 to 10 minutes, which is usually sufficient where emulsions are not likely to form. Methods for the design of decanters are given by Hooper (1997) and Signales (1975). The general approach taken is outlined here and illustrated by Example 16.4.

The decanter vessel is sized on the basis that the velocity of the continuous phase must be less than the settling velocity of the droplets of the dispersed phase. Plug flow is assumed, and the velocity of the continuous phase is calculated using the area of the interface:

$$u_c = \frac{L_c}{A_i} < u_d \quad (16.9)$$

where u_d = settling velocity of the dispersed phase droplets, m/s,

u_c = velocity of the continuous phase, m/s

L_c = continuous-phase volumetric flow rate, m³/s

A_i = area of the interface, m²

Stokes law is used to determine the settling velocity of the droplets:

$$u_d = \frac{d_d^2 g (\rho_d - \rho_c)}{18 \mu_c} \quad (16.10)$$

where d_d = droplet diameter, m

u_d = settling (terminal) velocity of the dispersed-phase droplets with diameter d , m/s

ρ_c = density of the continuous phase, kg/m³

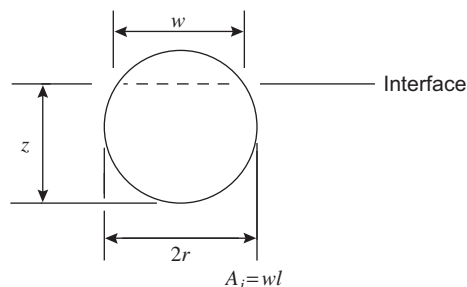
ρ_d = density of the dispersed phase, kg/m³

μ_c = viscosity of the continuous phase, N s/m²

g = gravitational acceleration, 9.81 m/s²

Equation 16.10 is used to calculate the settling velocity with an assumed droplet size of 150 μm, which is well below the droplet sizes normally found in decanter feeds. If the calculated settling velocity is greater than 4×10^{-3} m/s, then a figure of 4×10^{-3} m/s is used.

For a horizontal, cylindrical, decanter vessel, the interfacial area will depend on the position of the interface.



and

$$w = 2(2rs - z^2)^{1/2}$$

where w = width of the interface, m

z = height of the interface from the base of the vessel, m

l = length of the cylinder, m

r = radius of the cylinder, m

For a vertical, cylindrical decanter:

$$A_i = \pi r^2$$

The position of the interface should be such that the band of droplets that collect at the interface waiting to coalesce and cross the interface does not extend to the bottom (or top) of the vessel. [Ryon et al. \(1959\)](#) and Mizrahi and Barnea (1973) have shown that the depth of the dispersion band is a function of the liquid flow rate and the interfacial area. A value of 10% of the decanter height is usually taken for design purposes. If the performance of the decanter is likely to be critical to process performance, the design can be investigated using scale models. The model should be scaled to operate at the same Reynolds number as the proposed design so that the effect of turbulence can be investigated; see [Hooper \(1975\)](#).

Example 16.4

Design a decanter to separate a light oil from water. The oil is the dispersed phase.

Oil, flow rate 1000 kg/h, density 900 kg/m³, viscosity 3 mN s/m²

Water, flow rate 5000 kg/h, density 1000 kg/m³, viscosity 1 mN s/m²

Solution

Take $d_d = 150 \mu\text{m}$:

$$\begin{aligned} u_d &= \frac{(150 \times 10^{-6})^2 9.81(900 - 1000)}{18 \times 1 \times 10^{-3}} \\ &= -0.0012 \text{ m/s, } -1.2 \text{ mm/s (rising)} \end{aligned} \quad (16.10)$$

Because the flow rate is small, use a vertical, cylindrical vessel.

$$L_c = \frac{5000}{1000} \times \frac{1}{3600} = 1.39 \times 10^{-3} \text{ m}^3/\text{s}$$

$$u_c \leq u_d, \text{ and } u_c = \frac{L_c}{A_i}$$

hence

$$A_i = \frac{1.39 \times 10^{-3}}{0.0012} = 1.16 \text{ m}^2$$

$$r = \sqrt{\frac{1.16}{\pi}} = 0.61 \text{ m}$$

$$\text{diameter} = \underline{1.2 \text{ m}}$$

Take the height as twice the diameter, a reasonable value for a decanter:

$$\text{height} = \underline{2.4 \text{ m}}$$

Take the dispersion band as 10% of the height = 0.24 m.

Check the residence time of the droplets in the dispersion band:

$$\frac{0.24}{u_d} = \frac{0.24}{0.0012} = 200 \text{ s } (\sim 3 \text{ min})$$

This is satisfactory, a time of 2 to 5 minutes is normally recommended for control purposes. Check the size of the water (continuous, heavy phase) droplets that could be entrained with the oil (light phase).

$$\begin{aligned}\text{Velocity of oil phase} &= \frac{1000}{900} \times \frac{1}{3600} \times \frac{1}{1.16} \\ &= 2.7 \times 10^{-4} \text{ m/s (0.27 mm/s)}\end{aligned}$$

From Equation 16.10:

$$d_d = \left[\frac{u_d 18 \mu_c}{g(\rho_d - 2\rho_c)} \right]^{1/2}$$

so the entrained droplet size will be:

$$\begin{aligned}&= \left[\frac{2.7 \times 10^{-4} \times 18 \times 3 \times 10^{-3}}{9.81(1000 - 900)} \right]^{1/2} \\ &= 1.2 \times 10^{-4} \text{ m} = 120 \text{ } \mu\text{m}\end{aligned}$$

which is satisfactory; below 150 μm .

Piping arrangement

To minimize entrainment by the jet of liquid entering the vessel, the inlet velocity for a decanter should be kept below 1 m/s.

$$\text{Flow-rate} = \left[\frac{1000}{900} + \frac{5000}{1000} \right] \frac{1}{3600} = 1.7 \times 10^{-3} \text{ m}^3/\text{s}$$

$$\text{Area of pipe} = \frac{1.7 \times 10^{-3}}{1} = 1.7 \times 10^{-3} \text{ m}^2$$

$$\text{Pipe diameter} = \sqrt{\frac{1.7 \times 10^{-3} \times 4}{\pi}} = 0.047 \text{ m, say } \underline{\underline{50 \text{ mm}}}$$

Take the position of the interface as halfway up the vessel and the light liquid off-take as at 90% of the vessel height; then:

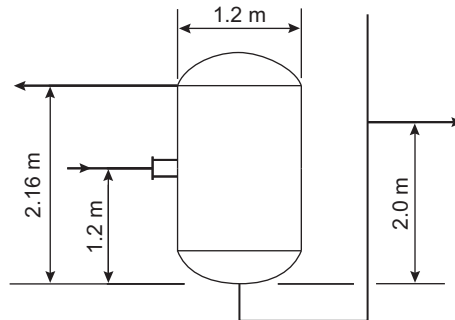
$$z_1 = 0.9 \times 2.4 = 2.16 \text{ m}$$

$$z_3 = 0.5 \times 2.4 = 1.2 \text{ m}$$

$$z_2 = \frac{(2.16 - 1.2)}{1000} \times 900 + 1.2 = \underline{\underline{2.06 \text{ m}}} \quad (16.8)$$

$$\text{say } \underline{\underline{2.0 \text{ m}}}$$

The proposed design:



Drain valves should be fitted at the interface so that any tendency for an emulsion to form can be checked and the emulsion accumulating at the interface drained off periodically as necessary.

16.4.2 Plate separators

Stacks of horizontal, parallel plates are used in some proprietary decanter designs to increase the interfacial area per unit volume and to reduce turbulence. They effectively convert the decanter volume into several smaller separators connected in parallel.

16.4.3 Coalescers

Proprietary equipment, in which the dispersion is forced through some form of coalescing medium, is often used for the coalescence and separation of finely dispersed droplets. A medium is chosen that is preferentially wetted by the dispersed phase; knitted wire or plastic mesh, beds of fibrous material, or special membranes are used. The coalescing medium works by holding up the dispersed droplets long enough for them to form globlets of sufficient size to settle. A typical unit is shown in Fig. 16.16; see Redmon (1963). Coalescing filters are suitable for separating small quantities of dispersed liquids from large throughputs.

Electrical coalescers, in which a high-voltage field is used to break down the stabilizing film surrounding the suspended droplets, are used for desalting crude oils and for similar applications; see Waterman (1965).

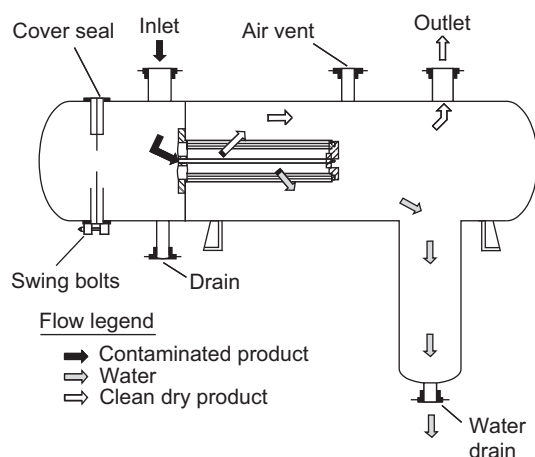


FIG. 16.16 Typical coalescer design.

16.4.4 Centrifugal separators

Liquid–liquid mixtures can be separated in the same centrifugal devices that are used for liquid–solid separations; see Section 18.6.

Sedimentation centrifuges

For difficult separations, where simple gravity settling is not satisfactory, sedimentation centrifuges should be considered. Centrifuging will give a cleaner separation than that obtainable by gravity settling. Centrifuges can be used where the difference in specific gravity between the liquids is very small, as low as 100 kg/m^3 , and they can handle high throughputs, up to around $100 \text{ m}^3/\text{h}$. Also, centrifuging will usually break any emulsion that may form. Bowl or disc centrifuges are normally used (see Section 18.6.3).

Hydrocyclones

Hydrocyclones are used for some liquid–liquid separations, but are not as effective in this application as in separating solids from liquids, as the high shear in a hydrocyclone can cause re-entrainment of droplets. The design of hydrocyclones is discussed in Section 18.6.4.

16.5 Separation of dissolved components

Liquid-phase solutions are present in almost every chemical process. The most commonly used techniques for the separation and purification of miscible liquids are distillation and solvent extraction. In recent years, adsorption, ion exchange, and chromatography have become practical alternatives to distillation or solvent extraction in many special applications.

Gases that are dissolved in a liquid can be removed by stripping; see Section 16.2.4.

Distillation is probably the most widely used separation technique in the chemical process industries and is covered in [Chapter 17](#) of this book. When a liquid mixture can be separated by distillation, this will usually be the most economical approach; however, distillation cannot be used for dissolved solids, salts, thermally sensitive compounds, or macromolecules that have low volatility. A large number of valuable product molecules, including many specialty chemicals and most active pharmaceutical ingredients and biochemicals, fall into these categories and require alternative separation processes.

Solvent extraction and the associated technique, leaching (solid–liquid extraction), are described later and in Section 17.16.

Components that will form a solid phase can be recovered from liquid solution by crystallization or precipitation. Crystallization and precipitation processes are usually followed by filtration, centrifugation, drying, and other solids-handling processes. The solids formation process usually controls particle size, shape, and strength, so the operations used to form and grow particles often dictate the requirements of the downstream equipment; see [Chapter 18](#).

Ion exchange can be used to recover charged particles. Ion exchange is used to substitute one ion for another in electrolyte solutions; for example, replacing a metal cation with H^+ to form an acid that can then be recovered by distillation.

Membrane separations can be used for large molecules, for reverse osmosis, and for concentration of slurries. Membranes can also be used when a solvent- or solute-selective membrane material can be developed.

Adsorption can be used to selectively remove a dissolved liquid or solid component from solution. When the adsorbent is regenerated by elution with a solvent, the process is known as preparative chromatography (for small-production volumes) or *production chromatography* (for large volumes), which are discussed in Section 16.5.8. Adsorption with regeneration by cycling temperature or pressure is less commonly used for liquids.

16.5.1 Evaporators

On an industrial scale, evaporation and crystallization are the main processes used for the recovery of dissolved solids from solutions. Evaporation is the removal of a solvent by vaporization from solids that are not volatile. It is normally used to produce a concentrated liquid, often before crystallization, but a dry solid product can be obtained with some specialized designs. The general subject of evaporation is covered in [Chhabra and Gurappa \(2019\)](#). The selection of the appropriate type of evaporator is discussed by [Cole \(1984\)](#) and [Gabelman \(2020\)](#). Evaporation is the subject of a book by [Billet \(1989\)](#).

Many evaporator designs have been developed for specialized applications in particular industries. The designs can be grouped into the following basic types.

Direct-heated evaporators

This type includes solar pans and submerged combustion units. Submerged combustion evaporators can be used for applications where contamination of the solution by the products of combustion is acceptable.

Long-tube evaporators (Fig. 16.17)

In this type, the liquid flows as a thin film on the walls of a long, vertical, heated tube. Both falling film and rising film types are used. They are high-capacity units suitable for low-viscosity solutions.

Forced-circulation evaporators (Fig. 16.18)

In forced-circulation evaporators the liquid is pumped through the tubes. They are suitable for use with materials that tend to foul the heat transfer surfaces and where crystallization can occur in the evaporator.

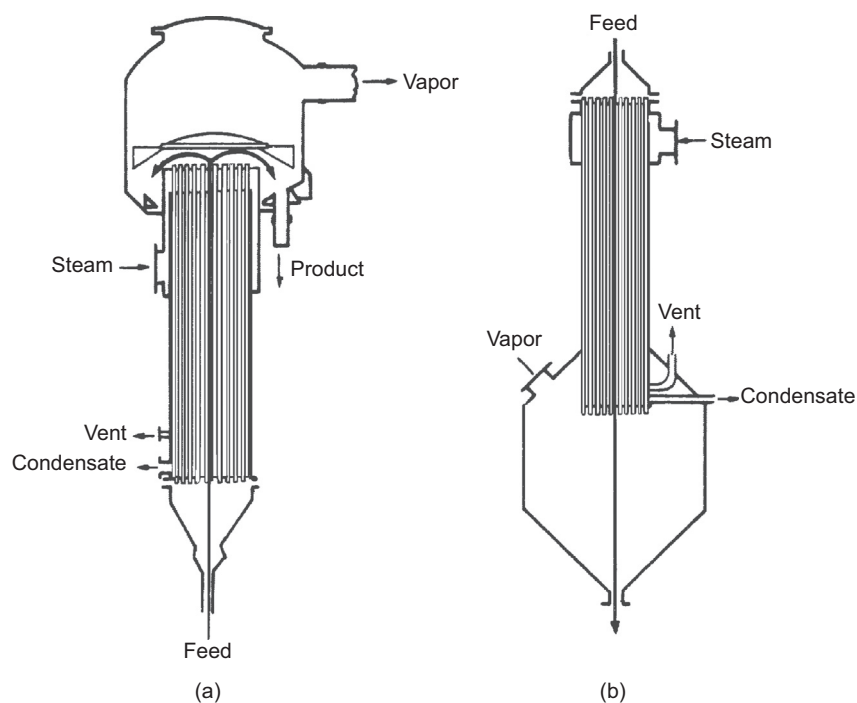


FIG. 16.17 Long-tube evaporators. (a) Rising film. (b) Falling film.

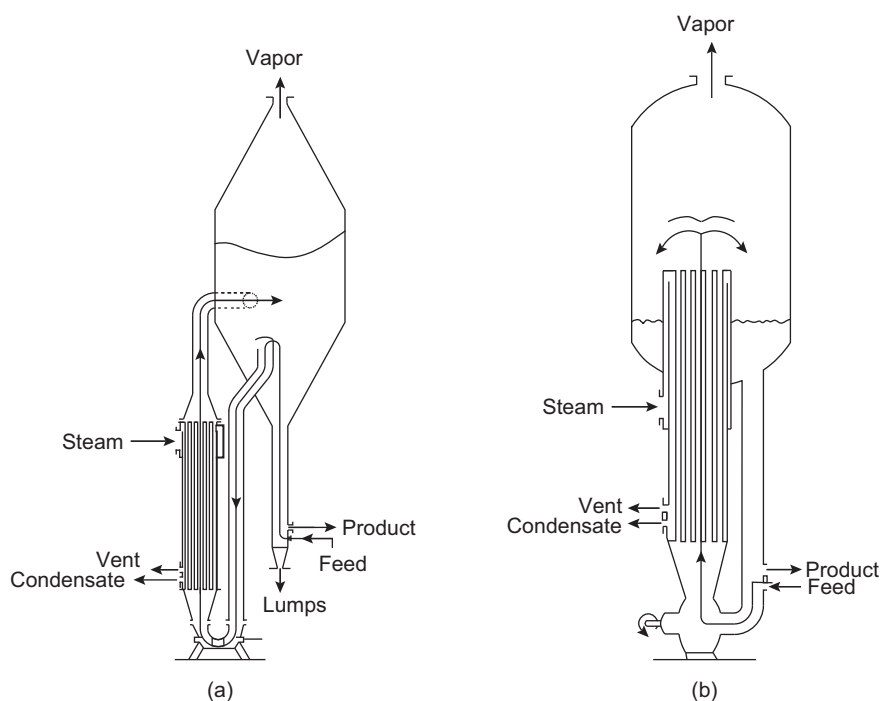


FIG. 16.18 Forced-circulation evaporators. (a) Submerged tube. (b) Boiling tube.

Wiped-film evaporators (Fig. 16.19)

In this design, a thin layer of solution is spread on the heating surface by mechanical means. Wiped-film evaporators are used for very viscous materials and for producing solid products. Wiped-film evaporators are sometimes referred to as agitated thin-film evaporators. The design and applications of this type of evaporator are discussed by [Mutzenburg \(1965\)](#), [Parker \(1965\)](#), and [Fischer \(1965\)](#).

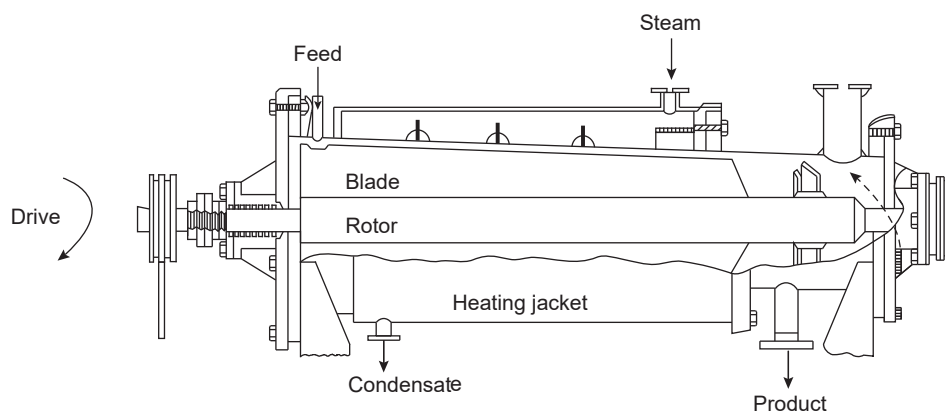


FIG. 16.19 Horizontal wiped-film evaporator.

Short-tube evaporators

Short-tube evaporators, also called calandria evaporators, are used in the sugar industry; see [Chhabra and Gurappa \(2019\)](#).

Evaporator selection

The selection of the most suitable evaporator type for a particular application will depend on the following factors:

1. The throughput required
2. The viscosity of the feed and the increase in viscosity during evaporation
3. The nature of the product required: solid, slurry, or concentrated solution
4. The heat sensitivity of the product
5. Whether the materials are fouling or nonfouling
6. Whether the solution is likely to foam
7. Whether direct heating can be used

A selection guide based on these factors is given in [Fig. 16.20](#); see also [Parker \(1963\)](#).

	Feed conditions							Suitable for heat-sensitive materials
	Viscosity, mN s/m ²							
Evaporator type	Very viscous > 1000	Medium viscosity < 1000 max	Low viscosity < 100	Foaming	Scaling or fouling	Crystals produced	Solids in suspension	
Recirculating Calandria (short vertical tube)		←						No
Forced circulation		←						Yes
Falling film			↔					No
Natural circulation			↔					No
Single pass wiped film	←							Yes
Tubular (long tube) Falling film			↔					Yes
Rising film			↔					Yes

FIG. 16.20 Evaporator selection guide.

Evaporator design

It can be seen from Figs. 16.17 and 16.18 that most evaporators consist of a shell and tube heat exchanger bundle combined with a vapor–liquid disengaging space. The major cost in these designs is the heat exchange tubes; hence the cost correlations for evaporators are based on surface area; see Table 7.2. The exception is for wiped-film evaporators, which must always be designed and costed in consultation with a vendor.

The heat transfer bundle of an evaporator can be designed using the methods given for boiling heat transfer in Section 19.11. The boiling fluid is always on the tube side, and the hot utility (usually steam) is placed on the shell side.

If the fraction of the feed that is evaporated is greater than about 30%, it may be advantageous to recirculate some of the product to the feed. This increases the liquid flow and gives more uniform heat transfer. When product recirculation is used in food applications, as in the production of condensed milk, care must be taken to ensure that the residence time in the evaporator is not so large as to damage the product or cause fouling of the tubes. The residence time can be estimated from the liquid holdup in the evaporator divided by the product flow rate. The falling-film type has the least liquid holdup and is preferred in food applications.

Auxiliary equipment

Condensers and vacuum pumps will be needed for evaporators operated under vacuum. For aqueous solutions, steam ejectors and jet condensers are normally used. Jet condensers are direct-contact condensers, where the vapor is condensed by contact with jets of cooling water. Indirect, surface condensers are used where it is necessary to keep the condensed vapor and cooling water effluent separate.

Example 16.5

Estimate the heat exchange area required for an evaporator that concentrates 4000 kg/h of apple juice by removing 40% of the water content. Low-pressure steam at 120 °C is available as heat source. The evaporator is run under a slight vacuum, such that the temperature of the boiling apple juice concentrate is maintained at 100 °C.

Solution

For an initial estimate we can assume the properties of apple juice are the same as those of water.

Latent heat of evaporation ~2200 kJ/kg at 100 °C

Heat exchange duty required = $4000 \times 0.4 \times 2200/3600 = 978$ kW

Because the fraction of feed evaporated is >30%, we can assume that the evaporator has product recirculation, in which case the tube-side temperature will be approximately constant at 100 °C.

Mean temperature difference = $120 - 100 = 20$ °C

An initial estimate of the heat transfer coefficient can be made using the nomograph in Fig. 19.1, which suggests that an overall heat transfer coefficient of about 1500 W/m²K will be reasonable.

Estimate area:

$$Q = UA\Delta T_m \quad (19.1)$$

where Q = heat transferred per unit time, W

U = the overall heat transfer coefficient, W/m²°C

A = heat transfer area, m²

ΔT_m = the mean temperature difference, the temperature driving force, °C

Hence:

$$\text{Area, } A = \frac{978 \times 10^3}{1500 \times 20} = \underline{\underline{32.6 \text{ m}^2}}$$

This estimate can be used to make an initial layout of the heat transfer bundle. The tube-side heat transfer coefficient can then be estimated using the methods given in Section 19.11, and the shell-side coefficient can be estimated using the methods given in Section 19.10 to arrive at a more accurate determination of the required evaporator area. For preliminary design purposes, this estimate would be adequate to generate an initial (Class 4) estimate of the evaporator cost using the correlation given in Table 7.2.

16.5.2 Crystallization

Crystallization is used for the production, purification, and recovery of solids. Crystalline products have an attractive appearance, are free flowing, and are easily handled and packaged. The process is used in a wide range of industries, from the small-scale production of specialized chemicals, such as pharmaceutical products, to the high-tonnage production of commodity products such as sugar, common salt, and fertilizers.

Crystallization theory is covered in Mullin (2019) and Jones (2002). Descriptions of the various crystallizers used commercially can be found in these texts and in handbooks: Mersmann (2001), Myerson et al. (2019), Green and Southard (2018), and Schweitzer (1997). Procedures for the scale-up and design of crystallizers are given by Mersmann (2001) and Mersham (1984, 1988).

Crystallization equipment can be classified by the method used to obtain supersaturation of the liquor and also by the method used to suspend the growing crystals. Supersaturation is obtained by cooling or evaporation. There are four basic types of crystallizer; these are described briefly next.

Tank crystallizers

Tank crystallizers are the simplest type of industrial crystallizing equipment. Crystallization is induced by cooling the mother liquor in tanks, which may be agitated and equipped with cooling coils or jackets. Tank crystallizers are operated batch-wise and are generally used for small-scale production.

Scraped-surface crystallizers

Scraped-surface crystallizers are similar in principle to the tank type, but the cooling surfaces are continually scraped or agitated to prevent fouling by deposited crystals and to promote heat transfer. They are suitable for processing high-viscosity liquors. Scraped-surface crystallizers can be operated batch-wise, with recirculation of the mother liquor, or continuously. A disadvantage of this type is that they tend to produce very small crystals.

Circulating magma crystallizers (Fig. 16.21)

In this type, both the liquor and growing crystals are circulated through the zone in which supersaturation occurs. Circulating magma crystallizers are probably the most important type of large-scale crystallizers used in the chemical process industry. Designs are available in which supersaturation is achieved by direct cooling, evaporation, or evaporative cooling under vacuum.

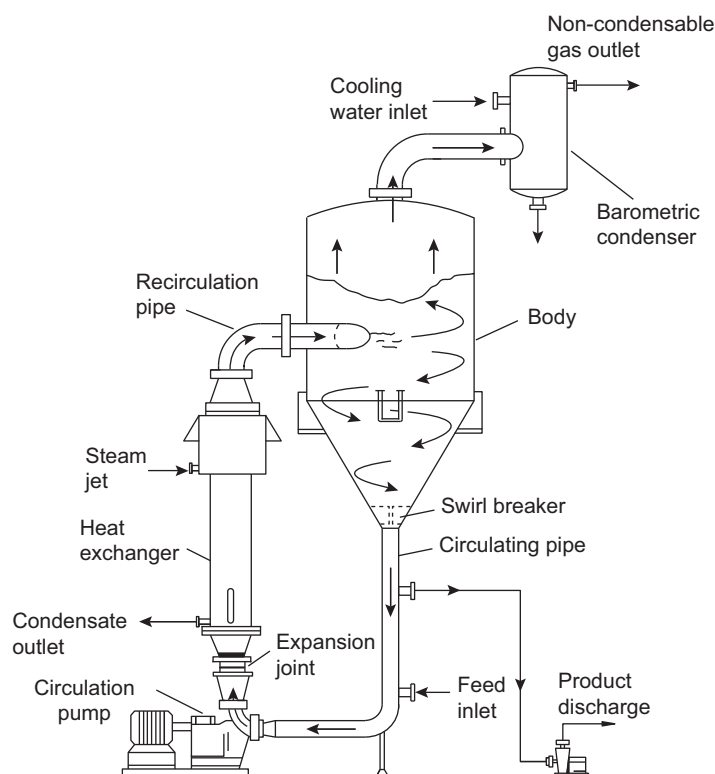


FIG. 16.21 Circulating magma crystallizer (evaporative type).

Circulating liquor crystallizers (Fig. 16.22)

In a circulating liquor crystallizer only the liquor is circulated through the heating or cooling equipment; the crystals are retained in suspension in the crystallizing zone by the up-flow of liquor. Circulating liquor crystallizers produce crystals of regular size. The basic design consists of three components: a vessel in which the crystals are suspended and grow and are removed; a means of producing supersaturation, by cooling or evaporation; and a means of circulating the liquor. The Oslo crystallizer (see Fig. 10.50) is the archetypal design for this type of crystallizing equipment.

Circulating liquor crystallizers and circulating magma crystallizers are used for the large-scale production of a wide range of crystal products. Typical applications of the main types of crystallizers are summarized in Table 16.2; see also Larson (1978).

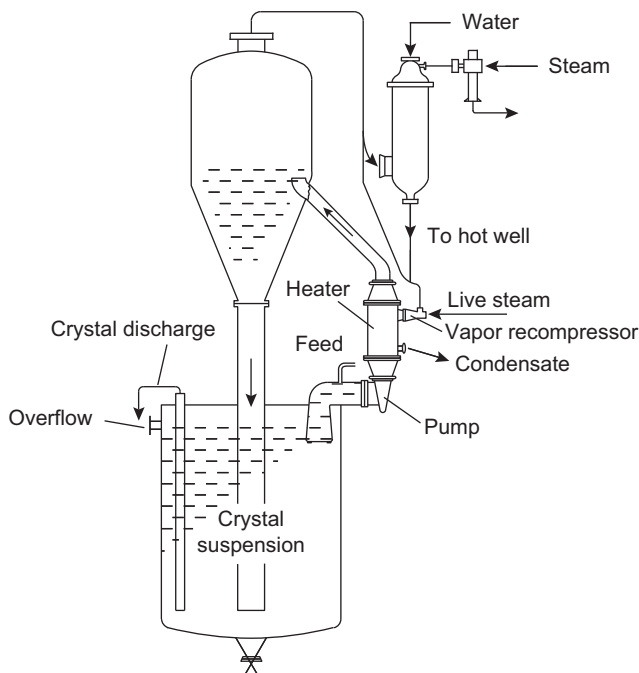


FIG. 16.22 Oslo evaporative crystallizer.

Crystallizer design

Crystallizers are normally sized in consultation with a specialist equipment vendor. The important design parameters are:

1. Process throughput
2. Feed concentration
3. Target solids yield (recovery)
4. Target particle size distribution
5. Product purity (particularly in fractional crystallization)
6. Heat addition or removal requirements, including latent heat as well as sensible heat

The yield of a crystallizer is limited by both solid–liquid equilibrium and the desire to maintain a low enough solids fraction to allow slurry flow of the product. If the solvent is removed by evaporation, the mother liquor can be recycled to the crystallizer after the crystals have been removed by filtration, as long as impurities do not accumulate in the recycle.

TABLE 16.2 Selection of crystallizers

Crystallizer type	Applications	Typical uses
Tank	Batch operation, small-scale production.	Fatty acids, vegetable oils, sugars
Scraped surface	Organic compounds, where fouling is a problem, viscous materials.	Chlorobenzenes, organic acids, paraffin waxes, naphthalene, urea
Circulating magma	Production of large-sized crystals. High throughputs.	Ammonium and other inorganic salts, sodium and potassium chlorides
Circulating liquor	Production of uniform crystals (smaller size than circulating silver magma). High throughputs.	Gypsum, inorganic salts, sodium and potassium nitrates, nitrates

16.5.3 Precipitation

Precipitation can be considered as a branch of crystallization, although the solid phase that is formed need not be crystalline.

The solubility of organic solutes can be influenced by temperature, composition, pH, solvent polarity, and ionic strength. If something is added to the solvent to change one or more of these properties, the solute can sometimes be precipitated out of solution. If the solvent volume is not significantly changed, the recovery of solute is equal to the change in solute solubility divided by the initial solubility:

$$\text{Solute recovery} = \frac{\text{solute precipitated}}{\text{total solute fed}} \approx \frac{\text{initial solubility} - \text{final solubility}}{\text{initial solubility}} \quad (16.11)$$

Precipitation is widely used in the recovery of large organic molecules, such as specialty chemicals, pharmaceutical and food compounds, proteins, and other biological products. Some of the common techniques that are used include:

1. Salting out, in which a salt such as calcium citrate, calcium chloride or ammonium sulfate is added to an aqueous solution to raise the ionic strength and cause precipitation.
2. Changing solution polarity by adding methanol, ethanol, acetone, acetonitrile or other suitable solvent.
3. Changing the pH by adding acid (sometimes known as acidulation) or base.
4. Heat treatment ("cooking") to thermally degrade an unwanted solute that is then precipitated.
5. Adsorption precipitation by adding diatomaceous earth, casein, gelatin, activated carbon, clay, or other large particles that can adsorb the organic species and then settle as precipitate.

Precipitation operations usually do not require evaporation of solvent or cooling of a saturated solution and so can be carried out in simpler equipment than crystallization. The process usually consists of a mixing tank or in-line mixer, followed by a solid–liquid separation device such as a hydrocyclone or centrifuge, as described in Section 18.6.

Precipitation is discussed in detail by [Sohnel and Garside \(1992\)](#).

16.5.4 Membrane separations

Membranes are widely used for concentration of solutions of dissolved solids, as well as suspensions of particulates. A tubular membrane that is permeable to the solvent but not the solute can be used to remove solvent from a solution in the same manner as cross-flow filtration (see Section 18.6.2). These processes are classified as microfiltration, ultrafiltration, or nanofiltration, depending on the size of particulate or molecule that is retained; see [Table 16.3](#). The use of a membrane for solvent removal is usually preferred over solvent evaporation if the solute is sensitive to high temperatures. This is often the case for biologically active large molecules such as proteins and enzymes and for flavor compounds found in foods and beverages.

Membranes can also be found that are selective for the solute over the solvent. Such membranes allow the solute to be transferred to a different solvent without intimate mixing of the solvents, which can be beneficial for solvent or product degradation or when the two solvents are mutually miscible. In the pharmaceuticals industry, the use of solute-selective membranes is known as *diafiltration*, because of the close analogy to dialysis in organisms. A recent variant on diafiltration is to use a charged membrane to increase the selectivity for solute; see [Mehta et al. \(2008\)](#) for a good example.

TABLE 16.3 Membrane filtration process

Process	Approximate size range (m)	Applications
Microfiltration	10^{-8} to 10^{-4}	pollen, bacteria, blood cells
Ultrafiltration	10^{-9} to 10^{-8}	proteins and virus
Nanofiltration	5×10^{-9} to 15×10^{-9}	water softening
Reverse osmosis	10^{-10} to 10^{-9}	desalination
Dialysis	10^{-9} to molecules	blood purification
Electrodialysis	10^{-9} to molecules	separation of electrolytes
Pervaporation	10^{-9} to molecules	dehydration of ethanol
Gas permeation	10^{-9} to molecules	hydrogen recovery, dehydration

When membranes are used for filtration or solute concentration, the limit on solvent recovery is set by the need to avoid fouling or scaling of the membrane and maintain a pumpable slurry. Mass transfer at the membrane wall causes the fluid closest to the membrane to be depleted in solvent; a phenomenon known as concentration polarization. Because the fluid at the membrane wall is depleted in solvent and enriched in solute, precipitation or crystallization can occur if the solute solubility is exceeded. Either of these processes can lead to membrane fouling and loss of throughput. Solvent recovery membranes are therefore usually operated at an outlet concentration well below saturation. Osmotic effects can also decrease the flux through a membrane.

The design of membranes for solute transfer is similar to the design of gas separation membranes, as described in Section 16.2.2, with the exception that a second solvent is usually introduced on the permeate side to remove the solute. Hollow-fiber or tubular membranes are most commonly used.

The design of membrane separations for liquids is discussed in [Scott and Hughes \(1996\)](#), [Cheryan \(1998\)](#), [McGregor \(1986\)](#), [Rautenbach and Albrecht \(1989\)](#), [Noble and Stern \(1995\)](#), [Mulder \(1996\)](#), [Porter \(1997\)](#), [Hoffman \(2003\)](#), [Sridhar and Moulik \(2018\)](#), and [Baker \(2012\)](#). Applications of membranes to biological systems are described in the book by [Wang \(2001\)](#) and the review paper by [van Reis and Zydney \(2007\)](#). The special case of recovery of purified water from salt solutions by reverse osmosis (RO) is discussed next.

Reverse osmosis

Reverse osmosis (RO) is by far the most widely used membrane process. In an RO plant, water passes through a membrane, while dissolved minerals and other solids are rejected in the retentate. Reverse osmosis is used to generate deionized process feed water, purify boiler feed water, recover water from waste streams, and desalinate sea water or brackish water for drinking and irrigation.

In an RO process, the feed water is pressurized to provide an adequate pressure gradient to overcome the difference in osmotic potential between the briny retentate and the purer permeate. Under the applied pressure gradient, water flows through the membrane against the concentration gradient. The membranes are usually designed as spiral-wound modules and are operated in cross-flow; see Section 16.2.2.

Reverse osmosis plants are usually purchased as modular plants designed by one of the major water treatment companies. Cost correlations have been developed and can be found in Aspen Capital Cost Estimator (ACCE), Cleopatra Enterprise, and other cost estimating programs.

The recovery of water from an RO plant depends on the feed water quality, the product specifications, and the need to prevent membrane fouling. Highly pure water is usually not obtained in a single stage. The membrane typically rejects 96% to 98% of the salts per stage, and several stages may be used to achieve the desired purity, with recycle of retentate as described in Section 16.2.2.

As with any solvent-rejection membrane, the designer must ensure that the retentate will not reach concentrations of solute that will cause the membrane to be fouled. Allowance must also be made for concentration polarization near the membrane. At high recovery of water, the osmotic pressure of the solution on the retentate side increases dramatically. The limit on recovery is often set by an economic trade-off between the cost of consuming additional feed and the cost of pumping to a higher pressure. [Table 16.4](#) gives the osmotic pressure of NaCl and sea salt solutions at different concentrations at 40 °C. Some guidelines on membrane fluxes and retentate flows are given by [Kucera \(2008\)](#). A typical product recovery when generating purified water from town water is 50% to 75%, but the recovery depends strongly on the factors described earlier and is often less than 30% in desalination plants.

TABLE 16.4 Osmotic pressure of sea salt and sodium chloride solutions at 40 °C

Molality of NaCl	Osmotic pressure
(mol/kg)	(atm)
0.01	0.49
0.10	4.76
0.50	23.60
1.0	48.08
1.50	73.93
2.00	101.3
3.00	161.6
4.00	230.5
5.00	309.4
Weight % sea salts	Osmotic pressure
(wt%)	(atm)
1.00	7.41
2.00	14.88
3.45*	26.17
5.00	38.96
7.50	61.40
10.00	86.46
15.00	146.6
20.00	225.1
25.00	331

* 3.45 wt% solids is taken as the value for standard sea water.
 From *Stoughton and Lietzke, 1965*.

When specifying the use of an RO system for process feed water, the design engineer must allow for the cost of the extra feed water that is needed. This is illustrated in Example 16.6.

The performance of RO plants can often be substantially improved by pretreatment of the water feed. Common pretreatment processes include filtration, softening of the water by cation exchange, activated carbon adsorption of chlorine and organics, and addition of chemicals to prevent biological fouling and suppress precipitation.

Because of its widespread use, there is an abundant literature on RO. It is covered in all of the general books on membranes listed earlier and in the specialist books by [Amjad \(1993\)](#), [Byrne \(2002\)](#), [Wilf et al. \(2007\)](#), [Kucera \(2010\)](#), and [AWWA \(2007\)](#). The use of RO for water recovery from waste water is discussed by [Aerts and Tong \(2009\)](#). Good introductory descriptions of the technology are given in the articles by [Stover \(2014\)](#) and [Kucera \(2019\)](#).

Example 16.6

A reverse osmosis plant is to be designed to produce 50 kg/s of boiler feed water with less than 20 ppmw of NaCl starting from sea water that contains 3.5 wt% NaCl. The 40 m² membrane modules operate at 60 atm and achieve a flux of 0.4 m³/m².day when the permeate pressure is 2 atm. If each membrane module has a rejection of 96% of the salt fed to it, determine the overall membrane sequence and the water feed rate required.

Solution

Target concentration = 20 ppmw = 0.002 wt%.

If the permeate pressure is 2 atm and the retentate is at 60 atm, then the pressure drop across the membrane is 60 – 2 = 58 atm.

From [Table 16.4](#), an osmotic pressure of 58 atm corresponds to a concentration of 7.44 wt%. We need to make an allowance for concentration polarization at the membrane, so assume that the retentate bulk concentration is 70% of this concentration (i.e., 5.21 wt%).

Mass balance on salt across first stage, basis 100 kg/s of feed, 96% rejection of salts:

$$\begin{aligned}\text{Feed} &= \text{Permeate} + \text{Retentate} \\ 3.5 &= 3.5 \times 0.04 + 3.5 \times 0.96 \\ &= 0.14 + 3.36\end{aligned}$$

So if the retentate water flow rate is x , then:

$$\frac{3.36}{x + 3.36} = 0.0521, \quad \text{hence } x = \frac{3.36}{0.0521} - 3.36 = 61.1 \text{ kg/s}$$

So, by difference, the water flow in the permeate is $96.5 - 61.1 = 35.4$ kg/s, and the salt concentration in the permeate is $0.14 / (0.14 + 61.1) = 0.39$ wt%, which does not meet the target.

If we assume a simple cascade with no retentate recycle, then the second stage can be calculated in the same way:

$$\text{Salt retained in permeate} = 0.14 \times 0.04 = 0.0056 \text{ kg/s}$$

$$\text{Salt rejected in retentate} = 0.14 \times 0.96 = 0.1344 \text{ kg/s}$$

$$\text{Retentate water flow} = \frac{0.1344}{0.0521} - 0.1344 = 2.44 \text{ kg/s}$$

$$\text{Permeate water flow} = 35.4 - 2.44 = 32.94 \text{ kg/s}$$

Permeate salt concentration = $0.0056 / (0.0056 + 32.94) = 0.017$ wt%, which still does not meet the target.

This calculation can be repeated for subsequent stages and is easily coded into a spreadsheet. The third-stage results are:

$$\text{Salt retained in permeate} = 0.0056 \times 0.04 = 0.000224 \text{ kg/s}$$

$$\text{Salt rejected in retentate} = 0.0056 \times 0.96 = 0.005376 \text{ kg/s}$$

$$\text{Retentate water flow} = \frac{0.005376}{0.0521} - 0.005376 = 0.098 \text{ kg/s}$$

$$\text{Permeate water flow} = 32.94 - 0.098 = 32.85 \text{ kg/s}$$

Permeate salt concentration = $0.000224 / (0.000224 + 32.85) = 0.00068$ wt%, which now exceeds the target specification, so a part of the second-stage product can bypass the third stage; however, for design purposes we will assume that all the stages are fully used, as this provides some additional safety factor for fouling, concentration polarization, etc.

The overall recovery of water is 32.85 kg/s out of 96.5 kg/s fed (i.e., 34%), so the feed rate to produce 50 kg/s of RO water is $50 / 0.34 = 147$ kg of water. Allowing for the salt in the feed, we also have $147 \times 0.035 / (1 - 0.035) = 5.33$ kg/s of salt, so the total feed is 152 kg/s of sea water.

Note that the retentate flow in the third stage has become very small. This is not a practical value, and a higher flow rate would be necessary. A higher flow rate can be accomplished by reducing the outlet retentate concentration and recycling the retentate to an earlier point in the network (see Fig. 16.10c). The network with recycles is more difficult to solve using hand calculations, but can easily be modeled using process simulation software. If the process simulation program does not have a membrane unit operation, then the RO membrane can be modeled as a fixed-split separator; see Section 4.5.3.

By examination of the permeate flows calculated earlier, we see that the permeate flow is essentially unchanged and roughly 32.9 kg/s from stage 2 onwards (≈ 50 kg/s on the design flow rate basis). The permeate from stage 1 is 35.4 kg/s, which when corrected to the 50 kg/s production basis would be $35.4 \times 152 / 100 = 53.8$ kg/s. So if we allow roughly 10% additional flow (relative to product rate) in each stage for recycles, then we can assume 59.2 kg/s flow in each stage:

$$59.2 \text{ kg/s} = \frac{59.2 \times 3600 \times 24}{1000} = 5114 \text{ m}^3/\text{day}$$

So if each 40 m² membrane module permeates 0.4 m³/m².day, then the total number of modules per stage:

$$= \frac{5114}{40 \times 0.4} = 320$$

Hence the total number of modules = $320 \times 3 = 960$.

With some allowance for spare modules, the sequence is then three stages, each containing 330 modules, with roughly 10% recycle from stages 2 and 3. These results would need to be confirmed by detailed simulation and discussion with a vendor.

16.5.5 Ion exchange

Ion exchange is used for water softening, demineralization and separation and recovery of salts, including salts of organic acids and bases. In an ion exchange process, the solution flows through a bed of resin beads. The resin is a polymer that has been functionalized by the addition of either acidic or basic groups. For example, sulfonated polystyrene contains $-\text{SO}_3^-$ groups that attach to cations from the solution, and hence can be used as a cation exchange resin. The particular choice of acidic or basic groups allows the designer to modify the strength of interaction and hence the selectivity of the resin.

When a solution is passed over a cation exchange resin, the cations in the solution equilibrate with the cations that were attached to the resin, and hence effectively become adsorbed onto the resin. When the resin nears breakthrough, it can be regenerated by washing with a solution of counter-ion: typically H^+ , Na^+ , or Ca^{2+} for cation exchange or Cl^- , HO^- , or NO_3^- for anion exchange.

Full deionization can be accomplished by carrying out cation exchange using H^+ as the counter-ion, followed by anion exchange using HO^- as the counter-ion.

The most common ion exchange process is water softening, in which Ca^{2+} and Mg^{2+} that occur naturally in hard water are exchanged with Na^+ using a cation exchange resin that is regenerated with NaCl . Water softening is used for boiler feed water and to prepare water for RO units. Small units are also used for domestic water softening in regions that have high water hardness.

The capacity of an ion exchange resin depends on the extent of functionalization of the polymer and is normally expressed in mmol/g or mmol/mL of resin. The units millimoles (mmol) are sometimes written as milliequivalents (meq). For cation exchange resins, the loading is per g or mL of dry hydrogen-form resin, and for anion exchange resins it is usually per g or mL of dry chlorine-form resin. Capacities for some of the more commonly used resins are given in [Green and Southard \(2018\)](#). A rough estimate of the total bed volume required can be made by assuming that the bed operates to 70% of breakthrough; a more detailed analysis would usually be made by a specialist designer.

As with adsorption (see Section 16.2.1), a continuous ion-exchange system requires at least two beds of resin, so that one can be in regeneration while the other is in process service.

When specifying an ion-exchange system, the design engineer must allow for the regenerant and effluent treatment systems. The regenerant is a salt of the counter-ion, usually in aqueous solution. The regenerant is required in an amount that is somewhat greater than the stoichiometric equivalent of the exchanged ion to provide an adequate difference in chemical potential to drive the exchanged ion off the resin and ensure that regeneration is completed. A good initial estimate is 150% to 200% of the stoichiometric equivalent. The regenerant is often fed at high concentration to minimize the amount of effluent formed during regeneration. The spent regenerant may require neutralization or other additional treatment before it can be sent to a wastewater plant.

Many pharmaceutical products and intermediates are organic salts that can be recovered by ion exchange. If the ion loaded on the resin is the desired product, then the regenerant should be chosen to give the product in a suitable form for further processing.

An introduction to the theory of ion exchange is given in [Chhabra and Gurappa \(2019\)](#). A more detailed discussion of the technology is given in the book by [Helfferich \(1995\)](#). [Wachinski \(2016\)](#) discusses the application of ion exchange to waste recovery.

Example 16.7

Thiocyanate ions can be removed from sodium carbonate/bicarbonate solution that is used for oxidative scrubbing of H_2S using a moderately basic anion exchange resin (see US 4,999,113). The flow rate of the scrubbing solution is $40 \text{ m}^3/\text{h}$, and the initial concentration of thiocyanate is 10 g/l . A polystyrene-based anion exchange resin suitable for this duty has capacity of 1.8 eq/L . The resin can be regenerated by flushing 1.5 equivalents of 4.0 wt% NaOH , preceded by the same volume of water. Two beds are used, with one in operation and one in regeneration. If the bed is designed to be 80% loaded at the end of a 2-hour operation cycle, estimate the bed volume. Also determine the flow rate of sodium hydroxide required during the regeneration.

Solution

After 2 hours, the amount of thiocyanate exchanged $= 40 \times 10 \times 2 = 800$ kg

Thiocyanate is SCN^- , so has a molecular weight $= 32 + 12 + 14 = 58$ g/mol

Molar loading $= 800/58 = 13.79$ kmol

Bed volume required for complete loading $= 13.79/1.8 = 7.66 \text{ m}^3$

So bed volume for 80% loading at end of 2 hours $= 7.66/0.8 = \underline{9.58 \text{ m}^3}$

If this were designed as a 4:1 cylindrical vessel, then volume $= \pi D_v^3$, so vessel diameter would be $= 1.45$ m and height $= 5.8$ m, allowing the vessels to be designed, sized, and costed using the methods given in [Chapter 14](#).

100 kg of 4.0 wt% NaOH has 4 kg of NaOH; hence $4 \times (17/40) = 1.7$ kg of HO^- anions

1.7 kg of HO^- $= 1700/17 = 100$ mol, so 4 wt% NaOH is 1 mol/kg

9.58 m^3 of resin has capacity $9.58 \times 1.8 = 17.24$ kmol

So, flush requires 1.5 equivalents $= 1.5 \times 17.24 = 25.87$ kmol, which must be completed in 1 hour (to allow for the preceding 1-hour flush with the same volumetric flow rate of water). Hence the flow rate of 4.0 wt% sodium hydroxide solution must be $\underline{25.87 \times 10^3 \text{ kg/hr}}$.

Note that the regeneration flow rate is less than the process flow rate, as a more concentrated solution is used for the regeneration.

16.5.6 Solvent extraction and leaching

Solvent extraction (liquid–liquid extraction)

Solvent extraction, also called liquid–liquid extraction, can be used to separate a substance from a solution by extraction into another solvent. It can be used either to recover a valuable substance from the original solution or to purify the original solvent by removing an unwanted component. Examples of solvent extraction are the extraction of uranium and plutonium salts from solution in nitric acid in nuclear fuel reprocessing using kerosene as solvent and the extraction of benzene from reformed naphtha using sulfolane as solvent.

The process depends on the substance being extracted, the solute, having a greater solubility in the solvent used for the extraction than in the original feed solvent. The two solvents must be essentially immiscible.

The solvents are mixed in a contactor to effect the transfer of solute, and then the phases are separated. The depleted feed solvent leaving the extractor is called the raffinate, and the solute-rich extraction solvent, the *extract*. The solute is normally recovered from the extraction solvent by distillation, and the extraction solvent recycled.

The simplest form of extractor is a mixer-settler, which consists of an agitated tank and a decanter. For multistage extraction processes, liquid–liquid contacting columns are used. The design of extraction columns is discussed in [Chapter 17](#), Section 17.16. See also [Chhabra and Gurappa \(2019\)](#), [Treybal \(1963\)](#), [Couper et al. \(2012\)](#), and [Green and Southard \(2018\)](#).

Leaching

Liquids can be extracted from solids by leaching. As the name implies, the soluble component contained in a solid is leached out by contacting the solid with a suitable solvent. A principal application of leaching is in the extraction of valuable oils from nuts and seeds such as palm oil and rapeseed oil.

The equipment used to contact the solids with the solvent is usually a special design to suit the type of solid being processed and is to an extent unique to the particular industry. General details of leaching equipment are given in [Green and Southard \(2018\)](#).

The leaching is normally done using a number of stages. In this respect, the process is similar to liquid–liquid extraction, and the methods used to determine the number of stages required are similar.

For a detailed discussion of the procedures used to determine the number of stages required for a particular process, see [Prabhudesai \(1997\)](#).

16.5.7 Adsorption

Fixed beds of adsorbent are occasionally used for removing small amounts of dissolved solids or liquids from a liquid stream. Common adsorbents are silica, alumina, activated carbon, zeolites, and clays.

When removing a dissolved solid, the process is usually treated as irreversible adsorption; see Section 16.2.1. Reversible adsorption of dissolved components with regeneration of the sorbent using a different solvent is a form of chromatography and is discussed in the following section.

16.5.8 Chromatography

The term chromatography is broadly applied to separation processes in which a fluid is separated into components by passing it over a bed of adsorbent in a continuous flow of carrier fluid. Gas chromatography (GC) is widely used as an analytical method, but is only rarely used for product recovery because of the high-volume flow rates and pressure drop requirements. Liquid-phase chromatography, on the other hand, is extensively used for product recovery and purification, particularly for fine chemicals and biological products; see [Harrison \(2014\)](#).

Most chromatographic separations are carried out in batch or semi-batch mode, but continuous chromatography can be carried out using flow schemes such as simulated moving bed (SMB) chromatography, described later.

The general principles of chromatography are discussed by [Ruthven \(1984\)](#), [Ganetsos and Barker \(1992\)](#), [Chhabra and Gurappa \(2019\)](#), [Harrison \(2014\)](#), and [Hagel et al. \(2007\)](#).

Chromatographic separations rely on different components in the feed having different adsorption equilibria with the solid phase. The solid-phase material can be an inorganic or organic adsorbent, resin, or gel, and is sometimes referred to as the *stationary phase*. The liquid phase is called the mobile phase and consists of the feed liquid, as well as the carrier liquid, which is also sometimes called the eluent or desorbent. The process performance is strongly influenced by the choice of stationary phase and mobile phase.

Batch chromatography

Batch chromatography processes operate in a similar mode to laboratory-scale chromatography. They are consequently favored by chemists for preparative chromatography and smaller-scale production chromatography, as the scale-up from lab methods is more straightforward than for continuous chromatography.

In batch chromatography, a pulse of feed is injected into a continuously flowing stream of mobile phase that passes over a long column packed with a suitable stationary phase, as shown in [Fig. 16.23\(a\)](#). The more strongly adsorbed species pass through the column more slowly than less strongly adsorbed species. If a long enough column is used, then the different species can be resolved into fractions (or “peaks”) that are enriched in different components of the feed (see [Fig. 16.23b](#)). If the column effluent composition is monitored, the first fractions recovered will contain material that adsorbs less strongly than the desired product (“lights”). This material can be sent for eluent recovery or discarded as waste. The next fractions are rich in product and are sent to product recovery. Finally, material that is more strongly adsorbed than the desired product (“heavies”) is eluted, and this material is also sent to solvent recovery or waste disposal.

If all the material can be desorbed from the column within a reasonable time and carrier flow rate, the column is then clean enough to accept a new injection of feed, and the process is repeated. Occasionally, the cost in time and desorbent to recover the heavier material is greater than the cost of emptying the column and reloading with fresh stationary phase, in which case the cycle is ended after collection of the product and the spent stationary phase is sent to waste disposal.

The eluent can be recovered from the product and waste streams and recycled to the feed; however, in many fine chemical and pharmaceutical processes, the spent solvent is discarded or sold as a by-product to eliminate concerns about components accumulating in the solvent recycle.

Batch chromatography columns can be designed as dynamic processes or by using an equilibrium stage analogy; see [Chapter 8](#) of [Ruthven \(1984\)](#).

When a batch column is scaled up to a larger diameter, there will be greater dispersion of the components, leading to broader peaks, and hence the column length must be increased. Batch chromatography is therefore inefficient in the use of stationary phase, particularly if the stationary phase is discarded frequently. When very long columns are used, the pressure drop can be very high; hence the process is sometimes known as high-pressure liquid chromatography or high-performance liquid chromatography (HPLC).

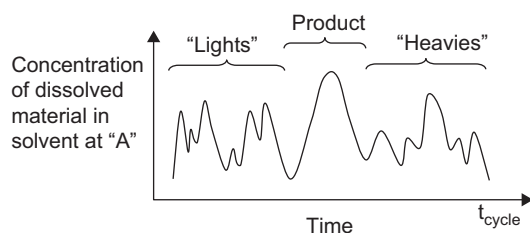
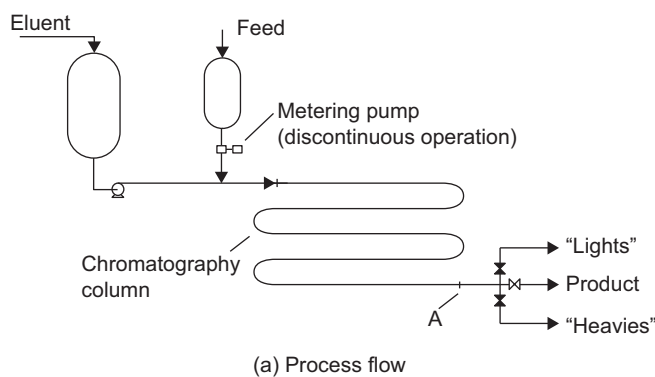


FIG. 16.23 Batch chromatography.

Gel permeation chromatography

Gel permeation chromatography is a variation of batch chromatography. The stationary phase is selected to have a pore structure that excludes the desired product. The larger molecules that cannot enter the pores are eluted first, and smaller molecules are eluted later. The separation order is thus reversed from typical batch chromatography. With gel permeation chromatography, it is much easier to completely regenerate the sorbent, and the cycle time can be reduced. Other aspects of the design are similar to conventional batch chromatography.

Affinity chromatography

Affinity chromatography is probably the most widely used preparative and production chromatography method in industry, particularly for recovery of biochemicals and large molecules.

In affinity chromatography, the stationary phase is selected or designed to have a highly specific interaction with the desired component. Commonly used interactions include enzyme–inhibitor, antibody–antigen, and lectin–cell wall. For example, monoclonal antibodies (mAb) can be synthesized that have very highly specific binding to a desired protein. If the mAb is chemically bound to beads of agarose, polyacrylamide, or other suitable material, the beads can be used to pack a column with specific selectivity for the target protein.

Affinity chromatography is in many respects more like an adsorption–desorption process (Section 16.2.1) or an ion exchange process (Section 16.5.5) than a chromatography process. The feed liquid can be passed through the bed without requiring additional eluent. The high selectivity of the stationary phase allows the bed to be kept on stream until the sorbent is fully loaded or nearly fully loaded. The sorbent is then regenerated by the eluent flow. The regeneration step usually involves a change in solvent properties to disrupt the affinity between the sorbent and adsorbed species, for example, a change in solvent polarity, pH, ionic strength, or occasionally even temperature. When multiple species are adsorbed, the eluent properties may be changed over time so that different species are eluted in sequence. This is known as applying a solvent gradient.

The design of affinity chromatography columns is similar to the design of ion exchange columns, as illustrated in Example 16.7. The volume of stationary-phase media is determined by the capacity of the media, the feed concentration and flow rate, and the desired time between regenerations. In batch processes, the operating cycle will usually be chosen to allow the column to be loaded, regenerated, and reconditioned within the process cycle time.

One of the most widely used forms of affinity chromatography is protein A chromatography, which can be used to purify a wide range of MAb and can achieve product purities higher than 99%. Applications and limitations of protein A chromatography are described by [Shukla et al. \(2007\)](#).

The cost of an affinity chromatography packing is usually much greater than the cost of conventional chromatography media, even though the adsorbent is used more efficiently and column sizes are generally smaller. The column is therefore more likely to be fully regenerated so that the stationary phase can be reused. Even so, the sorbent performance usually deteriorates over multiple cycles, and stationary-phase replacement can be a significant consumable cost in biochemical and pharmaceutical processes. For example, [Follman and Fahrner \(2004\)](#) state that protein A affinity chromatography accounts for 35% of the downstream purification costs for mAb production, although the cost of production of mAb is usually a small fraction of the sales price ([Kelley, 2009](#)).

A good introduction to affinity chromatography is given in the book by [Mohr and Pommerening \(1986\)](#). More recent work is described in [Ganetsos and Barker \(1992\)](#) and [Hagel et al. \(2007\)](#).

Continuous chromatography

A truly continuous chromatography process would employ countercurrent flow of the solid phase and the desorbent, as shown in [Fig. 16.24\(a\)](#). If the liquid feed is introduced at a height h_F above the base and desorbent liquid is introduced at the top, then a component that is more strongly adsorbed on the sorbent will tend to move up the column with the solid phase, giving the composition profile labeled A in [Fig. 16.24\(b\)](#). Conversely, the component or components that are less strongly adsorbed will tend to move down the column with the desorbent, giving a concentration profile like that labeled B in [Fig. 16.24\(b\)](#). At height h_E , above the feed point, the liquid phase is substantially free of component B, and a product stream containing only desorbent and A can be withdrawn. This stream is known as the extract. Similarly, at height h_R , below the feed point, all of component A has been adsorbed by the solid, and a liquid stream containing only desorbent and B can be recovered. This stream is known as the raffinate.

Above h_E , in the region labeled zone I, the desorbent flow washes A off the solid phase, regenerating a clean sorbent that can be recycled to the bottom of the column. Likewise, below h_R , in the region labeled zone IV, the adsorbent adsorbs remaining B from the liquid, generating a clean desorbent that can be recycled to the top of the column. In practice, this bottom zone is not always used, as it is often cheaper to separate B from desorbent by other methods such as distillation or crystallization.

A true countercurrent continuous chromatography process is analogous to a series of absorbers and strippers:

Zone I: desorption (stripping) of A

Zone II: desorption of B

Zone III: adsorption (absorption) of A, desorption of B

Zone IV: adsorption of B

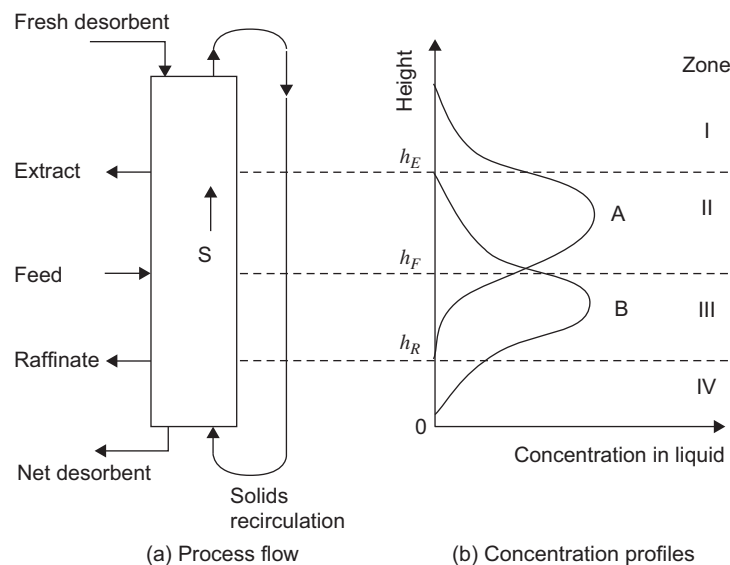


FIG. 16.24 Continuous countercurrent chromatography.

If the equilibrium constants are known, then the process can be approximately modeled in terms of theoretical stages using the Kremser equations or McCabe Thiele analysis; see [Ruthven \(1984\)](#).

A true moving-bed countercurrent chromatography process (“hypersorption”) was commercially demonstrated by Dow Chemical and Union Oil Co. in 1947; see [Kehde et al. \(1948\)](#). Unfortunately, most good sorbents are not strong enough to withstand circulation at high solids flow rates and suffer unacceptably high attrition, so this process is no longer practiced.

Instead of circulating the sorbent, the movement of solids can be simulated by using a large number of beds and periodically switching the bed location at which the net flows (feed, extract, raffinate, and desorbent) are fed or removed. In the UOP Sorbex process, illustrated in [Fig. 16.25](#), this is accomplished using a rotary valve; however, the same effect can be obtained using a large number of solenoid switching valves. When the rotary valve is moved to the next position, the net flow that was moving through the pipe between beds n and $n + 1$ is switched to move through the pipe between beds $n + 1$ and $n + 2$. Bed n thus effectively becomes bed $n - 1$ relative to the net flows, and movement of the bed is simulated.

Although the solid phase undergoes periodic discrete movement rather than continuous movement relative to the liquid and a steady state is never really established, the performance is nonetheless close to that of true countercurrent chromatography; see [Ruthven \(1984\)](#) and Menet and Thibaut (1999).

Simulated moving bed (SMB) chromatography can be used to obtain products with high purity and high recovery if a suitable combination of sorbent and desorbent can be found. The desorbent can be a solvent or a liquid that has similar adsorption properties to the target molecule and competes for sorbent sites. The sorbent is used much more efficiently than in batch chromatography, and the production rate per kg of sorbent is usually much higher.

Although SMB chromatography has been highly successful in a number of applications, its use is not as widespread as affinity chromatography. The development and scale-up of SMB processes can be tricky, as the ratio of liquid to solid flow rates in each zone has to be chosen to give the desired adsorption or desorption for that zone; see [Mazzotti et al. \(1997\)](#) and [Jupke et al. \(2002\)](#). If the effective stripping factor or absorption factor is too close to 1.0, then a large number of effective stages is needed, or separation will be poor. It is also much more difficult to apply a solvent gradient or vary solvent properties in an SMB process. Because of its higher process development costs, SMB chromatography tends to only be applied to products that are produced in high volume or that require a large amount of expensive sorbent. The largest applications are in recovery of paraxylene from mixed xylenes and

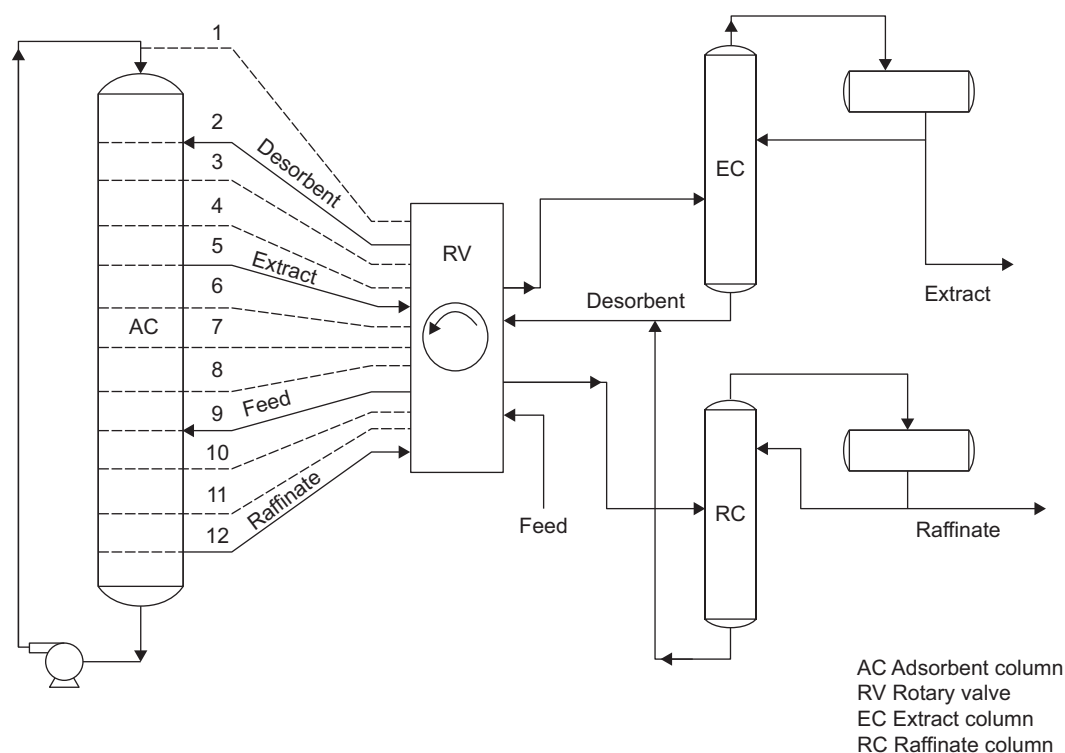


FIG. 16.25 UOP Sorbex process. (Reproduced with permission of UOP, LLC.)

production of high-fructose corn syrup, see [Ruthven \(1984\)](#) and [Meyers \(2016\)](#). There has also recently been a significant expansion in application of SMB chromatography to separation of enantiomers in the pharmaceutical industry, where the high product price and high recovery can justify the added development costs.

16.6 References

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16.7 Nomenclature

		Dimensions in MLT
A	Heat transfer area	L^2
A_i	Area of interface	L^2
A_v	Area for vapor flow	L^2
D_T	Tank or vessel diameter	L
D_v	Minimum vessel diameter for separator	L
d_d	Droplet diameter	L
F_1	Feed molar flow rate	MT^{-1}
F_2	Product molar flow rate	MT^{-1}
f_L	Fraction of bed that is fully loaded at end of adsorption phase of cycle	—
f_v	Fraction of cross-sectional area occupied by vapor	—
g	Gravitational acceleration	LT^{-2}

		Dimensions in MLT
h_E	Height of extract draw above base in continuous chromatography	L
h_F	Height of feed above base in continuous chromatography	L
h_L	Liquid depth	L
h_R	Height of raffinate draw above base in continuous chromatography	L
h_v	Height above liquid level	L
k	Equilibrium constant for adsorption	$M^{-n}L^nT^{2n}$
L_c	Continuous phase volumetric flow-rate	L^3T^{-1}
L_m	Length of membrane	L
L_v	Length of separator, vessel tangent-to-tangent length	L
l	Length of decanter vessel	L
M_a	Mass of adsorbent per bed	M
M_i	Molar flux of component i	$ML^{-2}T^{-1}$
$M_{i,ave}$	Average molar flux of component i	$ML^{-2}T^{-1}$
M_w	Molecular weight of adsorbed component	—
m	Adsorbent loading, g/g adsorbent	—
m_1	Maximum adsorbent loading (g/g adsorbent)	—
m_2	Minimum adsorbent loading (g/g adsorbent)	—
n	Exponent in equation 16.1	—
P_i	Permeability of membrane for component i (mol/m.s.bar)	T
P	Partial pressure of adsorbed component	$ML^{-1}T^{-2}$
p_{if}	Local partial pressure of component i on feed side	$ML^{-1}T^{-2}$
p_{ip}	Local partial pressure of component i on permeate side	$ML^{-1}T^{-2}$
Q	Heat transferred in unit time	ML^2T^{-3}
r	Radius of decanter vessel	L
S_{ij}	Selectivity of a membrane for component i over component j	—
T	Temperature	θ
ΔT_m	Mean temperature difference	θ
t_a	Time an adsorption bed is in the adsorption stage of the cycle	T
t_B	Breakthrough time	T
U	Overall heat-transfer coefficient	$MT^{-3}\theta^{-1}$
u_c	Velocity of continuous phase in a decanter	LT^{-1}
u_d	Settling (terminal) velocity of dispersed phase in a decanter	LT^{-1}
u_s	velocity in a separator	LT^{-1}
u_t	Settling velocity	LT^{-1}
u_v	Maximum allowable vapor velocity in a separating vessel	LT^{-1}
V_v	Gas, or vapor volumetric flow-rate	L^3T^{-1}
w	Width of interface in a decanter	L
x	Length	L
y_1	Feed mole fraction of adsorbed component	—

		Dimensions in MLT
y_2	Product mole fraction of adsorbed component	—
z	Height of the interface from the base of the vessel	L
z_1	Height to light liquid overflow from a decanter	L
z_2	Height to heavy liquid overflow from a decanter	L
z_3	Height to the interface in a decanter	L
δ	Membrane thickness	L
μ	Liquid viscosity	$\text{ML}^{-1}\text{T}^{-1}$
μ_c	Viscosity of continuous phase	$\text{ML}^{-1}\text{T}^{-1}$
ρ	Liquid density	ML^{-3}
ρ_c	Density of the continuous phase	ML^{-3}
ρ_d	Density of the dispersed phase	ML^{-3}
ρ_L	Liquid density	ML^{-3}
ρ_v	Vapor density	ML^{-3}
ρ_1	Light liquid density in a decanter	ML^{-3}
ρ_2	Heavy liquid density in a decanter	ML^{-3}

16.8 Problems

- 16.1** An electronics plant uses an adsorption system to recover solvent from a vent gas stream and prevent volatile organic compound (VOC) emissions. The vent gas flow rate is $20 \text{ m}^3/\text{s}$ of dry air at 293 K, 1.5 atm. The initial solvent loading is 1.5 mol%, which must be reduced to 20 ppm to comply with emissions permits. Activated carbon is used as adsorbent and has a capacity of 20 mol/kg of adsorbent and a heat of adsorption of 8 kcal/mol solvent. The sorbent can be regenerated by raising the temperature to 363 K. Design a suitable TSA system for this process. Estimate the amount of adsorbent needed, the vessel volumes, and the minimum regenerant heat requirement for your design. The activated carbon adsorbent has average bulk density 120 kg/m^3 and heat capacity $0.7 \text{ J/g}\cdot^\circ\text{C}$. The solvent flammability limits in air are 2.5 to 12.0 vol% at 293 K and 1.2 to 16.0 vol% at 363 K.
- 16.2** It has been suggested that carbon dioxide can be recovered from flue gas using a membrane process. The flue gas has the following composition: nitrogen 73.9 mol%, oxygen 3.1 mol%, carbon dioxide 7.7 mol%, water vapor 15.3 mol%. Using the data in [Table 16.1](#), determine the best membrane material for this process. What are the advantages and disadvantages of the proposed membrane process?
- 16.3** In a process for the production of acrylic fibers by the emulsion polymerization of acrylonitrile, the unreacted monomer is recovered from water by distillation. Acrylonitrile forms an azeotrope with water, and the overhead product from the column contains around 5 mol% water. The overheads are condensed and the recovered acrylonitrile separated from the water in a decanter. The decanter operating temperature will be 20°C .
Size a suitable decanter for a feed rate of 3000 kg/h .
- 16.4** In the production of aniline by the hydrogenation of nitrobenzene, the reactor products are separated from unreacted hydrogen in a condenser. The condensate, which is mainly water and aniline, together with a small amount of unreacted nitrobenzene and cyclohexylamine, is fed to a decanter to separate the water and aniline.

The separation will not be complete, as aniline is slightly soluble in water, and water in aniline. A typical material balance for the decanter is given here:

Basis 100 kg feed

	feed	aqueous stream	organic stream
water	23.8	21.4	2.4
aniline	72.2	1.1	71.1
nitrobenzene	3.2	trace	3.2
cyclohexylamine	0.8	0.8	trace
total	100	23.3	76.7

Design a decanter for this duty for a feed rate of 3500 kg/h. Concentrate on the separation of the water and aniline. The densities of water–aniline solutions are given in Appendix F, Problem F8, available in the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793. The decanter will operate at a maximum temperature of 30 °C.

- 16.5** Water droplets are to be separated from air in a simple separation drum. The flow rate of the air is 1000 m³/h, at s.t.p., and it contains 75 kg of water. The drum will operate at 1.1 bara pressure and 20 °C. Size a suitable liquid–vapor separator.
- 16.6** The vapor from a chlorine vaporizer will contain some liquid droplets. The vaporizer consists of a vertical, cylindrical, vessel with a submerged bundle for heating. A vapor rate of 2500 kg/h is required, and the vaporizer will operate at 6 bara. Size the vessel to restrict the carryover of liquid droplets. The liquid hold-up time need not be considered, as the liquid level will be a function of the thermal design.

Separation columns (distillation, absorption, and extraction)

KEY LEARNING OBJECTIVES

- How to design distillation columns
- How to size distillation columns and select and design distillation column trays
- How to design distillation columns using packing instead of trays
- How to design absorption and stripping columns
- How to design liquid–liquid extraction columns

17.1 Introduction

This chapter covers the design of separating columns. Though the emphasis is on distillation processes, the basic construction features, and many of the design methods, also apply to other multistage processes, such as stripping, absorption, and extraction. Only a brief review of the fundamental principles that underlie the design procedures will be given; a fuller discussion can be found in [Chhabra and Gurappa \(2019\)](#) and in other textbooks: King (1980), [Hengstebeck \(1976\)](#), [Kister \(1992\)](#), [Doherty and Malone \(2001\)](#) and [Luyben \(2013\)](#).

Distillation is probably the most widely used separation process in the chemical and allied industries; its applications ranging from the rectification of alcohol, which has been practiced since antiquity, to the fractionation of crude oil. A good understanding of methods used for correlating vapor–liquid equilibrium data is essential to the understanding of distillation and other equilibrium-staged processes; this subject was covered in [Chapter 4](#).

In recent years, much of the work done to develop reliable design methods for distillation equipment has been carried out by a commercial organization, Fractionation Research, Inc. (FRI), an organization set up with the resources to carry out experimental work on full-size columns. Because their work is proprietary, it is not published in the open literature and it has not been possible to refer to their methods in this book. Fractionation Research's design manuals will, however, be available to design engineers whose companies are subscribing members of the organization. FRI has also produced an excellent training video that shows the physical phenomena that occur when a plate column is operated in different hydraulic regimes. This video can be ordered from FRI at www.fri.org.

Distillation column design

The design of a distillation column can be divided into the following steps:

1. Specify the degree of separation required: set product specifications.
2. Select the operating conditions: batch or continuous; operating pressure.
3. Select the type of contacting device: plates or packing.

4. Determine the stage and reflux requirements: the number of equilibrium stages.
5. Size the column: diameter, number of real stages.
6. Design the column internals: plates, distributors, packing supports.
7. Mechanical design: vessel and internal fittings.

The principal step is to determine the stage and reflux requirements. This is a relatively simple procedure when the feed is a binary mixture but can be complex when the feed contains more than two components (multicomponent systems).

Almost all distillation design is carried out using commercial process simulation software, as introduced in [Chapter 4](#). The process simulation programs allow the designer to determine the stage and reflux requirements that are needed to attain the desired separation and then size the column and design the column internals. Once the column size is known, the shell can be designed as a pressure vessel (see [Chapter 14](#)) and the condenser and reboiler can be designed as heat exchangers (see [Chapter 19](#)). The whole design can then be costed and optimized. An example of distillation column optimization was given in [Chapter 12](#).

17.2 Continuous distillation: Process description

The separation of liquid mixtures by distillation depends on differences in volatility between the components. The greater the relative volatilities, the easier is the separation. The basic equipment required for continuous distillation is shown in [Fig. 17.1](#). Vapor flows up the column and liquid flows countercurrently down the column. The vapor and liquid are brought into contact on plates or packing. Part of the condensate from the condenser is returned to the top of the column to provide liquid flow above the feed point (reflux), and part of the liquid from the base of the column is vaporized in the reboiler and returned to provide the vapor flow.

In the section below the feed, the more volatile components are stripped from the liquid, and this is known as the *stripping section*. Above the feed, the concentration of the more volatile components is increased, and this is called the *enrichment*, or more commonly, the *rectifying section*. [Fig. 17.1\(a\)](#) shows a column producing two product streams, referred to as *distillate* and *bottoms*, from a single feed. Columns are occasionally used with more than one feed and with side streams withdrawn at points up the column (see [Fig. 17.1b](#)). This does not alter the basic operation, but complicates the analysis of the process, to some extent.

If the process requirement is to strip a volatile component from a relatively nonvolatile solvent, the rectifying section may be omitted, and the column would then be called a *stripping column*.

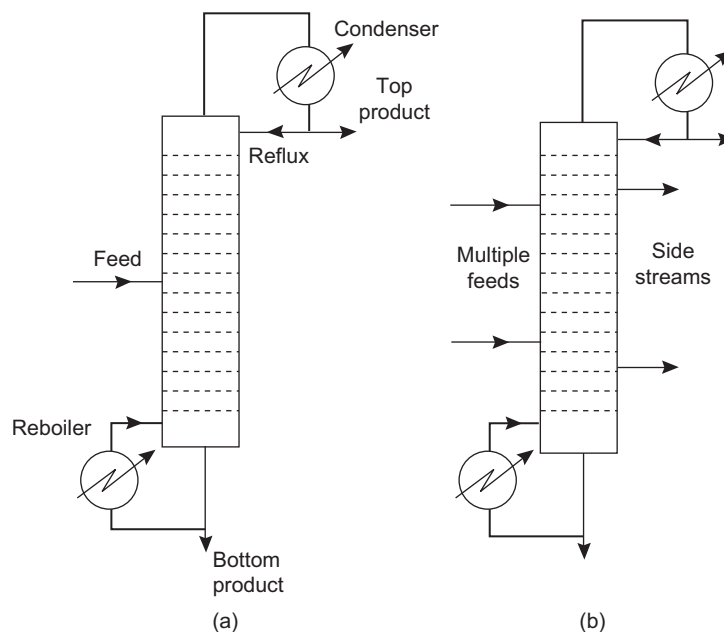


FIG. 17.1 Distillation column. (a) Basic column. (b) Multiple feeds and side streams.

In some operations, where some or all of the top product is required as a vapor, only sufficient liquid is condensed to provide the reflux flow to the column, and the condenser is referred to as a partial condenser. When the liquid is totally condensed, the liquid returned to the column will have the same composition as the top product. In a partial condenser the reflux will be in equilibrium with the vapor leaving the condenser. Virtually pure top and bottom products can be obtained in a single column from a binary feed if no azeotrope is formed, but where the feed contains more than two components, only a single “pure” product can be produced, either from the top or bottom of the column. Several columns will be needed to separate a multicomponent feed into its constituent parts.

17.2.1 Reflux considerations

The reflux ratio, R , is normally defined as:

$$R = \frac{\text{flow returned as reflux}}{\text{flow of top product taken off}}$$

The number of stages required for a given separation will depend on the reflux ratio used.

In an operating column, the effective reflux ratio will be increased by vapor condensed within the column due to heat leakage through the walls. With a well-lagged column, the heat loss will be small, and no allowance is normally made for this increased flow in design calculations. If a column is poorly insulated, changes in the internal reflux due to sudden changes in the external conditions, such as a sudden rain storm, can have a noticeable effect on the column operation and control.

Total reflux

Total reflux is the condition when all the condensate is returned to the column as reflux: No product is taken off, and there is no feed.

At total reflux, the number of stages required for a given separation is the minimum at which it is theoretically possible to achieve the separation. Though not a practical operating condition, it is a useful guide to the likely number of stages that will be needed.

Columns are often started up with no product take-off and operated at total reflux until steady conditions are attained. The testing of columns is also conveniently carried out at total reflux.

Minimum reflux

As the reflux ratio is reduced, a *pinch point* will occur at which the separation can only be achieved with an infinite number of stages. This sets the minimum possible reflux ratio for the specified separation.

Optimum reflux ratio

Practical reflux ratios will lie somewhere between the minimum for the specified separation and total reflux. The designer must select a value at which the specified separation is achieved at minimum cost. Increasing the reflux reduces the number of stages required, and hence the capital cost, but increases the utility requirements (steam and cooling water) and the operating costs. The optimum reflux ratio will be that which gives the lowest total annualized cost or greatest net present value. No hard and fast rules can be given for the selection of the design reflux ratio, but for many systems the optimum will lie between 1.1 and 1.3 times the minimum reflux ratio. As a first approximation, 1.15 times minimum reflux is often used.

For new designs, where the ratio cannot be decided on from past experience, the effect of reflux ratio on the number of stages can be investigated using a process simulation model.

At low reflux ratios, the calculated number of stages will be highly dependent on the accuracy of the vapor–liquid equilibrium data available. If the data or phase equilibrium model is suspect, the designer should select a higher-than-normal ratio to give more confidence in the design.

17.2.2 Feed-point location

The precise location of the feed point will affect the number of stages required for a specified separation and the subsequent operation of the column. As a general rule, the feed should enter the column at the point that gives the best match between the feed composition (vapor and liquid if two phases) and the vapor and liquid streams in the column. In practice, it is wise to provide two or three feed-point nozzles located near the predicted feed point to allow for uncertainties in the design calculations and data and possible changes in the feed composition after start-up.

17.2.3 Selection of column pressure

Except when distilling heat-sensitive materials, the main consideration when selecting the column operating pressure will be to ensure that the dew point of the distillate is above the temperature that can be easily obtained with plant cooling water. The maximum, summer, temperature of cooling water is usually taken as 30 °C. If this means that high pressures will be needed, the provision of refrigerated cooling should be considered. Vacuum operation is used to reduce the column temperatures for the distillation of heat-sensitive materials and where very high temperatures would otherwise be needed to distil relatively nonvolatile materials.

When calculating the stage and reflux requirements using shortcut methods, it is usual to take the operating pressure as constant throughout the column. In vacuum columns, the column pressure drop will be a significant fraction of the total pressure, and the change in pressure up the column should be allowed for when calculating the stage temperatures. When using rigorous simulation methods, a rough initial estimate of column pressure drop can be made by assuming a pressure drop per tray equal to twice the liquid static head on the tray, that is, $2 \rho_L g h_w$, where ρ_L is the liquid density (kg/m³), g is the gravitational acceleration (m/s²), and h_w is the weir height (m).

17.3 Continuous distillation: Basic principles

17.3.1 Stage equations

Material and energy balance equations can be written for any stage in a multistage process.

Fig. 17.2 shows the material flows into and out of a typical stage n in a distillation column. The equations for this stage are set out here, for any component i .

Material balance

$$V_{n+1}y_{n+1} + L_{n-1}x_{n-1} + F_nz_n = V_ny_n + L_nx_n + S_nx_n \quad (17.1)$$

Energy balance

$$V_{n+1}H_{n+1} + L_{n-1}h_{n-1} + Fh_f + q_n = V_nH_n + L_nh_n + S_nh_n \quad (17.2)$$

where V_n = vapor flow from the stage

V_{n+1} = vapor flow into the stage from the stage below

L_n = liquid flow from the stage

L_{n-1} = liquid flow into the stage from the stage above

F_n = any feed flow into the stage

S_n = any side stream from the stage

q_n = heat flow into, or removal from, the stage

n = any stage, numbered from the top of the column

z = mol fraction of component i in the feed stream (note, feed may be two-phase)

x = mol fraction of component i in the liquid streams

y = mol fraction component i in the vapor streams

H = specific enthalpy vapor phase

h = specific enthalpy liquid phase

h_f = specific enthalpy feed (vapor + liquid)

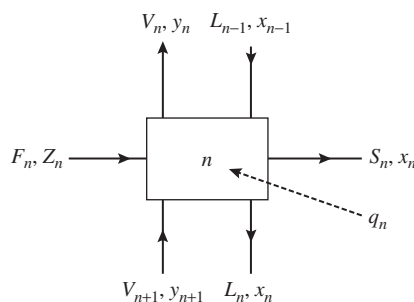


FIG. 17.2 Stage flows.

All flows are the total stream flows (mol/unit time), and the specific enthalpies are also for the total stream (J/mol).

It is convenient to carry out the analysis in terms of “equilibrium stages.” In an equilibrium stage (theoretical plate), the liquid and vapor streams leaving the stage are taken to be in equilibrium, and their compositions are determined by the vapor–liquid equilibrium relationship for the system; see [Chapter 4](#). In terms of equilibrium constants:

$$y_i = K_i x_i \quad (17.3)$$

The performance of real stages is related to an equilibrium stage by the concept of plate or stage efficiencies for plate contactors and “height equivalent to a theoretical plate” for packed columns.

In addition to the equations arising from the material and energy balances over a stage, and the equilibrium relationships, there will be a fourth relationship, the summation equation for the liquid and vapor compositions:

$$\sum x_{i,n} = \sum y_{i,n} = 1.0 \quad (17.4)$$

These four equations are the so-called MESH equations for the stage: Material balance, Equilibrium, Summation, and Heat (energy) balance equations. MESH equations can be written for each stage and for the reboiler and condenser. The solution of this set of equations forms the basis of the rigorous methods that have been developed for the analysis of staged separation processes and that are solved in the process simulation programs.

17.3.2 Dew point and bubble point

To estimate the stage, condenser, and reboiler temperatures, procedures are required for calculating dew and bubble points. By definition, a saturated liquid is at its bubble point (any rise in temperature will cause a bubble of vapor to form) and a saturated vapor is at its dew point (any drop in temperature will cause a drop of liquid to form).

Dew points and bubble points can be calculated from the vapor–liquid equilibrium for the system. In terms of equilibrium constants, the bubble point is defined by the equation:

$$\text{bubble point: } \sum y_i = \sum K_i x_i = 1.0 \quad (17.5a)$$

$$\text{and dew point: } \sum x_i = \sum \frac{y_i}{K_i} = 1.0 \quad (17.5b)$$

For multicomponent mixtures, the temperature that satisfies these equations, at a given system pressure, must be found by iteration. For binary systems, the equations can be solved more readily because the component compositions are not independent; fixing one fixes the other.

$$y_a = 1 - y_b \quad (17.6a)$$

$$x_a = 1 - x_b \quad (17.6b)$$

17.3.3 Equilibrium flash calculations

In an equilibrium flash process, a feed stream is separated into liquid and vapor streams at equilibrium. The composition of the streams depends on the quantity of the feed vaporized (flashed). The equations used for equilibrium flash calculations are developed later, and a typical calculation is shown in Example 17.1.

Flash calculations are often needed to determine the condition of the feed to a distillation column and, occasionally, to determine the flow of vapor from the reboiler, or condenser if a partial condenser is used.

Single-stage flash distillation processes are used to make a coarse separation of the light components in a feed, often as a preliminary step before a multicomponent distillation column.

Fig. 17.3 shows a typical equilibrium flash process. The equations describing this process are:

Material balance, for any component, i

$$Fz_i = Vy_i + Lx_i \quad (17.7)$$

Energy balance, total stream enthalpies:

$$Fh_f = VH + Lh \quad (17.8)$$

If the vapor–liquid equilibrium relationship is expressed in terms of equilibrium constants, Equation 17.7 can be written in a more useful form:

$$\begin{aligned} Fz_i &= VK_i x_i + Lx_i \\ &= Lx_i \left[\frac{V}{L} K_i + 1 \right] \end{aligned}$$

from which

$$L = \sum_i \frac{Fz_i}{\left[\frac{VK_i}{L} + 1 \right]} \quad (17.9)$$

and, similarly,

$$V = \sum_i \frac{Fz_i}{\left[\frac{L}{VK_i} + 1 \right]} \quad (17.10)$$

The groups incorporating the liquid and vapor flow rates and the equilibrium constants have a general significance in separation process calculations.

The group L/VK_i is known as the absorption factor, A_i , and is the ratio of the moles of any component in the liquid stream to the moles in the vapor stream.

The group VK_i/L is called the stripping factor, S_i , and is the reciprocal of the absorption factor.

Efficient techniques for the solution of the trial and error calculations necessary in multicomponent flash calculations are given by several authors, including Hengstebeck (1976) and King (1980). Flash models are available in all the commercial process simulation programs and are easy to configure. It is often a good idea to use flash models to check that the phase equilibrium model that has been selected makes an accurate prediction of any experimental data that are available. Flash models are also useful for checking for changes in volatility order or formation of second liquid phases within a distillation column.

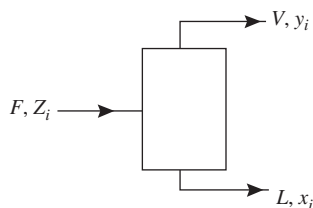


FIG. 17.3 Flash distillation.

Example 17.1

A feed to a column has the composition given in the following table and is at a pressure of 14 bar and a temperature of 60 °C. Calculate the flow and composition of the liquid and vapor phases. Equilibrium data can be taken from De Priester charts (Dadyburjor, 1978).

	kmol/h	z_i
Feed ethane (C ₂)	20	0.25
propane (C ₃)	20	0.25
isobutane (iC ₄)	20	0.25
n-pentane (nC ₅)	20	0.25

Solution

For two phases to exist, the flash temperature must lie between the bubble point and dew point of the mixture.

From Equations 17.5a and 17.5b:

$$\sum K_i z_i > 1.0$$

$$\sum \frac{z_i}{K_i} > 1.0$$

Check the feed condition:

	K_i	$K_i z_i$	z_i/K_i
C ₂	3.8	0.95	0.07
C ₃	1.3	0.33	0.19
iC ₄	0.43	0.11	0.58
nC ₅	0.16	0.04	1.56
		$\Sigma 1.43$	$\Sigma 2.40$

Therefore the feed is a two-phase mixture.

Flash calculation:

Try $L/V = 1.5$				Try $L/V = 3.0$	
	K_i	$A_i = L/VK_i$	$V_i = Fz_i/(1 + A_i)$	A_i	V_i
C ₂	3.8	0.395	14.34	0.789	11.17
C ₃	1.3	1.154	9.29	2.308	6.04
iC ₄	0.43	3.488	4.46	6.977	2.51
nC ₅	0.16	9.375	1.93	18.750	1.01
$V_{\text{calc}} = 30.02$			$V_{\text{calc}} = 20.73$		
$L/V = \frac{80-30.02}{30.02} = 1.67$			$L/V = 2.80$		

Hengstebeck's method is used to find the third trial value for L/V . The calculated values are plotted against the assumed values, and the intercept with a line at 45 degrees (calculated = assumed) gives the new trial value: 2.4.

Try $L/V = 2.4$				
	A_i	V_i	$y_i = V_i/V$	$x_i = (Fz_i - V_i)/L$
C ₂	0.632	12.26	0.52	0.14
C ₃	1.846	7.03	0.30	0.23
iC ₄	5.581	3.04	0.13	0.30
nC ₅	15.00	1.25	0.05	0.33
	$V_{\text{cal}} = 23.58$		1.00	1.00

$$L = 80 - 23.58 = 56.42 \text{ kmol/h}$$

$$L/V \text{ calculated} = 56.42/23.58 = 2.39 \text{ close enough to the assumed value of } 2.4$$

Adiabatic flash

In many flash processes the feed stream is at a higher pressure than the flash pressure and the heat for vaporization is provided by the enthalpy of the feed. In this situation the flash temperature will not be known and must be found by iteration. A temperature must be found at which both the material and energy balances are satisfied. This is easily solved using process simulation software by specifying the flash outlet pressure and specifying zero heat input. The program then calculates the temperature and stream flow rates that satisfy the MESH equations.

17.4 Design variables in distillation

It was shown in [Chapter 1](#) that to carry out a design calculation, the designer must specify values for a certain number of independent variables to define the problem completely and that the ease of calculation will often depend on the judicious choice of these design variables. The choice of design variables is particularly important in distillation, as the problem must be sufficiently well defined to find a feasible solution when simulated using a computer.

The total number of variables and equations required to describe a multicomponent distillation can be very large, because the MESH equations must be solved for every stage, including the reboiler and condenser. It becomes difficult for the designer to keep track of all the variables and equations, and mistakes are likely to be made, as the number of degrees of freedom will be the difference between two large numbers. Instead, a simpler procedure known as the “description rule” given by [Hanson et al. \(1962\)](#) can be used. The description rule states that to determine a separation process completely, the number of independent variables that must be set (by the designer) will equal the number that are set in the construction of the column or that can be controlled by external means in its operation. The method is best illustrated by considering the operation of the simplest type of column: with one feed, no side streams, a total condenser, and a reboiler. The construction will fix the number of stages above and below the feed point (two variables). The feed rate, column pressure, and condenser and reboiler duties (cooling water and steam flows) will be controlled (four variables). There are therefore six variables in total.

To design the column this number of variables must be specified, but the same variables need not be selected. Typically, in a design situation the feed rate will be fixed by the upstream design. The column pressure will also usually be fixed early in the design. Distillation processes are usually operated at low pressure, where relative volatility is high, but the pressure is usually constrained to be high enough for the condenser to operate using cooling water rather than refrigeration. If the feed rate and pressure are specified, then four degrees of freedom remain. Rigorous column models in process simulation programs require the designer to specify the number of stages above and below the feed, leaving the designer with two degrees of freedom. If two additional independent parameters are specified, then the problem is completely defined and has a single solution. For example, if the designer specifies a reflux ratio and a boil-up ratio or a reflux ratio and a distillate rate, then there will be a corresponding unique set of distillate and bottoms compositions for a given feed composition. If the designer chooses to specify the compositions of two key components in either the distillate or the bottoms, then there will be a required reflux rate, boil-up rate, distillate flow rate, etc. Similarly, specifying the purity and recovery of a single component in one of the products will completely specify the problem.

When replacing variables identified by the application of the description rule, it is important to ensure that those selected are truly independent and that the values assigned to them lie within the range of possible, practical values. For example, if the distillate mass flow rate is specified, then the bottoms flow rate is fixed by overall material balance and cannot be specified independently. Proper attention to the specification of variables is particularly important when using purity or composition specifications in multicomponent distillation. It would clearly not be possible to obtain 99% purity of the light key component in the distillate if the feed contained 2% of components that boiled at lower temperatures than the light key component. The selection of key components and product specifications for multicomponent distillation are discussed in more detail in Section 17.6.

The number of independent variables that have to be specified to define a problem will depend on the type of separation process being considered. Some examples of the application of the description rule to more complex columns are given by [Hanson et al. \(1962\)](#).

17.5 Design methods for binary systems

The distillation of binary mixtures is a relatively simple problem. With a binary mixture, fixing the composition of one component fixes the composition of the other. The stage and reflux requirements can be determined using simple graphical methods developed in the 1920s, and iterative calculations are not required.

It must be emphasized, however, that the graphical methods for binary distillation are no longer used in any practical context. Very few industrial distillation problems involve true binary mixtures. Other components will usually be present even if the two main components constitute more than 99.9 mol% of the mixture. The design engineer will usually need to know how these other components distribute between the distillate and bottoms to ensure that product specifications can be met and to determine the contaminant loads on downstream operations. Furthermore, the distillation design problem is seldom solved in isolation from the overall process design, and the widespread use of process simulation programs has made the graphical methods obsolete. The initialization of a rigorous simulation of a binary distillation uses the same methods used for a multicomponent distillation described in Section 4.5.2. The graphical methods give no insight into the solution procedures used for multicomponent distillation.

Despite these considerations, many educators find the graphical methods for binary distillation to be useful as a means of explaining some of the phenomena that can occur in multistage separations, and these methods are part of the required chemical engineering curriculum in most countries. The graphical methods can be used to illustrate some problems that are common to binary and multicomponent distillation. Graphical methods are still useful in understanding and initializing other staged separation processes such as absorption, stripping, and extraction.

The discussion of binary distillation methods in this chapter has been limited to a brief overview with emphasis on the insights that can be obtained from the graphs. For more details of the classical binary distillation methods, see [Chhabra and Gurappa \(2019\)](#) and earlier editions of this book.

17.5.1 Basic equations

[Sorel \(1899\)](#) first derived and applied the basic stage equations to the analysis of binary systems. [Fig. 17.4\(a\)](#) shows the flows and compositions in the top part of a column. Taking the system boundary to include the stage n and the condenser gives the following equations:

Material balance

total flows

$$V_{n+1} = L_n + D \quad (17.11)$$

where D is the distillate flow rate, and for either component:

$$V_{n+1}y_{n+1} = L_nx_n + Dx_d \quad (17.12)$$

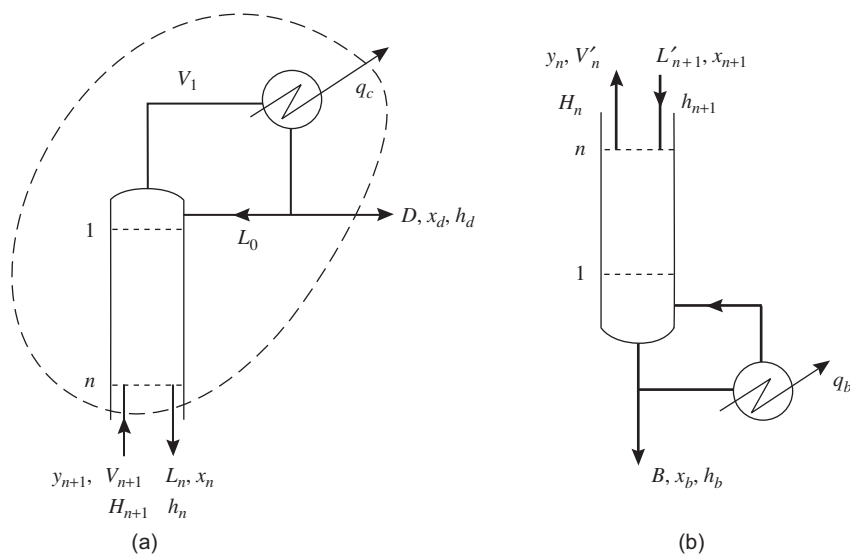


FIG. 17.4 Column flows and compositions. (a) Above feed. (b) Below feed.

Energy balance

total stream enthalpies

$$V_{n+1}H_{n+1} = L_n h_n + D h_d + q_c \quad (17.13)$$

where q_c is the heat removed in the condenser.

Combining Equations 17.11 and 17.12 gives

$$y_{n+1} = \frac{L_n}{L_n + D} x_n + \frac{D}{L_n + D} x_d \quad (17.14)$$

Combining Equations 17.11 and 17.13 gives

$$V_{n+1}H_{n+1} = (L_n + D)H_{n+1} = L_n h_n + D h_d + q_c \quad (17.15)$$

Analogous equations can be written for the stripping section (Fig. 17.4b).

$$x_{n+1} = \frac{V'_n}{V'_n + B} y_n + \frac{B}{V'_n + B} x_b \quad (17.16)$$

and

$$L'_{n+1} h_{n+1} = (V'_n + B) h_{n+1} = V'_n H_n + B h_b - q_b \quad (17.17)$$

where B is the bottoms flow rate.

At constant pressure, the stage temperatures will be functions of the vapor and liquid compositions only (dew and bubble points), and the specific enthalpies will therefore also be functions of composition

$$H = f(y) \quad (17.18a)$$

$$h = f(x) \quad (17.18b)$$

Lewis–Sorel method (equimolar overflow)

For most distillation problems a simplifying assumption, first proposed by Lewis (1909), can be made that eliminates the need to solve the stage energy-balance equations. The molar liquid and vapor flow rates are taken as constant in the stripping and rectifying sections. This condition is referred to as equimolar overflow: the molar vapor and liquid flows from each stage are constant. This will only be true where the component molar latent heats of vaporization are the same and, together with the specific heats, are constant over the range of temperature in the column; there is no significant heat of mixing; and the heat losses are negligible. These conditions are substantially true for practical systems when the components form near-ideal liquid mixtures.

Even when the latent heats are substantially different, the error introduced by assuming equimolar overflow to calculate the number of stages is often small compared with the error in the stage efficiency and is acceptable.

With equimolar overflow, Equations 17.14 and 17.16 can be written without the subscripts to denote the stage number:

$$y_{n+1} = \frac{L}{L + D} x_n + \frac{D}{L + D} x_d \quad (17.19)$$

$$x_{n+1} = \frac{V'}{V' + B} y_n + \frac{B}{V' + B} x_b \quad (17.20)$$

where L = the constant liquid flow in the rectifying section = the reflux flow, L_0 , and V' is the constant vapor flow in the stripping section.

Equations 17.19 and 17.20 can be written in an alternative form:

$$y_{n+1} = \frac{L}{V} x_n + \frac{D}{V} x_d \quad (17.21)$$

$$y_n = \frac{L'}{V'} x_{n+1} - \frac{B}{V'} x_b \quad (17.22)$$

where V is the constant vapor flow in the rectifying section $= (L + D)$, and L' is the constant liquid flow in the stripping section $= V' + B$.

These equations are linear, with slopes L/V and L'/V' . They are referred to as *operating lines* and give the relationship between the liquid and vapor compositions between stages. For an equilibrium stage, the compositions of the liquid and vapor streams leaving the stage are given by the equilibrium relationship.

17.5.2 McCabe–Thiele method

Equations 17.21 and 17.22 and the equilibrium relationship are conveniently solved by the graphical method developed by [McCabe and Thiele \(1925\)](#). A simple procedure for the construction of the diagram is given next and illustrated in Example 17.2.

Procedure

Refer to [Fig. 17.5](#); all compositions are those of the more volatile component.

1. Plot the vapor–liquid equilibrium curve from data available at the column operating pressure. In terms of relative volatility:

$$y = \frac{\alpha x}{(1 + (\alpha - 1)x)} \quad (17.23)$$

where α is the geometric average relative volatility of the lighter (more volatile) component with respect to the heavier component (less volatile).

It is usually more convenient, and less confusing, to use equal scales for the x - and y -axes.

2. Make a material balance over the column to determine the top and bottom compositions, x_d and x_b , from the data given.
3. The upper and lower operating lines intersect the diagonal at x_d and x_b , respectively; mark these points on the diagram.
4. The point of intersection of the two operating lines is dependent on the phase condition of the feed. The line on which the intersection occurs is called the *q line*. The *q* line is found as follows:
 - i. Calculate the value of the ratio q given by

$$q = \frac{\text{heat to vaporize 1 mol of feed}}{\text{molar latent heat of feed}}$$

- ii. Plot the *q* line, slope $= q/(q - 1)$, intersecting the diagonal at z_f (the feed composition).

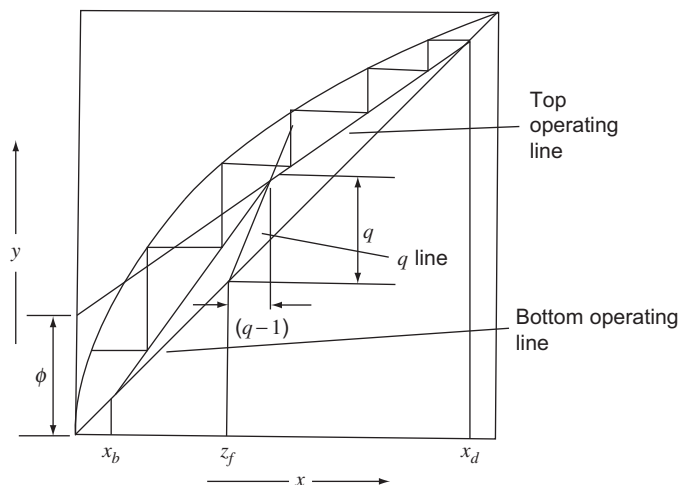


FIG. 17.5 McCabe–Thiele diagram.

5. Select the reflux ratio and determine the point where the upper operating line extended cuts the y-axis:

$$\phi = \frac{x_d}{1 + R} \quad (17.24)$$

6. Draw in the upper operating line (UOL) from x_d on the diagonal to ϕ .
7. Draw in the lower operating line (LOL) from x_b on the diagonal to the point of intersection of the top operating line and the q line.
8. Starting at x_d or x_b , step off the number of stages.

Note: The feed point should be located on the stage closest to the intersection of the operating lines.

The reboiler, and a partial condenser if used, act as equilibrium stages; however, when designing a column there is little point in reducing the estimated number of stages to account for this. Not counting the reboiler as a stage can be considered an additional design margin.

It can be seen from Equation 17.24 and Fig. 17.5 that as R increases, ϕ decreases until the limit is reached where $\phi = 0$ and the upper and lower operating lines both lie along the diagonal, as in Fig. 17.6. This is the total reflux condition in which the minimum number of stages is needed for the separation.

Similarly, as R is reduced, the intersection between the upper and lower operating lines moves away from the diagonal until it reaches the equilibrium line, as illustrated in Fig. 17.7. This is the minimum reflux condition. If the reflux ratio were to be reduced further, then there would be no feasible intersection of the operating lines.

It can also be seen that at minimum reflux the space between the operating and equilibrium lines becomes very small at the intersection point, which is known as a “pinched” condition. An infinite number of trays is required at minimum reflux because of these pinch points, as can be seen in Fig. 17.7. Pinch points also often occur when the relative volatility of the mixture is not constant, particularly when azeotropes or near azeotropes form, as illustrated in Fig. 17.9. Pinch points also occur at the top or bottom of the column if very stringent purity specifications must be met.

Pinch points are also found in multicomponent distillation and are easily visualized as regions where the composition profiles appear to be varying only very slightly from stage to stage. When a pinch point occurs, the solution is usually to increase the reflux or change column pressure to obtain a more favorable equilibrium.

The efficiency of real contacting stages can be accounted for by reducing the height of the steps on the McCabe–Thiele diagram (Fig. 17.8). Stage efficiencies are discussed in Section 17.10.

The McCabe–Thiele method can be used for the design of columns with side streams and multiple feeds. The liquid and vapor flows in the sections between the feed and take-off points are calculated and operating lines drawn for each section.

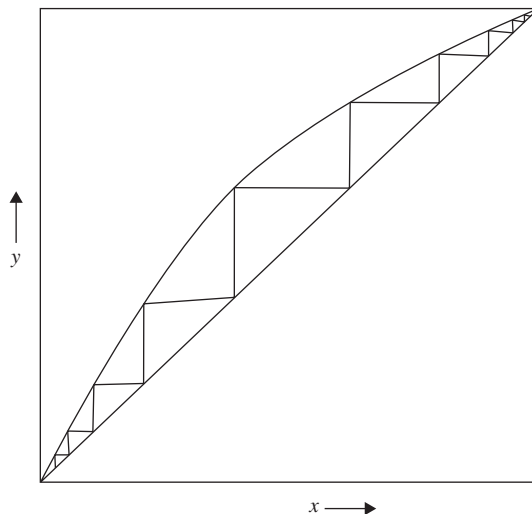


FIG. 17.6 Total reflux.

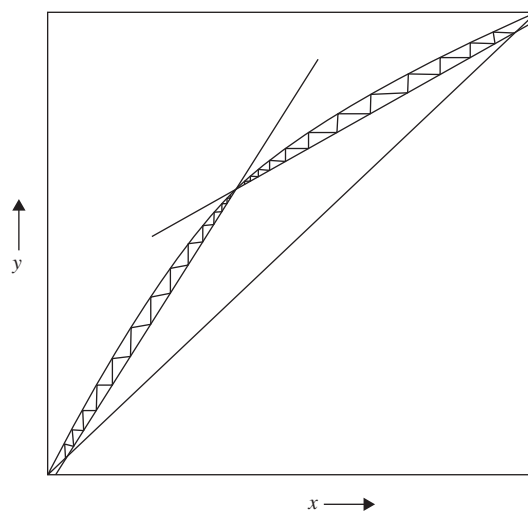


FIG. 17.7 Minimum reflux.

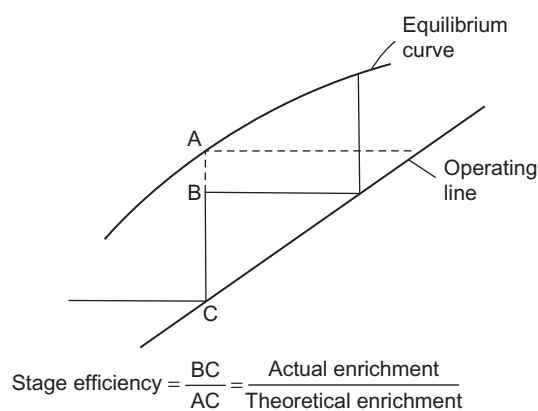


FIG. 17.8 Stage efficiency.

Example 17.2

Acetone is to be recovered from an aqueous waste stream by continuous distillation. The feed contains 10 mol% acetone. Acetone of at least 95 mol% purity is wanted, and the aqueous effluent must not contain more than 1 mol% acetone. The feed will be a saturated liquid. Estimate the number of ideal stages required.

Solution

There is no point in operating this column at other than atmospheric pressure. The equilibrium data of [Kojima et al. \(1968\)](#) will be used.

Mol fraction x , liquid	0.00	0.05	0.10	0.15	0.20	0.25	0.30
Acetone y , vapor	0.00	0.6381	0.7301	0.7716	0.7916	0.8034	0.8124
Bubble point °C	100.0	74.80	68.53	65.26	63.59	62.60	61.87
x	0.35	0.40	0.45	0.50	0.55	0.60	0.65
y	0.8201	0.8269	0.8376	0.8387	0.8455	0.8532	0.8615
°C	61.26	60.75	60.35	59.95	59.54	59.12	58.71
x	0.70	0.75	0.80	0.85	0.90	0.95	
y	0.8712	0.8817	0.8950	0.9118	0.9335	0.9627	
°C	58.29	57.90	57.49	57.08	56.68	56.30	

The equilibrium curve can be drawn with sufficient accuracy to determine the stages above the feed by plotting the concentrations at increments of 0.1.

Following the procedure given here, we can mark the product compositions.

Because the feed is a saturated liquid, $q = 1$, and the slope of the q line is $1/(1 - 1) = \infty$. The q line is thus plotted as a vertical line through the feed composition.

For this problem, the condition of minimum reflux occurs where the top operating line just touches the equilibrium curve at the point where the q line cuts the curve.

From Fig. 17.9,

$$\Phi \text{ for the operating line at minimum reflux} = 0.59$$

From Equation 17.24, $R_{\min} = 0.95/0.59 - 1 = 0.62$

Take $R = R_{\min} \times 2 = 1.24$

As the flows above the feed point will be small, a high reflux ratio is justified; the condenser duty will be small.

$$\phi = \frac{0.95}{1 + 1.24} = 0.42$$

We can then plot the UOL and LOL.

Stepping off from the bottom, we require two stages in the stripping section and an additional eight stages in the rectifying section. The feed should be on the third stage from the bottom. Note that the number of stages below the feed is small, and because the reboiler counts as an equilibrium stage, if we feed onto the third plate from the bottom, we would actually be feeding onto the stage marked 4 in Fig. 17.9. This feed would be badly mismatched with the vapor and liquid compositions in the column, so it would be advisable to allow for possible feeds to the second and third plates from the bottom so that the feed point can be moved. Note also that the column is close to pinched at the top of the rectifying section. It would be prudent to add extra trays in this section.

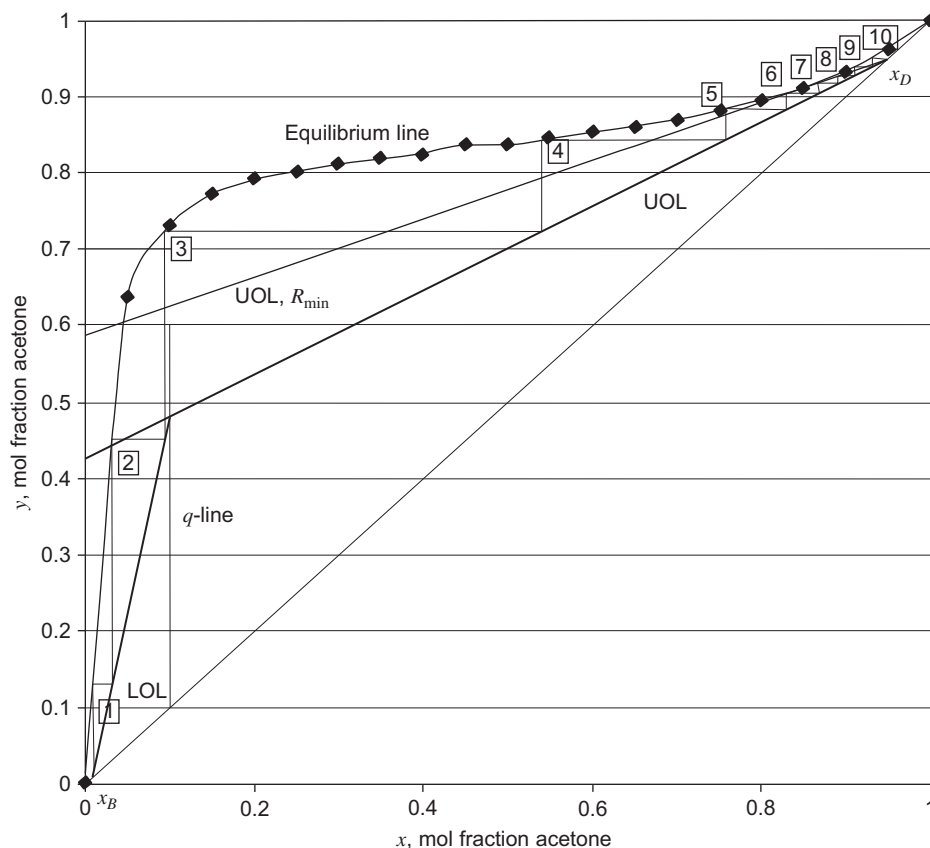


FIG. 17.9 McCabe–Thiele plot, Example 17.2.

17.6 Multicomponent distillation: General considerations

The problem of determining the stage and reflux requirements for multicomponent distillations is much more complex than for binary mixtures. With a multicomponent mixture, fixing one component composition does not uniquely determine the other component compositions and the stage temperature. Also when the feed contains more than two components, it is not possible to specify the complete composition of the top and bottom products independently. The separation between the top and bottom products is usually specified by setting limits on two “key” components, between which it is desired to make the separation.

The complexity of multicomponent distillation calculations can be appreciated by considering a typical problem. The normal procedure is to solve the MESH equations (Section 17.3.1), stage by stage, from the top or bottom of the column. For such a calculation to converge to a solution, the compositions obtained from either the bottom-up or top-down calculations must match the composition at the other end of the column predicted from an overall material balance. But the calculated compositions will depend on the compositions assumed for the top or bottom products at the start of the calculations. Though it is possible to match the key components, the other components will not match unless the designer was particularly fortunate in choosing the trial top and bottom compositions. For a completely rigorous solution, the compositions must be adjusted and the calculations repeated until satisfactory convergence is obtained. Clearly, the greater the number of components considered, the more difficult the problem. As was shown in Section 17.3.2, iterative calculations will be needed to determine the stage temperatures. For other than ideal mixtures, the calculations will be further complicated by the fact that the component volatilities will be functions of the unknown stage compositions and temperatures. The relationship between volatility, composition, and temperature may be highly nonlinear, as discussed in [Chapter 4](#). If more than a few stages are required, stage-by-stage calculations are complex and tedious, and even with sophisticated process simulation programs convergence cannot be guaranteed.

Before the widespread availability of digital computers, various “shortcut” methods were developed to simplify the task of designing multicomponent columns. A comprehensive summary of the methods used for hydrocarbon systems is given by Edmister (1947 to 1949) in a series of articles in the journal *The Petroleum Engineer*. Though computer programs will normally be available for the rigorous solution of the MESH equations, shortcut methods are still useful in preliminary design work and as an aid in initializing problems for computer solution.

17.6.1 Key components

Before starting the column design, the designer must select the two *key* components between which it is desired to make the separation. The light key will be the component that it is desired to keep out of the bottom product, and the heavy key the component to be kept out of the top product. The keys are known as *adjacent keys* if they are “adjacent” in a listing of the components in order of volatility, and *split keys* or *nonadjacent keys* if some other component lies between them in the volatility order. A separation between adjacent keys is known as a *sharp split*, whereas a separation with nonadjacent keys is known as a *sloppy split*.

The choice of key components will normally be clear, but sometimes, particularly if close boiling isomers are present, judgment must be used in their selection. If any uncertainty exists, trial calculations should be made using different components as the keys to determine the pair that requires the largest number of stages for separation (the worst case). The Fenske equation can be used for these calculations; see Section 17.7.1.

The *nonkey* components that appear in both top and bottom products are known as *distributed* components; and those that are not present, to any significant extent, in one or other product, are known as *nondistributed* components.

17.6.2 Product specifications

Specifications for the column will normally be set in terms of the purity or recovery of the key components.

A purity specification sets the mole (or mass) fraction of a component in one of the product streams. Purity specifications are easily understood and are easy to relate to the required product specifications. For example, if the standard specification for the desired grade of product is 99.5% pure, then the designer could specify 99.5% purity of the product in the distillate of a finishing column. Similarly, if the purity of the heavy key component in the distillate must be less than 50 ppm, then this can be used as the specification.

Although purity specifications are intuitively obvious, their use often leads to infeasible column specifications. The designer must check carefully to ensure that the amounts of lighter-than-light-key (or heavier-than-heavy-key) components are not large enough to render the purity specification infeasible. Consider, for example, a feed that contains 0.5 mol% A, 49.5 mol% B, and 50 mol% C, where A is most volatile and C is least volatile. The highest purity of B that can be obtained in the distillate is 99%, and that would require complete recovery of B and complete rejection of C from the distillate, with a very high reflux ratio. If only 99% of the C is recovered in the bottoms product and only 99% of the B is recovered overhead, then the maximum feasible purity of B is $0.99(49.5) / [0.5 + 0.99(49.5) + 0.5] = 98\%$.

When a purity specification must be met in a column that forms part of a multicomponent sequence of columns, it is essential to set the specifications of the other columns so that the desired purity specification is feasible.

Instead of specifying purity, the designer can specify the recovery of one or more of the key components in either the distillate or bottoms. The recovery of a component in a product stream is defined as the fraction of the feed molar flow rate of the component that is recovered in that product stream. The relationship between purity and recovery is often not simple, particularly when many components are present or the key components are nonadjacent.

Recovery specifications are easily related to economic trade-offs, because the value of recovering an additional 0.1% or 0.01% of the desired product is easily assessed and can be traded off against the additional capital and operating costs of the column. Recovery of the desired product is usually set at 99% or greater. Recovery specifications for a distillation column are less likely to be infeasible than purity specifications; however, their use does not guarantee that the product will meet the specifications required for sale.

In general, it is better to use purity specifications for the final column that produces product (usually termed the *finishing column*) and recovery specifications elsewhere in the distillation sequence. A combination of purity and recovery specifications can also be used for a single column. For example, in a finishing column in which the desired product is taken as distillate, the designer could specify the purity and recovery of the desired (light key) component in the distillate. No specifications on the heavy key or other components are needed, but the same feasibility checks must be made for the purity specification.

In mixtures that form azeotropes, the volatility order changes with composition. This creates additional problems when setting product specifications. The design of azeotropic distillation sequences is discussed in Section 17.6.5.

Most process simulation programs allow the designer to select two specifications for a distillation column corresponding to the two remaining degrees of freedom once the feed rate, pressure, number of stages, and feed stage have been selected. If the column is to be designed to achieve a given purity or recovery, then it obviously makes sense to use this as a specification if the simulation program allows it, but the designer may need to provide an initial estimate of other parameters such as reflux ratio or distillate rate to ensure good convergence. Estimates of these parameters can be made using shortcut calculations or shortcut column models, as described in Section 17.7. The use of shortcut models to initialize a rigorous simulation model was discussed in Section 4.5.2 and illustrated in Example 4.3. Example 4.4 showed the effect of changing to a recovery specification in a rigorous solution of the same problem.

In some cases, the simulation program or model may not allow the use of purity or recovery specifications, in which case the designer must adjust other variables such as reflux rate, boil-up rate and distillate, or bottoms flow rate until the specifications are met.

17.6.3 Number and sequencing of columns

In multicomponent distillation the production of a pure product usually requires at least two distillation columns. A common arrangement is to remove all components lighter than the desired product in a first column, then separate the desired product from heavier components in a second column. This arrangement is illustrated in Fig. 17.10, and

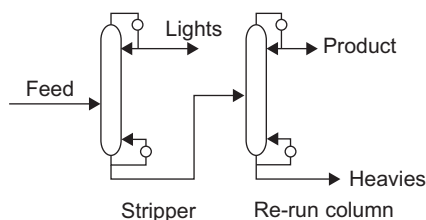


FIG. 17.10 Stripper and re-run column.

is known as a stripper and re-run column sequence. Because almost all processes produce some by-products that are lighter and some that are heavier than the desired product, this scheme is widely used.

If additional pure products are to be produced, additional columns will be needed. The recovery of an additional pure component from the lights stream in Fig. 17.10 would require one more column if the component were the least volatile component in the lights (i.e., the light key component of the first column in Fig. 17.10), or two more columns if there were any additional components that boiled between the two desired products.

If a mixture contained N components and the designer wanted to separate it into pure components, then $N - 1$ columns would be needed, as each component could be removed in order of volatility until the final binary pair remained. If only M pure products are required, then the number of columns needed is generally bounded between $M + 1$ and either $2M$ or $N - 1$, whichever is least. It might be expected that the minimum number of columns would be M , but the presence of small amounts of light or heavy components invariably requires the designer to use at least one extra column to enable the product purity specification to be achieved.

The distillation sequence in which the components are separated in order of decreasing volatility is known as a *direct sequence* and is shown in Fig. 17.11(a). There are many other possible distillation sequences. Fig. 17.11(b) shows an *indirect sequence* in which the heaviest component is removed first and the distillate is fed to the second column. Components are then removed in order from least volatile to most volatile. Fig. 17.11(c) shows a mixed sequence in which the first separation is between components in the middle of the volatility order.

With five components, there are 14 possible column configurations. As the number of components increases, the number of possible column sequences increases combinatorially. With 10 components there are nearly 5000 possible schemes. The optimum scheme will be the one that has the best overall economic performance.

Various methods have been proposed for screening alternative designs to determine the optimum sequence; see Doherty and Malone (2001), Smith (2016), and Kumar (1982). These methods usually use shortcut column models and approximate costing relationships, so it is often worthwhile to complete detailed designs for a few of the best candidate schemes identified. There can also be strong interactions between the column sequence and the associated heat integration that can influence the final scheme that is selected.

Although distillation column sequencing is an interesting research problem, in practice there are very few processes that make more than two or three pure products. The optimum sequence can often be determined using heuristic rules such as:

1. Remove corrosive components first to avoid using expensive metallurgy throughout the sequence.
2. Remove the heaviest component first if solids are present in the feed. The presence of solids requires the use of special plates that are designed to resist plugging and have very low-stage efficiency. It is best to get the solids out of the way as early as possible.
3. Split any components that cannot be condensed using cooling water from those that can be condensed early in the sequence. The lighter components can then be compressed to higher pressure, separated using absorption or adsorption, or separated in refrigerated columns. This rule avoids the use of refrigerated condensers, higher pressures, or partial condensers elsewhere in the sequence.
4. Postpone the most difficult separation, such as between close-boiling compounds, until late in the sequence. A difficult separation will require many stages and high reflux, and so the feed rate to that column should be made as low as possible so that the column handles less material.
5. Take the desired products as distillates whenever practical to avoid any flushing of dirt or debris into the desired product. The same rule also applies to recycle streams.
6. Remove any components that are present in large excess early in the sequence to make the downstream columns cheaper.

Tall columns

Where a large number of stages is required, it may be necessary to split a column into two separate columns to reduce the height of the column, even though the required separation could, theoretically, have been obtained in a single column. This may also be done in vacuum distillations to reduce the column pressure drop and limit the bottom temperatures.

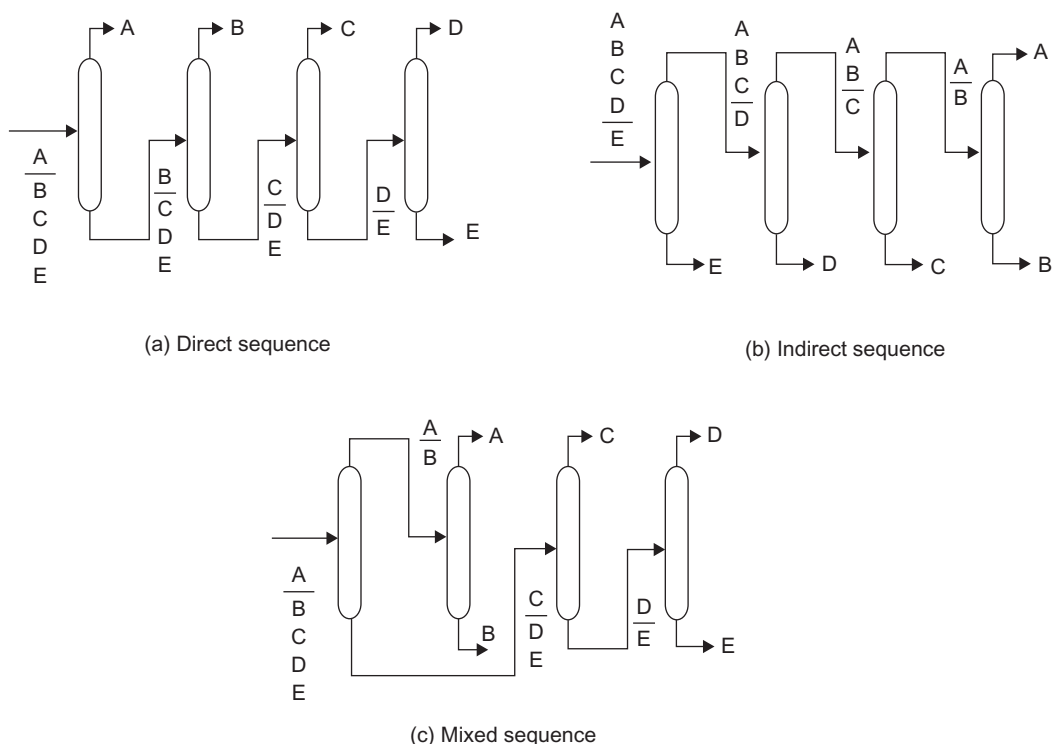


FIG. 17.11 Column sequences for a five-component mixture.

17.6.4 Complex columns

It is relatively easy to withdraw side streams from plate columns and to supply additional feeds to the column. If a liquid side stream is withdrawn from a tray above the feed, as shown in Fig. 17.12(a), it will be depleted in the heavier components of the feed (which preferentially stayed in the liquid phase and went down in the stripping section) and will also be depleted in the lighter components of the feed (which are preferentially in the vapor phase). Although the side stream will not be pure, it will be enriched in some of the components of midrange volatility. In some cases, the purity of the side stream may be adequate, for example, if the side stream is a process recycle.

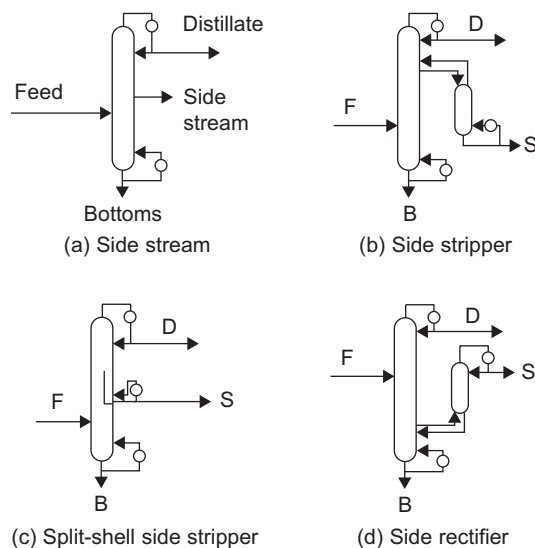


FIG. 17.12 Side streams and side columns.

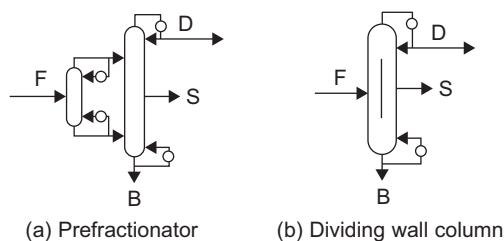


FIG. 17.13 Complex column designs.

The purity of a desired component in the side stream can be increased by sending the side stream to a small side stripper column that strips out any lighter components, as shown in Fig. 17.12(b). The vapor from the side stripper is returned to the main column. A side stripper can be constructed as part of the main column by using a partitioned section of the main column, as shown in Fig. 17.12(c). Side rectifiers are also used; see Fig. 17.12(d).

Side strippers and rectifiers allow up to three pure products to be made in one and a half columns, or in a single shell if a partition wall is used. Other complex column configurations are also possible, such as prefractionators and dividing wall columns, illustrated in Fig. 17.13. These complex columns generally have lower capital and operating costs than sequences of simple columns. More degrees of freedom are introduced into the design, so more care is needed in optimizing the columns. Smith (2016) gives an excellent introduction to the design of complex columns. Greene (2001), Schultz et al. (2002), Kaibel (2002), and Parkinson (2007) describe industrial applications of dividing wall columns. Side strippers are widely used in petroleum refining; see Watkins (1979). Most process simulators allow the designer to add side strippers and rectifiers or select from a set of prebuilt complex column models.

17.6.5 Distillation column sequencing for azeotropic mixtures

When a mixture forms an azeotrope, determining the best column sequence is not straightforward. Homogeneous azeotropes are mixtures of two or more components that have the same vapor- and liquid-phase composition at the boiling point. Heterogeneous azeotropes have two liquid phases that are in equilibrium with a vapor that has the same composition as the combined liquid composition at the boiling point. Different strategies are used for separation depending on the type of azeotrope.

The design of azeotropic distillation sequences has been the subject of much academic research, and there is not sufficient space here to describe all of the techniques that have been developed. The reader should refer to Smith (2016) and Doherty and Malone (2001) for a more detailed treatment of the subject. The general strategy for separating an azeotropic mixture can be summarized as:

1. If the azeotrope is heterogeneous, use a liquid–liquid split (decanter). The two liquid phases will usually have compositions on either side of the azeotrope. Each of these liquids can be distilled to give a pure product and the azeotrope, and the azeotropic mixtures can be recycled to the decanter. In some cases a third component, known as an entrainer, is added to cause the formation of a heterogeneous azeotropic mixture. The degree of separation in the liquid–liquid split can often be increased by lowering the temperature, which tends to increase the size of the two-phase region in the composition space.

For example, Fig. 17.14 shows the separation of an ethanol–water mixture using benzene as entrainer. The ethanol–water mixture is distilled to give a low-boiling azeotrope, which is then sent to a first column that is refluxed with the oil phase from a decanter. The first column produces ethanol as bottom product and a heterogeneous azeotrope as distillate. The distillate is sent to a decanter and separated into oil-rich and water-rich phases. The water phase is sent to a second column that produces water as bottoms product and heterogeneous azeotrope as distillate. The distillate from the second column is also sent to the decanter. This flow scheme was widely used for ethanol dehydration until cheaper and safer processes based on adsorbing the water using molecular sieves were introduced.

This flow scheme is sometimes known as *azeotropic distillation*; however, the term *heterogeneous azeotropic distillation* is better, as all of the other methods also involve distilling azeotropes.

2. If the azeotrope is homogeneous, then the effect of varying pressure should be investigated. The composition of the azeotrope is always pressure dependent. If there is a large change in composition over a reasonable range of pressure, two columns at different pressures can be used. Each column produces a pure product and a mixture corresponding to the azeotrope at the pressure of that column. The azeotropic mixture is then fed to the other

column, as illustrated in Fig. 17.15. The mixture from the low-pressure column must be pumped back up to the pressure of the high-pressure column. Note that the feed can be to either column and that it is also possible to produce the products as distillates if the azeotrope is maximum-boiling rather than minimum-boiling.

The pressure-swing distillation flow scheme is relatively simple and does not require any additional components to be added, but if the azeotrope composition is only weakly sensitive to pressure, the recycle from the low-pressure column to the high-pressure column will be large. The recycled material must be vaporized in the low-pressure column, so the low-pressure column can become quite expensive.

3. If pressure-swing distillation is not economically attractive, consider adding an entrainer. The preferred entrainers are usually those that form heterogeneous azeotropes, as discussed earlier, but homogeneous entrainers can also be used, in which case the process is known as extractive distillation.

The most commonly used type of entrainer is a higher boiling compound that does not form an azeotrope with either component of the azeotropic pair. If a high-boiling entrainer is used, it depresses the volatility of one component of the azeotrope, allowing the other component to be recovered in the distillate. The bottoms from the first column are then sent to a second column in which the other pure component is recovered as distillate and the entrainer is recovered as bottoms for recycle to the first column, as shown in Fig. 17.16. Other schemes with low-boiling or mid-boiling entrainers are also possible, as described by Doherty and Malone (2001).

The entrainer should be selected from compounds that are already present within the process whenever possible. The use of compounds that are already present reduces consumables costs and waste formation and usually makes it easier to reach product specifications. The other compounds that are present in the process can be screened for suitability as entrainers by looking at boiling points and checking for the formation of additional azeotropes. If nothing suitable is found, the more sophisticated methods for evaluating entrainers described by Doherty and Malone (2001) should be used.

4. If the azeotropic composition is close to the required purity specification, consider removing the minor component by adsorption using a selective sorbent. If a regenerable sorbent can be found, this process may be cheaper than a multicolumn distillation. Pressure-swing adsorption using a molecular sieve sorbent as drying agent is now the most widely used method for breaking the ethanol–water azeotrope. Adsorption processes are discussed in Chapter 16.
5. If a suitable membrane material can be found that is permeable to one component of the mixture but impermeable to the other, then membrane separation can be used in combination with distillation. A typical flowsheet for the case where the light component is permeable is shown in Fig. 17.17. A distillation column is used to separate the mixture of A and B into pure heavy component B and azeotrope. The azeotropic mixture is sent to the membrane unit, where pure light component A is recovered. It is usually not economical to operate a membrane unit at high recovery of permeate, so the retentate still contains a significant fraction of A and is recycled to the distillation column. If the membrane is not impermeable to component B, a permeate stream that is enriched in A can be sent to a second distillation column that then produces pure A and an azeotropic mixture for recycle to the membrane unit. The design of membrane processes is described in Chapter 16.

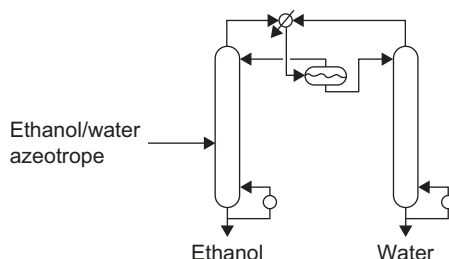


FIG. 17.14 Dehydration of ethanol using benzene as entrainer.

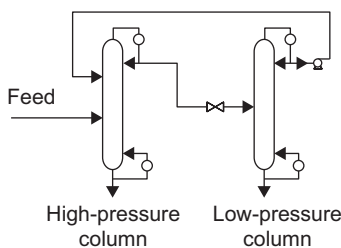


FIG. 17.15 Pressure-swing distillation.

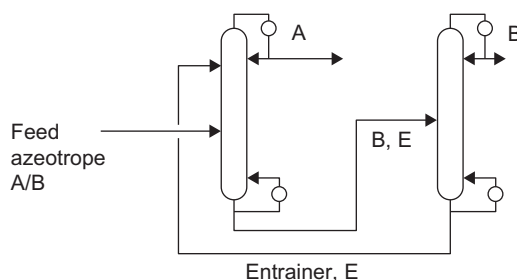


FIG. 17.16 Extractive distillation.

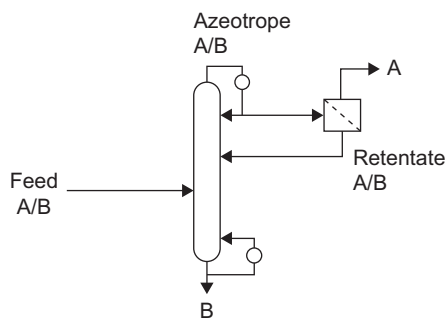


FIG. 17.17 Membrane distillation.

17.7 Multicomponent distillation: Shortcut methods for stage and reflux requirements

Some of the more useful shortcut procedures that can be used to estimate stage and reflux requirements without the aid of computers are given in this section. Most of the shortcut methods were developed for the design of separation columns for hydrocarbon systems in the petroleum and petrochemical industries, and caution must be exercised when applying them to other systems. They usually depend on the assumption of constant relative volatility and should not be used for severely nonideal systems. Shortcut methods for nonideal and azeotropic systems are given by Featherstone (1971, 1973).

Although the shortcut methods were developed for hand calculations, they are easily coded into spreadsheets and are available as subroutines in all the commercial process simulation programs. The shortcut methods are useful when configuring rigorous distillation models, as described in Section 4.5.2.

The two most frequently used empirical methods for estimating the stage requirements for multicomponent distillations are the correlations published by Gilliland (1940) and by Erbar and Maddox (1961). These relate the number of ideal stages required for a given separation, at a given reflux ratio, to the number at total reflux (minimum possible) and the minimum reflux ratio (infinite number of stages). The Erbar–Maddox correlation is given in this section, as it is now generally considered to give more reliable predictions than Gilliland’s correlation. The Erbar–Maddox correlation is shown in Fig. 17.18, which gives the ratio of number of stages required to the number at total reflux, as a function of the reflux ratio, with the minimum reflux ratio as a parameter. To use Fig. 17.18, estimates of the number of stages at total reflux and the minimum reflux ratio are needed.

17.7.1 Minimum number of stages (Fenske equation)

The Fenske equation (Fenske, 1932) can be used to estimate the minimum stages required at total reflux. The derivation of this equation for a binary system is given in Chhabra and Gurappa (2019). The equation applies equally to multicomponent systems and can be written as:

$$\left[\frac{x_i}{x_r} \right]_d = \alpha_i^{N_{\min}} \left[\frac{x_i}{x_r} \right]_b \quad (17.25)$$

where $[x_i/x_r]$ = the ratio of the concentration of any component i to the concentration of a reference component r , and the suffixes d and b denote the distillate (d) and the bottoms (b)

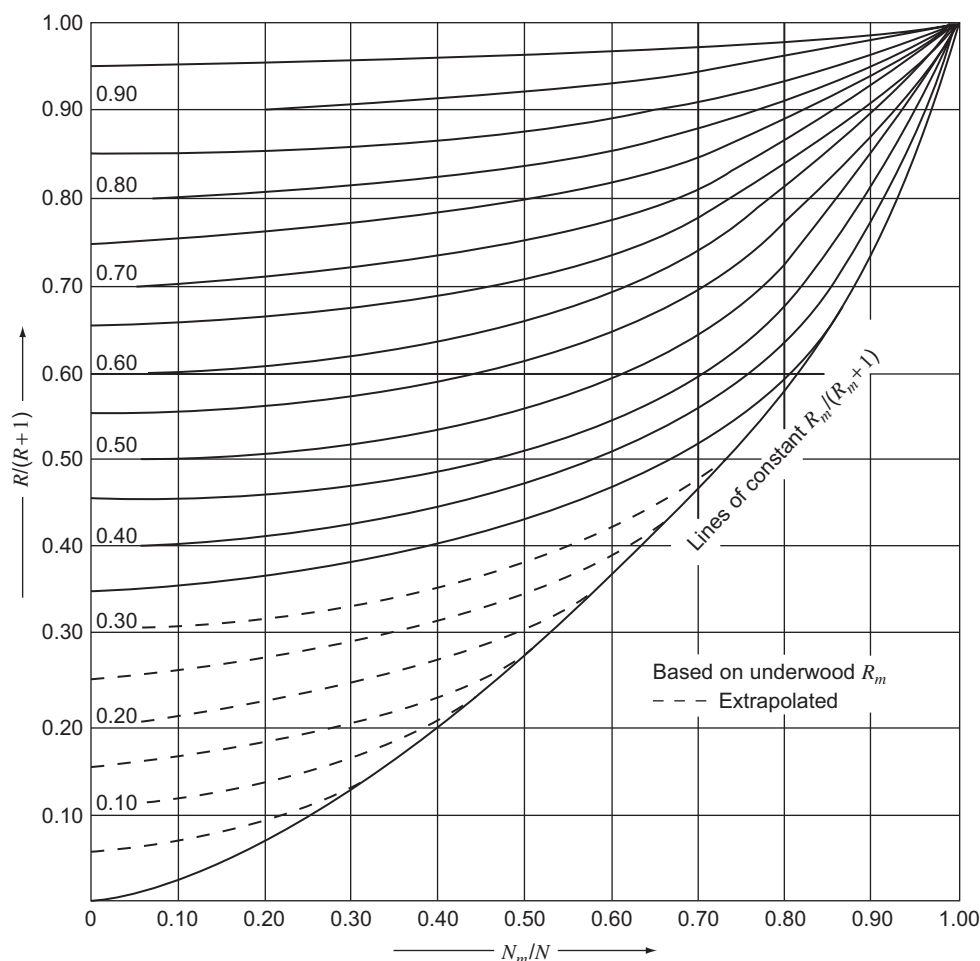


FIG. 17.18 Erbar-Maddox correlation. (From Erbar & Maddox, 1961.)

N_{min} = minimum number of stages at total reflux, including the reboiler

α_i = average relative volatility of the component i with respect to the reference component

Normally the separation required will be specified in terms of the key components, and Equation 17.25 can be rearranged to give an estimate of the number of stages.

$$N_{min} = \frac{\log \left[\frac{x_{LK}}{x_{HK}} \right]_d \left[\frac{x_{HK}}{x_{LK}} \right]_b}{\log \alpha_{LK}} \quad (17.26)$$

where α_{LK} is the average relative volatility of the light key with respect to the heavy key, and x_{LK} and x_{HK} are the light and heavy key concentrations. The relative volatility is taken as the geometric mean of the values at the column top and bottom temperatures. To calculate these temperatures, initial estimates of the compositions must be made, so the calculation of the minimum number of stages by the Fenske equation is a trial-and-error procedure. The procedure is illustrated in Example 17.3. If there is a wide difference between the relative volatilities at the top and bottom of the column, the use of the average value in the Fenske equation will underestimate the number of stages. In these circumstances, a better estimate can be made by calculating the number of stages in the rectifying and stripping sections separately; taking the feed concentration as the base concentration for the rectifying section and as the top concentration for the stripping section, and estimating the average relative volatilities separately for each section. This procedure will also give an estimate of the feed point location.

Winn (1958) has derived an equation for estimating the number of stages at total reflux, which is similar to the Fenske equation, but which can be used when the relative volatility cannot be taken as constant.

If the number of stages is known, Equation 17.25 can be used to estimate the split of components between the top and bottom of the column at total reflux. It can be written in a more convenient form for calculating the split of components:

$$\frac{d_i}{b_i} = \alpha_i^{N_{\min}} \left[\frac{d_r}{b_r} \right] \quad (17.27)$$

where d_i and b_i are the flow rates of the component i in the distillate and bottoms and d_r and b_r are the flow rates of the reference component in the distillate and bottoms.

Note: From the column material balance:

$$d_i + b_i = f_i$$

where f_i is the flow rate of component i in the feed.

17.7.2 Minimum reflux ratio

Colburn (1941) and Underwood (1948) have derived equations for estimating the minimum reflux ratio for multicomponent distillations. As the Underwood equation is more widely used, it is presented in this section. The equation can be stated in the form:

$$\sum \frac{\alpha_i x_{i,d}}{\alpha_i - \theta} = R_{\min} + 1 \quad (17.28)$$

where α_i = the relative volatility of component i with respect to some reference component, usually the heavy key

$R_{\min}$ = the minimum reflux ratio

$x_{i,d}$ = concentration of component i in the distillate at minimum reflux

θ = the root of the equation:

$$\sum \frac{\alpha_i x_{i,f}}{\alpha_i - \theta} = 1 - q \quad (17.29)$$

where $x_{i,f}$ = the concentration of component i in the feed, and q depends on the condition of the feed and was defined in Section 17.5.2.

The value of θ must lie between the values of the relative volatility of the light and heavy keys and is found by trial and error.

In the derivation of Equations 17.28 and 17.29, the relative volatilities are taken as constant. The geometric average of values estimated at the top and bottom temperatures should be used. This requires an estimate of the top and bottom compositions. Though the compositions should strictly be those at minimum reflux, the values determined at total reflux, from the Fenske equation, can be used. A better estimate can be obtained by replacing the number of stages at total reflux in Equation 17.27 by an estimate of the actual number; a value equal to $N_{\min}/0.6$ is often used. The Erbar–Maddox method of estimating the stage and reflux requirements, using the Fenske and Underwood equations, is illustrated in Example 17.3.

17.7.3 Feed-point location

A limitation of the Erbar–Maddox and similar empirical methods is that they do not give the feed-point location. An estimate can be made by using the Fenske equation to calculate the number of stages in the rectifying and stripping sections separately, but this requires an estimate of the feed-point temperature. An alternative approach is to use the empirical equation given by Kirkbride (1944):

$$\log \left[\frac{N_r}{N_s} \right] = 0.206 \log \left[\left(\frac{B}{D} \right) \left(\frac{x_{f,\text{HK}}}{x_{f,\text{LK}}} \right) \left(\frac{x_{b,\text{LK}}}{x_{d,\text{HK}}} \right)^2 \right] \quad (17.30)$$

where N_r = number of stages above the feed, including any partial condenser

N_s = number of stages below the feed, including the reboiler

$x_{f,\text{HK}}$ = concentration of the heavy key in the feed

$x_{f,\text{LK}}$ = concentration of the light key in the feed

$x_{d,\text{HK}}$ = concentration of the heavy key in the top product

$x_{b,\text{LK}}$ = concentration of the light key if in the bottom product

The use of this equation is illustrated in Example 17.4.

Example 17.3

Estimate the minimum number of ideal stages needed in the butane–pentane splitter defined by the compositions given in the following table. The column will operate at a pressure of 8.3 bar. Evaluate the effect of changes in reflux ratio on the number of stages required. This is an example of the application of the Erbar–Maddox method. The feed is at its boiling point.

		Feed (<i>f</i>)	Distillate (<i>d</i>)	Bottoms (<i>b</i>)
Propane, C ₃	5	5	0	
i-Butane, iC ₄	15	15	0	
n-Butane, nC ₄	25	24	1	
i-Pentane, iC ₅	20	1	19	
n-Pentane, nC ₅	35	0	35	
		100	45	55 kmol

Solution

The top and bottom temperatures (dew points and bubble points) were calculated by the method given in Section 17.3.2. Relative volatilities are given by:

$$\alpha_i = \frac{K_i}{K_{HK}}$$

Equilibrium constants were taken from De Priester charts (Dadyburjor, 1978).

Relative volatilities

	Top	Bottom	Average
Temp. °C	65	120	
C ₃	5.5	4.5	5.0
iC ₄	2.7	2.5	2.6
(LK) nC ₄	2.1	2.0	2.0
(HK) iC ₅	1.0	1.0	1.0
nC ₅	0.84	0.85	0.85

Minimum number of stages; Fenske equation, Equation 17.26:

$$N_{\min} = \frac{\log \left[\frac{24}{1} \right] \left[\frac{19}{1} \right]}{\log 2} = \underline{\underline{8.8}}$$

Minimum reflux ratio; Underwood Equations 17.28 and 17.29.

This calculation is best tabulated.

As the feed is at its boiling point $q = 1$:

$$\sum \frac{\alpha_i x_{i,f}}{\alpha_i - \theta} = 0 \quad (17.29)$$

Try

$x_{i,f}$	α_i	$\alpha_i x_{i,f}$	$\theta = 1.5$	$\theta = 1.3$	$\theta = 1.35$
0.05	5	0.25	0.071	0.068	0.068
0.15	2.6	0.39	0.355	0.300	0.312
0.25	2.0	0.50	1.000	0.714	0.769
0.20	1	0.20	−0.400	−0.667	−0.571
0.35	0.85	0.30	−0.462	−0.667	−0.600
			$\Sigma = 0.564$	−0.252	0.022
					close enough

$\theta = 1.35$

Equation 17.28:

$x_{i,d}$	α_i	$\alpha_i x_{i,d}$	$\alpha_i x_{i,d} / (\alpha_i - \theta)$
0.11	5	0.55	0.15
0.33	2.6	0.86	0.69
0.53	2.0	1.08	1.66

$x_{i,d}$	α_i	$\alpha_i x_{i,d}$	$\alpha_i x_{i,d}/(\alpha_i - \theta)$
0.02	1	0.02	-0.06
0.01	0.85	0.01	-0.02
			$\Sigma = 2.42$

$$R_m + 1 = 2.42$$

$$R_m = \underline{1.42}$$

$$\frac{R_m}{(R_m + 1)} = \frac{1.42}{2.42} = 0.59$$

Specimen calculation for $R = 2.0$:

$$\frac{R}{(R + 1)} = \frac{2}{3} = 0.66$$

From Fig. 17.18:

$$\frac{N_{\min}}{N} = 0.56$$

$$N = \frac{8.8}{0.56} = \underline{15.7}$$

For other reflux ratios:

R	2	3	4	5	6
N	15.7	11.9	10.7	10.4	10.1

Note: The number of stages should be rounded up to the nearest integer. Above a reflux ratio of 4 there is little change in the number of stages required, but given the low number of theoretical stages needed, the optimum reflux ratio is probably less than 2.0.

Example 17.4

Estimate the position of the feed point for the separation considered in Example 17.3 for a reflux ratio of 3.

Solution

Use the Kirkbride equation, Equation 17.30. Product distributions are taken from Example 17.3, though they could be confirmed using Equation 17.27:

$$x_{b,LK} = \frac{1}{55} = 0.018$$

$$x_{d,HK} = \frac{1}{45} = 0.022$$

$$\log \left(\frac{N_r}{N_s} \right) = 0.206 \log \left[\frac{55}{45} \left(\frac{0.2}{0.25} \right) \left(\frac{0.018}{0.022} \right)^2 \right]$$

$$\log \left(\frac{N_r}{N_s} \right) = 0.206 \log (0.65)$$

$$\left(\frac{N_r}{N_s} \right) = \underline{0.91}$$

For $R = 3$, $N = 12$

Number of stages, excluding the reboiler = 11 :

$$N_r + N_s = 11$$

$$N_s = 11 - N_r = 11 - 0.91N_s$$

$$N_s = \frac{11}{1.91} = 5.76, \text{ say } \underline{6}$$

17.8 Multicomponent distillation: Rigorous solution procedures (computer methods)

The rigorous column models in the commercial process simulation programs solve the full set of MESH equations (Section 17.3.1). A considerable amount of work has been done to develop efficient and reliable computer-aided design procedures for distillation and other staged processes. A detailed discussion of this work is beyond the scope of this book, and the reader is referred to the specialist books that have been published on the subject: [Smith \(1963\)](#), [Holland \(1997\)](#), and [Kister \(1992\)](#), and to the numerous papers that have appeared in the chemical engineering literature. A good summary of the present state of the art is given by [Haas \(1992\)](#). In this section only a brief outline of the methods that have been developed will be given.

The basic steps in any rigorous solution procedure will be:

1. Specification of the problem; complete specification is essential for computer methods.
2. Selection of values for the iteration variables; for example, estimated stage temperatures and liquid and vapor flows (the column temperature and flow profiles).
3. A calculation procedure for the solution of the stage equations.
4. A procedure for the selection of new values for the iteration variables for each set of trial calculations.
5. A procedure to test for convergence; to check if a satisfactory solution has been achieved.

Rating and design methods

All the methods described here require the specification of the number of stages below and above the feed point. They are therefore not directly applicable to design: where the designer wants to determine the number of stages required for a specified separation. They are strictly what are referred to as “rating methods” and are used to determine the performance of existing, or specified, columns. Given the number of stages, they can be used to determine product compositions. Iterative procedures are necessary to apply rating methods to the design of new columns. Shortcut models can be used to generate initial estimates of the number of stages and feed stage, as described earlier and in Section 4.5.2. If a good initial estimate is provided, the rigorous model should converge faster and can be used to size and optimize the column.

17.8.1 Linear algebra (simultaneous) methods

If the equilibrium relationships and flow rates are known (or assumed), the set of material balance equations for each component is linear in the component compositions. [Amundson and Pontinen \(1958\)](#) developed a method in which these equations are solved simultaneously and the results used to provide improved estimates of the temperature and flow profiles. The set of equations can be expressed in matrix form and solved using standard inversion routines. Convergence can usually be achieved after a few iterations and can be improved by use of Newton’s method.

This approach has been further developed by other workers, notably [Wang and Henke \(1966\)](#) and [Naphtali and Sandholm \(1971\)](#). The Naphtali and Sandholm method for solving rigorous columns is available in many commercial simulation programs.

17.8.2 Inside-out algorithms

The inside-out algorithms accelerate convergence by decomposing the solution of the MESH equations into two nested iteration loops. The method was initially proposed by [Boston and Sullivan \(1974\)](#) and has undergone many improvements; see [Boston \(1980\)](#).

The outer iteration loop determines local estimates of K values and stream enthalpies using models that depend on composition and temperature. The local model parameters are the iteration variables for the outside loop. The initial estimates for the outside loop come from the initial estimate of composition and temperature profile supplied by the user.

The inner iteration loop contains the MESH equations, expressed in terms of the local physical property parameters obtained from the outer loop. With simplified physical property models, the inner loop can be converged more quickly. Convergence methods such as bounded Wegstein and Broyden quasi-Newton are typically used, as described in Section 4.7.2.

When the inside loop is converged, the new estimates of composition and temperature are used to update the outer loop parameters. The convergence tolerance of the inside loop is usually tightened at each iteration of the outside loop. The outer loop converges when the changes in local model parameters are within a satisfactory tolerance from one iteration to the next.

All of the commercial process simulation programs offer inside-out algorithms, and some offer several variants that use different convergence methods. Inside-out algorithms are very effective if good initial estimates are provided. Because of their robust and rapid convergence, they are usually the default methods recommended by the simulation program.

Inside-out algorithms can be difficult to converge if no estimate of the temperature profile is provided, so the design engineer should always enter an estimated temperature profile. Shortcut methods can be used to obtain initial estimates of composition and temperature profiles. Another effective strategy is to initialize the model using specifications that are easily met, such as reflux ratio and bottoms flow, and then use the resulting temperature and composition profiles as initial estimates for a simulation with the required purity or recovery specifications.

17.8.3 Relaxation methods

With the exception of this method, all the methods described solve the stage equations for the steady-state design conditions. In an operating column other conditions will exist at start-up, and the column will approach the “design” steady-state conditions after a certain period. The stage material balance equations can be written in a finite difference form, and procedures for the solution of these equations will model the unsteady-state behavior of the column.

[Rose et al. \(1958\)](#) and [Hanson and Somerville \(1963\)](#) have applied “relaxation methods” to the solution of the unsteady-state equations to obtain the steady-state values. The application of this method to the design of multistage columns is described by [Hanson and Somerville \(1963\)](#). They give a program listing and worked examples for a distillation column with side streams and for a reboiled absorber.

Relaxation methods are not competitive with the “steady-state” methods in the use of computer time because of slow convergence. However, because they model the actual operation of the column, convergence should be achieved for all practical problems. Relaxation methods are used for dynamic simulation of distillation and for rate-based models such as Aspen Plus RateFrac and BatchFrac. Dynamic models are very useful when attempting to understand the control and operation of distillation columns.

17.9 Other distillation processes

17.9.1 Batch distillation

In batch distillation the mixture to be distilled is charged as a batch to the still and the distillation carried out until a satisfactory top or bottom product is achieved. The still usually consists of a vessel surmounted by a packed or plate column. The heater may be incorporated in the vessel or a separate reboiler used. Batch distillation should be considered under the following circumstances:

1. Where the quantity to be distilled is small
2. Where a range of products has to be produced
3. Where the feed is produced at irregular intervals
4. Where batch integrity is important
5. Where the feed composition varies over a wide range

When the choice between batch and continuous distillation is uncertain, an economic evaluation of both systems should be made.

Batch distillation is an unsteady-state process, the composition in the still (bottoms) varying as the batch is distilled.

Two modes of operation are used:

1. Fixed reflux, where the reflux rate is kept constant. The compositions will vary as the more volatile component is distilled off, and the distillation is stopped when the average composition of the distillate collected, or the bottoms left, meet the specification required.
2. Variable reflux, where the reflux rate is varied throughout the distillation to produce a fixed overhead composition. The reflux ratio will need to be progressively increased as the fraction of the more volatile component in the base of the still decreases.

The basic theory of batch distillation is given in [Chhabra and Gurappa \(2019\)](#) and in several other texts: [Hart \(1997\)](#), [Green and Southard \(2018\)](#), and [Couper et al. \(2012\)](#). In the simple theoretical analysis of batch distillation columns, the liquid hold-up in the column is usually ignored. This hold-up can have a significant effect on the separating efficiency and should be taken into account when designing batch distillation columns. The practical design of batch distillation columns is covered by [Hengstebeck \(1976\)](#), [Ellerbe \(1997\)](#), and [Hart \(1997\)](#).

17.9.2 Vacuum distillation

Components that boil at high temperatures or suffer thermal degradation are sometimes distilled under vacuum to reduce the temperature required for distillation. Vacuum distillation is more expensive than steam distillation, but can be used for compounds that are miscible with water or for processes where the introduction of water might lead to problems such as the formation of azeotropes. The vacuum is usually generated using a vacuum pump or an ejector system on the column overhead product. Selection and design of vacuum pumps and ejectors are described in [Chapter 20](#).

Vacuum columns have high capital and operating costs for the following reasons:

1. Low pressure decreases vapor density, so the column diameter is increased.
2. The vacuum production equipment has high capital and operating costs.
3. The column must be designed to withstand an external pressure. Thicker walls are required for vessels subject to external pressure, as described in Section 14.7.
4. Additional safety precautions and inspection are needed to ensure air cannot enter the equipment if the process fluids are flammable.

Because vacuum columns need low pressure drop per tray, low weir heights are used for plate columns, leading to low stage efficiency and a need for more trays. Packings are therefore often preferred for vacuum service.

17.9.3 Steam distillation

In steam distillation, steam is introduced into the column to lower the partial pressure of the volatile components. Steam distillation is used for the distillation of heat-sensitive products and for compounds with a high boiling point. It is an alternative to vacuum distillation. The products must be immiscible with water. Some steam will normally be allowed to condense to provide the heat required for the distillation. Live steam can be injected directly into the column base, or the steam can be generated by a heater in the still or in an external boiler.

The design procedure for columns employing steam distillation is essentially the same as that for conventional columns, making allowance for the presence of steam in the vapor.

Steam distillation is used extensively in the extraction of essential oils from plant materials.

17.9.4 Reactive distillation

Reactive distillation is the name given to a process where the chemical reaction and product separation are carried out simultaneously in one vessel. Carrying out the reaction, with separation and purification of the product by distillation, gives the following advantages:

1. Chemical equilibrium restrictions can be overcome if a product is removed as it is formed.
2. Energy savings can be obtained if the heat of reaction can be used for the distillation.
3. Capital costs are reduced, as only one vessel is required.

The design of reactive distillation columns is complicated by the complex interactions between the reaction and separation processes. Detailed discussion of reactive distillation is given by [Towler and Frey \(2002\)](#) and [Sundmacher and Kiene \(2003\)](#).

Reactive distillation is used in the production of MTBE (methyl tertiary butyl ether) and methyl acetate.

17.9.5 Petroleum fractionation

Petroleum mixtures such as crude oil and the products of oil refining processes contain from 10^2 to greater than 10^5 components, typically including almost every possible hydrocarbon isomer in the boiling range of the mixture. It is usually neither necessary nor desirable to separate these mixtures into pure components, as the processing goal is to form mixtures with suitable properties, such as volatility and viscosity, for use as fuels. The mixture is distilled into *fractions* or *cuts* that have a suitable boiling range for blending into a fuel or sending for additional processing. The distillation of mineral oils is therefore known as *fractionation*, although this term is sometimes also applied to conventional multicomponent distillation.

The specifications for fractionation columns are usually not set in terms of key components. Instead, the designer specifies cut points for the product streams. The cut points are points on the product stream boiling curve, typically at 5% and 95% of the total material distilled. The sharpness of separation between two fractions is then measured by the overlap between the 95% cut temperature of the lighter fraction and the 5% cut temperature of the heavy fraction.

A good introduction to petroleum fractionation is given by [Watkins \(1979\)](#).

The simulation of petroleum fractionation columns is discussed in Sections 4.4.2 and 4.5.2. Most of the commercial process simulation programs incorporate prebuilt complex column configurations for petroleum fractionation and also have standard sets of pseudocomponents that can be used to fit the feed and product boiling curves.

17.10 Plate efficiency

The designer is concerned with real contacting stages, not the theoretical equilibrium stage assumed for convenience in the mathematical analysis of multistage processes. Equilibrium will rarely be attained in a real stage. The concept of stage efficiency is used to link the performance of practical contacting stages to the theoretical equilibrium stage.

Three principal definitions of efficiency are used:

1. Murphree plate efficiency ([Murphree, 1925](#)), defined in terms of the vapor compositions by:

$$E_{mV} = \frac{y_n - y_{n-1}}{y_e - y_{n-1}} \quad (17.31)$$

where y_e is the composition of the vapor that would be in equilibrium with the liquid leaving the plate. The Murphree plate efficiency is the ratio of the actual separation achieved to that which would be achieved in an equilibrium stage (see [Fig. 17.8](#)). In this definition of efficiency, the liquid and the vapor stream are taken to be perfectly mixed; the compositions in Equation 17.31 are the average composition values for the streams.

2. Point efficiency (Murphree point efficiency). If the vapor and liquid compositions are taken at a point on the plate, Equation 17.31 gives the local or point efficiency, E_{mv} .
3. Overall column efficiency. This is sometimes confusingly referred to as the overall plate efficiency.

$$E_o = \frac{\text{number of ideal stages}}{\text{number of real stages}} \quad (17.32)$$

An estimate of the overall column efficiency will be needed when the design method used gives an estimate of the number of ideal stages required for the separation.

In some methods, the Murphree plate efficiencies can be incorporated into the procedure for calculating the number of stages and the number of real stages determined directly.

For the idealized situation where the operating and equilibrium lines are straight, the overall column efficiency and the Murphree plate efficiency are related by an equation derived by Lewis (1936):

$$E_0 = \frac{\log \left[1 + E_{mV} \left(\frac{mV}{L} - 1 \right) \right]}{\log \left(\frac{mV}{L} \right)} \quad (17.33)$$

where m = slope of the equilibrium line

V = molar flow rate of the vapor

L = molar flow rate of the liquid

Equation 17.33 is not of much practical use in distillation, as the slopes of the operating and equilibrium lines will vary throughout the column. It can be used by dividing the column into sections and calculating the slopes over each section. For most practical purposes, providing the plate efficiency does not vary too much, a simple average of the plate efficiency calculated at the column top, bottom, and feed points will be sufficiently accurate.

17.10.1 Prediction of plate efficiency

Whenever possible, the plate efficiencies used in design should be based on measured values for similar systems obtained on full-sized columns. There is no entirely satisfactory method for predicting plate efficiencies from the system physical properties and plate design parameters; however, the methods given in this section can be used to make a rough estimate where no reliable experimental values are available. They can also be used to extrapolate data obtained from small-scale experimental columns. If the system properties are at all unusual, experimental confirmation of the predicted values should always be obtained. The small, laboratory-scale, glass sieve plate column developed by Oldershaw (1941) has been shown to give reliable values for scale-up. The use of Oldershaw columns is described in papers by Swanson and Gester (1962), Veatch et al. (1960), and Fair et al. (1983).

Some typical values of plate efficiency for a range of systems are given in Table 17.1. More extensive compilations of experimental data are given by Vital et al. (1984) and Kister (1992).

Plate, and overall column, efficiencies will normally be between 30% and 80%, and as a rough guide a figure of 70% can be assumed for preliminary designs. Efficiencies will be lower for vacuum distillations, as low weir heights are used to keep the pressure drop small (see Section 17.10.4).

Multicomponent systems

The prediction methods given in the following sections, and those available in the open literature, are usually restricted to binary systems. It is clear that in a binary system the efficiency obtained for each component must be the same. This is not so for a multicomponent system; the heavier components will usually exhibit lower efficiencies than the lighter components.

The following guide rules, adapted from a paper by Toor and Burchard (1960), can be used to estimate the efficiencies for a multicomponent system from binary data:

1. If the components are similar, the multicomponent efficiencies will be similar to the binary efficiency.
2. If the predicted efficiencies for the binary pairs are high, the multicomponent efficiency will be high.
3. If the resistance to mass transfer is mainly in the liquid phase, the difference between the binary and multicomponent efficiencies will be small.
4. If the resistance is mainly in the vapor phase, as it normally will be, the difference between the binary and multicomponent efficiencies can be substantial.

The prediction of efficiencies for multicomponent systems is also discussed by Chan and Fair (1984b). For mixtures of dissimilar compounds, the efficiency can be quite different from that predicted for each binary pair, and laboratory or pilot-plant studies should be made to confirm any predictions.

17.10.2 O'Connell's correlation

A quick estimate of the overall column efficiency can be obtained from the correlation given by O'Connell (1946), which is shown in Fig. 17.19. The overall column efficiency is correlated with the product of the relative volatility of the light key component (relative to the heavy key) and the molar average viscosity of the feed, estimated at the

TABLE 17.1 Representative efficiencies, sieve plates

System	Column dia., m	Pressure kPa, abs	Efficiency %	
			E_{mV}	E_o
Water–methanol	1.0	—	80	
Water–ethanol	0.2	101	90	
Water–isopropanol	—	—		70
Water–acetone	0.15	90	80	
Water–acetic acid	0.46	101	75	
Water–ammonia	0.3	101	90	
Water–carbon dioxide	0.08	—	80	
Toluene–propanol	0.46	—	65	
Toluene–ethylene dichloride	0.05	101		75
Toluene–methylethylketone	0.15	—		85
Toluene–cyclohexane	2.4	—		70
Toluene–methylcyclohexane	—	27		90
Toluene–octane	0.15	101		40
Heptane–cyclohexane	1.2	165	95	85
	2.4	165		75
Propane–butane	—	—		100
Isobutane–n-butane	—	2070		110
Benzene–toluene	0.13	—	75	
Benzene–methanol	0.18	690	94	
Benzene–propanol	0.46	—	55	
Ethylbenzene–styrene	—	—	75	

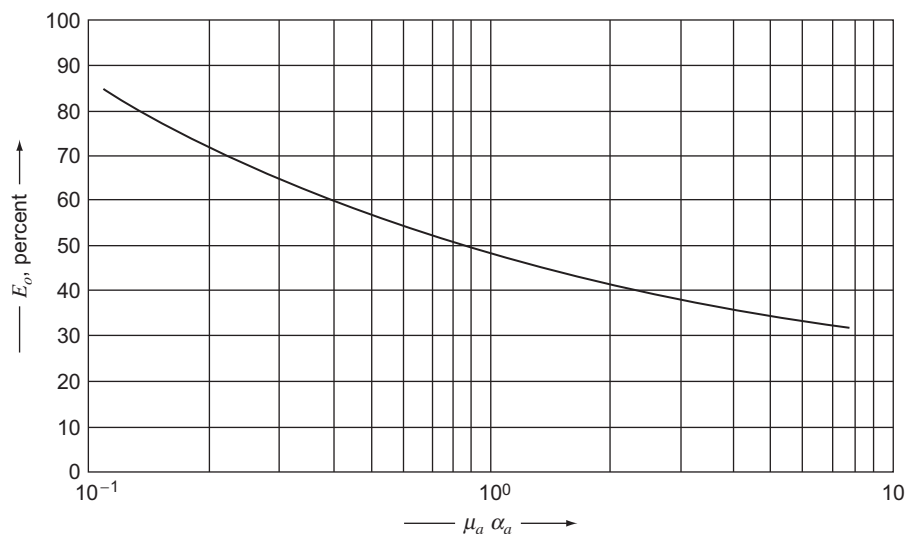
 E_{mV} = Murphree plate efficiency. E_o = Overall column efficiency.

FIG. 17.19 Distillation column efficiencies (bubble-caps). (From O'Connell, 1946.)

average column temperature. The correlation was based mainly on data obtained with hydrocarbon systems, but includes some values for chlorinated solvents and water–alcohol mixtures. It has been found to give reliable estimates of the overall column efficiency for hydrocarbon systems and can be used to make an approximate estimate of the efficiency for other systems. The method takes no account of the plate design parameters and includes only two physical property variables.

The O'Connell correlation is the most widely used method for estimating stage efficiency in industrial practice. The calculation is much simpler than the more sophisticated methods that follow, and the results are good enough for most design purposes. The relationship between the O'Connell correlation and the more theoretical methods is discussed by [Duss and Taylor \(2018\)](#). They also discuss situations where the O'Connell correlation is not applicable.

[Edujlee \(1958\)](#) has expressed the O'Connell correlation in the form of an equation:

$$E_o = 51 - 32.5 \log(\mu_a \alpha_a) \quad (17.34)$$

where μ_a = the molar average liquid viscosity, mNs/m²

α_a = average relative volatility of the light key

Absorbers

O'Connell gave a similar correlation for the plate efficiency of absorbers ([Fig. 17.20](#)). Appreciably lower plate efficiencies are obtained in absorption than in distillation.

In O'Connell's paper, the plate efficiency is correlated with a function involving Henry's constant, the total pressure, and the solvent viscosity at the operating temperature.

To convert the original data to SI units, it is convenient to express this function in the following form:

$$x = 0.062 \left[\frac{\rho_s P}{\mu_s H M_s} \right] = 0.062 \left[\frac{\rho_s}{\mu_s K M_s} \right] \quad (17.35)$$

where H = the Henry's law constant, Nm⁻²/mol fraction

P = total pressure, N/m²

μ_s = solvent viscosity, mNs/m²

M_s = molecular weight of the solvent

ρ_s = solvent density, kg/m³

K = equilibrium constant for the solute

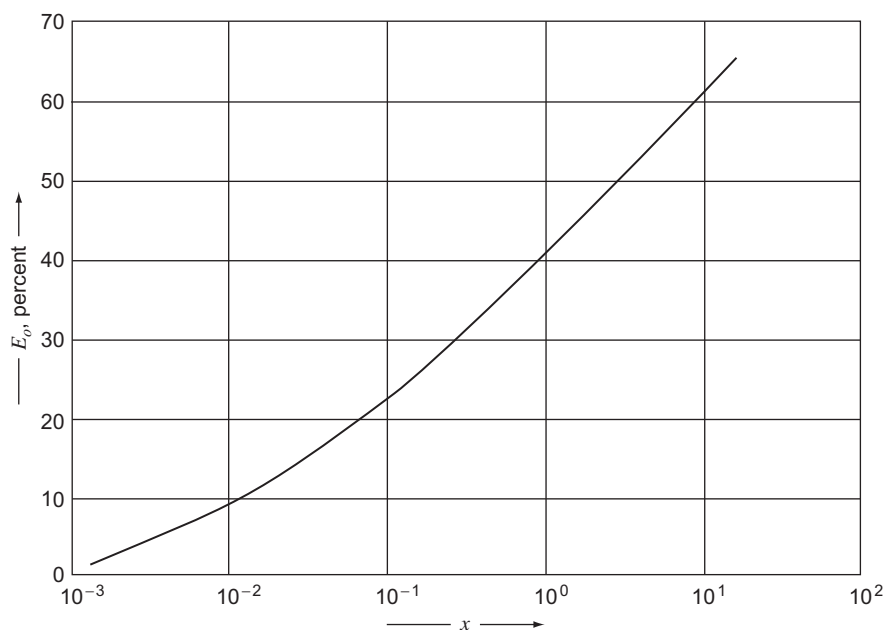


FIG. 17.20 Absorber column efficiencies (bubble-caps). (From O'Connell, 1946.)

Example 17.5

Using O'Connell's correlation, estimate the overall column efficiency and the number of real stages required for the separation given in Example 17.3 when the reflux ratio is 2.0.

Solution

From Example 17.3, feed composition, mol fractions:

Propane 0.05, i-butane 0.15, n-butane 0.25, i-pentane 0.20, n-pentane 0.35

Column-top temperature 65 °C, bottom temperature 120 °C

Average relative volatility light key = 2.0

Take the viscosity at the average column temperature, 93 °C:

viscosities, propane = 0.03 mNs/m²

butane = 0.12 mNs/m²

pentane = 0.14 mNs/m²

For feed composition, molar average viscosity = $0.03 \times 0.05 + 0.12(0.15 + 0.25)$:

$$\begin{aligned} &+ 0.14(0.20 + 0.35) \\ &= 0.13 \text{ mNs/m}^2 \\ \alpha_a \mu_a &= 2.0 \times 0.13 = 0.26 \end{aligned}$$

From Fig. 17.19, $E_o = 70\%$

From Example 17.3, when the reflux ratio is 2.0, number of ideal stages = 16, one ideal stage will be the reboiler, so number of actual stages (rounding up):

$$= \frac{(16 - 1)}{0.7} = \underline{\underline{22}}$$

17.10.3 Van Winkle's correlation

Van Winkle et al. (1972) have published an empirical correlation for the plate efficiency that can be used to predict plate efficiencies for binary systems. Their correlation uses dimensionless groups that include those system variables and plate parameters that are known to affect plate efficiency. They give two equations, the simplest, and that which they consider the most accurate, is given here. The data used to derive the correlation covered both bubble-cap and sieve plates.

$$E_{mV} = 0.07 Dg^{0.14} Sc^{0.25} Re^{0.08} \quad (17.36)$$

where Dg = surface tension number = $(\sigma_L / \mu_L u_v)$

u_v = superficial vapor velocity

σ_L = liquid surface tension

μ_L = liquid viscosity

Sc = liquid Schmidt number = $(\mu_L / \rho_L D_{LK})$

ρ_L = liquid density

D_{LK} = liquid diffusivity, light key component

Re = Reynolds number = $(h_w u_v \rho_v / \mu_L (FA))$

h_w = weir height

ρ_v = vapor density

$$(FA) = \text{fractional area} = \frac{(\text{area of holes or risers})}{(\text{total column cross-sectional area})}$$

The use of this method is illustrated in Example 17.8.

17.10.4 AIChE method

This method of predicting plate efficiency, published in 1958, was the result of a 5-year study of bubble-cap plate efficiency directed by the Research Committee of the American Institute of Chemical Engineers. The AIChE method is the most detailed method for predicting plate efficiencies that is available in the open literature. It takes into account all the major factors that are known to affect plate efficiency, including:

- The mass transfer characteristics of the liquid and vapor phases
- The design parameters of the plate
- The vapor and liquid flow rates
- The degree of mixing on the plate

The method is well established, and in the absence of experimental values, or proprietary prediction methods, should be used when more than a rough estimate of efficiency is needed.

The approach taken is semi-empirical. Point efficiencies are estimated making use of the “two-film theory” and the Murphree efficiency estimated allowing for the degree of mixing likely to be obtained on real plates.

The procedure and equations are given in this section without discussion of the theoretical basis of the method. The reader should refer to the AIChE manual (1958) or to [Smith \(1963\)](#), who gives a comprehensive account of the method and extends its use to sieve plates. [Chan and Fair \(1984a\)](#) published an alternative method for point efficiencies on sieve plates, which they demonstrate gives closer predictions than the AIChE method. The Chan and Fair method follows the same overall methodology as the AIChE method but uses an improved correlation for vapor phase mass transfer, given later.

AIChE method

The mass transfer resistances in the vapor and liquid phases are expressed in terms of the number of transfer units, N_G and N_L . The point efficiency is related to the number of transfer units by the equation:

$$\frac{1}{\ln(1 - E_{mv})} = - \left[\frac{1}{N_G} + \frac{mV}{L} \times \frac{1}{N_L} \right] \quad (17.37)$$

where m is the slope of the equilibrium line and V and L the vapor and liquid molar flow rates.

Equation 17.37 is plotted in [Fig. 17.21](#).

The number of gas-phase transfer units in the AIChE method is given by:

$$N_G = \frac{(0.776 + 4.57 \times 10^{-3} h_w - 0.24F_v + 105L_p)}{\left(\frac{\mu_v}{\rho_v D_v} \right)^{0.5}} \quad (17.38)$$

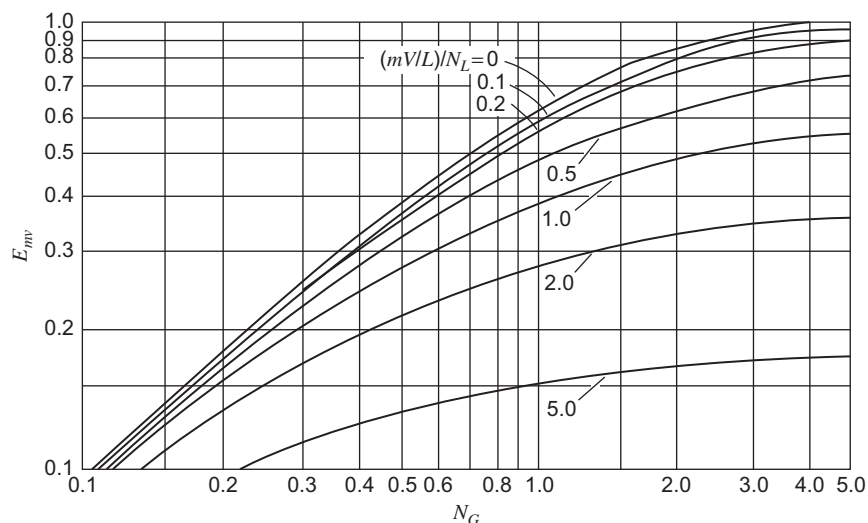


FIG. 17.21 Relationship between point efficiency and number of liquid and vapor transfer units (Equation 17.37).

where h_w = weir height, mm.

F_v = the column vapor "F" factor = $u_a \rho_v^{0.5}$.

u_a = vapor velocity based on the active tray area (bubbling area), see Section 17.13.2, m/s.

L_p = the volumetric liquid flow rate across the plate, divided by the average width of the plate, m^3/sm . The average width can be calculated by dividing the active area by the length of the liquid path Z_L .

μ_v = vapor viscosity, Ns/m^2 .

ρ_v = vapor density; kg/m^3 .

D_v = vapor diffusivity, m^2/s .

In the alternative method proposed by [Chan and Fair \(1984a\)](#), the number of gas-phase mass transfer units is given by:

$$N_G = \frac{D_v^{0.5}(1030f - 867f^2) \bar{t}_v}{h_L^{0.5}} \quad (17.39)$$

where h_L = liquid hold-up on tray, cm

$\bar{t}_v$ = average vapor residence time, s

$f = u_a / u_{af}$ = fractional approach to the vapor velocity based on active area at flooding, u_{af}

The remainder of the Chan and Fair method is the same as the AIChE method.

In both methods, the number of liquid-phase transfer units is given by:

$$N_L = (4.13 \times 10^8 D_L)^{0.5} (0.21F_v + 0.15)t_L \quad (17.40)$$

where D_L = liquid-phase diffusivity, m^2/s

t_L = liquid contact time, s

given by:

$$t_L = \frac{Z_c Z_L}{L_p} \quad (17.41)$$

where Z_L = length of the liquid path from inlet downcomer to outlet weir, m

Z_c = liquid hold-up on the plate, m^3 per m^2 active area

given by:

for bubble-cap plates:

$$Z_c = 0.042 + 0.19 \times 10^{-3} h_w - 0.014 F_v + 2.5 L_p \quad (17.42)$$

for sieve plates:

$$Z_c = 0.006 + 0.73 \times 10^{-3} h_w - 0.24 \times 10^{-3} F_v h_w + 1.22 L_p \quad (17.43)$$

The Murphree efficiency E_{mV} is only equal to the point efficiency E_{mv} if the liquid on the plate is perfectly mixed. On a real plate this will not be so, and to estimate the plate efficiency from the point efficiency, some means of estimating the degree of mixing is needed. The dimensionless Peclet number characterizes the degree of mixing in a system. For a plate, the Peclet number is given by:

$$\text{Pe} = \frac{Z_L^2}{D_e t_L} \quad (17.44)$$

where D_e is the "eddy diffusivity", m^2/s .

A Peclet number of zero indicates perfect mixing, and a value of ∞ indicates plug flow.

For bubble-cap and sieve plates, the eddy diffusivity can be estimated from the equation:

$$D_e = (0.0038 + 0.017 u_a + 3.86 L_p + 0.18 \times 10^{-3} h_w)^2 \quad (17.45)$$

The relation between the plate efficiency and point efficiency with the Peclet number as a parameter is shown in [Fig. 17.22](#). The application of the AIChE method is illustrated in Example 17.7.

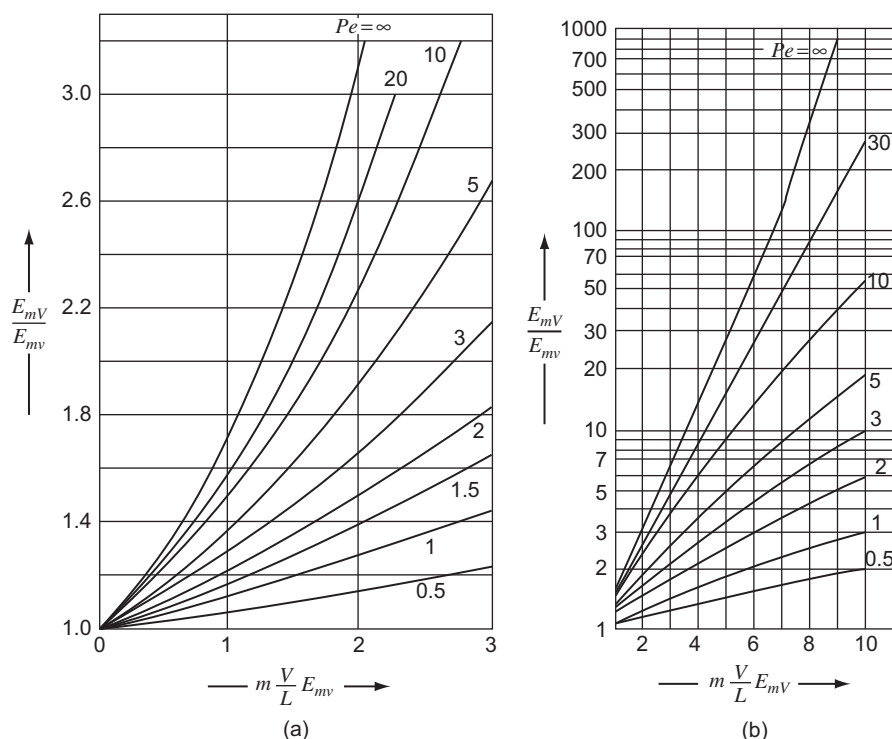


FIG. 17.22 Relationship between plate and point efficiency.

Estimation of physical properties

To use the AIChE method or Van Winkle's correlation, estimates of the physical properties are required. It is unlikely that experimental values will be found in the literature for all systems that are of practical interest. The prediction methods given in [Chapter 4](#), and in the references given in that chapter, can be used to estimate values.

The AIChE design manual recommends the [Wilke and Chang \(1955\)](#) equation for liquid diffusivities (see Section 15.3.4) and the [Wilke and Lee \(1955\)](#) modification to the Hirschfelder, Bird, and Spotz equation for gas diffusivities.

Plate design parameters

The significance of the weir height in the AIChE equations should be noted. The weir height was the plate parameter found to have the strongest effect on plate efficiency. Increasing weir height will increase the plate efficiency, but at the expense of an increase in pressure drop and entrainment. Weir heights will normally be in the range 40 to 100 mm for columns operating at and above atmospheric pressure, but will be as low as 6 mm for vacuum columns. This largely accounts for the lower plate efficiencies obtained in vacuum columns.

The length of the liquid path Z_L is taken into account when assessing the plate mixing performance. The mixing correlation given in the AIChE method was not tested on large-diameter columns, and [Smith \(1963\)](#) states that the correlation should not be used for large-diameter plates; however, on a large plate the liquid path will normally be subdivided, and the value of Z_L will be similar to that in a small column. The assumption that the vapor space is well-mixed across the tray may also not be valid for large column diameters.

The vapor " F " factor F_v is a function of the active tray area. Increasing F_v decreases the number of gas-phase transfer units. The liquid flow term L_p is also a function of the active tray area and the liquid path length. It will only have a significant effect on the number of transfer units if the path length is long. In practice, the range of values for F_v , the active area, and the path length will be limited by other plate design considerations.

Multicomponent systems

The AIChE method was developed from measurements on binary systems. The AIChE manual should be consulted for advice on its application to multicomponent systems. See also the comments in Section 17.10.1.

17.10.5 Entrainment

The AIChE method and that of Van Winkle predict the “dry” Murphree plate efficiency. In operation some liquid droplets will be entrained and carried up the column by the vapor flow, and this will reduce the actual, operating, efficiency.

The dry-plate efficiency can be corrected for the effects of entrainment using the equation proposed by Colburn (1936):

$$E_a = \frac{E_{mV}}{1 + E_{mV} \left[\frac{\psi}{1 - \psi} \right]} \quad (17.46)$$

where E_a = actual plate efficiency, allowing for entrainment

$$\psi = \text{the fractional entrainment} = \frac{\text{entrained liquid}}{\text{gross liquid flow}}$$

A method for predicting the entrainment from sieve plates is given in Section 17.13.5, Fig. 17.36; a similar method for bubble-cap plates is given by Bolles (1963).

17.11 Approximate column sizing

An approximate estimate of the overall column size can be made once the number of real stages required for the separation is known. This is often needed to make a rough estimate of the capital cost for project evaluation.

Plate spacing

The overall height of the column will depend on the plate spacing. Plate spacings from 0.15 m (6 in) to 1 m (36 in) are normally used. The spacing chosen will depend on the column diameter and operating conditions. Close spacing is used with small-diameter columns, and where head room is restricted; as it will be when a column is installed in a building. For columns above 1 m in diameter, plate spacings of 0.3 to 0.6 m will normally be used, and 0.5 m (18 in) can be taken as an initial estimate. This would be revised, as necessary, when the detailed plate design is made.

A larger spacing will be needed between certain plates to accommodate feed and side-stream arrangements and for manways.

Column diameter

The principal factor that determines the column diameter is the vapor flow rate. The vapor velocity must be below that which would cause excessive liquid entrainment or a high pressure drop. The equation given here, which is based on the well-known Souders and Brown equation, Lowenstein (1961), can be used to estimate the maximum allowable superficial vapor velocity, and hence the column area and diameter:

$$\hat{u}_v = \left(-0.171l_t^2 + 0.27l_t - 0.047 \right) \left[\frac{\rho_L - \rho_v}{\rho_v} \right]^{1/2} \quad (17.47)$$

where $\hat{u}_v$ = maximum allowable vapor velocity, based on the gross (total) column cross-sectional area, m/s
 l_t = plate spacing, m (range 0.5 to 1.5)

The column diameter, D_c , can then be calculated:

$$D_c = \sqrt{\frac{4\hat{V}_w}{\pi\rho_v\hat{u}_v}} \quad (17.48)$$

where $\hat{V}_w$ is the maximum vapor rate, kg/s.

This approximate estimate of the diameter would be revised when the detailed plate design is undertaken.

The column diameter estimated should then be rounded up to the nearest standard head size so that preformed heads can be used as vessel closures; see Section 14.5.2. The column sizing programs in most commercial process simulation programs use North American standard head sizes, which are available in 6-in (152.4-mm) increments.

17.12 Plate contactors

Cross-flow plates are the most common type of plate contactor used in distillation and absorption columns. In a cross-flow plate the liquid flows across the plate and the vapor up through the plate. A typical layout is shown in Fig. 17.23. The flowing liquid is transferred from plate to plate through vertical channels called “downcomers.” A pool of liquid is retained on the plate by an outlet weir.

Other types of plate are used that have no downcomers (non-cross-flow plates), with the liquid showering down the column through large openings in the plates (sometimes called shower plates). These and other proprietary non-cross-flow plates are used for special purposes, particularly when a low-pressure drop is required.

Four principal types of cross-flow tray are used, classified according to the method used to contact the vapor and liquid.

1. Sieve plate (perforated plate) (Fig. 17.24)

This is the simplest type of cross-flow plate. The vapor passes up through perforations in the plate, and the liquid is retained on the plate by the vapor flow. There is no positive vapor–liquid seal, and at low flow rates liquid will “weep” through the holes, reducing the plate efficiency. The perforations are usually small holes, but larger holes and slots can be used.

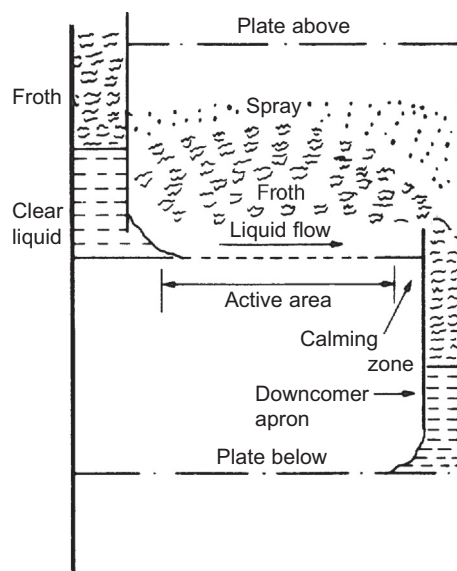


FIG. 17.23 Typical cross-flow plate (sieve).

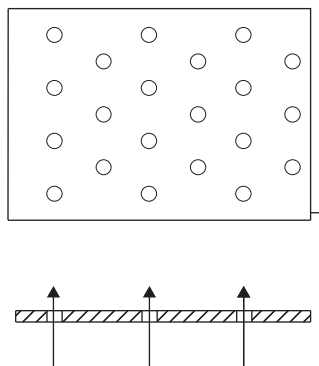


FIG. 17.24 Sieve plate.

2. Bubble-cap plates (Fig. 17.25)

Bubble-cap plates are plates in which the vapor passes up through short pipes, called risers, covered by a cap with a serrated edge, or slots. The bubble-cap plate is the traditional, oldest type of cross-flow plate, and many different designs have been developed. Standard cap designs would now be specified for most applications.

The most significant feature of the bubble-cap plate is that the use of risers ensures that a level of liquid is maintained on the tray at all vapor flow rates. Bubble-caps therefore have good turndown performance at low flow rates. They are more expensive than sieve plates and more prone to corrosion, fouling, and plugging, and so are usually only found on older columns.

3. Valve plates (floating-cap plates) (Fig. 17.26)

Valve plates are proprietary designs. They are essentially sieve plates with large-diameter holes covered by movable flaps, which lift as the vapor flow increases.

As the area for vapor flow varies with the flow rate, valve plates can operate efficiently at lower flow rates than sieve plates: the valves closing at low vapor rates. The cost of valve plates is intermediate between sieve plates and bubble-cap plates.

Some very elaborate valve designs have been developed, but the simple type shown in Fig. 17.26 is satisfactory for most applications.

4. Valve plates (fixed valve plates) (Fig. 17.27)

A fixed valve plate is similar to a sieve plate, except the holes are only partially punched out, so that the hole remains partially covered, as shown in Fig. 17.27. Fixed valve trays are almost as inexpensive as sieve trays and have improved turndown performance. The relatively small cost difference between fixed valve trays and sieve trays can usually be justified by the improved turndown performance, and fixed valve trays are the most common type specified in nonfouling applications.

Many different proprietary designs of fixed and floating valves have been developed. Performance details can be obtained from the tray vendors.

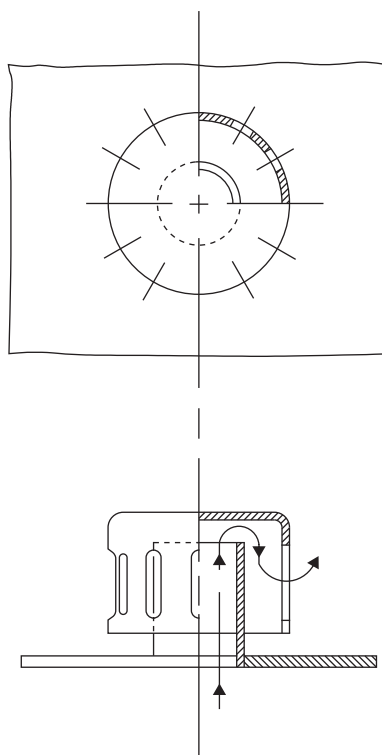


FIG. 17.25 Bubble-cap.

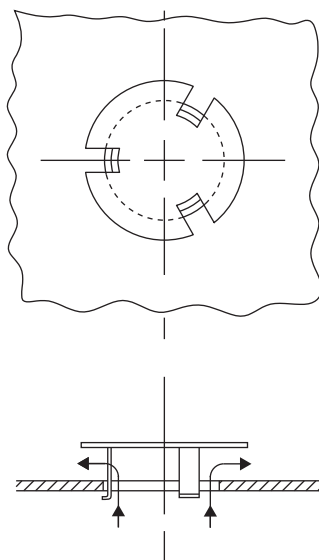


FIG. 17.26 Simple valve.

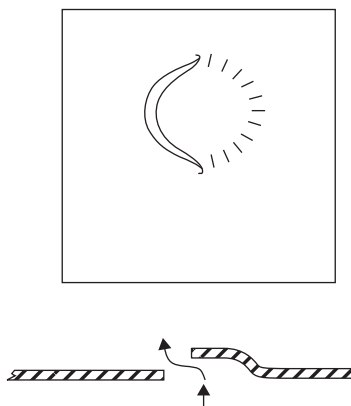


FIG. 17.27 Fixed valve.

Liquid flow pattern

Cross-flow trays are also classified according to the number of liquid passes on the plate. The design shown in Fig. 17.28(a) is a single-pass plate. For low liquid flow rates reverse-flow plates are used (see Fig. 17.28b). In this type the plate is divided by a low central partition and inlet and outlet downcomers are on the same side of the plate. Multiple-pass plates, in which the liquid stream is subdivided by using several downcomers, are used for high liquid flow rates and large-diameter columns. A double-pass plate is shown in Fig. 17.28(c).

Selection of the liquid flow pattern is discussed in Section 17.13.4. An approximate criterion for selecting the liquid flow pattern is the liquid volumetric flow rate per unit weir length, which should ideally be in the range 5 to 8 liters/s per m (2 to 3 gpm/in). Weir length is discussed in more detail in Section 17.13.8.

17.12.1 Selection of plate type

The principal factors to consider when comparing the performance of bubble-cap, sieve, and valve plates are cost, capacity, operating range, efficiency, and pressure drop.

Cost. Bubble-cap plates are appreciably more expensive than sieve or valve plates. The relative cost will depend on the material of construction used; for mild steel, the ratios bubble-cap:valve:fixed valve:sieve are approximately 3.0:1.2:1.1:1.0.

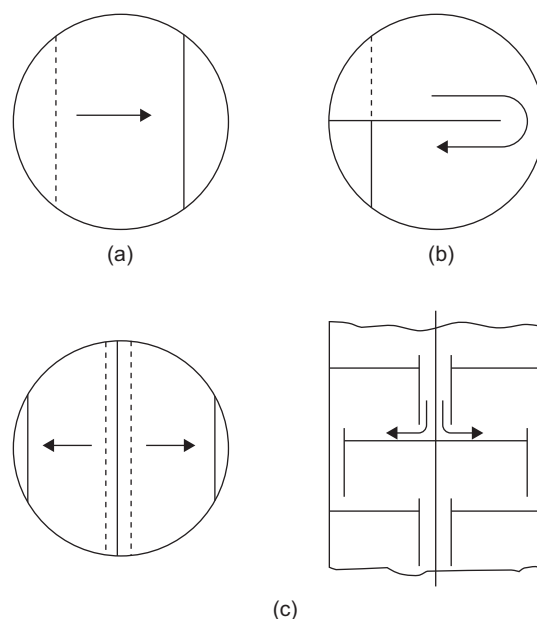


FIG. 17.28 Liquid flow patterns on cross-flow trays. (a) Single pass. (b) Reverse flow. (c) Double pass.

Capacity. There is little difference in the capacity rating of the three types (the diameter of the column required for a given flow rate); the ranking from best to worst is sieve, valve, and bubble-cap.

Operating range. This is the most significant factor. By operating range is meant the range of vapor and liquid rates over which the plate will operate satisfactorily (the stable operating range). Some flexibility will always be required in an operating plant to allow for changes in production rate and to cover start-up and shut-down conditions. The ratio of the highest to the lowest flow rates is often referred to as the “turn-down” ratio. Bubble-cap plates have a positive liquid seal and can therefore operate efficiently at very low vapor rates.

Sieve plates and fixed valve plates rely on the flow of vapor through the holes to hold the liquid on the plate and cannot operate at very low vapor rates. With good design, sieve plates can give a satisfactory operating range; typically, from 50% to 120% of design capacity. Fixed valve plates have somewhat better turndown performance. Valve plates are intended to give greater flexibility than sieve plates at a lower cost than bubble-caps.

Efficiency. The Murphree efficiency of the three types of plate will be virtually the same when operating over their design flow range, and no real distinction can be made between them; see [Zuiderweg et al. \(1960\)](#).

Pressure drop. The pressure drop over the plates can be an important design consideration, particularly for vacuum columns. The plate pressure drop will depend on the detailed design of the plate, but, in general, sieve plates give the lowest pressure drop, followed by valves, with bubble-caps giving the highest.

Summary. Sieve plates are the cheapest and least prone to fouling and are satisfactory for most applications. Fixed valve plates are almost as cheap as sieve plates and have improved turndown behavior. The improved performance usually justifies the increased cost, and this type is most commonly selected for nonfouling applications. Moving valve plates should be considered if the specified turndown ratio cannot be met with sieve plates or fixed valve plates. Bubble-caps should only be used where very low vapor (gas) rates have to be handled and a positive liquid seal is essential at all flow rates.

17.12.2 Plate construction

The mechanical design features of sieve plates are described in this section. The same general construction is also used for bubble-cap and valve plates. Details of the various types of bubble-cap used and the preferred dimensions of standard cap designs can be found in the books by [Smith \(1963\)](#) and [Coker \(2010\)](#). The manufacturers’ design manuals should be consulted for details of valve plate design.

Two different types of plate construction are used. Large-diameter plates are normally constructed in sections, supported on beams. Small plates are installed in the column as a stack of preassembled plates.

Sectional construction

A typical plate is shown in Fig. 17.29. The plate sections are supported on a ring welded around the vessel wall and on beams. The beams and ring are about 50 mm wide, with the beams set at around 0.6-m spacing. The beams are usually angle or channel sections, constructed from folded sheet. Special fasteners are used so the sections can be assembled from one side only. One section is designed to be removable to act as a manway. This reduces the number of manways needed on the vessel wall, which reduces the vessel cost.

Stacked plates (cartridge plates)

The stacked type of construction is used where the column diameter is too small for a worker to enter to assemble the plates, say less than 1.2 m (4 ft). Each plate is fabricated complete with the downcomer and joined to the plate above and below using screwed rods (spacers) (Fig. 17.30). The plates are installed in the column shell as an assembly (stack) of 10 or so plates. Tall columns have to be divided into flanged sections so that plate assemblies can be easily installed and removed. The weir and downcomer supports are usually formed by turning up the edge of the plate.

The plates are not fixed to the vessel wall, as they are with sectional plates, so there is no positive liquid seal at the edge of the plate, and a small amount of leakage will occur. In some designs the plate edges are turned up around the circumference to make better contact at the wall. This can make it difficult to remove the plates for cleaning and maintenance without damage.

Downcomers

The segmental, or chord downcomer, shown in Fig. 17.31(a), is the simplest and cheapest form of construction and is satisfactory for most purposes. The downcomer channel is formed by a flat plate, called an apron, which extends down from the outlet weir. The apron is usually vertical, but may be sloped (Fig. 17.31b) to increase the plate area available for perforation. This design is commonly in high-capacity trays. If a more positive seal is required at the downcomer at the outlet, an inlet weir can be fitted (Fig. 17.31c) or a recessed seal pan used (Fig. 17.31d). Circular downcomers (pipes) are sometimes used for small liquid flow rates. Curved downcomers are often used in high-capacity trays for large columns. Truncated downcomers (Fig. 17.31e) can be used to increase the plate area available for perforation and are also commonly used for high-capacity trays.

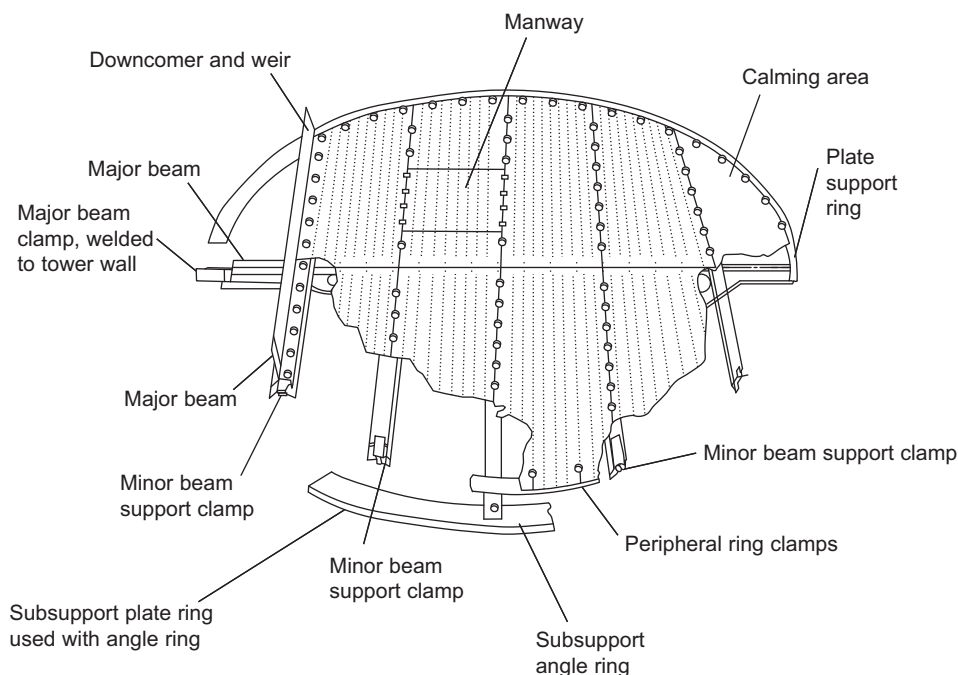


FIG. 17.29 Typical sectional plate construction.

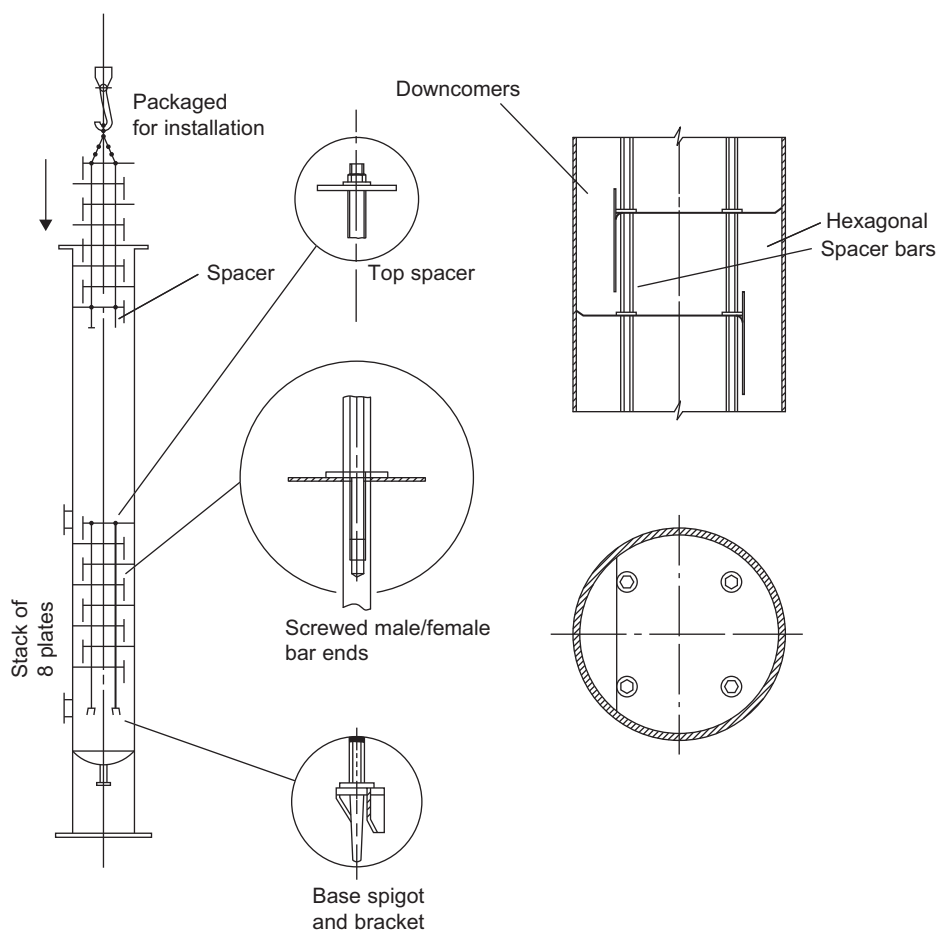


FIG. 17.30 Typical stacked-plate construction.

Side stream and feed points

Where a side stream is withdrawn from the column, the plate design must be modified to provide a liquid seal at the take-off pipe. A typical design is shown in Fig. 17.32(a). Side-draw pipes and run-down lines must be sized for self-venting flow, and provision must be made for vapor to vent from the line in case vapor is entrained from the column or formed by flashing in the line. Sewell (1975) gives a correlation for the minimum pipe diameter that will allow self-venting flow.

When the feed stream is liquid, it will be normally introduced into the downcomer leading to the feed plate and the plate spacing increased at this point (see Fig. 17.32b). This design should not be used if the feed is at the bubble point or is two-phase, as the feed may flash on entering the column, in which case downcomer flooding could occur.

Structural design

The plate structure must be designed to support the hydraulic loads on the plate during operation and the loads imposed during construction and maintenance. Typical design values used for these loads are:

Hydraulic load: 600 N/m^2 live load on the plate, plus 3000 N/m^2 over the downcomer seal area.

Erection and maintenance: 1500 N concentrated load on any structural member.

It is important to set close tolerances on the weir height, downcomer clearance, and plate flatness, to ensure an even flow of liquid across the plate. The tolerances specified will depend on the dimensions of the plate but will typically be about 3 mm .

The plate deflection under load is also important and will normally be specified as not greater than 3 mm under the operating conditions for plates greater than 2.5 m and proportionally less for smaller diameters.

The mechanical specification of bubble-cap, sieve, and valve plates is covered in a series of articles by Glitsch (1960), McClain (1960), Thrift (1960a,b), and Patton and Pritchard (1960).

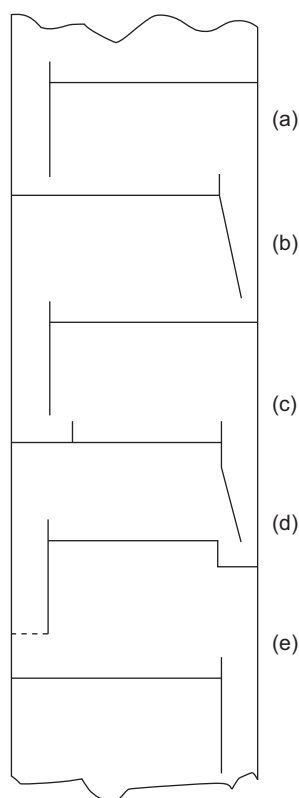


FIG. 17.31 Segment (chord) downcomer designs. (a) Vertical apron. (b) Inclined apron. (c) Inlet weir. (d) Recessed well. (e) Truncated downcomer.

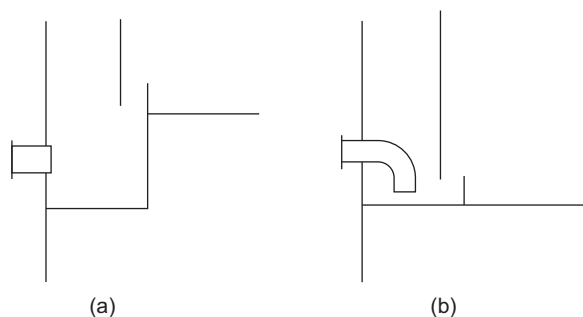


FIG. 17.32 Feed and take-off nozzles.

17.13 Plate hydraulic design

The basic requirements of a plate contacting stage are that it should:

Provide good vapor–liquid contact.

Provide sufficient liquid hold-up for good mass transfer (high efficiency).

Have sufficient area and spacing to keep the entrainment and pressure drop within acceptable limits.

Have sufficient downcomer area for the liquid to flow freely from plate to plate.

Plate design, like most engineering design, is a combination of theory and practice. The design methods use semi-empirical correlations derived from fundamental research work combined with practical experience obtained from the operation of commercial columns. Proven layouts are used, and the plate dimensions are kept within the range of values known to give satisfactory performance.

A short procedure for the hydraulic design of sieve plates is given in this section. Design methods for bubble-cap plates are given by Bolles (1963) and Coker (2010). Valve plates are proprietary designs and will be designed in consultation with the vendors. Design manuals are available from some vendors.

A detailed discussion of the extensive literature on plate design and performance will not be given. Chase (1967) and Zuideweg (1982) give critical reviews of the literature on sieve plates.

Several design methods have been published for sieve plates: Kister (1992), Barnicki and Davies (1989), Koch and Kuzniar (1966), Fair (1963), and Huang and Hodson (1958); see also the book by Lockett (1986).

Operating range

Satisfactory operation will only be achieved over a limited range of vapor and liquid flow rates. A typical performance diagram for a sieve plate is shown in Fig. 17.33.

The upper limit to vapor flow is set by the condition of flooding. At flooding there is a sharp drop in plate efficiency and an increase in pressure drop. Flooding is caused by either the excessive carryover of liquid to the next plate by entrainment (entrainment or jet flooding) or by liquid backing up in the downcomers.

The lower limit of the vapor flow is set by the condition of weeping. Weeping occurs when the vapor flow is insufficient to maintain a level of liquid on the plate. "Coning" occurs at low liquid rates and is the term given to the condition where the vapor pushes the liquid back from the holes and jets upward, with poor liquid contact.

In the following sections, gas can be taken as synonymous with vapor when applying the method to the design of plates for absorption columns.

17.13.1 Plate design procedure

A trial-and-error approach is necessary in plate design: starting with a rough plate layout, checking key performance factors, and revising the design as necessary until a satisfactory design is achieved. A typical design procedure is set out here and discussed in the following sections. The normal range of each design variable is given in the discussion, together with recommended values that can be used to start the design.

Most of the commercial process simulation programs offer tray design modules. These programs can be used for preliminary tray layout for costing purposes, but the default dimensions selected or calculated by these programs often do not give the best performance over the intended range of operation. An experienced designer will run several cases to confirm the tray performance is satisfactory over the whole range of operation. Hand calculations using the methods given in this section can also be used to guide the process simulation programs to a better design.

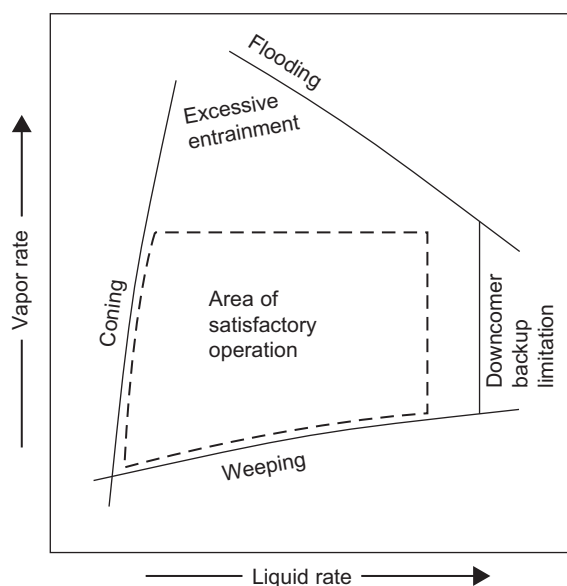


FIG. 17.33 Sieve plate performance diagram.

Procedure

1. Calculate the maximum and minimum vapor and liquid flow rates for the turn-down ratio required.
2. Collect, or estimate, the system physical properties.
3. Select a trial plate spacing (Section 17.11).
4. Estimate the column diameter, based on flooding considerations (Section 17.13.3).
5. Decide the liquid flow arrangement (Section 17.13.4).
6. Make a trial plate layout: downcomer area, active area, hole area, hole size, weir height (Sections 17.13.8 to 17.13.10).
7. Check the weeping rate (Section 17.13.6); if unsatisfactory, return to step 6.
8. Check the plate pressure drop (Section 17.13.14); if too high, return to step 6.
9. Check downcomer backup; if too high, return to step 3 or 6 (Section 17.13.15).
10. Decide plate layout details: calming zones, unperforated areas. Check hole pitch; if unsatisfactory, return to step 6 (Section 17.13.11).
11. Recalculate the percentage flooding based on chosen column diameter.
12. Check entrainment; if too high, return to step 4 (Section 17.13.5).
13. Optimize design: repeat steps 3 to 12 to find smallest diameter and plate spacing acceptable (lowest cost).
14. Finalize design: draw up the plate specification and sketch the layout.

This procedure is illustrated in Example 17.6.

17.13.2 Plate areas

The following area terms are used in the plate design procedure:

A_c = total column cross-sectional area

A_d = cross-sectional area of downcomer

A_n = net area available for vapor–liquid disengagement, normally equal to $A_c - A_d$ for a single-pass plate

A_a = active, or bubbling, area, equal to $A_c - 2A_d$ for single-pass plates

A_h = hole area, the total area of all the active holes

A_p = perforated area (including blanked areas)

A_{ap} = the clearance area under the downcomer apron

17.13.3 Diameter

The flooding condition fixes the upper limit of vapor velocity. A high vapor velocity is needed for high plate efficiencies, and the velocity will normally be between 70% and 90% of that which would cause flooding. For design, a value of 80% to 85% of the flooding velocity should be used.

The flooding velocity can be estimated from the correlation given by Fair (1961):

$$u_f = K_1 \sqrt{\frac{\rho_L - \rho_v}{\rho_v}} \quad (17.49)$$

where u_f = flooding vapor velocity, m/s, based on the net column cross-sectional area A_n (see Section 17.13.2)

K_1 = a constant obtained from Fig. 17.34

The liquid–vapor flow factor F_{LV} in Fig. 17.34 is given by:

$$F_{LV} = \frac{L_w}{V_w} \sqrt{\frac{\rho_v}{\rho_L}} \quad (17.50)$$

where L_w = liquid mass flow rate, kg/s

V_w = vapor mass flow rate, kg/s

The following restrictions apply to the use of Fig. 17.34:

1. Hole size less than 6.5 mm. Entrainment may be greater with larger hole sizes.
2. Weir height less than 15% of the plate spacing.
3. Nonfoaming systems.

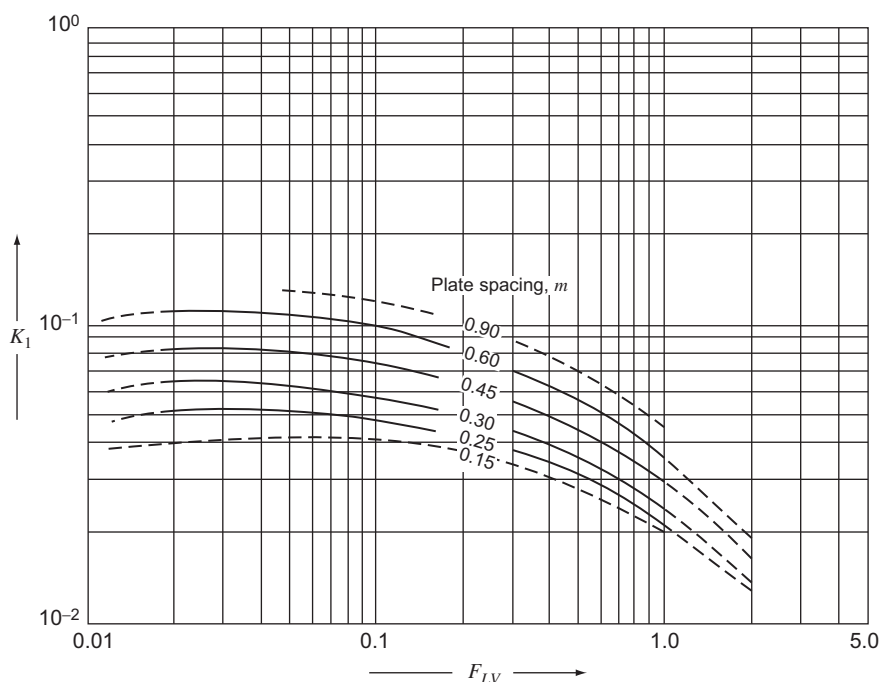


FIG. 17.34 Flooding velocity, sieve plates.

4. Hole: active area ratio greater than 0.10; for other ratios, apply the following corrections:

Hole: active area	Multiply K_1 by
0.10	1.0
0.08	0.9
0.06	0.8

5. Liquid surface tension 0.02 N/m; for other surface tensions, σ , multiply the value of K_1 by $[\sigma/0.02]^{0.2}$

To calculate the column diameter, an estimate of the net area A_n is required. As a first trial take the downcomer area as 12% of the total, and assume that the hole active area is 10%.

Where the vapor and liquid flow rates or physical properties vary significantly throughout the column, a plate design should be made for several points up the column. For distillation, it will usually be sufficient to design for the conditions above and below the feed points. Changes in the vapor flow rate will normally be accommodated by adjusting the hole area, often by blanking off some rows of holes. Different column diameters would only be used where there is a considerable change in flow rate. Changes in liquid rate can be allowed for by adjusting the liquid downcomer areas.

17.13.4 Liquid–flow arrangement

The choice of plate type (reverse, single pass, or multiple pass) will depend on the liquid flow rate and column diameter. An initial selection can be made using Fig. 17.35, which has been adapted from a similar figure given by Huang and Hodson (1958).

17.13.5 Entrainment

Entrainment can be estimated from the correlation given by Fair (1961) (Fig. 17.36), which gives the fractional entrainment ψ (kg/kg gross liquid flow) as a function of the liquid–vapor factor F_{LV} , with the percentage approach to flooding as a parameter.

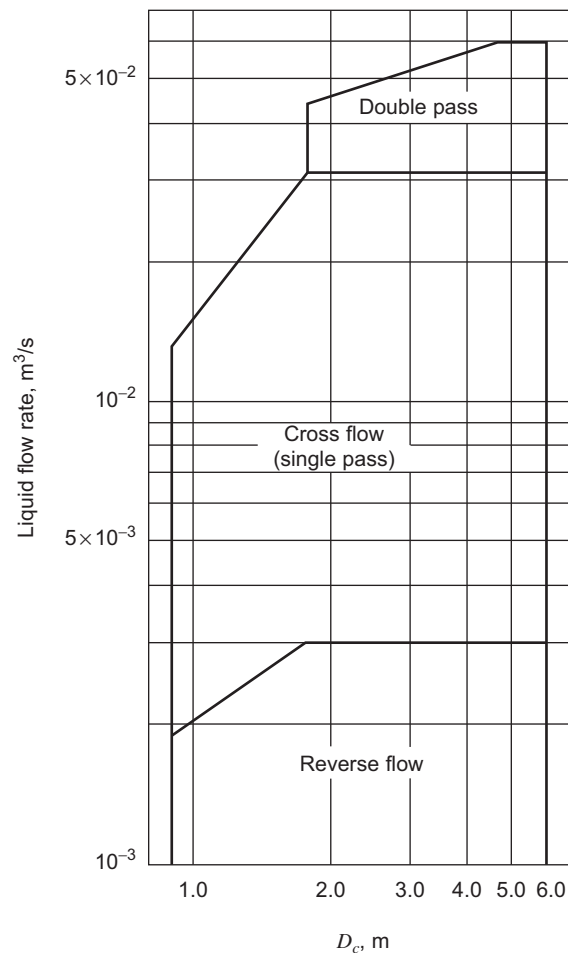


FIG. 17.35 Selection of liquid-flow arrangement.

The percentage flooding is given by:

$$\text{percentage flooding} = \frac{u_n \text{ (actual velocity based on net area)}}{u_f \text{ (from Equation 17.49)}} \quad (17.51)$$

The effect of entrainment on plate efficiency can be estimated using Equation 17.46.

As a rough guide the upper limit of ψ can be taken as 0.1; below this figure the effect on efficiency will be small. The optimum design value may be above this figure; see Fair (1963).

17.13.6 Weep point

The lower limit of the operating range occurs when liquid leakage through the plate holes becomes excessive. This is known as the weep point. The vapor velocity at the weep point is the minimum value for stable operation. The hole area must be chosen so that at the lowest operating rate the vapor flow velocity is still well above the weep point.

Several correlations have been proposed for predicting the vapor velocity at the weep point; see Chase (1967). That given by Eduljee (1959) is one of the simplest to use and has been shown to be reliable.

The minimum design vapor velocity is given by:

$$u_h = \frac{[K_2 - 0.90(25.4 - d_h)]}{(\rho_v)^{1/2}} \quad (17.52)$$

where u_h = minimum vapor velocity through the holes (based on the hole area), m/s

d_h = hole diameter, mm

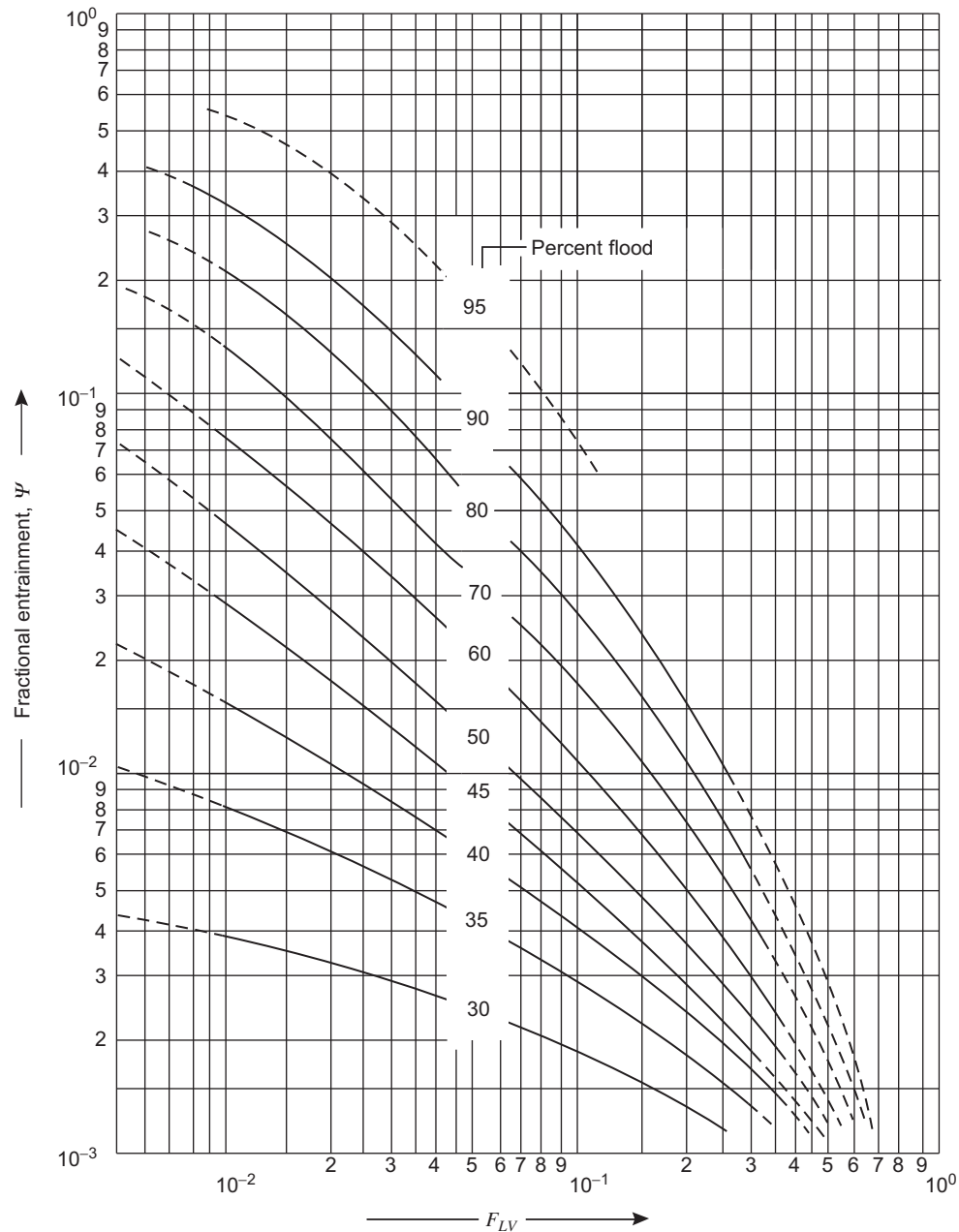


FIG. 17.36 Entrainment correlation for sieve plates. (From Fair, 1961.)

K_2 = a constant, dependent on the depth of clear liquid on the plate, obtained from Fig. 17.37

The clear liquid depth is equal to the height of the weir h_w plus the depth of the crest of liquid over the weir h_{ow} ; this is discussed in the next section.

17.13.7 Weir liquid crest

The height of the liquid crest over the weir can be estimated using the Francis weir formula (see Chhabra & Shankar, 2017). For a segmental downcomer this can be written as:

$$h_{ow} = 750 \left[\frac{L_w}{\rho_L l_w} \right]^{2/3} \quad (17.53)$$

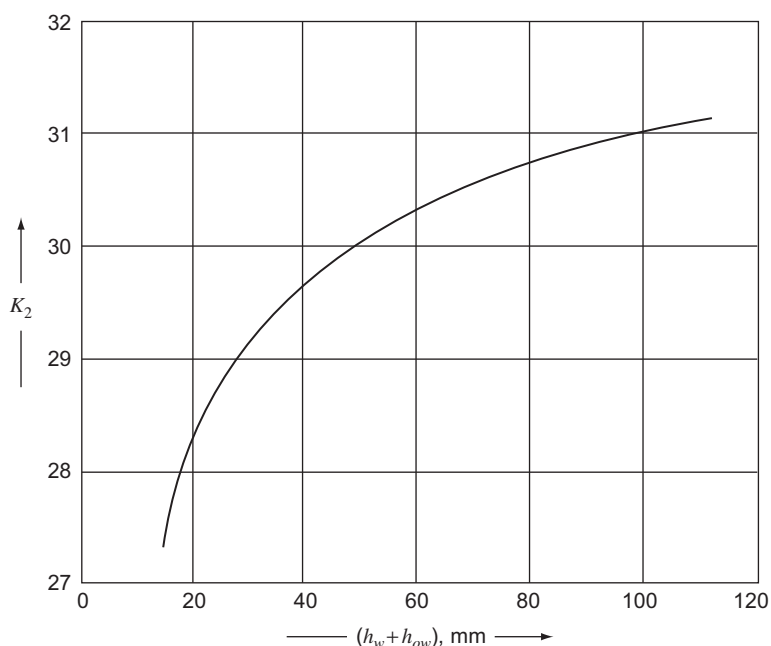


FIG. 17.37 Weep-point correlation. (From Eduljee, 1959.)

where l_w = weir length, m
 h_{0w} = weir crest, mm liquid
 L_w = liquid flow rate, kg/s

With segmental downcomers the column wall constricts the liquid flow, and the weir crest will be higher than that predicted by the Francis formula for flow over an open weir. The constant in Equation 17.53 has been increased to allow for this effect.

To ensure an even flow of liquid along the weir, the crest should be at least 10 mm at the lowest liquid rate. Serrated weirs known as picket-fence weirs are sometimes used for very low liquid rates, as illustrated in Fig. 17.38.

17.13.8 Weir dimensions

Weir height

The height of the weir determines the volume of liquid on the plate and is an important factor in determining the plate efficiency (see Section 17.10.4). A high weir will increase the plate efficiency but at the expense of a higher plate pressure drop. For columns operating above atmospheric pressure, the weir heights will normally be between 40 mm and 90 mm (1.5 and 3.5 in); 40 and 50 mm is recommended. For vacuum operation, lower weir heights are used to reduce the pressure drop; 6 to 12 mm ($\frac{1}{4}$ to $\frac{1}{2}$ in) is recommended.

Inlet weirs

Inlet weirs, or recessed pans, are sometimes used to improve the distribution of liquid across the plate but are seldom needed with segmental downcomers.

Weir length

With segmental downcomers the length of the weir fixes the area of the downcomer. The chord length will normally be between 0.6 and 0.85 of the column diameter. A good initial value to use is 0.77, equivalent to a downcomer

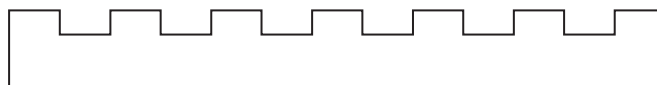


FIG. 17.38 Picket-fence weir.

area of 12%. The liquid flow rate over the weir should ideally be in the range 5 to 8 liters/s per m (2 to 3 gpm/in). If this is not feasible with a single-pass tray, then reverse-flow or multiple-pass trays should be considered, as illustrated in Fig. 17.28. If the liquid flow is too low, then a picket-fence weir can be specified.

The relationship between weir length and downcomer area for segmental downcomers is given in Fig. 17.39.

For double-pass plates, the width of the central downcomer is normally 200 to 250 mm (8 to 10 in).

17.13.9 Perforated area

The area available for perforation will be reduced by the obstruction caused by structural members (the support rings and beams) and by the use of calming zones.

Calming zones are unperforated strips of plate at the inlet and outlet sides of the plate. The width of each zone is usually made the same; recommended values are below 1.5 m diameter, 75 mm; above, 100 mm.

The width of the support ring for sectional plates will normally be 50 to 75 mm: the support ring should not extend into the downcomer area. A strip of unperforated plate will be left around the edge of cartridge-type trays to stiffen the plate.

The unperforated area can be calculated from the plate geometry. The relationship between the weir chord length, the chord height, and the angle subtended by the chord is given in Fig. 17.40.

17.13.10 Hole size

The hole sizes used vary from 2.5 to 19 mm; 5 mm is the preferred size for nonfouling applications. Larger holes are recommended for fouling systems. The holes are drilled or punched. Punching is cheaper, but the minimum size of hole that can be punched will depend on the plate thickness. For carbon steel, hole sizes approximately equal to the plate thickness can be punched, but for stainless steel the minimum hole size that can be punched is about twice the plate thickness. Typical plate thicknesses used are 5 mm ($\frac{3}{16}$ in) for carbon steel and 3 mm (12 gauge) for stainless steel.

When punched plates are used, they should be installed with the direction of punching upward. Punching forms a slight nozzle, and reversing the plate will increase the pressure drop.

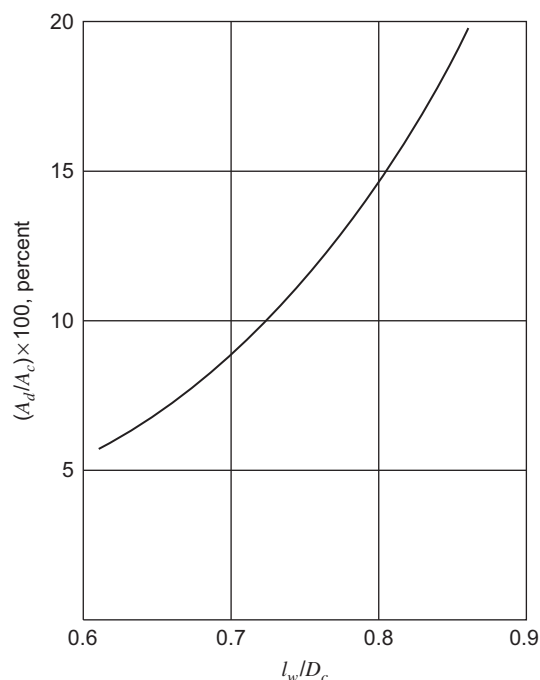


FIG. 17.39 Relation between downcomer area and weir length.

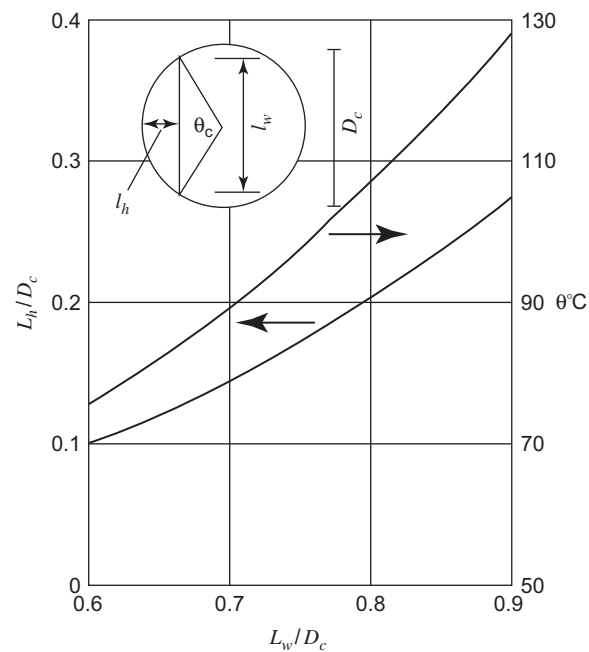


FIG. 17.40 Relation between angle subtended by chord, chord height, and chord length.

17.13.11 Hole pitch

The hole pitch (distance between the hole centers) l_p should not be less than 2.0 hole diameters, and the normal range will be 2.5 to 4.0 diameters. Within this range, the pitch can be selected to give the number of active holes required for the total hole area specified.

Square and equilateral triangular patterns are used; triangular is preferred. The total hole area as a fraction of the perforated area A_p is given by the following expression for an equilateral triangular pitch:

$$\frac{A_h}{A_p} = 0.9 \left[\frac{d_h}{l_p} \right]^2 \quad (17.54)$$

This equation is plotted in Fig. 17.41.

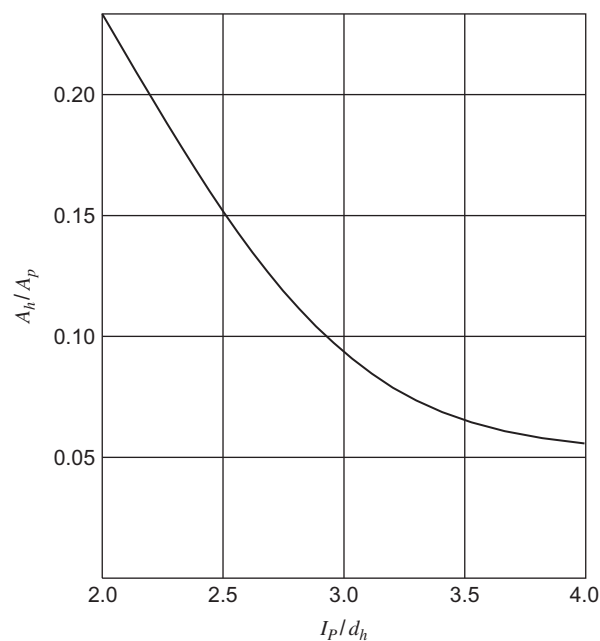


FIG. 17.41 Relation between hole area and pitch.

17.13.12 Hydraulic gradient

The hydraulic gradient is the difference in liquid level needed to drive the liquid flow across the plate. On sieve plates, unlike bubble-cap plates, the resistance to liquid flow will be small, and the hydraulic gradient is usually ignored in sieve-plate design. It can be significant in vacuum operation, as with the low weir heights used, the hydraulic gradient can be a significant fraction of the total liquid depth. Methods for estimating the hydraulic gradient are given by Fair (1963).

17.13.13 Liquid throw

The liquid throw is the horizontal distance traveled by the liquid stream flowing over the downcomer weir. It is only an important consideration in the design of multiple-pass plates. Bolles (1963) gives a method for estimating the liquid throw. If the liquid throw is excessive, antijump baffles can be used to ensure that liquid flows down and does not jump to the adjacent section.

17.13.14 Plate pressure drop

The pressure drop over the plates is an important design consideration. There are two main sources of pressure loss: that due to vapor flow through the holes (an orifice loss) and that due to the static head of liquid on the plate.

A simple additive model is normally used to predict the total pressure drop. The total is taken as the sum of the pressure drop calculated for the flow of vapor through the dry plate (the dry plate drop h_d); the head of clear liquid on the plate ($h_w + h_{ow}$); and a term to account for other, minor, sources of pressure loss, the so-called residual loss h_r . The residual loss is the difference between the observed experimental pressure drop and the simple sum of the dry plate drop and the clear-liquid height. It accounts for the two effects: the energy to form the vapor bubbles and the fact that on an operating plate the liquid head will not be clear liquid but a head of "aerated" liquid froth, and the froth density and height will be different from that of the clear liquid.

It is convenient to express the pressure drops in terms of millimeters of liquid. In pressure units:

$$\Delta P_t = 9.81 \times 10^{-3} h_t \rho_L \quad (17.55)$$

where ΔP_t = total plate pressure drop, Pa (N/m²)
 h_t = total plate pressure drop, mm liquid

Dry plate drop

The pressure drop through the dry plate can be estimated using expressions derived for flow through orifices:

$$h_d = 51 \left[\frac{u_h}{C_0} \right]^2 \frac{\rho_v}{\rho_L} \quad (17.56)$$

where the orifice coefficient C_0 is a function of the plate thickness, hole diameter, and the hole-to-perforated area ratio. C_0 can be obtained from Fig. 17.42, which has been adapted from a similar figure by Liebson et al. (1957). u_h is the velocity through the holes, m/s.

Residual head

Methods have been proposed for estimating the residual head as a function of liquid surface tension, froth density, and froth height; however, as this correction term is small, the use of an elaborate method for its estimation is not justified, and the simple equation proposed by Hunt et al. (1955) can be used:

$$h_r = \frac{12.5 \times 10^3}{\rho_L} \quad (17.57)$$

Equation 17.57 is equivalent to taking the residual drop as a fixed value of 12.5 mm of water ($\frac{1}{2}$ in).

Total drop

The total plate drop is given by:

$$h_t = h_d + (h_w + h_{ow}) + h_r \quad (17.58)$$

If the hydraulic gradient is significant, half its value is added to the clear liquid height.

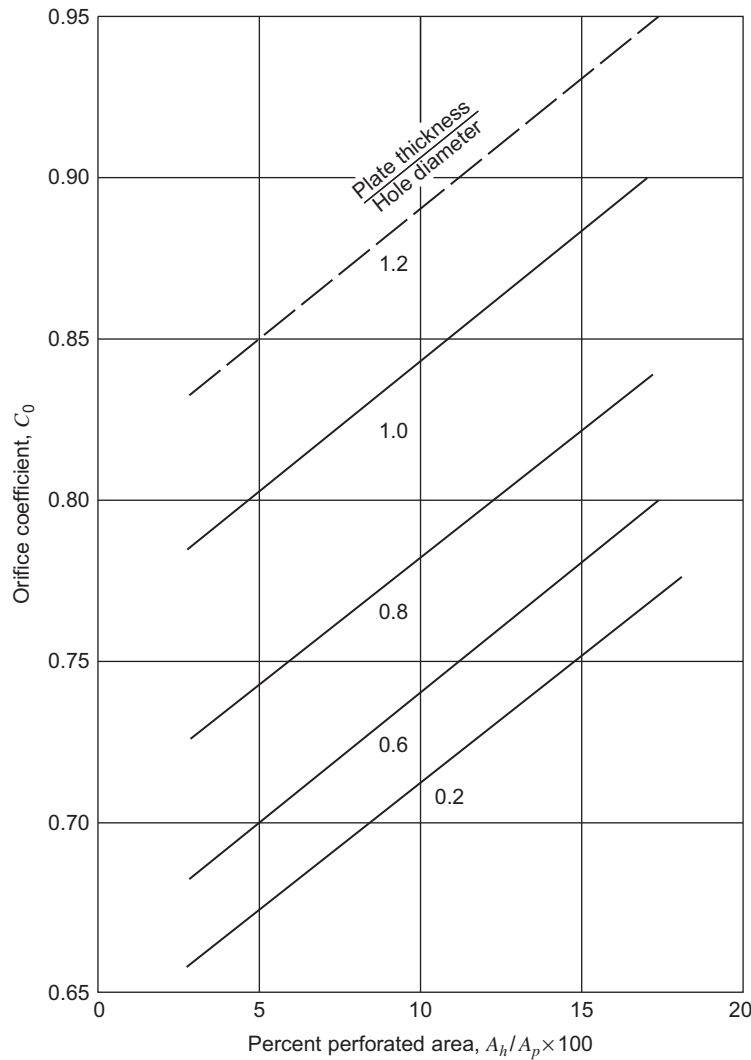


FIG. 17.42 Discharge coefficient, sieve plates. (From Liebson et al., 1957.)

17.13.15 Downcomer design (backup)

The downcomer area and plate spacing must be such that the level of the liquid and froth in the downcomer is well below the top of the outlet weir on the plate above. If the level rises above the outlet weir, the column will flood.

The backup of liquid in the downcomer is caused by the pressure drop over the plate (the downcomer in effect forms one leg of a U-tube) and the resistance to flow in the downcomer itself (Fig. 17.43).

In terms of clear liquid, the downcomer backup is given by:

$$h_b = (h_w + h_{ow}) + h_t + h_{dc} \quad (17.59)$$

where h_b = downcomer backup, measured from plate surface, mm

h_{dc} = head loss in the downcomer, mm

The main resistance to flow will be caused by the constriction at the downcomer outlet, and the head loss in the downcomer can be estimated using the equation given by Cicalese et al. (1947):

$$h_{dc} = 166 \left[\frac{L_{wd}}{\rho_L A_m} \right]^2 \quad (17.60)$$

where L_{wd} = liquid flow rate in downcomer, kg/s

A_m = either the downcomer area A_d or the clearance area under the downcomer A_{ap} , whichever is smaller, m^2

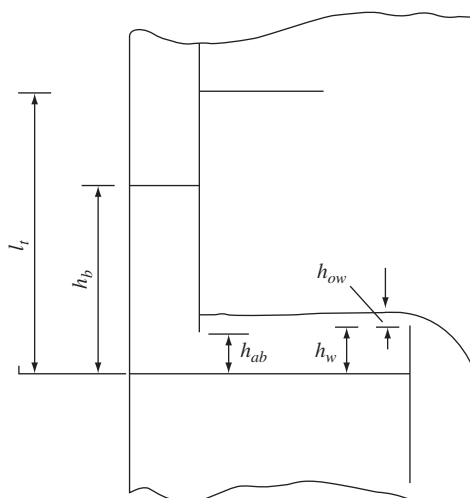


FIG. 17.43 Downcomer back-up.

The clearance area under the downcomer is given by:

$$A_{ap} = h_{ap} l_w \quad (17.61)$$

where h_{ap} is height of the bottom edge of the apron above the plate. This height is normally set at 5 to 10 mm ($\frac{1}{4}$ to $\frac{1}{2}$ in) below the outlet weir height:

$$h_{ap} = h_w - (5 \text{ to } 10 \text{ mm})$$

Froth height

To predict the height of “aerated” liquid on the plate and the height of froth in the downcomer, some means of estimating the froth density is required. The density of the “aerated” liquid will normally be between 0.4 and 0.7 times that of the clear liquid. A number of correlations have been proposed for estimating froth density as a function of the vapor flow rate and the liquid physical properties; see [Chase \(1967\)](#); however, none is particularly reliable, and for design purposes it is usually satisfactory to assume an average value of 0.5 of the liquid density.

This value is also taken as the mean density of the fluid in the downcomer, which means that for safe design the clear liquid backup, calculated from Equation 17.59, should not exceed half the plate spacing l_t , to avoid flooding.

Allowing for the weir height:

$$h_b \leq \frac{1}{2}(l_t + h_w) \quad (17.62)$$

This criterion is, if anything, oversafe, and where close plate spacing is desired, a better estimate of the froth density in the downcomer should be made. The method proposed by [Thomas and Shah \(1964\)](#) is recommended. [Kister \(1992\)](#) recommends that the froth height in the downcomer should not be greater than 80% of the tray spacing.

Downcomer residence time

Sufficient residence time must be allowed in the downcomer for the entrained vapor to disengage from the liquid stream, to prevent heavily “aerated” liquid being carried under the downcomer. A time of at least 3 seconds is recommended.

The downcomer residence time is given by:

$$t_r = \frac{A_d h_{bc} \rho_L}{L_{wd}} \quad (17.63)$$

where t_r = residence time, s

h_{bc} = clear liquid backup, m

Example 17.6

Design the plates for the column specified in Example 17.2. Take the minimum feed rate as 70% of the maximum (maximum feed 10,000 kg/h). Use sieve plates.

Solution

As the liquid and vapor flow rates and compositions will vary up the column, plate designs should be made above and below the feed point. Only the bottom plate will be designed in detail in this example.

From the McCabe–Thiele diagram, Example 17.2:

Number of stages = 10

Top composition 95 mol%, bottom composition 1 mol%

Reflux ratio = 1.24

Flow rates

Mol. weight feed = $0.1 \times 58 + (1 - 0.1)18 = 22$

Feed = $10,000/22 = 454.5$ kmol/h

Overall mass balance: $D + B = 454.5$

A mass balance on acetone gives:

$$0.95D + 0.01B = 0.1(454.5)$$

Hence: $D = 43.5$ kmol/h, $B = 411.0$ kmol/h

Vapor rate, $V = D(1 + R) = 43.5(1 + 1.24) = 97.5$ kmol/h

The feed is saturated liquid, so:

liquid flow above feed, $L = RD = 1.24(43.52) = 54.0$ kmol/h

liquid flow below feed, $L = RD + F = 454.5 + 54 = 508.5$ kmol/h

Physical properties

Estimate base pressure, assume column efficiency of 60%, ignore reboiler.

$$\text{Number of real stages} = \frac{10}{0.6} = 17$$

Assume pressure drop per plate is 100 mm water.

$$\text{Column pressure drop} = 100 \times 10^{-3} \times 1000 \times 9.81 \times 17 = 16,677 \text{ Pa}$$

$$\text{Top pressure, } 1 \text{ atm}(14.7 \text{ lb/in}^2) = 101.4 \times 10^3 \text{ Pa}$$

$$\text{Estimated bottom pressure} = 101.4 \times 10^3 + 16,677$$

$$118,077 \text{ Pa} = \underline{\underline{1.18 \text{ bar}}}$$

From UniSim Design, base temperature 96.0 °C:

$$\rho_v = 0.693 \text{ kg/m}^3, \rho_L = 944 \text{ kg/m}^3$$

$$\text{Molecular weight} = 18.4, \text{ surface tension} = 58.9 \times 10^{-3} \text{ N/m}$$

Distillate, 95 mol% acetone, 56 °C:

$$\rho_v = 2.07 \text{ kg/m}^3, \rho_L = 748 \text{ kg/m}^3$$

$$\text{Molecular weight} = 56.1, \text{ surface tension} = 22.7 \times 10^{-3} \text{ N/m}$$

Column diameter

Neglecting differences in molecular weight between vapor and liquid:

$$F_{LV} \text{ bottom} = \frac{508.5}{97.5} \sqrt{\frac{0.693}{944}} = 0.141 \quad (17.50)$$

$$F_{LV} \text{ top} = \frac{54}{97.5} \sqrt{\frac{2.07}{748}} = 0.0291 \quad (17.50)$$

Take plate spacing as 0.5 m

From Fig. 17.34:

$$\text{base } K_1 = 7.5 \times 10^{-2}$$

$$\text{top } K_1 = 9.0 \times 10^{-2}$$

Correction for surface tensions:

$$\text{base } K_1 = \left(\frac{59}{20}\right)^{0.2} \times 7.5 \times 10^{-2} = 9.3 \times 10^{-2}$$

$$\text{top } K_1 = \left(\frac{23}{20}\right)^{0.2} \times 9.0 \times 10^{-2} = 9.3 \times 10^{-2}$$

$$\text{base } u_f = 9.3 \times 10^{-2} \sqrt{\frac{944 - 0.693}{0.693}} = 3.43 \text{ m/s} \quad (17.49)$$

$$\text{top } u_f = 9.3 \times 10^{-2} \sqrt{\frac{748 - 2.07}{2.07}} = 1.77 \text{ m/s} \quad (17.49)$$

Design for 85% flooding at maximum flow rate:

$$\text{base } u_n = 3.43 \times 0.85 = 2.92 \text{ m/s}$$

$$\text{top } u_n = 1.77 \times 0.85 = 1.50 \text{ m/s}$$

Maximum volumetric flow rate:

$$\text{base} = \frac{97.5 \times 18.4}{0.693 \times 3600} = 0.719 \text{ m}^3/\text{s}$$

$$\text{top} = \frac{97.5 \times 56.1}{2.07 \times 3600} = 0.734 \text{ m}^3/\text{s}$$

Net area required:

$$\text{base} = \frac{0.719}{2.92} = 0.246 \text{ m}^2$$

$$\text{top} = \frac{0.734}{1.50} = 0.489 \text{ m}^2$$

As first trial, take downcomer area as 12% of total:

Column cross-sectioned area:

$$\text{base} = \frac{0.246}{0.88} = 0.280 \text{ m}^2$$

$$\text{top} = \frac{0.489}{0.88} = 0.556 \text{ m}^2$$

Column diameter:

$$\text{base} = \sqrt{\frac{0.28 \times 4}{\pi}} = 0.60 \text{ m}$$

$$\text{top} = \sqrt{\frac{0.556 \times 4}{\pi}} = 0.84 \text{ m}$$

Use same diameter above and below feed, reducing the perforated area for plates above the feed.

This is too large to use standard pipe, so round up to nearest standard head size; inside diameter 914.4 mm (36 in).

Liquid flow pattern

$$\text{Maximum volumetric liquid rate} = \frac{508.5}{3600} \times \frac{18.4}{944} = 2.75 \times 10^{-3} \text{ m}^3/\text{s}$$

The plate diameter is outside the range of Fig. 17.35, but it is clear that a single-pass plate can be used.

Provisional plate design

Column diameter $D_c = 0.914 \text{ m}$

Column area $A_c = 0.556 \text{ m}^2$

Downcomer area $A_d = 0.12 \times 0.556 = 0.067 \text{ m}^2$, at 12%

Net area $A_n = A_c - A_d = 0.556 - 0.067 = 0.489 \text{ m}^2$

Active area $A_a = A_c - 2A_d = 0.556 - 0.134 = 0.422 \text{ m}^2$

Hole area A_h take 10% A_a as first trial = 0.042 m^2

Weir length (from Fig. 17.39) = $0.76 \times 0.914 = 0.695 \text{ m}$

Take weir height	50 mm
Hole diameter	5 mm
Plate thickness	5 mm

Check weeping

$$\text{Maximum liquid rate} = \frac{508.5}{3600} \times \frac{18.4}{944} = 2.60 \text{ kg/s}$$

$$\text{Minimum liquid rate at 70\% turndown} = 0.7 \times 2.6 = 1.82 \text{ kg/s}$$

$$\text{Maximum } h_{ow} = \left(\frac{2.6}{944 \times 0.695} \right)^{2/3} = 25.0 \text{ mm liquid} \quad (17.53)$$

$$\text{Minimum } h_{ow} = \left(\frac{1.82}{944 \times 0.695} \right)^{2/3} = 19.7 \text{ mm liquid} \quad (17.53)$$

$$\text{At minimum rate } h_w + h_{ow} = 50 + 19.7 = 69.7 \text{ mm}$$

From Fig. 17.37,

$$K_2 = 30.6$$

$$u_h (\text{min}) = \frac{[30.6 - 0.90(25.4 - 5)]}{(0.693)^{1/2}} = 14.7 \text{ m/s} \quad (17.52)$$

$$\text{Actual minimum vapor velocity} = \frac{\text{minimum vapour rate}}{A_h} = \frac{0.7 \times 0.719}{0.042} = 12.0 \text{ m/s}$$

So the minimum operating rate will lead to weeping at the bottom of the column. Reduce hole area to 7% of active area = $0.422 \times 0.07 = 0.0295 \text{ m}^2$.

$$\text{New actual minimum vapor velocity} = \frac{0.7 \times 0.719}{0.0295} = 17.1 \text{ m/s}$$

which is now well above the weep point.

Plate pressure drop

Dry plate drop:

Maximum vapor velocity through holes:

$$u_h (\text{max}) = \frac{0.719}{0.0295} = 24.4 \text{ m/s}$$

From Fig. 17.42, for plate thickness/hole diameter = 1 and $A_h/A_p \approx A_h/A_a = 0.07$, $C_0 = 0.82$:

$$h_d = 51 \left[\frac{24.4}{0.82} \right]^2 \frac{0.693}{944} = 33.1 \text{ mm liquid} \quad (17.56)$$

residual head:

$$h_r = \frac{12.5 \times 10^3}{944} = 13.2 \text{ mm liquid} \quad (17.55)$$

total plate pressure drop:

$$h_t = 33 + (50 + 25) + 13 = 118 \text{ mm liquid}$$

Note: 100 mm was assumed to calculate the base pressure. The calculation could be repeated with a revised estimate, but the small change in physical properties will have little effect on the plate design. 118 mm per plate is considered acceptable.

Downcomer liquid backup

Downcomer pressure loss:

Take $h_{ap} = h_w - 10 = 40 \text{ mm}$.

Area under apron, $A_{ap} = 0.695 \times 40 \times 10^{-3} = 0.028 \text{ m}^2$.

As this is less than $A_d = 0.067 \text{ m}^2$ use A_{ap} in Equation 17.60:

$$h_{dc} = 166 \left[\frac{2.60}{944 \times 0.028} \right]^2 = 1.61 \text{ mm} \quad (17.60)$$

say 2 mm.

Backup in downcomer:

$$h_b = (50 + 25) + 118 + 2 = 195 \text{ mm} \quad (17.59)$$

$$195 \text{ mm} < \frac{1}{2} (\text{plate spacing} + \text{weir height}), \text{ so plate spacing is acceptable}$$

Check residence time:

$$t_r = \frac{0.067 \times 0.195 \times 944}{2.60} = 4.7 \text{ s} \quad (17.63)$$

>3 s, satisfactory.

Check entrainment

$$u_v = \frac{0.719}{0.489} = 1.47 \text{ m/s}$$

$$\text{percent flooding} = \frac{1.47}{3.43} = 42.8\%$$

$F_{LV} = 0.14$, so from Fig. 17.36, $\Psi = 0.0038$, well below 0.1.

As the percent flooding is well below the design figure of 85, the column diameter could be reduced, but this would increase the pressure drop.

Trial layout

Use cartridge-type construction. Allow 50-mm unperforated strip around plate edge; 50-mm-wide calming zones.

Perforated area

From Fig. 17.40, at $l_w/D_c = 0.695/0.914 = 0.76$, $\theta_c = 99^\circ$

angle subtended by the edge of the plate = $180 - 99 = 81^\circ$

mean length, unperforated edge strips = $(0.914 - 50 \times 10^{-3})\pi \times 81/180 = 1.22 \text{ m}$

area of unperforated edge strips = $50 \times 10^{-3} \times 1.22 = 0.061 \text{ m}^2$

mean length of calming zone, approx. = weir length + width of unperforated strip
 $= 0.695 + 50 \times 10^{-3} = 0.745 \text{ m}$

area of calming zones = $2(0.745 \times 50 \times 10^{-3}) = 0.0745 \text{ m}^2$

total area for perforations, $A_p = 0.422 - 0.061 - 0.075 = 0.286 \text{ m}^2$

$A_h/A_p = 0.0295/0.286 = 0.103$

From Fig. 17.41, $l_p/d_h = 2.9$; satisfactory, within 2.5 to 4.0.

Number of holes

Area of one hole = $1.964 \times 10^{-5} \text{ m}^2$

Number of holes = $\frac{0.0295}{1.964 \times 10^{-5}} = 1502$

Plate specification

The final plate specification is shown in Fig. 17.44.

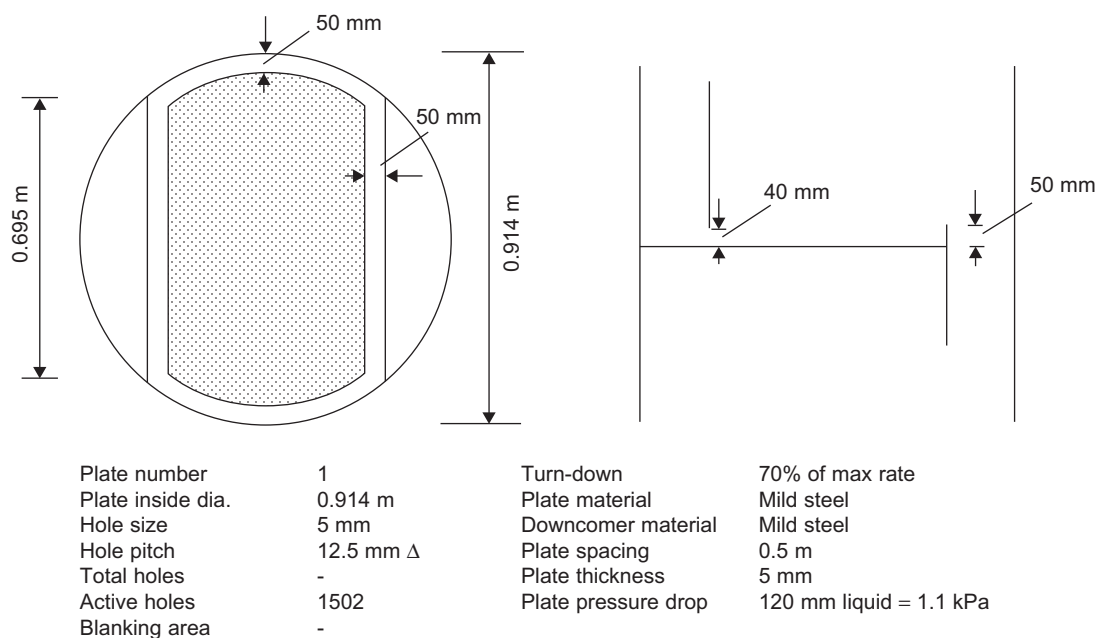


FIG. 17.44 Plate specification for Example 17.6.

Example 17.7

For the plate design in Example 17.6, estimate the plate efficiency for the plate on which the concentration of acetone is 5 mol%. Use the AIChE method.

Solution

Plate will be in the stripping section (see Fig. 17.9).

Plate dimensions:

$$\text{active area} = 0.422 \text{ m}^2$$

$$\text{length between downcomers (Fig. 17.40) (liquid path, } Z_L) = 0.914 (1 - 2 \times 0.175) = 0.594 \text{ m}$$

$$\text{weir height} = 50 \text{ mm}$$

Flow rates, check efficiency at minimum rates, at column base:

$$\text{Vapor} = 0.7 \times \frac{97.5}{3600} = 0.019 \text{ kmol/s}$$

$$\text{Liquid} = 0.7 \times \frac{508.5}{3600} = 0.099 \text{ kmol/s}$$

From the McCabe–Thiele diagram (Fig. 17.9) at $x = 0.05$, assuming 60% plate efficiency, $y \approx 0.35$. The liquid composition, $x = 0.05$, will occur on around the third plate from the bottom (allowing for the reboiler and 60% efficiency per stage). The pressure on this plate will be approximately:

$$101.4 \times 10^3 + (14 \times 0.118 \times 9.81 \times 944) = 116.7 \text{ kPa}$$

$$\text{say, } 1.17 \text{ bar}$$

At this pressure, the plate temperature will be about 92 °C, and the liquid and vapor physical properties, from UniSim Design are:

Liquid:

$$\text{molar weight} = 20, \rho_L = 932.7 \text{ kg/m}^3, \mu_L = 0.3544 \times 10^{-3} \text{ Nm}^{-2}\text{s}, \sigma = 60.2 \times 10^{-3} \text{ N/m}$$

Vapor:

$$\text{molar weight} = 32, \rho_v = 1.233 \text{ kg/m}^3, \mu_v = 9.17 \times 10^{-6} \text{ Nm}^{-2}\text{s}$$

$$D_L = 4.16 \times 10^{-9} \text{ m}^2/\text{s}$$

$$D_v = 17.4 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\text{Vapor volumetric flow rate} = \frac{0.019 \times 32}{1.233} = 0.493 \text{ m}^3/\text{s}$$

$$\text{Liquid volumetric flow rate} = \frac{0.099 \times 20}{932.7} = 2.12 \times 10^{-3} \text{ m}^3/\text{s}$$

$$u_a = \frac{0.493}{0.422} = 1.17 \text{ m/s}$$

$$F_v = u_a \sqrt{\rho_v} = 2.365 \text{ kg}^{0.5} \text{ m}^{-0.5} \text{ s}^{-1}$$

$$\text{Average width over active surface} = 0.422/0.594 = 0.71 \text{ m}$$

$$L_p = \frac{2.12 \times 10^{-3}}{0.71} = 2.99 \times 10^{-3} \text{ m}^2/\text{s}$$

$$N_G = \frac{(0.776 + 4.57 \times 10^{-3} \times 50 - 0.24 \times 2.365 + 105 \times 2.99 \times 10^{-3})}{\left(\frac{9.17 \times 10^{-6}}{1.233 \times 17.4 \times 10^{-6}}\right)^{0.5}} = 1.15 \quad (17.38)$$

$$Z_c = 0.006 + 0.73 \times 10^{-3} \times 50 - 0.24 \times 10^{-3} \times 2.365 \times 50 + 1.22 \times 2.99 \times 10^{-3}$$

$$= 17.8 \times 10^{-3} \quad (17.43)$$

$$t_L = \frac{17.8 \times 10^{-3} \times 0.594}{2.99 \times 10^{-3}} = 3.54 \text{ s} \quad (17.41)$$

$$N_L = (4.13 \times 10^8 \times 4.16 \times 10^{-9})^{0.5} (0.21 \times 2.365 + 0.15) \times 3.54 = 3.00 \quad (17.40)$$

$$D_e = (0.0038 + 0.017 \times 1.17 + 3.86 \times 2.99 \times 10^{-3} + 0.18 \times 10^{-3} \times 50)^2$$

$$= 1.96 \times 10^{-3} \quad (17.45)$$

$$\text{Pe} = \frac{(0.594)^2}{1.96 \times 10^{-3} \times 3.54} = 50.8 \quad (17.44)$$

From the McCabe–Thiele diagram, at $x = 0.05$, the slope of the equilibrium line ≈ 12.0 , so:

$$\frac{mV}{L} = \frac{12 \times 0.019}{0.099} = 2.30$$

$$\frac{\left(\frac{mV}{L}\right)}{N_L} = \frac{2.30}{3.00} = 0.767$$

From Fig. 17.21, $E_{mv} = 0.43$:

$$\frac{mV}{L} \times E_{mv} = 2.30 \times 43 = 0.989$$

From Fig. 17.22, $E_{mV}/E_{mv} = 1.62$:

$$E_{mV} = 0.43 \times 1.62 = 0.697$$

So plate efficiency = 70%.

Note: The slope of the equilibrium line is difficult to determine at $x = 0.05$, but any error will not greatly affect the value of E_{mV} .

Example 17.8

Calculate the plate efficiency for the plate design considered in Examples 17.6 and 17.7, using Van Winkle's correlation.

Solution

From Examples 17.6 and 17.7:

$$\rho_L = 932.7 \text{ kg/m}^3, \mu_L = 0.3544 \times 10^{-3} \text{ Nm}^{-2}\text{s}, D_{LK} = D_L = 4.16 \times 10^{-9} \text{ m}^2/\text{s}, \sigma = 60.2 \times 10^{-3} \text{ N/m}$$

$$\rho_v = 1.233 \text{ kg/m}^3, \mu_v = 9.17 \times 10^{-6} \text{ Nm}^{-2}\text{s}$$

$$h_w = 50 \text{ mm}$$

$$FA(\text{fractional area}) = A_h/A_c = 0.0295/0.556 = 0.053$$

$$u_v = \text{superficial vapor velocity} = 0.493/0.556 = 0.887 \text{ m/s}$$

$$Dg = \left(\frac{0.0602}{0.3544 \times 10^{-3} \times 0.887} \right) = 191.6$$

$$Sc = \left(\frac{0.3544 \times 10^{-3}}{932.7 \times 4.16 \times 10^{-9}} \right) = 91.3$$

$$Re = \left(\frac{50 \times 10^{-3} \times 0.887 \times 932.7}{0.3544 \times 0.053} \right) = 2.2 \times 10^3$$

$$E_{mV} = 0.07(191.6)^{0.14}(91.3)^{0.25}(2.2 \times 10^3)^{0.08} \\ = \underline{0.836} \text{ (84\%)} \quad (17.36)$$

This seems rather large compared with the value found using the AIChE method, so the value calculated in Example 17.7 is preferred.

17.14 Packed columns

Packed columns are used for distillation, gas absorption, and liquid–liquid extraction; only distillation and absorption will be considered in this section. Stripping (desorption) is the reverse of absorption and the same design methods apply.

The gas–liquid contact in a packed column is continuous, not stage-wise like a plate column. The liquid flows down the column over the packing surface, and the gas or vapor flows countercurrently, up the column. In some gas absorption columns co-current flow is used. The performance of a packed column is highly dependent on the maintenance of good liquid and gas distribution throughout the packed bed, and this is an important consideration in packed-column design.

A schematic diagram, showing the main features of a packed absorption column, is given in Fig. 17.45. A packed distillation column will be similar to the plate columns shown in Fig. 17.1, with the plates replaced by packed sections.

The design of packed columns using random packings is covered in books by Kister (1992), Strigle (1994), and Billet (1995).

Choice of plates or packing

The choice between a plate or packed column for a particular application can only be made with complete assurance by costing each design; however, the choice can usually be made on the basis of experience by considering the main advantages and disadvantages of each type, listed here:

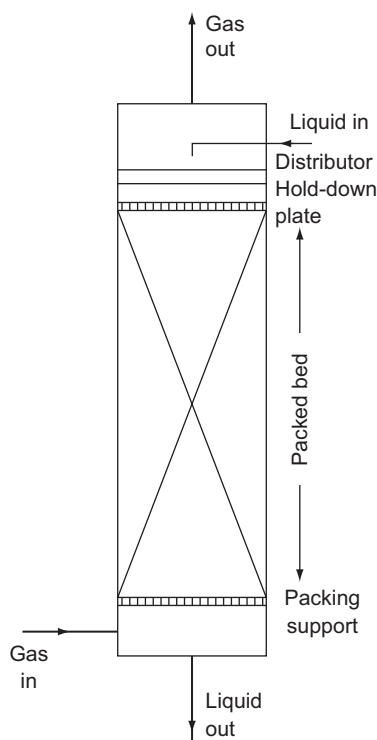


FIG. 17.45 Packed absorption column.

1. Plate columns can be designed to handle a wider range of liquid and gas flow rates than packed columns.
2. Packed columns are not suitable for very low liquid rates.
3. The efficiency of a plate can be predicted with more certainty than the equivalent term for packing (HETP or HTU).
4. Plate columns can be designed with more assurance than packed columns. There is always some doubt that good liquid distribution can be maintained throughout a packed column under all operating conditions, particularly in large columns.
5. It is easier to make provision for cooling in a plate column; coils can be installed on the plates.
6. It is easier to make provision for the withdrawal of side streams from plate columns.
7. If the liquid causes fouling or contains solids, it is easier to make provision for cleaning in a plate column; manways can be installed on the plates. With small-diameter columns, it may be cheaper to use packing and replace the packing when it becomes fouled.
8. For corrosive liquids, a packed column will usually be cheaper than the equivalent plate column.
9. The liquid hold-up is appreciably lower in a packed column than a plate column. This can be important when the inventory of toxic or flammable liquids must be minimized for safety reasons.
10. Packed columns are more suitable for handling foaming systems.
11. The pressure drop per equilibrium stage (HETP) can be lower for packing than plates, and packing should be considered for vacuum columns.
12. Packing should always be considered for small-diameter columns, say less than 0.6 m, where plates would be difficult to install and expensive.

Packed-column design procedures

The design of a packed column involves the following steps:

1. Select the type and size of packing.
2. Determine the column height required for the specified separation.
3. Determine the column diameter (capacity) to handle the liquid and vapor flow rates.
4. Select and design the column internal features: packing support, liquid distributor, redistributors.

These steps are discussed in the following sections, and a packed-column design is illustrated in Example 17.9.

17.14.1 Types of packing

The principal requirements of a packing are that it should:

- Provide a large surface area: a high interfacial area between the gas and liquid.
- Have an open structure: low resistance to gas flow.
- Promote uniform liquid distribution on the packing surface.
- Promote uniform vapor or gas flow across the column cross-section.

Many types and shapes of packing have been developed to satisfy these requirements. They can be divided into two broad classes:

1. Packings with a regular geometry, such as stacked rings, grids, and proprietary structured packings.
2. Random packings: rings, saddles, and proprietary shapes, which are dumped into the column and take up a random arrangement.

Grids have an open structure and are used for high gas rates, where low pressure drop is essential; for example, in cooling towers. Random packings and structured packing elements are more commonly used in the process industries.

Random packing

The principal types of random packings are shown in Fig. 17.46. Design data for these packings are given in Table 17.2. The design methods and data given in this section can be used for the preliminary design of packed columns, but for detailed design, it is advisable to consult the packing manufacturer's technical literature to obtain data for the particular packing that will be used. The packing manufacturers should be consulted for details of the many special types of packing that are available for special applications.

Raschig rings (Fig. 17.46a) are one of the oldest specially manufactured types of random packing and are still in general use. Pall rings (see Fig. 17.46b) are essentially Raschig rings in which openings have been made by folding strips of the surface into the ring. This increases the free area and improves the liquid distribution characteristics. Berl saddles (see Fig. 17.46c) were developed to give improved liquid distribution compared with Raschig rings. INTALOX saddles (see Fig. 17.46d) can be considered to be an improved type of Berl saddle; their shape makes them easier to manufacture than Berl saddles. The HY-PAK and SUPER INTALOX packings shown in Fig. 17.46(e and f) can be considered improved types of Pall ring and INTALOX saddle, respectively.

INTALOX saddles, SUPER INTALOX, and HY-PAK packings are proprietary designs and registered trademarks of Koch-Glitsch, LP.

Ring and saddle packings are available in a variety of materials: ceramics, metals, plastics, and carbon. Metal and plastics (polypropylene) rings are more efficient than ceramic rings, as it is possible to make the walls thinner.

Raschig rings are cheaper per unit volume than Pall rings or saddles but are less efficient, and the total cost of the column will usually be higher if Raschig rings are specified. For new columns, the choice will normally be between Pall rings and Berl or INTALOX saddles.

The choice of material will depend on the nature of the fluids and the operating temperature. Ceramic packing will be the first choice for corrosive liquids, but ceramics are unsuitable for use with strong alkalis. Packings made of plastics are attacked by some organic solvents and can only be used up to moderate temperatures, so are unsuitable for distillation columns. Where the column operation is likely to be unstable, metal rings should be specified, as ceramic packing is easily broken. The choice of packings for distillation and absorption is discussed in detail by Eckert (1963), Strigle (1994), Kister (1992), and Billet (1995).

Packing size

In general, the largest size of packing that is suitable for the size of column should be used, up to 50 mm. Small sizes are appreciably more expensive than the larger sizes.

Above 50 mm, the lower cost per cubic meter does not normally compensate for the lower mass transfer efficiency. Use of too large a size in a small column can cause poor liquid distribution.

Recommended size ranges are:

Column diameter	Use packing size
<0.3 m (1 ft)	<25 mm (1 in)
0.3 to 0.9 m (1 to 3 ft)	25 to 38 mm (1 to 1.5 in)
>0.9 m	50 to 75 mm (2 to 3 in)

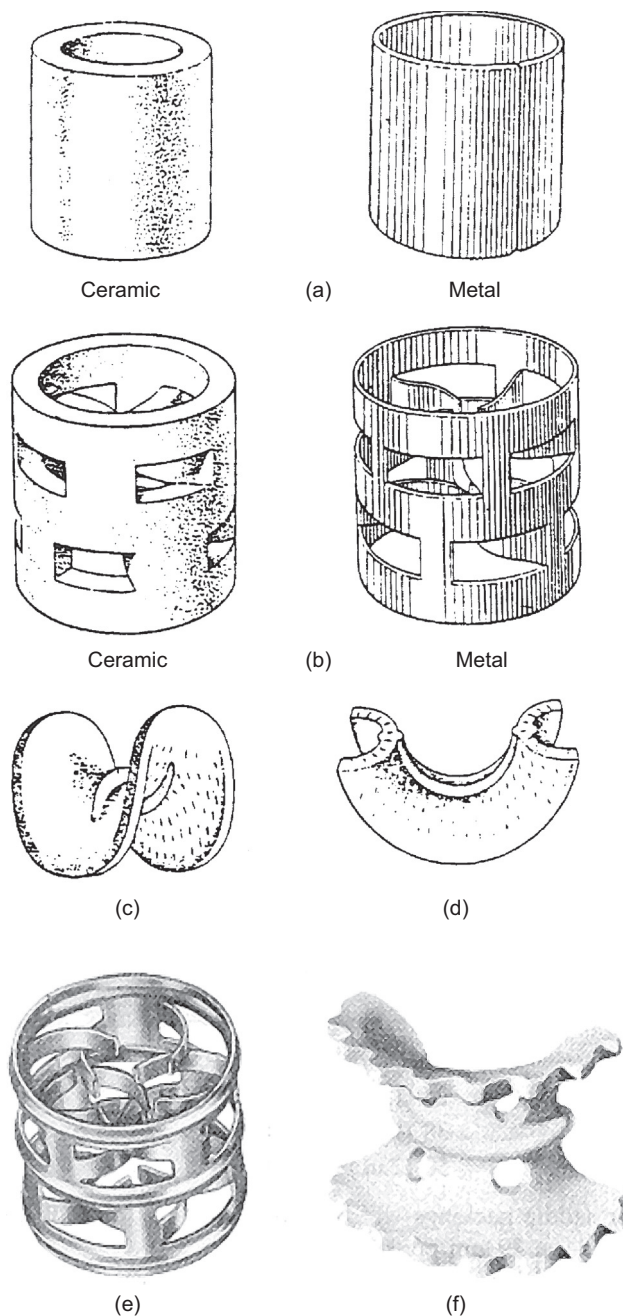


FIG. 17.46 Types of packing (Koch-Glitsch, LP). (a) Raschig rings. (b) Pall rings. (c) Berl saddle ceramic. (d) INTALOX saddle ceramic. (e) Metal HY-PAK. (f) Ceramic, SUPER INTALOX.

Structured packing

The term *structured packing* refers to packing elements made up from wire mesh or perforated metal sheets. The material is folded and arranged with a regular geometry to give a high surface area with a high void fraction. A typical example is shown in Fig. 17.47.

Different designs of structured packing are produced by a number of manufacturers. The basic construction and performance of the various proprietary types available are similar. They are available in metal, plastics, and stoneware. The advantage of structured packing over random packing is their low HETP (typically less than 0.5 m) and low pressure drop (around 100 Pa/m). They are being increasingly used in the following applications:

TABLE 17.2 Design data for various packings

	Size		Bulk density (kg/m ³)	Surface area a (m ² /m ³)	Packing factor F_p m ⁻¹
	in.	mm			
Raschig rings ceramic	0.50	13	881	368	2100
	1.0	25	673	190	525
	1.5	38	689	128	310
	2.0	51	651	95	210
	3.0	76	561	69	120
Metal (density for carbon steel)	0.5	13	1201	417	980
	1.0	25	625	207	375
	1.5	38	785	141	270
	2.0	51	593	102	190
	3.0	76	400	72	105
Pall rings metal (density for carbon steel)	0.625	16	593	341	230
	1.0	25	481	210	160
	1.25	32	385	128	92
	2.0	51	353	102	66
	3.5	76	273	66	52
Plastics (density for polypropylene)	0.625	16	112	341	320
	1.0	25	88	207	170
	1.5	38	76	128	130
	2.0	51	68	102	82
	3.5	89	64	85	52
INTALOX saddles ceramic	0.5	13	737	480	660
	1.0	25	673	253	300
	1.5	38	625	194	170
	2.0	51	609	108	130
	3.0	76	577		72

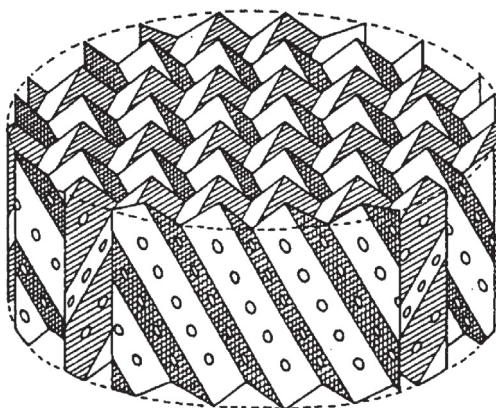


FIG. 17.47 Make-up of structured packing. (Reproduced from Butcher [1988] with permission.)

1. For difficult separations, requiring many stages, such as the separation of isomers.
2. High vacuum distillation.
3. For column revamps, to increase capacity and reduce reflux ratio requirements.

The applications have mainly been in distillation, but structured packing can also be used in absorption, in applications where high efficiency and low pressure drop are needed.

The cost of structured packing per cubic meter will be significantly higher than that of random packing, but this is offset by their higher efficiency.

The manufacturers' technical literature should be consulted for design data. A review of the types available is given by [Butcher \(1988\)](#). Generalized methods for predicting the capacity and pressure drop of structured packing are given by [Fair and Bravo \(1990\)](#) and [Kister and Gill \(1992\)](#). The use of structured packing in distillation is discussed in detail in the book by [Kister \(1992\)](#).

Structured packing has a high surface area of thin metal and consequently can be susceptible to ignition of trapped hydrocarbons or pyrophoric corrosion products, as heat is not easily conducted away from local hot spots. The FRI Design Practices Committee has provided guidelines on the design and maintenance of packed columns to reduce the likelihood of packing fires; see FRI Design Practices [Committee \(2007\)](#).

17.14.2 Packed-bed height

Distillation

For the design of packed distillation columns, it is simpler to treat the separation as a staged process and use the concept of the height of an equivalent equilibrium stage to convert the number of ideal stages required to a height of packing. The methods for estimating the number of ideal stages given in Sections 17.5 to 17.8 can then be applied to packed columns.

The height of an equivalent equilibrium stage, usually called the height equivalent to a theoretical plate (HETP), is the height of packing that will give the same separation as an equilibrium stage. It has been shown by [Eckert \(1975\)](#) that in distillation, the HETP for a given type and size of packing is essentially constant and independent of the system physical properties, providing good liquid distribution is maintained and the pressure drop is at least above 17 mm water per meter of packing height. The following values for Pall rings can be used to make an approximate estimate of the bed height required:

Size, mm	HETP, m
25 (1 in)	0.4–0.5
38 (1½ in)	0.6–0.75
50 (2 in)	0.75–1.0

The HETP for saddle packings will be similar to that for Pall rings, providing the pressure drop is at least 29 mm per m.

The HETP for Raschig rings will be higher than for Pall rings or saddles, and the values given earlier will only apply at an appreciably higher pressure drop, greater than 42 mm per m.

The methods for estimating the heights of transfer units (HTU), given in Section 17.14.3, can be used for distillation. The relationship between transfer units and the HETP is given by:

$$\text{HETP} = \frac{H_{OG} \ln \left(\frac{mG_m}{L_m} \right)}{\left(\frac{mG_m}{L_m} - 1 \right)} \quad (17.64)$$

where H_{OG} is the height of an overall gas-phase transfer unit.

The slope of the equilibrium line m will normally vary throughout a distillation column, so it will be necessary to calculate the HETP for each plate or a series of plates.

Absorption

Though packed absorption and stripping columns can also be designed as staged processes, it is usually more convenient to use the integrated form of the differential equations set up by considering the rates of mass transfer at a point in the column. The derivation of these equations is given in [Chhabra and Gurappa \(2019\)](#).

Where the concentration of the solute is small, say less than 10%, the flow of gas and liquid will be essentially constant throughout the column, and the height of packing required, Z , is given by:

$$Z = \frac{G_m}{K_G a P} \int_{y_2}^{y_1} \frac{dy}{y - y_e} \quad (17.65)$$

in terms of the overall gas-phase mass transfer coefficient K_G and the gas composition.

Or,

$$Z = \frac{L_m}{K_L a C_t} \int_{x_2}^{x_1} \frac{dx}{x_e - x} \quad (17.66)$$

in terms of the overall liquid-phase mass transfer coefficient K_L and the liquid composition,

where G_m = molar gas flow rate per unit cross-sectional area

L_m = molar liquid flow rate per unit cross-sectional area

a = interfacial surface area per unit volume

P = total pressure

C_t = total molar concentration

y_1 and y_2 = the mol fractions of the solute in the gas at the bottom and top of the column, respectively

x_1 and x_2 = the mol fractions of the solute in the liquid at the bottom and top of the column, respectively

x_e = the mole fraction in the liquid that would be in equilibrium with the gas concentration at any point

y_e = the mole fraction in the gas that would be in equilibrium with the liquid concentration at any point

The relationship between the equilibrium concentrations and actual concentrations is shown in [Fig. 17.48](#).

For design purposes, it is convenient to write Equations 17.65 and 17.66 in terms of “transfer units” (HTU); where the value of the integral is the number of transfer units and the group in front of the integral sign, which has units of length, is the height of a transfer unit.

$$Z = H_{OG} N_{OG} \quad (17.67a)$$

or

$$Z = H_{OL} N_{OL} \quad (17.67b)$$

where H_{OG} is the height of an overall gas-phase transfer unit:

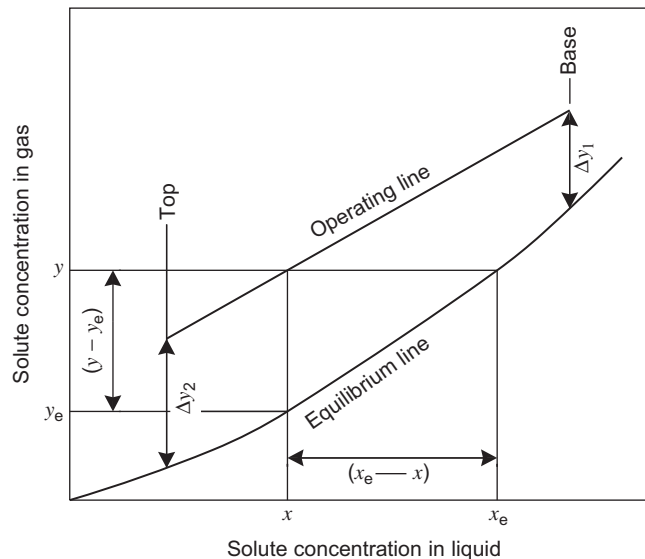


FIG. 17.48 Gas absorption concentration relationships.

$$= \frac{G_m}{K_G a P} \quad (17.68)$$

N_{OG} is the number of overall gas-phase transfer units:

$$= \int_{y_2}^{y_1} \frac{dy}{y - y_e} \quad (17.69)$$

H_{OL} is the height of an overall liquid-phase transfer unit:

$$= \frac{L_m}{K_L a C_t} \quad (17.70)$$

N_{OL} is the number of overall liquid-phase transfer units:

$$= \int_{x_2}^{x_1} \frac{dx}{x_e - x} \quad (17.71)$$

The number of overall gas-phase transfer units is often more conveniently expressed in terms of the partial pressure of the solute gas.

$$N_{OG} = \int_{p_1}^{p_2} \frac{dp}{p - p_e} \quad (17.72)$$

The relationship between the overall height of a transfer unit and the individual film transfer units H_L and H_G , which are based on the concentration driving force across the liquid and gas films, is given by:

$$H_{OG} = H_G + m \frac{G_m}{L_m} H_L \quad (17.73)$$

$$H_{OL} = H_L + \frac{L_m}{m G_m} H_G \quad (17.74)$$

where m is the slope of the equilibrium line and G_m/L_m the slope of the operating line.

The number of transfer units is obtained by graphical or numerical integration of Equations 17.69, 17.71, or 17.72.

Where the operating and equilibrium lines are straight, and they can usually be considered to be so for dilute systems, the number of transfer units is given by:

$$N_{OG} = \frac{y_1 - y_2}{\Delta y_{1m}} \quad (17.75)$$

where Δy_{1m} is the log mean driving force, given by:

$$y_{1m} = \frac{\Delta y_1 - \Delta y_2}{\ln \left(\frac{\Delta y_1}{\Delta y_2} \right)} \quad (17.76)$$

where $\Delta y_1 = y_1 - y_e$

$\Delta y_2 = y_2 - y_e$

If the equilibrium curve and operating lines can be taken as straight and the solvent feed is essentially solute free, the number of transfer units is given by:

$$N_{OG} = \frac{1}{1 - \left(\frac{m G_m}{L_m} \right)} \ln \left[\left(1 - \frac{m G_m}{L_m} \right) \frac{y_1}{y_2} + \frac{m G_m}{L_m} \right] \quad (17.77)$$

This equation is plotted in Fig. 17.49, which can be used to make a quick estimate of the number of transfer units required for a given separation.

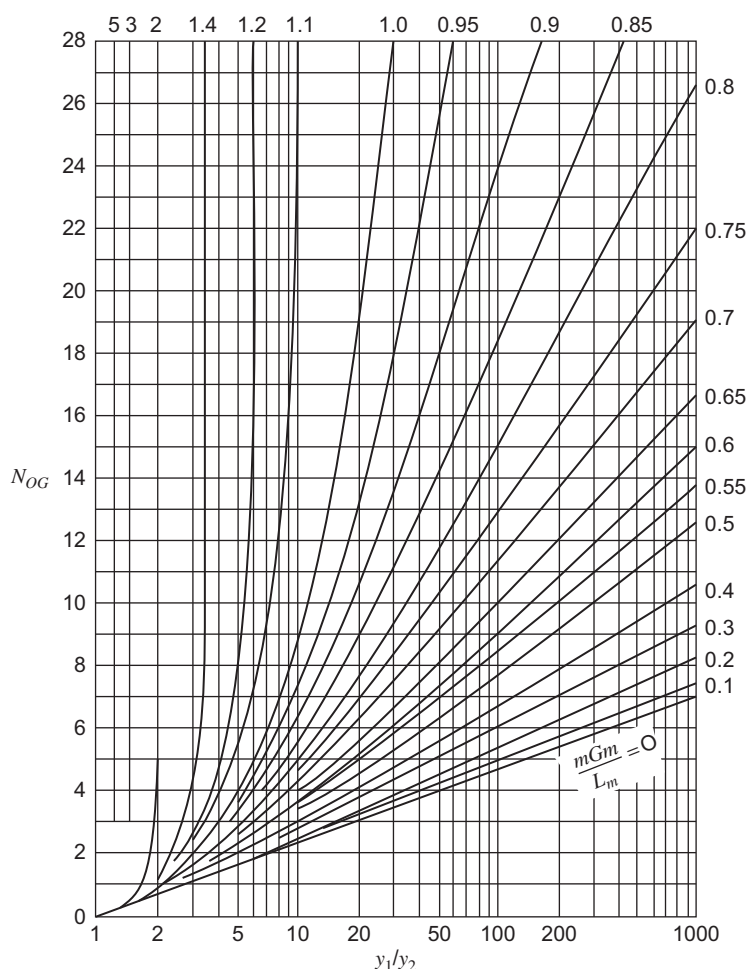


FIG. 17.49 Number of transfer units N_{OG} as a function of y_1/y_2 with mG_m/L_m as parameter.

It can be seen from Fig. 17.49 that the number of stages required for a given separation is highly dependent on the flow rate L_m . If the solvent rate is not set by other process considerations, Fig. 17.49 can be used to make quick estimates of the column height at different flow rates to find the most economic value. Colburn (1939) has suggested that the optimum value for the term mG_m/L_m will lie between 0.7 and 0.8.

Only physical absorption from dilute gases has been considered in this section. For a discussion of absorption from concentrated gases and absorption with chemical reaction, the reader should refer to Chhabra and Gurappa (2019) or to the book by Treybal (1980). The special case of absorption of acid gases is discussed extensively in the book by Kohl and Nielsen (1997). If the inlet gas concentration is not too high, the equations for dilute systems can be used by dividing the operating line up into two or three straight sections.

Stripping

In a stripping column, an absorbed solute is removed from a liquid solvent by countercurrent contact with a vapor. Stripping and absorption are usually used together, with a stripping column regenerating the solvent for an absorber column. The analysis of stripping is similar to that for absorption; see Green and Southard (2018).

Because the objective in stripping is to achieve a desired outlet liquid concentration, it is more customary to work in overall liquid-phase transfer units:

$$N_{OL} = \int_{x_2}^{x_1} \frac{dx}{x_e - x} \quad (17.71)$$

If the equilibrium and operating lines can be taken as straight and the stripping vapor is essentially solute free, the number of transfer units is given by:

$$N_{OL} = \frac{1}{\left(1 - \frac{L}{mG}\right)} \ln \left[\left(\frac{L}{mG} \right) + \left(1 - \frac{L}{mG} \right) \frac{x_2}{x_1} \right] \quad (17.78)$$

The parameter (L/mG) is known as the stripping factor.

When there are strong chemical interactions between the solute and solvent, the effect is usually to make the equilibrium line curve so that it is concave upwards, as shown in Fig. 17.50. The operating line for stripping must lie below the equilibrium line, so if a low outlet concentration of solute is needed, as is often the case, then L/G must be less than the slope of the equilibrium line, m .

To achieve $L/G < m$, the designer can raise the stripping vapor rate, G , or increase temperature or reduce pressure to increase m .

As the equilibrium line becomes more concave, it obviously becomes more difficult to achieve $L/G < m$ and obtain completely regenerated solvent. Absorber–stripper processes are therefore used for bulk separations, as it is too expensive to completely regenerate the solvent. If necessary, residual amounts of solute in the vapor leaving the absorber can be recovered using adsorption; see Section 16.2.1.

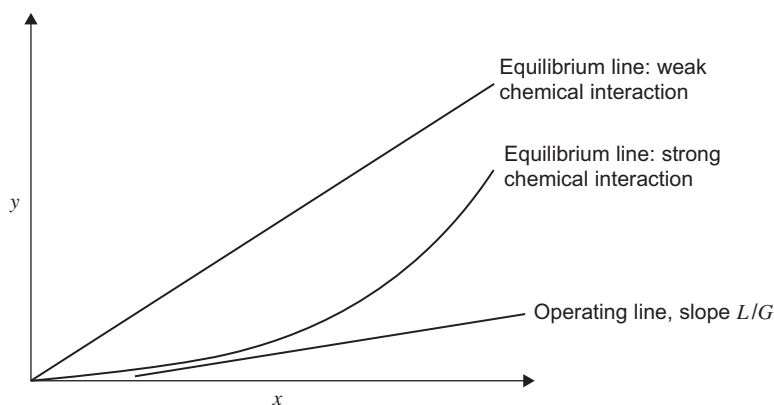


FIG. 17.50 Equilibrium and operating lines for stripping.

17.14.3 Prediction of the height of a transfer unit

There is no entirely satisfactory method for predicting the height of a transfer unit. In practice, the value for a particular packing will depend not only on the physical properties and flow rates of the gas and liquid but also on the uniformity of the liquid distribution throughout the column, which is dependent on the column height and diameter. This makes it difficult to extrapolate data obtained from small-size laboratory and pilot plant columns to industrial-scale columns. Whenever possible, estimates should be based on actual values obtained from operating columns of similar size to that being designed.

Experimental values for several systems are given by Cornell et al. (1960), Eckert (1963), and Vital et al. (1984). A selection of values for a range of systems is given in Table 17.3. The composite mass transfer term $K_G a$ is normally used when reporting experimental mass transfer coefficients for packing, as the effective interfacial area for mass transfer will be less than the actual surface area a of the packing.

Many correlations have been published for predicting the height of a transfer unit and the mass transfer coefficients; several are reviewed in Chhabra and Gurappa (2019). The two methods given in this section have been found to be reliable for preliminary design work and, in the absence of practical values, can be used for the final design with a suitable factor of safety.

The approach taken by the authors of the two methods is fundamentally different, and this provides a useful cross-check on the predicted values. Judgment must always be used when using predictive methods in design, and it is always worthwhile trying several methods and comparing the results.

TABLE 17.3 Typical packing efficiencies

System	Pressure kPa	Column dia, m	Packing		HTU m	HETP m
			Type	Size, mm		
<i>Absorption</i>						
Hydrocarbons	6000	0.9	Pall	50		0.85
NH ₃ –Air–H ₂ O	101	—	Berl	50	0.50	
Air–water	101	—	Berl	50	0.50	
Acetone–water	101	0.6	Pall	50		0.75
<i>Distillation</i>						
Pentane–propane	101	0.46	Pall	25		0.46
IPA–water	101	0.46	Int.	25	0.75	0.50
Methanol–water	101	0.41	Pall	25	0.52	
	101	0.20	Int.	25		0.46
Acetone–water	101	0.46	Pall	25		0.37
	101	0.36	Int.	25		0.46
Formic acid–water	101	0.91	Pall	50		0.45
Acetone–water	101	0.38	Pall	38	0.55	0.45
	101	0.38	Int.	50	0.50	0.45
	101	1.07	Int.	38		1.22
MEK–toluene	101	0.38	Pall	25	0.29	0.35
	101	0.38	Int.	25	0.27	0.23
	101	0.38	Berl	25	0.31	0.31

Pall = Pall rings, Berl = Berl saddles, Int. = INTALOX saddles.

Typical values for the HTU of random packings are:

25 mm (1 in)	0.3 to 0.6 m (1 to 2 ft)
38 mm (1½ in)	0.5 to 0.75 m (1½ to 2½ ft)
50 mm (2 in)	0.6 to 1.0 m (2 to 3 ft)

Cornell's method

Cornell et al. (1960) reviewed the previously published data and presented empirical equations for predicting the height of the gas and liquid film transfer units. Their correlation takes into account the physical properties of the system, the gas and liquid flow rates, and the column diameter and height. Equations and figures are given for a range of sizes of Raschig rings and Berl saddles. Only those for Berl saddles are given here, as it is unlikely that Raschig rings would be considered for a new column. Though the mass transfer efficiency of Pall rings and INTALOX saddles will be higher than that of the equivalent size Berl saddle, the method can be used to make conservative estimates for these packings.

Bolles and Fair (1982) have extended the correlations given in the earlier paper to include metal Pall rings.

Cornell's equations are:

$$H_G = 0.011 \psi_h (Sc)_v^{0.5} \left(\frac{D_c}{0.305} \right)^{1.11} \left(\frac{Z}{3.05} \right)^{0.33} / (L_w^* f_1 f_2 f_3)^{0.5} \quad (17.79)$$

$$H_L = 0.305 \phi_h (Sc)_L^{0.5} K_3 \left(\frac{Z}{3.05} \right)^{0.15} \quad (17.80)$$

where H_G = height of a gas-phase transfer unit, m

H_L = height of a liquid-phase transfer unit, m

$(Sc)_v$ = gas Schmidt number = $(\mu_L/\rho_L D_{LK})$

$(Sc)_L$ = liquid Schmidt number = $(\mu_L/\rho_L D_L)$

D_c = column diameter, m

Z = column height, m

K_3 = percentage flooding correction factor, from Fig. 17.51

ψ_h = H_G factor from Fig. 17.52

ϕ_h = H_L factor from Fig. 17.53

L_w^* = liquid mass flow rate per unit area column cross-sectional area, kg/m²s

f_1 = liquid viscosity correction factor = $(\mu_L/\mu_w)^{0.16}$

f_2 = liquid density correction factor = $(\rho_w/\rho_L)^{1.25}$

f_3 = surface tension correction factor = $(\sigma_w/\sigma_L)^{0.8}$

where the suffix w refers to the physical properties of water at 20 °C; all other physical properties are evaluated at the column conditions.

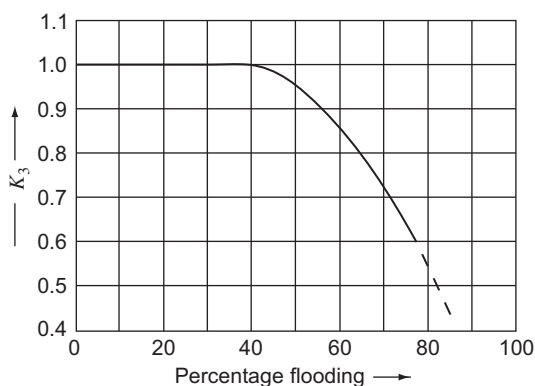


FIG. 17.51 Percentage flooding correction factor.

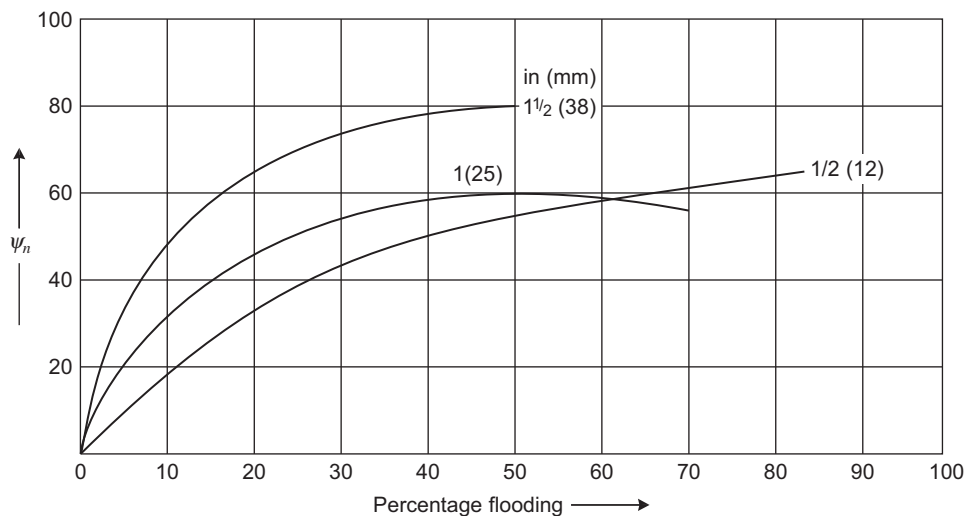
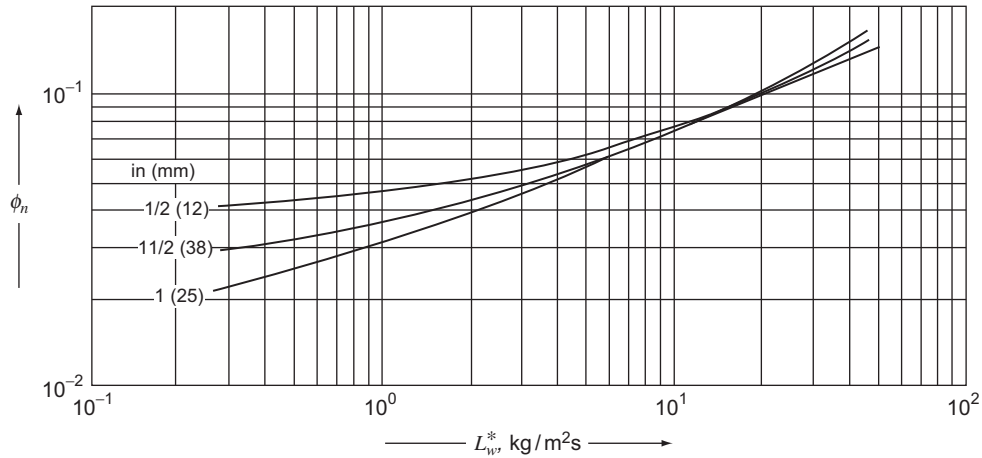


FIG. 17.52 Factor for H_G for Berl saddles.

FIG. 17.53 Factor for H_L for Berl saddles.

The terms $(D_c/0.305)$ and $(Z/3.05)$ are included in the equations to allow for the effects of column diameter and packed-bed height. The “standard” values used by Cornell were 1 ft (0.305 m) for diameter and 10 ft (3.05 m) for height. These correction terms will clearly give silly results if applied over too wide a range of values. For design purposes, the diameter correction term should be taken as a fixed value of 2.3 for columns above 0.6 m (2 ft) diameter, and the height correction should only be included when the distance between liquid redistributors is greater than 3 m. To use Figs. 17.51 and 17.52, an estimate of the column percentage flooding is needed. This can be obtained from Fig. 17.54, where a flooding line has been included with the lines of constant pressure drop.

$$\text{Percentage flooding} = \left[\frac{K_4 \text{ at design pressure drop}}{K_4 \text{ at flooding}} \right]^{1/2} \quad (17.81)$$

Onda's method

Onda et al. (1968) published useful correlations for the film mass transfer coefficients k_G and k_L and the effective wetted area of the packing a_{wv} which can be used to calculate H_G and H_L .

Their correlations were based on a large amount of data on gas absorption and distillation, with a variety of packings that included Pall rings and Berl saddles. Their method for estimating the effective area of packing can also be used with experimentally determined values of the mass transfer coefficients and values predicted using other correlations.

The equation for the effective area is:

$$\frac{a_w}{a} = 1 - \exp \left[-1.45 \left(\frac{\sigma_c}{\sigma_L} \right)^{0.75} \left(\frac{L_w^*}{a \mu_L} \right)^{0.1} \left(\frac{L_w^{*2} a}{\rho_L^2 g} \right)^{-0.05} \left(\frac{L_w^{*2}}{\rho_L \sigma_L a} \right)^{0.2} \right] \quad (17.82)$$

and for the mass coefficients:

$$k_L \left(\frac{\rho_L}{\mu_L g} \right)^{1/3} = 0.0051 \left(\frac{L_w^*}{a_w \mu_L} \right)^{2/3} \left(\frac{\mu_L}{\rho_L D_L} \right)^{-1/2} (ad_p)^{0.4} \quad (17.83)$$

$$\frac{k_G}{a} \frac{RT}{D_v} = K_5 \left(\frac{V_w^*}{a \mu_v} \right)^{0.7} \left(\frac{\mu_v}{\rho_v D_v} \right)^{1/3} (ad_p)^{-2.0} \quad (17.84)$$

where $K_5 = 5.23$ for packing sizes above 15 mm and 2.00 for sizes below 15 mm

L_w^* = liquid mass flow rate per unit cross-sectional area, $\text{kg/m}^2\text{s}$

V_w^* = gas mass flow rate per unit column cross-sectional area, $\text{kg/m}^2\text{s}$

a_w = effective interfacial area of packing per unit volume, m^2/m^3

a = actual area of packing per unit volume (see Table 17.2), m^2/m^3

d_p = packing size, m

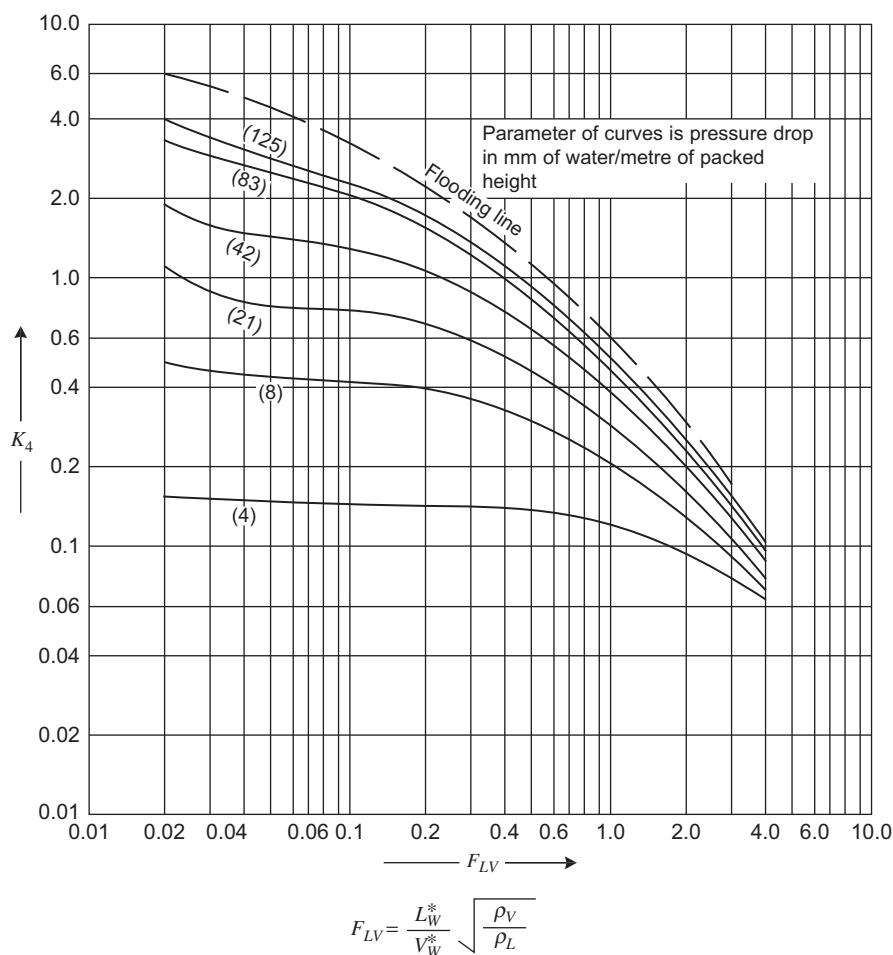


FIG. 17.54 Generalized pressure drop correlation. (Adapted from a Fig. by Koch-Glitsch, LP, with permission.)

σ_c = critical surface tension for the particular packing material given here:

Material	σ_c mN/m
Ceramic	61
Metal (steel)	75
Plastic (polyethylene)	33
Carbon	56

σ_L = liquid surface tension, N/m

k_G = gas film mass transfer coefficient, kmol/m²s atm or kmol/m²s bar

k_L = liquid film mass transfer coefficient, kmol/m²s (kmol/m³) = m/s

Note: All the groups in the equations are dimensionless.

The units for k_G will depend on the units used for the gas constant:

$$R = 0.08206 \text{ atm m}^3/\text{kmol K} \text{ or}$$

$$0.08314 \text{ bar m}^3/\text{kmol K}$$

The film transfer unit heights are given by:

$$H_G = \frac{G_m}{k_G a_w P} \quad (17.85)$$

$$H_L = \frac{L_m}{k_L a_w C_t} \quad (17.86)$$

where P = column operating pressure, atm or bar

C_t = total concentration, $\text{kmol/m}^3 = \rho_L / \text{molecular weight solvent}$

G_m = molar gas flow rate per unit cross-sectional area, $\text{kmol/m}^2\text{s}$

L_m = molar liquid flow rate per unit cross-sectional area, $\text{kmol/m}^2\text{s}$

17.14.4 Column diameter (capacity)

The capacity of a packed column is determined by its cross-sectional area. Normally, the column will be designed to operate at the highest economical pressure drop to ensure good liquid and gas distribution. For random packings, the pressure drop will not normally exceed 80 mm of water per meter of packing height. At this value, the gas velocity will be about 80% of the flooding velocity. Recommended design values, mm water per m packing, are:

Absorbers and strippers	15 to 50
Distillation, atmospheric and moderate pressure	40 to 80

Where the liquid is likely to foam, these values should be halved.

For vacuum distillations, the maximum allowable pressure drop will be determined by the process requirements, but for satisfactory liquid distribution, the pressure drop should not be less than 8 mm water per m. If very low bottom pressures are required, special low-pressure-drop gauze packings should be considered such as Hyperfil, Multifil or Dixon rings.

The column cross-sectional area and diameter for the selected pressure drop can be determined from the generalized pressure drop correlation given in Fig. 17.54. Fig. 17.54 correlates the liquid and vapor flow rates, system physical properties and packing characteristics, with the gas mass flow rate per unit cross-sectional area; with lines of constant pressure drop as a parameter.

The term K_4 in Fig. 17.54 is the function:

$$K_4 = \frac{13.1 (V_w^*)^2 F_p \left(\frac{\mu_L}{\rho_L} \right)^{0.1}}{\rho_v (\rho_L - \rho_v)} \quad (17.87)$$

where V_w^* = gas mass flow rate per unit column cross-sectional area, $\text{kg/m}^2\text{s}$

F_p = packing factor, characteristic of the size and type of packing, see Table 17.2, m^{-1}

μ_L = liquid viscosity, Ns/m^2

ρ_L, ρ_v = liquid and vapor densities, kg/m^3

The values of the flow factor F_{LV} given in Fig. 17.54 cover the range that will generally give satisfactory column performance.

The ratio of liquid to gas flow will be fixed by the reflux ratio in distillation, and in gas absorption will be selected to give the required separation with the most economic use of solvent.

A new generalized correlation for pressure drop in packed columns, similar to Fig. 17.54, has been published by Leva (1992, 1995). The new correlation gives a better prediction for systems where the density of the irrigating fluid is appreciably greater than that of water. It can also be used to predict the pressure drop over dry packing.

Example 17.9

Sulfur dioxide produced by the combustion of sulfur in air is absorbed in water. Pure SO_2 is then recovered from the solution by steam stripping. Make a preliminary design for the absorption column. The feed will be 5000 kg/h of gas containing 8% v/v SO_2 . The gas will be cooled to 20 °C. A 95% recovery of the sulfur dioxide is required.

Solution

As the solubility of SO_2 in water is high, operation at atmospheric pressure should be satisfactory. The feed-water temperature will be taken as 20 °C, a reasonable design value.

Solubility data

From Green and Southard (2018):

SO ₂	% w/w solution	0.05	0.1	0.15	0.2	0.3	0.5	0.7	1.0	1.5
	Partial press. gas mmHg	1.2	3.2	5.8	8.5	14.1	26	39	59	92

These data are plotted in Fig. 17.55.

Number of stages

Partial pressure in the exit gas at 95% recovery = $60.8 \times 0.05 = 3.04$ mmHg

Over this range of partial pressure, the equilibrium line is essentially straight, so Fig. 17.49 can be used to estimate the number of stages needed.

The use of Fig. 17.49 will slightly overestimate the number of stages, and a more accurate estimate would be made by graphical integration of Equation 17.72, but this is not justified in view of the uncertainty in the prediction of the transfer unit height.

Molecular weights: SO₂ = 64, H₂O = 18, air = 29

Slope of equilibrium line

From the data: partial pressure at 1.0% w/w SO₂ = 59 mmHg.

$$\text{Mole fraction in vapor} = \frac{59}{760} = 0.0776$$

$$\text{Mole fraction in liquid} = \frac{\frac{1}{64}}{\frac{1}{64} + \frac{99}{18}} = 0.0028$$

$$m = \frac{0.0776}{0.0028} = 27.4$$

To decide the most economic water flow rate, the stripper design should be considered together with the absorption design, but for the purpose of this example, the absorption design will be considered alone. Using Fig. 17.49, the number of stages required at different water rates will be determined and the "optimum" rate chosen:

$$\frac{y_1}{y_2} = \frac{p_1}{p_2} = \frac{60.8}{3.04} = 20$$

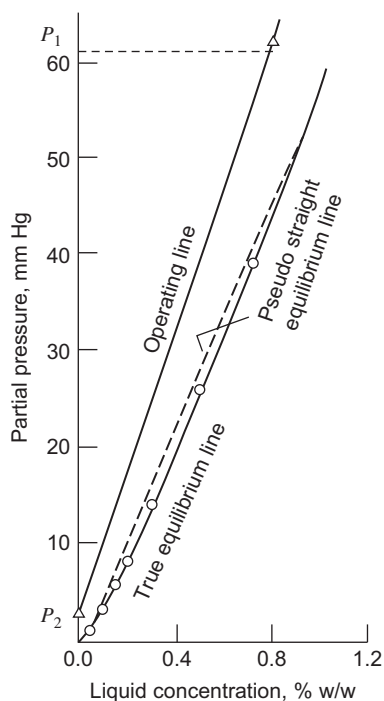


FIG. 17.55 SO₂ absorber design (Example 17.9).

$m \frac{G_m}{L_m}$	0.5	0.6	0.7	0.8	0.9	1.0
N_{OG}	3.7	4.1	6.3	8	10.8	19.0

It can be seen that the “optimum” will be between $mG_m/L_m = 0.6$ and 0.8 , as would be expected. Below 0.6 there is only a small decrease in the number of stages required with increasing liquid rate; above 0.8 the number of stages increases rapidly with decreasing liquid rate.

Check the liquid outlet composition at 0.6 and 0.8 :

$$\text{Material balance } L_m x_1 = G_m (y_1 - y_2)$$

$$\text{so } x_1 = \frac{G_m}{L_m} (0.08 \times 0.95) = \frac{m}{27.4} \frac{G_m}{L_m} (0.076)$$

$$\text{at } \frac{mG_m}{L_m} = 0.6, x_1 = 1.66 \times 10^{-3} \text{ mol fraction}$$

$$\text{at } \frac{mG_m}{L_m} = 0.8, x_1 = 2.22 \times 10^{-3} \text{ mol fraction}$$

Use 0.8 , as the higher concentration will favor the stripper design and operation without significantly increasing the number of stages needed in the absorber.

$$N_{OG} = \underline{\underline{8}}$$

Column diameter

The physical properties of the gas can be taken as those for air, as the concentration of SO_2 is low.

$$\text{Gas flow - rate} = \frac{5000}{3600} = 1.39 \text{ kg/s}, = \frac{1.39}{29} = 0.048 \text{ kmol/s}$$

$$\begin{aligned} \text{Liquid flow - rate} &= \frac{27.4}{0.8} \times 0.048 = 1.64 \text{ kmol/s} \\ &= 29.5 \text{ kg/s.} \end{aligned}$$

Select 38 mm ($1\frac{1}{2} \text{ in}$) ceramic INTALOX saddles.

From Table 17.2, $F_p = 170 \text{ m}^{-1}$:

$$\text{Gas density at } 20^\circ \text{C} = \frac{29}{22.4} \times \frac{273}{293} = 1.21 \text{ kg/m}^3$$

$$\text{Liquid density} \approx 1000 \text{ kg/m}^3$$

$$\text{Liquid viscosity} = 10^{-3} \text{ Ns/m}^2$$

$$\frac{L_W^*}{V_W^*} \sqrt{\frac{\rho_v}{\rho_L}} = \frac{29.5}{1.39} \sqrt{\frac{1.21}{10^3}} = 0.74$$

Design for a pressure drop of $20 \text{ mm H}_2\text{O/m}$ packing:

From Fig. 17.54,

$$K_4 = 0.35$$

$$\text{At flooding } K_4 = 0.8$$

$$\text{Percentage flooding} = \sqrt{\frac{0.35}{0.8}} \times 100 = 66 \%, \text{ satisfactory.}$$

From Equation 17.87:

$$V_W^* = \left[\frac{K_4 \rho_V (\rho_L - \rho_V)}{13.1 F_p (\mu_L / \rho_L)^{0.1}} \right]^{1/2}$$

$$= \left[\frac{0.35 \times 1.21 (1000 - 1.21)}{13.1 \times 170 (10^{-3} / 10^3)^{0.1}} \right]^{1/2} = 0.87 \text{ kg/m}^2\text{s}$$

$$\text{Column area required} = \frac{1.39}{0.87} = 1.6 \text{ m}^2$$

$$\text{Diameter} = \sqrt{\frac{4}{\pi} \times 1.6} = 1.43 \text{ m}$$

Round off to 1.50 m

$$\text{Column area} = \frac{\pi}{4} \times 1.5^2 = 1.77 \text{ m}^2$$

$$\text{Packing size to column diameter ratio} = \frac{1.5}{38 \times 10^{-3}} = 39$$

A larger packing size could be considered.

Percentage flooding at selected diameter:

$$= 66 \times \frac{1.6}{1.77} = 60 \%$$

Could consider reducing column diameter to the nearest standard pipe size.

Estimation of H_{OG}

Cornell's method

$$D_L = 1.7 \times 10^{-9} \text{ m}^2/\text{s}$$

$$D_v = 1.45 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\mu_v = 0.018 \times 10^{-3} \text{ Ns/m}^2$$

$$(Sc)_v = \frac{0.018 \times 10^{-3}}{1.21 \times 1.45 \times 10^{-5}} = 1.04$$

$$(Sc)_L = \frac{10^{-3}}{1000 \times 1.7 \times 10^{-9}} = 588$$

$$L_W^* = \frac{29.5}{1.77} = 16.7 \text{ kg/s m}^2$$

From Fig. 17.51, at 60% flooding, $K_3 = 0.85$.

From Fig. 17.52, at 60% flooding, $\psi_h = 80$.

From Fig. 17.53, at $L_W^* = 16.7$, $\phi_h = 0.1$.

H_{OG} can be expected to be around 1 m, so as a first estimate Z can be taken as 8 m. The column diameter is greater than 0.6 m, so the diameter correction term will be taken as 2.3.

$$H_L = 0.305 \times 0.1 (588)^{0.5} \times 0.85 \left(\frac{8}{3.05} \right)^{0.15} = 0.7 \text{ m} \quad (17.80)$$

As the liquid temperature has been taken as 20 °C and the liquid is water:

$$f_1 = f_2 = f_3 = 1$$

$$H_G = 0.011 \times 80(1.04)^{0.5}(2.3) \left(\frac{8}{3.05} \right)^{0.33} / (16.7)^{0.5} = 0.7 \text{ m} \quad (17.79)$$

$$H_{OG} = 0.7 + 0.8 \times 0.7 = 1.3 \text{ m} \quad (17.73)$$

$$Z = 8 \times 1.3 = 10.4 \text{ m, close enough to the estimated value.}$$

Onda's method

$$R = 0.08314 \text{ bar m}^3/\text{kmol.K.}$$

Surface tension of liquid, taken as water at 20 °C = $70 \times 10^{-3} \text{ N/m}$

$$g = 9.81 \text{ m/s}^2$$

$$d_p = 38 \times 10^{-3} \text{ m}$$

From Table 17.3, for 38 mm INTALOX saddles:

$$a = 194 \text{ m}^2/\text{m}^3$$

$$\sigma_c \text{ for ceramics} = 61 \times 10^{-3} \text{ N/m}$$

$$\begin{aligned} \frac{a_W}{a} = 1 - \exp \left[-1.45 \left(\frac{61 \times 10^{-3}}{70 \times 10^{-3}} \right)^{0.75} \left(\frac{17.6}{194 \times 10^{-3}} \right)^{0.1} \left(\frac{17.6^2 \times 194}{1000^2 \times 9.81} \right)^{-0.05} \right. \\ \left. \times \left(\frac{17.6^2}{1000 \times 70 \times 10^{-3} \times 194} \right)^{0.2} \right] = 0.71 \end{aligned} \quad (17.82)$$

$$a_W = 0.71 \times 194 = 138 \text{ m}^2/\text{m}^3$$

$$k_L \left(\frac{10^3}{10^{-3} \times 9.81} \right)^{1/3} = 0.0051 \left(\frac{17.6}{138 \times 10^{-3}} \right)^{2/3} \left(\frac{10^{-3}}{10^3 \times 1.7 \times 10^{-9}} \right)^{-1/2} \times (194 \times 38 \times 10^{-3})^{0.4}$$

$$k_L = 2.5 \times 10^{-4} \text{ m/s} \quad (17.83)$$

$$V_W^* \text{ on actual column diameter } \frac{1.39}{1.77} = 0.79 \text{ kg/m}^2\text{s}$$

$$k_G \frac{0.08314 \times 293}{194 \times 1.45 \times 10^{-5}} = 5.23 \left(\frac{0.79}{194 \times 0.018 \times 10^{-3}} \right)^{0.7} \times \left(\frac{0.018 \times 10^{-3}}{1.21 \times 1.45 \times 10^{-5}} \right)^{1/3} (194 \times 38 \times 10^{-3})^{-2.0}$$

$$k_G = 5.0 \times 10^{-4} \text{ kmol/sm}^2 \text{ bar} \quad (17.84)$$

$$G_m = \frac{0.79}{29} = 0.027 \text{ kmol/m}^2\text{s}$$

$$L_m = \frac{16.7}{18} = 0.93 \text{ kmol/m}^2\text{s}$$

$$H_G = \frac{0.027}{5.0 \times 10^{-4} \times 138 \times 1.013} = 0.39 \text{ m}$$

$$C_T = \text{total concentration, as water,} \quad (17.85)$$

$$= \frac{1000}{18} = 55.5 \text{ kmol/m}^3$$

$$H_L = \frac{0.93}{2.5 \times 10^{-4} \times 138 \times 55.6} = 0.49 \text{ m} \quad (17.86)$$

$$H_{OG} = 0.39 + 0.8 \times 0.49 = \underline{\underline{0.78 \text{ m}}} \quad (17.73)$$

Use higher value, estimated using Cornell's method, and round up packed-bed height to 11 m.

17.14.5 Column internals

The internal fittings in a packed column are simpler than those in a plate column but must be carefully designed to ensure good performance. As a general rule, the standard fittings developed by the packing manufacturers should be specified. Some typical designs are shown in Figs. 17.56 to 17.65, and their use is discussed in the following paragraphs.

Packing support

The function of the support plate is to carry the weight of the wet packing while allowing free passage of the gas and liquid. These requirements conflict; a poorly designed support will give a high pressure drop and can cause local flooding. Simple grid and perforated plate supports are used, but in these designs the liquid and gas have to vie for the same openings. Wide-spaced grids are used to increase the flow area, with layers of larger-size packing stacked on the grid to support the small-size random packing (Fig. 17.56).

The best design of packing support is one in which gas inlets are provided above the level where the liquid flows from the bed, such as the gas-injection type shown in Figs. 17.57 and 17.58. These designs have a low pressure drop and no tendency to flooding. They are available in a wide range of sizes and materials: metals, ceramics, and plastics.

Liquid distributors

The satisfactory performance of a plate column is dependent on maintaining a uniform flow of liquid throughout the column, and good initial liquid distribution is essential. Various designs of distributors are used. For small-diameter columns a central open feed pipe, or one fitted with a spray nozzle, may well be adequate, but for larger columns more elaborate designs are needed to ensure good distribution at all liquid flow rates. The two most commonly used designs are the orifice type, shown in Fig. 17.59, and the weir type, shown in Fig. 17.60. In the orifice

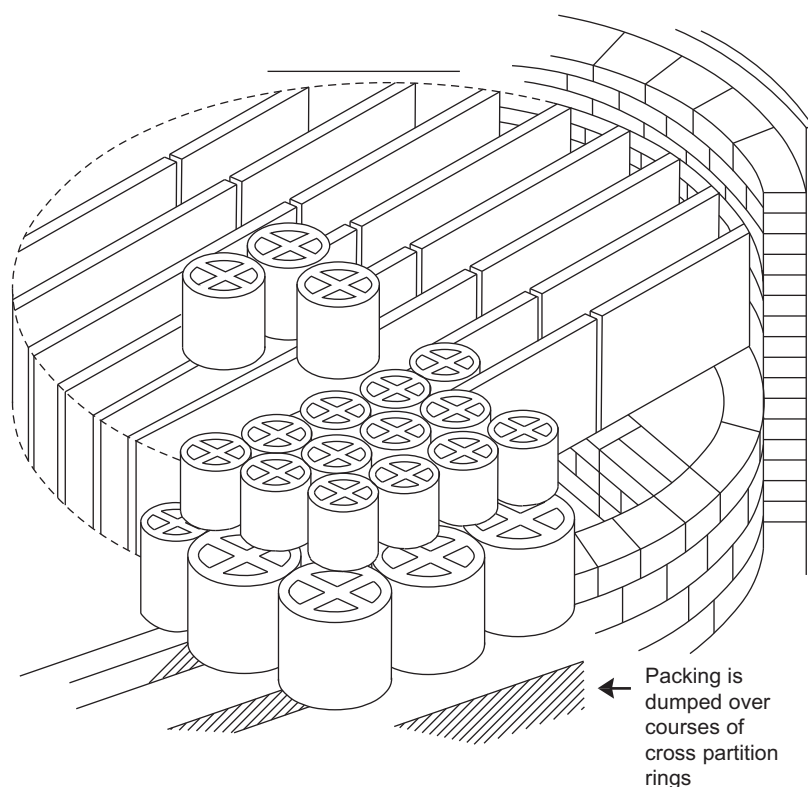


FIG. 17.56 Stacked packing used to support random packing.

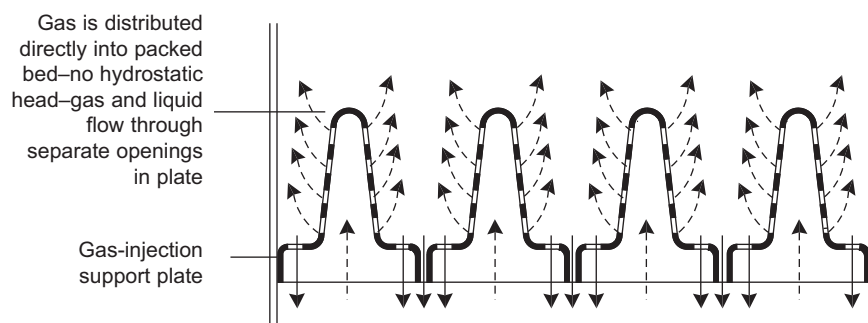


FIG. 17.57 The principle of the gas-injection packing support.

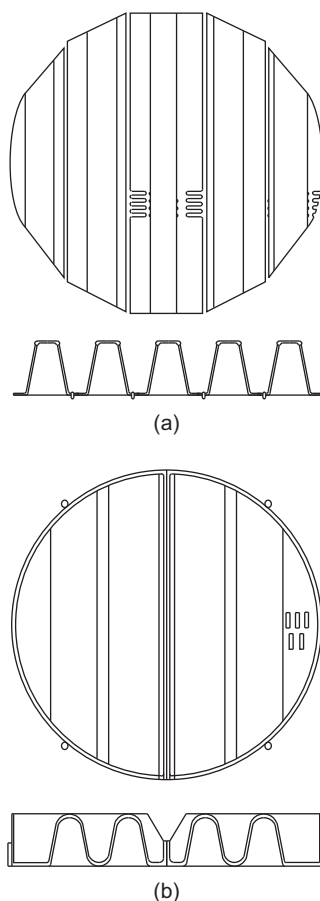


FIG. 17.58 Typical designs of gas-injection supports (Koch-Glitsch, LP). (a) Small-diameter columns. (b) Large-diameter columns.

type the liquid flows through holes in the plate and the gas through short stand pipes. The gas pipes should be sized to give sufficient area for gas flow without creating a significant pressure drop; the holes should be small enough to ensure that there is a level of liquid on the plate at the lowest liquid rate but large enough to prevent the distributor overflowing at the highest rate. In the weir type the liquid flows over notched weirs in the gas stand pipes. This type can be designed to cope with a wider range of liquid flow rates than the simpler orifice type.

For large-diameter columns, the trough-type distributor shown in Fig. 17.61 can be used and will give good liquid distribution with a large free area for gas flow.

All distributors that rely on the gravity flow of liquid must be installed in the column level, or maldistribution of liquid will occur.

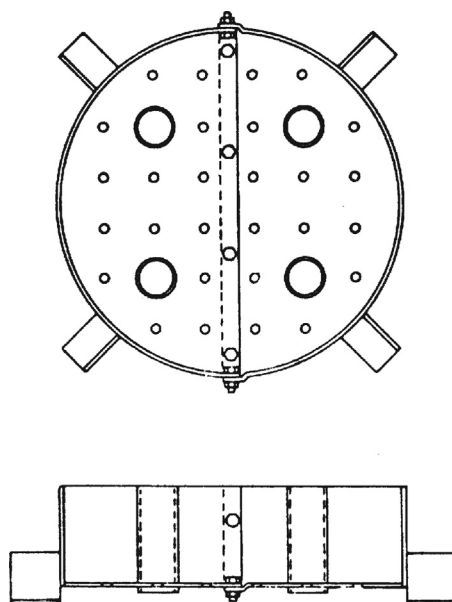


FIG. 17.59 Orifice-type distributor (Koch-Glitsch, LP).

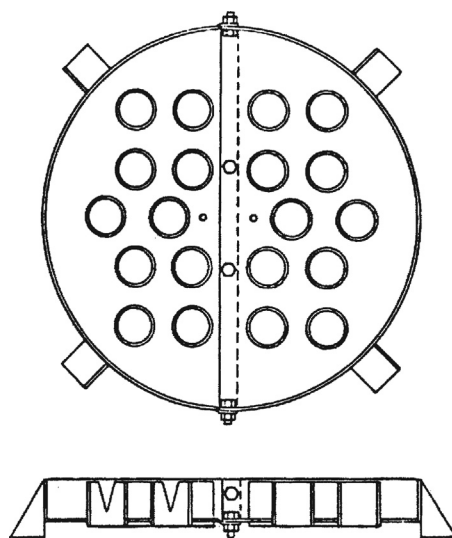


FIG. 17.60 Weir-type distributor (Koch-Glitsch, LP).

A pipe manifold distributor (Fig. 17.62) can be used when the liquid is fed to the column under pressure and the flow rate is reasonably constant. The distribution pipes and orifices should be sized to give an even flow from each element.

Liquid redistributors

Redistributors are used to collect liquid that has migrated to the column walls and redistribute it evenly over the packing. They will also even out any maldistribution that has occurred within the packing.

A full redistributor combines the functions of a packing support and a liquid distributor; a typical design is shown in Fig. 17.63.

The “wall-wiper” type of redistributor, in which a ring collects liquid from the column wall and redirects it into the center packing, is occasionally used in small-diameter columns, less than 0.6 m. Care should be taken when specifying this type to select a design that does not unduly restrict the gas flow and cause local flooding. A good design is that shown in Fig. 17.64.

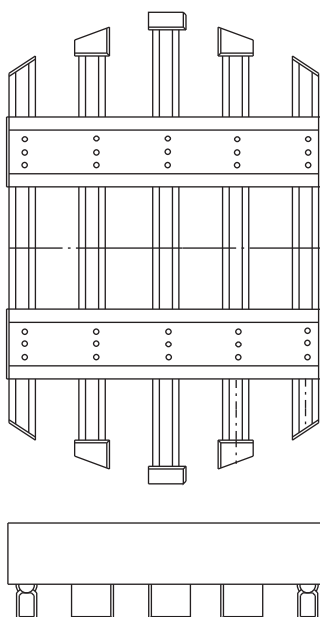


FIG. 17.61 Weir-trough distributors (Koch-Glitsch, LP).

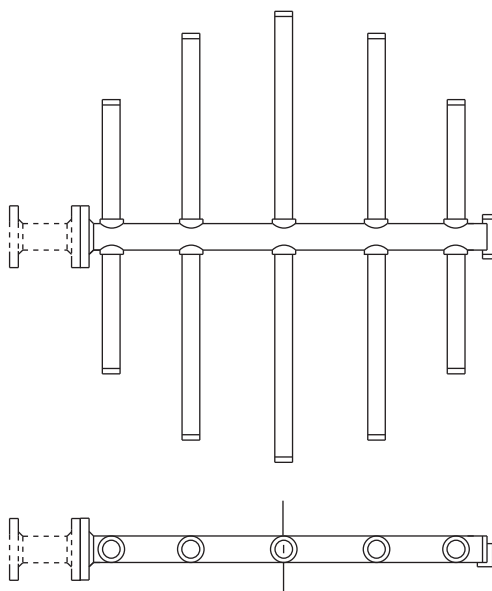


FIG. 17.62 Pipe distributor (Koch-Glitsch, LP).

The maximum bed height that should be used without liquid redistribution depends on the type of packing and the process. Distillation is less susceptible to maldistribution than absorption and stripping. As a general guide, the maximum bed height should not exceed 3 column diameters for Raschig rings and 8 to 10 for Pall rings and saddles. In a large-diameter column the bed height will also be limited by the maximum weight of packing that can be supported by the packing support and column walls; this will be around 8 m.

Hold-down plates

At high gas rates, or if surging occurs through misoperation, the top layers of packing can be fluidized. Under these conditions ceramic packing can break up and the pieces filter down the column and plug the packing; metal and plastic packing can be blown out of the column. Hold-down plates are used with ceramic packing to weigh down the top layers and prevent fluidization; a typical design is shown in Fig. 17.65. Bed limiters are sometimes used with plastic and metal

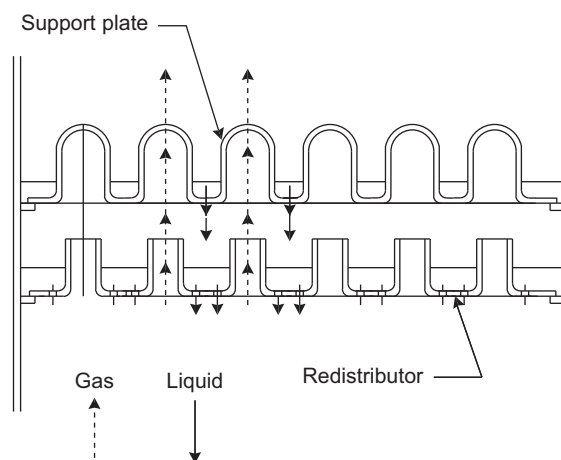


FIG. 17.63 Full redistributor.

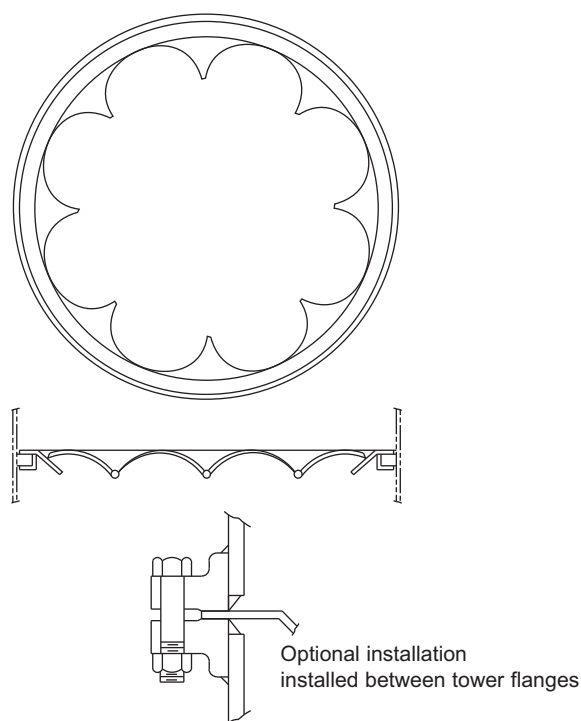


FIG. 17.64 "Wall-wiper" redistributor (Koch-Glitsch, LP).

packings to prevent expansion of the bed when operating at a high pressure drop. They are similar to hold-down plates but are of lighter construction and are fixed to the column walls. The openings in hold-down plates and bed limiters should be small enough to retain the packing, but should not restrict the gas and liquid flow.

Installing packing

Ceramic and metal packings are normally dumped into the column "wet" to ensure a truly random distribution and prevent damage to the packing. The column is partially filled with water and the packing dumped into the water. A height of water must be kept above the packing at all times.

If the columns must be packed dry, for instance, to avoid contamination of process fluids with water, the packing can be lowered into the column in buckets or other containers. Ceramic packings should not be dropped from a height of more than half a meter.

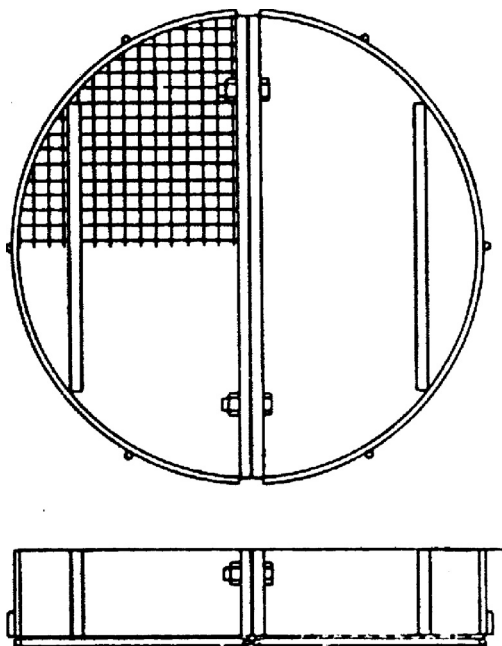


FIG. 17.65 Hold-down plate design (Koch-Glitsch, LP).

Liquid hold-up

An estimate of the amount of liquid held up in the packing under operating conditions is needed to calculate the total load carried by the packing support. The liquid hold-up depends on the liquid rate and, to some extent, on the gas flow rate. The packing manufacturer's design literature should be consulted to obtain accurate estimates. As a rough guide, a value of about 25% of the packing weight can be taken for ceramic packings.

17.14.6 Wetting rates

If very low liquid rates must be used, outside the range of F_{LV} given in Fig. 17.54, the packing wetting rate should be checked to make sure it is above the minimum recommended by the packing manufacturer.

Wetting rate is defined as:

$$\text{wetting rate} = \frac{\text{volumetric liquid rate per unit cross-sectional area}}{\text{packing surface area per unit volume}}$$

A nomograph for the calculation of wetting rates is given in Chhabra and Gurappa (2019).

Wetting rates are frequently expressed in terms of mass or volume flow rate per unit column cross-sectional area. Kister (1992) gives values for minimum wetting rates of 0.5 to 2 gpm/ft² (0.35×10^{-3} to 1.4×10^{-3} m³ s⁻¹/m²) for random packing and 0.1 to 0.2 gpm/ft² (0.07×10^{-3} to 0.14×10^{-3} m³ s⁻¹/m²) for structured packing. Norman (1961) recommends that the liquid rate in absorbers should be kept above 2.7 kg/m²s.

If the design liquid rate is too low, the diameter of the column should be reduced. For some processes liquid can be recycled to increase the flow over the packing.

A substantial factor of safety should be applied to the calculated bed height for processes where the wetting rate is likely to be low, and less space should be allowed between liquid redistributors.

17.15 Column auxiliaries

Intermediate storage tanks are sometimes used to smooth out fluctuations in column operation and process upsets, for example, when a batch reactor section feeds a continuous distillation section. These tanks should be sized

to give sufficient hold-up time for smooth operation and control. The hold-up time required will depend on the nature of the process and on how critical the operation is; some typical values for distillation processes are given here:

Operation	Time, minutes
Feed to a train of columns	10–20
Between columns	5–10
Feed to a column from storage	2–5
Reflux drum	5–15

The time given is that for the level in the tank to fall from the normal operating level to the minimum operating level if the feed ceases.

Horizontal or vertical tanks are used, depending on the size and duty. Where only a small hold-up volume is required, this can be provided by adding volume to the sump of the upstream column by extending the column base or, for reflux accumulators, by extending the bottom header of the condenser.

The specification and sizing of surge tanks and accumulators are discussed in more detail by [Mehra \(1979\)](#) and [Evans \(1980\)](#).

The design of condensers is discussed in Section 19.10, and the design of reboilers is covered in Section 19.11. The control of distillation columns is addressed in Section 5.4.7.

17.16 Solvent extraction (liquid–liquid extraction)

Extraction should be considered as an alternative to distillation in the following situations:

1. When the components in the feed have close boiling points. Extraction in a suitable solvent may be more economic if the relative volatility is below 1.2.
2. If the feed components form an azeotrope.
3. If the solute is heat sensitive and can be extracted into a lower-boiling solvent to reduce the thermal history during recovery.

Solvent selection

The following factors must be considered when selecting a suitable solvent for a given extraction:

1. Phase equilibrium: The distribution coefficient or partition coefficient, K , is the ratio of the mole fractions in each phase: $x_\alpha = K x_\beta$. The distribution coefficient is analogous to the equilibrium constant in absorption. The relative separation or selectivity is a measure of the distribution of two solutes between the two solvents and is the ratio of the distribution coefficients for the two solutes. Selectivity is analogous to relative volatility in distillation. The greater the difference in solubility of the preferred solute between the two solvents, the easier it will be to extract. This is particularly important when trying to preferentially extract one component from a mixture, as often occurs in the recovery of fermentation products and specialty chemicals.
2. Partition ratio: This is the weight fraction of the solute in the extract divided by the weight fraction in the raffinate. This determines the quantity of solvent and number of stages needed. The less solvent needed, the lower the solvent and solvent recovery costs.
3. Density: The greater the density difference between the feed and extraction solvents, the easier it will be to separate the solvents.
4. Miscibility: Ideally, the two solvents should be immiscible. The greater the solubility of the extraction solvent in the feed solvent, the more difficult it will be to recover the solvent from the raffinate and the higher the cost.
5. Safety: Whenever it is possible and economical, a solvent should be chosen that is not toxic or dangerously flammable.
6. Cost: The purchase cost of the solvent is important but should not be considered in isolation from the total process costs. It may be worth considering a more expensive solvent if it is more effective and easier to recover.

17.16.1 Extraction equipment

Extraction equipment can be divided into two broad groups:

1. Stage-wise extractors, in which the liquids are alternately contacted (mixed) and then separated in a series of stages. The “mixer–settler” contactor is an example of this type. Several mixer–settlers are often used in series to increase the effectiveness of the extraction.

- Differential extractors, in which the phases are continuously in contact in the extractor and are only separated at the exits, for example, in packed-column extractors.

Extraction columns can be further subdivided according to the method used to promote contact between the phases: packed, plate, mechanically agitated, or pulsed columns. Various types of proprietary centrifugal extractors are also used.

The following factors should be taken into consideration when selecting an extractor for a particular application:

- The number of stages required
- The throughputs
- The settling characteristics of the phases
- The available floor area and headroom

Hanson (1968) has given a selection guide based on these factors, which can be used to select the type of equipment most likely to be suitable (Fig. 17.66). Additional guidance is given in the paper by Koch and Shivelor (2015).

The basic principles of liquid–liquid extraction are covered in several specialist texts: Treybal (1980), Robbins (1997), and Humphrey and Keller (1997).

17.16.2 Extractor design

Number of stages

The primary task in the design of an extractor for a liquid–liquid extraction process is the determination of the number of stages needed to achieve the separation required.

The stages may be arranged in three ways:

- Fresh solvent fed to each stage, the raffinate passing from stage to stage (cross-flow)
- The extracting solvent fed co-currently with the raffinate, from stage to stage (co-current flow)
- The extracting solvent fed countercurrent to the raffinate (countercurrent flow)

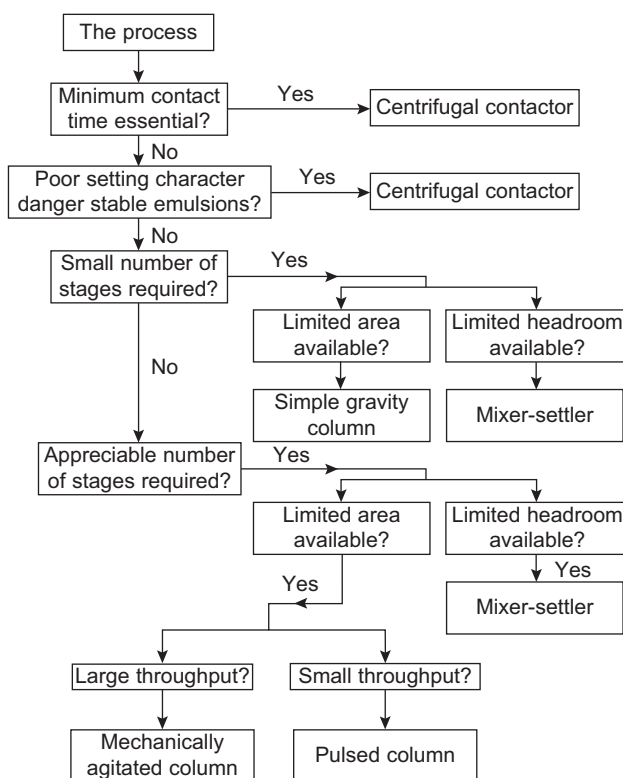


FIG. 17.66 Selection guide for liquid–liquid contactors. From Hanson, 1968.

Countercurrent flow is the most efficient method and the most commonly used. It will give the greatest concentration of the solute in the extract and the least use of solvent.

Equilibrium data

To determine the number of stages, it is best to plot the equilibrium data on a triangular diagram (Fig. 17.67). Each corner of the triangle represents 100% of the feed solvent, solute or extraction solvent. Each side represents the composition of one of the binary pairs. The ternary compositions are shown in the interior of the triangle. Mixtures within the region bounded by the curve will separate into two phases. The tie lines link the equilibrium compositions of the separate phases. The tie lines reduce in length toward the top of the curve. The point where they disappear is called the *plait* point.

A fuller discussion of the various classes of diagrams used to represent liquid–liquid equilibria is given in Chhabra and Gurappa (2019); see also Treybal (1980) and Humphrey et al. (1984).

The most comprehensive source of equilibrium data for liquid–liquid systems is the DECHEMA data series (Sørensen & Arlt, 1979). Equilibrium data for some systems are also given by Green and Southard (2018). Most commercial process simulation programs can be used to make initial estimates of liquid–liquid equilibria using a three-phase flash operation. Care is needed in selecting an appropriate phase equilibrium model. The UNIQUAC and UNIFAC equations can be used to estimate liquid–liquid equilibria; see Chapter 4. The size of the two-phase region can be highly sensitive to temperature, and it is usually a good idea to use a process simulation program to generate sufficient data to plot the triangular diagram at several temperatures. Care must be taken to ensure that the simulation model is tuned to accurately predict any available experimental data.

Number of stages

The number of stages required for a given separation can be determined from the triangular diagram using a method analogous to the McCabe–Thiele method for distillation. The method set out here is for countercurrent extraction.

Procedure

Refer to Figs. 17.67 and 17.68.

Let the flow rates be:

F = feed of the solution to be extracted

E = extract

R = raffinate

S = the extracting solvent

and the compositions:

r = raffinate

e = extract

s = solvent

f = feed

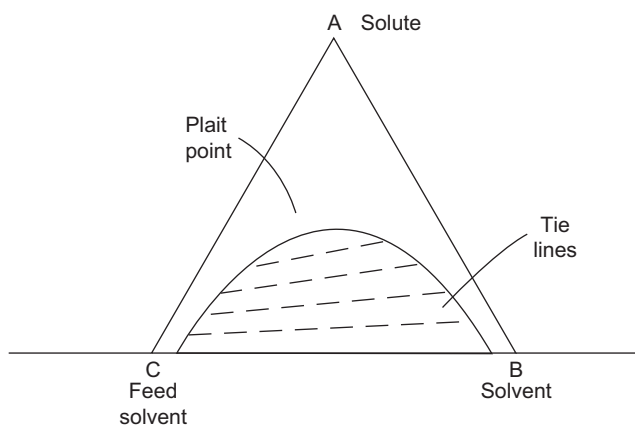


FIG. 17.67 Equilibrium diagram solute distributed between two solvents.

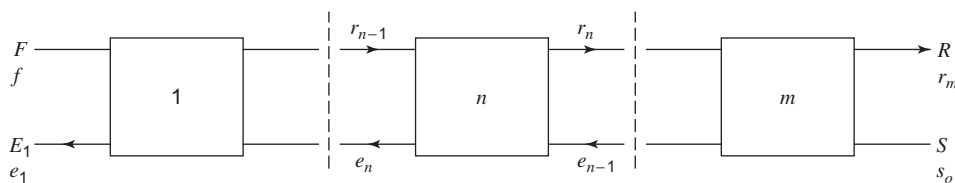


FIG. 17.68 Countercurrent extraction.

Then a material balance over stage n gives:

$$F + E_{n+1} = R_n + E_1$$

It can be shown that the difference in flow rate between the raffinate leaving any stage, R_n , and the extract entering the stage, E_n , is constant. It can also be shown that the difference between the amounts of each component entering and leaving a stage is constant. This means that if lines are drawn on the triangular diagram linking the composition of the raffinate from a stage and the extract entering from the next stage, they will pass through a common pole when extrapolated. The number of stages needed can be found by making use of this construction and the equilibrium compositions given by the tie lines.

Construction

1. Draw the liquid–liquid equilibrium data on triangular graph paper. Show sufficient tie lines to enable the equilibrium compositions to be determined at each stage.
2. Mark the feed and extraction-solvent compositions on the diagram. Join them with a line. The composition of a mixture of the feed and solvent will lie on this line.
3. Calculate the composition of the mixture given by mixing the feed with the extraction solvent. Mark this point, O, on the line drawn in step 2.
4. Mark the final raffinate composition, r_m , on the equilibrium curve.
5. Draw a line from r_m through point O. This will cut the curve at the final extract composition, e_1 .
Note: If the extract composition is specified rather than the raffinate, draw the line from e_1 through O to find r_m .
6. Draw a line from the solvent composition, S_0 , through r_m and extend it beyond r_m .
7. Draw a line from e_1 through f and extend it to cross the line drawn in step 6 at the pole point, P.
8. Find the composition of the raffinate leaving the first stage, r_1 , by judging the position of the tie line from e_1 . Draw a line from the pole point, P, through r_1 to cut the curve at e_2 , the extract leaving stage 2.
9. Repeat this procedure until sufficient stages have been drawn to reach the desired raffinate final composition.

If an extended tie line passes through the pole point, P, an infinite number of stages will be needed. This condition sets the minimum flow of extraction-solvent required. It is analogous to a pinch point in distillation.

The method is illustrated in Example 17.10.

Example 17.10

Acetone is to be extracted from a solution in water using 1,1,2-trichloroethane. The feed concentration is 45.0% w/w acetone. Determine the number of stages required to reduce the concentration of acetone to below 10% using 32 kg of extraction solvent per 100 kg feed.

The equilibrium data for this system are given by [Treybal et al. \(1946\)](#).

Solution

Composition of feed + solvent, point O = $0.45 \times 100 / (100 + 32) = 0.34 = 34\%$.

Draw line from TCE (trichloroethane) = 100%, point s_0 , to feed composition, f , 45% acetone.

Mark point O on this line at 34% acetone.

Mark required final raffinate composition, r_m , on the equilibrium curve at 10%.

Draw line from this point through point O to find final extract composition, e_1 .

Draw line from this point through the feed composition, f , and extend this line to cut a line extended from s_0 through r_m at P.

Using the tie lines plotted on the figure, judge the position that a tie line would have from e_1 and mark it in to find the point on the curve giving the composition of the raffinate leaving the first stage, r_1 .

Draw a line through from the pole point, P, through r_1 , to find the point on the curve giving the extract composition leaving the second stage, e_2 .

Repeat these steps until the raffinate composition found is below 10%.

From Fig. 17.69, it can be seen that five stages are needed.

In this example, the fact that the raffinate composition from stage 5 passes through the specified raffinate composition of 10% is fortuitous. As the construction, particularly the judgment of the position of the tie lines, is approximate, the number of stages should be increased to six. This should ensure that the specified raffinate composition of below 10% is met.

Immiscible solvents

If the solvents are immiscible, the procedure for determining the number of stages required is simplified. The equilibrium curve can be drawn on regular, orthogonal graph paper. An operating line giving the relationship between the compositions of the raffinate and extracts entering and leaving each stage can then be drawn and the stages stepped off. The procedure is similar to the McCabe–Thiele construction for determining the number of stages in distillation; see Section 17.5.2. The slope of the operating line is the ratio of the final raffinate to fresh solvent flow rates. Alternatively, the methods given in Section 17.14.2 for design of absorption columns can be used, where G is the flow rate of the initial solvent phase, L is the flow rate of the extraction solvent, y is the mole fraction of solute in the initial solvent, and m is the partition coefficient.

For a full discussion of the methods that can be used to determine the stage requirements in liquid–liquid extraction, refer to Treybal (1980), Green and Southard (2018), and Robbins (1997). Computer programs are available for the design of extraction processes and are included in the various commercial process simulation packages available; see Chapter 4. It is usually a good practice to make a few hand calculations to determine a good initialization for the process simulation model. At a minimum, the designer should estimate the partition coefficient and ensure that the absorption factor $mG/L \leq 1$ if the goal is to obtain a high recovery of solute.

17.16.3 Extraction columns

The simplest form of extractor is a spray column. The column is empty; one liquid forms a continuous phase and the other liquid flows up or down the column in the form of droplets. Mass transfer takes place to or from the droplets to the continuous phase. The efficiency of a spray tower will be low, particularly with large-diameter columns, due to back-mixing. The efficiency of the basic, empty spray column can be improved by installing plates or packing.

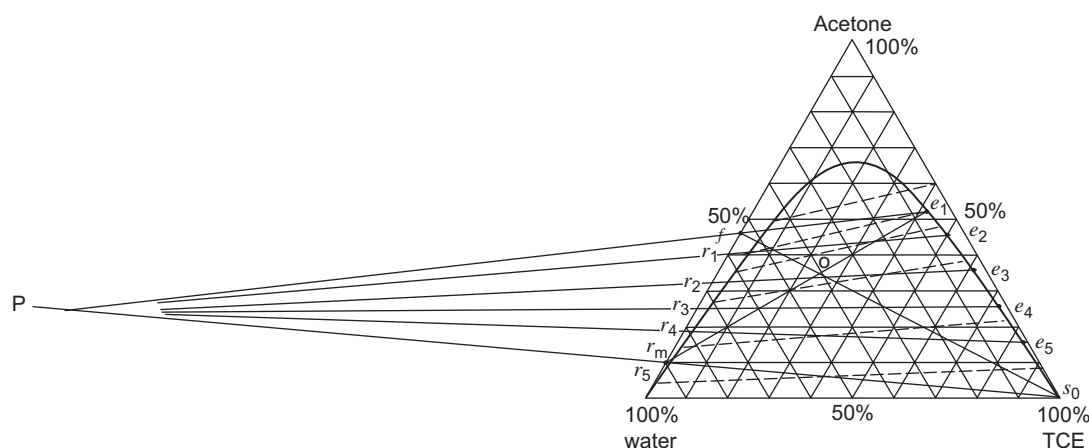


FIG. 17.69 Example 17.10.

Sieve plates are used, similar to those used for distillation and absorption. The plates can be designed so that either the heavy phase is dispersed in the light phase (sometimes called “rain decks”) or the light phase is dispersed in the heavy phase (“reverse rain decks”).

Random packings are also used; they are the same as those used in packed distillation and absorption columns. The properties of random packings are given in [Tables 17.2 and 17.3](#). Proprietary structured packings are also used. Criteria for selection of column internals are given by [Koch and Shivelor \(2015\)](#).

Mass transfer in packed columns is a continuous, differential process, so the transfer unit method should be used to determine the column height, as used in absorption; see Section 17.14.2. It is often convenient to treat packed extraction columns as staged processes and use the HETP for the packing employed. For random packings, the HETP in extraction will typically range from 0.5 to 1.5 m, depending on the type and size of packing used.

Flooding

No simple correlation is available to predict the flooding velocities in extraction columns, and hence the column diameter needed. The more specialized texts should be consulted to obtain guidance on the appropriate method to use for a particular problem; see [Treybal \(1980\)](#), [Green and Southard \(2018\)](#), and [Humphrey and Keller \(1997\)](#).

17.16.4 Supercritical fluid extraction

A recent development in liquid–liquid extraction has been the use of supercritical fluids as the extraction solvent. Carbon dioxide at high pressure is the most commonly used fluid. It is used in processes for the decaffeination of coffee and tea. The solvent can be recovered from the extract solution as a gas by reducing the pressure. Supercritical extraction processes are discussed by [Humphrey and Keller \(1997\)](#).

17.17 Capital cost of separation columns

Distillation, absorption, and extraction columns are always pressure vessels and must be designed and costed using the methods set out in [Chapter 14](#). Separation columns are often tall structures whose wall thickness can be governed by wind or earthquake loads, so a combined loading analysis is necessary; see Section 14.8. The column diameter and height can be estimated using the methods described in Section 17.11. The maximum design temperature and pressure for the column will usually be calculated from the conditions at the reboiler with a suitable additional margin; see Section 14.4. If the column operates below ambient temperature, then the minimum temperature will be calculated from the condenser temperature with a suitable design margin.

Column internals such as plates or packing are costed separately and added to the vessel cost.

Correlations for preliminary estimates of vessel and internals costs are given in [Table 7.2](#). An example of costing a trayed distillation column is given in Example 7.3.

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17.19 Nomenclature

		Dimensions in MLT θ
A_a	Active area of plate	L^2
A_{ap}	Clearance area under apron	L^2
A_c	Total column cross-sectional area	L^2
A_d	Downcomer cross-sectional area	L^2
A_h	Total hole area	L^2
A_i	Absorption factor	—
A_m	Area term in equation 17.60	L^2
A_n	Net area available for vapor-liquid disengagement	L^2
A_p	Perforated area	L^2
a	Packing surface area per unit volume	L^{-1}
a_w	Effective interfacial area of packing per unit volume	L^{-1}
B	Moles of bottom product per unit time	MT^{-1}
b_i	Moles of component i in bottom product	M
C_o	Orifice coefficient in Equation 17.56	—
C_T	Total molar concentration	ML^{-3}
D	Moles of distillate per unit time	MT^{-1}
D_c	Column diameter	L
D_e	Eddy diffusivity	L^2T^{-1}
D_L	Liquid diffusivity	L^2T^{-1}
D_{LK}	Diffusivity of light key component in liquid phase	L^2T^{-1}
D_v	Diffusivity of vapor	L^2T^{-1}
d_h	Hole diameter	L
d_i	Moles of component i in distillate per unit time	MT^{-1}
d_p	Size of packing	L
E	Extract flow rate	MT^{-1}
E_a	Actual plate efficiency, allowing for entrainment	—
E_{mV}	Murphree plate efficiency	—
E_{mv}	Murphree point efficiency	—
E_o	Overall column efficiency	—
e	Extract composition	—
FA	Fractional area, equation 17.69	—
F	Feed, of the solution to be extracted	MT^{-1}
F_n	Feed rate to stage n	MT^{-1}
F_p	Packing factor	L^{-1}
F_v	Column "F" factor = $u_a \sqrt{\rho_v}$	$M^{1/2}L^{-1/2}T^{-1}$
F_{LV}	Column liquid-vapor factor in Fig. 17.34	—

		Dimensions in MLT θ
F	Feed composition (in extraction)	—
F	Fractional approach to flooding = u_a / u_{af}	—
f_i	Moles of component i in feed per unit time	MT^{-1}
f_1	Viscosity correction factor in Equation 17.79	—
f_2	Liquid density correction factor in Equation 17.79	—
f_3	Surface tension correction factor in Equation 17.79	—
G_m	Molar flow rate of gas per unit area	$\text{ML}^{-2}\text{T}^{-1}$
g	Gravitational acceleration	LT^{-2}
H	Specific enthalpy of vapor phase	L^2T^{-2}
H_G	Height of gas film transfer unit	L
H_L	Height of liquid film transfer unit	L
H_{OG}	Height of overall gas phase transfer unit	L
H_{OL}	Height of overall liquid phase transfer unit	L
$\mathcal{H}$	Henry's constant	$\text{ML}^{-1}\text{T}^{-2}$
h	Specific enthalpy of liquid phase	L^2T^{-2}
h_{ap}	Apron clearance	L
h_b	Height of liquid backed-up in downcomer	L
h_{bc}	Downcomer back-up in terms of clear liquid head	L
h_d	Dry plate pressure drop, head of liquid	L
h_{dc}	Head loss in downcomer	L
h_f	Specific enthalpy of feed stream	L^2T^{-2}
h_L	Liquid hold-up on tray	L
h_{ow}	Height of liquid crest over downcomer weir	L
h_r	Plate residual pressure drop, head of liquid	L
h_t	Total plate pressure drop, head of liquid	L
h_w	Weir height	L
K	Equilibrium constant	—
K_G	Overall gas phase mass transfer coefficient	L^{-1}T
K_i	Equilibrium constant for component i	—
K_L	Overall liquid phase mass transfer coefficient	LT^{-1}
K_1	Constant in Equation 17.49	LT^{-1}
K_2	Constant in Equation 17.52	—
K_3	Percentage flooding factor in Equation 17.80	—
K_4	Parameter in Fig. 17.54, defined by Equation 17.81	—
K_5	Constant in Equation 17.84	—
k_G	Gas film mass transfer coefficient	L^{-1}T
k_L	Liquid film mass transfer coefficient	LT^{-1}
L	Liquid flow rate, mols per unit time	MT^{-1}
L_m	Molar flow rate of liquid per unit area	$\text{ML}^{-2}\text{T}^{-1}$

		Dimensions in MLT θ
L_p	Volumetric flow rate across plate divided by average plate width	L^2T^{-1}
L_w	Liquid mass flow rate	MT^{-1}
L_w^*	Liquid mass flow rate per unit area	$MT^{-2}T^{-1}$
L_{wd}	Liquid mass flow rate through downcomer	MT^{-1}
l_h	Weir chord height	L
l_p	Pitch of holes (distance between centers)	L
l_t	Plate spacing in column	L
l_w	Weir length	L
M	Number of pure products required	—
M_s	Molecular weight of solvent	—
m	Slope of equilibrium line	—
N	Number of components	—
N	Number of stages	—
N_G	Number of gas-film transfer units	—
N_L	Number of liquid-film transfer units	—
N_{min}	Number of stages at total reflux	—
N_{OG}	Number of overall gas-phase transfer units	—
N_{OL}	Number of overall liquid-phase transfer units	—
N_r	Number of equilibrium stages above feed	—
N_s	Number of equilibrium stages below feed	—
n	Stage number	—
P	Total pressure	$ML^{-1}T^{-2}$
ΔP_t	Total plate pressure drop	$ML^{-1}T^{-2}$
p	Partial pressure	$ML^{-1}T^{-2}$
q	Heat to vaporize one mol of feed divided by molar latent heat	—
q_b	Heat supplied to reboiler	ML^2T^{-3}
q_c	Heat removed in condenser	ML^2T^{-3}
q_n	Heat supplied to or removed from stage n	ML^2T^{-3}
R	Universal gas constant	$L^2T^{-2}\theta^{-1}$
R	Reflux ratio	—
R	Raffinate flow rate (in extraction)	MT^{-1}
R_{min}	Minimum reflux ratio	—
r	Raffinate composition	—
S	Extracting solvent flow rate	MT^{-1}
S_i	Stripping factor	—
S_n	Side stream flow from stage n	MT^{-1}
s	Solvent composition	—
t_L	Liquid contact time	T

		Dimensions in MLT θ
t_r	Residence time in downcomer	T
$\bar{t}_v$	Average vapor residence time	T
u_a	Vapor velocity based on active area	LT ⁻¹
u_{af}	Vapor velocity at flooding point based on active area A_a	LT ⁻¹
u_f	Vapor velocity at flooding point based on net area A_n	LT ⁻¹
u_h	Vapor velocity through holes	LT ⁻¹
u_n	Vapor velocity based on net cross-sectional area	LT ⁻¹
u_v	Superficial vapor velocity (based on total cross-sectional area)	LT ⁻¹
V	Vapor flow rate mols per unit time	MT ⁻¹
V_w	Vapor mass flow rate	MT ⁻¹
V_w^*	Vapor mass flow rate per unit area	ML ⁻² T ⁻¹
x	Mol fraction of component in liquid phase	—
x_b	Mol fraction of component in bottom product	—
x_d	Mol fraction of component in distillate	—
x_e	Equilibrium mol fraction in liquid phase	—
x_i	Mol fraction of component i	—
x_r	Mol fraction of reference component (Equation 17.25)	—
x_1	Mol fraction of solute in solution at column base	—
x_2	Mol fraction of solute in solution at column top	—
x_α	Mol fraction in first liquid phase	—
x_β	Mol fraction in second liquid phase	—
y	Mol fraction of component in vapor phase	—
y_e	Equilibrium mol fraction in vapor phase	—
y_i	Mol fraction of component i	—
Δy	Concentration driving force in the gas phase	—
Δy_{lm}	Log mean concentration driving force	—
y_1	Concentration of solute in gas phase at column base	—
y_2	Concentration of solute in gas phase at column top	—
Z	Height of packing	L
Z_c	Liquid hold-up on plate	L
Z_L	Length of liquid path	L
z_i	Mol fraction of component i in feed stream	—
z_f	Mol fraction of component in feed stream	—
α	Relative volatility	—
α_i	Relative volatility of component i	—
α_a	Average relative volatility of light key	—
θ	Root of Equation 17.29	—

		Dimensions in MLT θ
μ	Dynamic viscosity	$\text{ML}^{-1}\text{T}^{-1}$
μ_a	Molar average liquid viscosity	$\text{ML}^{-1}\text{T}^{-1}$
μ_s	Viscosity of solvent	$\text{ML}^{-1}\text{T}^{-1}$
μ_w	Viscosity of water at 20 °C	$\text{ML}^{-1}\text{T}^{-1}$
ρ	Density	ML^{-3}
ρ_s	Density of solvent	ML^{-3}
ρ_w	Density of water at 20 °C	ML^{-3}
σ	Surface tension	MT^{-2}
σ_c	Critical surface tension for packing material	MT^{-2}
σ_w	Surface tension of water at 20 °C	MT^{-2}
Φ	Intercept of operating line on Y axis	—
Φ_n	Factor in Equation 17.80	—
ψ	Fractional entrainment	—
ψ_h	Factor in Equation 17.79	—
Dg	Surface tension number	
Pe	Peclet number	
Re	Reynolds number	
Sc	Schmidt number	
Subscripts		
b	Bottoms	
d	Distillate (Distillate)	
e	At equilibrium with the other phase	
f	Feed	
HK	Heavy key	
l	Component number	
L	Liquid	
LK	Light key	
m	Last stage (in extraction)	
min	Minimum	
n	Stage number	
r	Reference component	
s	Solvent	
v	Vapor	
w	Water	
α	First liquid phase	
β	Second liquid phase	
1	Base of packed column	
2	Top of packed column	
Superscripts		
'	Stripping section of column	

17.20 Problems

All of the problems in this chapter can be solved using either hand calculations or process simulation programs. Where data are given, they can be used to confirm the validity of the phase equilibrium model used in a simulation.

- 17.1. At a pressure of 10 bar, determine the bubble and dew point of a mixture of hydrocarbons, composition, mol%: n-butane 21, n-pentane 48, n-hexane 31. The equilibrium K factors can be estimated using the De Priester charts in [Dadyburjor \(1978\)](#) or found using a process simulation program.
- 17.2. The feed to a distillation column has the following composition: mol%: propane 5.0, isobutane 15, n-butane 25, isopentane 20, n-pentane 35. The feed is pre-heated to a temperature of 90 °C, at 8.3 bar pressure. Estimate the proportion of the feed that is vapor.
- 17.3. Propane is separated from propylene by distillation. The compounds have close boiling points, and the relative volatility will be low. For a feed composition of 10% w/w propane, 90% w/w propylene, estimate the number of theoretical plates needed to produce propylene overhead with a minimum purity of 99.5 mol%. The column will operate with a reflux ratio of 20. The feed will be at its boiling point. Take the relative volatility as constant at 1.1.
- 17.4. The composition of the feed to a debutanizer is given here. Make a preliminary design for a column to recover 98% of the n-butane overhead and 95% of the isopentane from the column base. The column will operate at 14 bar, and the feed will be at its boiling point. Use the shortcut methods and follow the procedure set out below. Use the De Priester charts in [Dadyburjor \(1978\)](#) to determine the relative volatility if solving by hand. The liquid viscosity can be estimated using the data given in Appendix C (available in the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793).
 - a. Investigate the effect of reflux ratio on the number of theoretical stages.
 - b. Select the optimum reflux ratio.
 - c. Determine the number of stages at this reflux ratio.
 - d. Estimate the stage efficiency.
 - e. Determine the number of real stages.
 - f. Estimate the feed point.
 - g. Estimate the column diameter.

Feed composition:

		kg/h
propane	C ₃	910
isobutane	i-C ₄	180
n-butane	n-C ₄	270
isopentane	i-C ₅	70
normal pentane	n-C ₅	90
normal hexane	n-C ₆	20

- 17.5. In a process for the manufacture of acetone, acetone is separated from acetic acid by distillation. The feed to the column is 60 mol% acetone, the balance acetic acid. The column is to recover 95% of the acetone in the feed with a purity of 99.5 mol% acetone. The column will operate at a pressure of 760 mmHg, and the feed will be pre-heated to 70 °C. For this separation, determine:
 - a. The number of minimum number of stages required
 - b. The minimum reflux ratio
 - c. The number of theoretical stages for a reflux ratio 1.5 times the minimum
 - d. The number of actual stages if the plate efficiency can be taken as 60%

Equilibrium data for the system acetone–acetic acid, at 760 mmHg, mol fractions acetone:

liquid phase	0.10	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
Vapor phase	0.31	0.56	0.73	0.84	0.91	0.95	0.97	0.98	0.99
Boiling point °C	103.8	93.1	85.8	79.7	74.6	70.2	66.1	62.6	59.2

Reference: *Othmer (1943)*.

- 17.6.** In the manufacture of absolute alcohol by fermentation, the product is separated and purified using several stages of distillation. In the first stage, a mixture of 5 mol% ethanol in water, with traces of acetaldehyde and fusel oil, is concentrated to 50 mol%. The concentration of alcohol in the wastewater is reduced to less than 0.1 mol%.

Design a sieve plate column to perform this separation for a feed rate of 10,000 kg/h. Treat the feed as a binary mixture of ethanol and water.

Take the feed temperature as 20 °C. The column will operate at 1 atmosphere.

Determine:

- The number of theoretical stages
- An estimate of the stage efficiency
- The number of actual stages needed

Design a suitable sieve plate for conditions below the feed point.

Equilibrium data for the system ethanol–water, at 760 mmHg, mol fractions ethanol:

liquid phase	0.019	0.072	0.124	0.234	0.327	0.508	0.573	0.676	0.747	0.894
vapor phase	0.170	0.389	0.470	0.545	0.583	0.656	0.684	0.739	0.782	0.894
Boiling point °C	95.5	89.0	85.3	82.7	81.5	79.8	79.3	78.7	78.4	78.2

Reference: *Carey and Lewis (1932)*.

- 17.7.** In the manufacture of methyl ethyl ketone from butanol, the product is separated from unreacted butanol by distillation. The feed to the column consists of a mixture of 0.90 mol fraction MEK and 0.10 mol fraction 2-butanol, with a trace of trichloroethane.

The feed rate to the column is 20 kmol/h and the feed temperature 35 °C. The specifications required are top product 0.99 mol fraction MEK and bottom product 0.99 mol fraction butanol.

Design a column for this separation. The column will operate at essentially atmospheric pressure. Use a reflux ratio 1.5 times the minimum.

- Determine the minimum reflux ratio
- Determine the number of theoretical stages
- Estimate the stage efficiency
- Determine the number of actual stages needed
- Design a suitable sieve plate for conditions below the feed point

Equilibrium data for the system MEK –2-butanol, mol fractions MEK:

liquid phase	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
Vapor phase	0.23	0.41	0.53	0.64	0.73	0.80	0.86	0.91	0.95
boiling point °C	97	94	92	90	87	85	84	82	80

- 17.8.** A column is required to recover acetone from an aqueous solution. The feed contains 5 mol% acetone. A product purity of 99.5% w/w is required, and the effluent water must contain less than 100 ppm acetone.

The feed temperature will range from 10 to 25 °C. The column will operate at atmospheric pressure. For a feed of 7500 kg/h, compare the designs for a sieve plate and packed column for this duty. Use a reflux ratio of 3. Compare the capital and utility cost for the two designs.

No reboiler is required for this column; live steam can be used.

Equilibrium data for the system acetone–water is given in Example 17.2.

- 17.9. In the manufacture of methyl ethyl ketone (MEK), the product MEK is extracted from a solution in water using 1,1,2 trichloroethane as the solvent.

For a feed rate 2000 kg/h of solution, composition 30% w/w MEK, determine the number of stages required to recover 95% of the dissolved MEK using 700 kg/h TCE, with countercurrent flow.

Tie-line data for the system MEK–water–TCE percentages w/w, from [Newman et al. \(1949\)](#).

Water-rich phase		Solvent-rich phase	
MEK	TCE	MEK	TCE
18.15	0.11	75.00	19.92
12.78	0.16	58.62	38.65
9.23	0.23	44.38	54.14
6.00	0.30	31.20	67.80
2.83	0.37	16.90	82.58
1.02	0.41	5.58	94.42

- 17.10. Chlorine is to be removed from a vent stream by scrubbing with a 5% w/w aqueous solution of sodium hydroxide. The vent stream is essentially nitrogen, with a maximum concentration of 5.5% w/w chlorine. The concentration of chlorine leaving the scrubber must be less than 50 ppm by weight. The maximum flow rate of the vent stream to the scrubber will be 4500 kg/h. Design a suitable packed column for this duty. The column will operate at 1.1 bar and ambient temperature. If necessary, the aqueous stream may be recirculated to maintain a suitable wetting rate.

Note: The reaction of chlorine with the aqueous solution will be rapid, and there will be essentially no back-pressure of chlorine from the solution.

Specification and design of solids-handling equipment

KEY LEARNING OBJECTIVES

- The importance of solids-handling operations in the process industries
- How to size and design equipment for sizing, handling, transporting, mixing, and storing solids
- How to size and design equipment to recover solids from gas and liquid streams
- How to design dryers
- How to design equipment for forming solid particles with the desired shape, size, and properties
- Hazards intrinsic to solids processing

18.1 Introduction

Solid-phase materials are encountered in most chemical processes. Foods, fertilizers, polymers, and most pharmaceutical products are finished as solid-phase products. Minerals, coal, biomass, and recycled materials are solid-phase feeds. Most specialty chemical and pharmaceutical processes use crystallization as a means of achieving high-purity products. Even processes that handle only vapor and liquid streams usually use solid-phase catalysts and adsorbents.

Solids are, in general, more difficult to process than gases or liquids. The equipment used for processing solids consumes more power (electricity) and is more likely to suffer mechanical failure. Solids-handling plants can be prone to shutdowns due to caking or plugging of equipment or interruptions of solids flow caused by bridging of poorly designed hoppers and feeders. Solids can form dusts during processing, which cause safety, industrial hygiene, and environmental hazards that must be carefully controlled and mitigated.

The design of solids-handling plant is rarely given adequate attention in undergraduate chemical engineering courses, and most graduating engineers are only superficially aware of solids-handling operations. A wide range of specialized equipment is used in solids processing, and there can be large differences between different vendors' designs. Although there is insufficient space in this volume to give detailed design principles for all solids-handling operations, the following sections introduce the most commonly encountered unit operations and provide references that go into greater detail. A great deal of additional information can be found by searching manufacturers' websites on the Internet.

As with all process operations that use proprietary equipment, final equipment selection and costing should always be carried out in consultation with a vendor.

18.2 Properties of granular materials

This section gives an overview of the physical properties of solid materials that affect their handling, flow, and storage behavior and that are important in the design of solids-processing equipment.

A bulk solid is an aggregate of many individual particles. Some processing operations are primarily affected by individual particle properties, whereas others are designed based on properties of the bulk material or distributions of individual particle properties.

Several books have been written on the characterization and classification of particles and powders: Allen (1996, 1997), Merkus (2009), and Stanley-Wood and Lines (1992). These discuss the various measurement techniques used and go into considerably more detail than is needed for most chemical engineering design purposes. Kaye (1997), Rhodes (2008), and Higashitani et al. (2019) provide good overviews of these subjects.

18.2.1 Properties of solid particles

Very few of the solid materials encountered in the process industries consist of uniform dense spheres. Particle properties are affected by the intrinsic properties of the material, as well as the processes by which solid particles are formed, shaped, sized, and handled.

Individual particle properties can be very important for end-use applications and are often required as product specifications for solid products. For example, the size and shape of tablets must be highly uniform to ensure that patients receive the correct dose and release rate of an active pharmaceutical ingredient.

Particle size and shape

Particle size is one of the most important characteristics for processing of solids. The effective particle diameter occurs in the design equations for many types of solids-handling equipment. Unfortunately, for irregular-shaped particles there is no simple definition of the effective diameter, and different definitions may be most suitable in different circumstances.

An irregular particle can be characterized in several ways. Consider Fig. 18.1. If the particle is lying flat in a stable position, the longest dimension is obvious and is defined as the length, L . The dimension in the direction perpendicular to the longest axis in the plane of stability is defined as the width, W , and the dimension in the direction normal to the plane of stability is defined as the thickness, t .

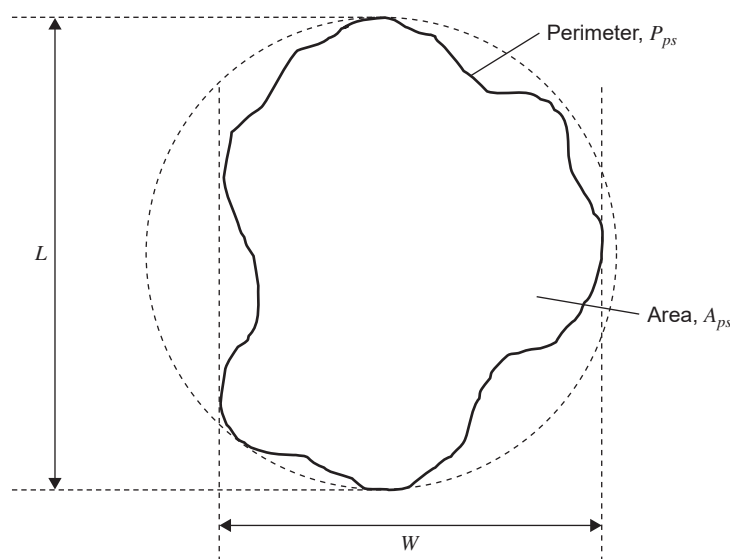


FIG. 18.1 Characterization of an irregular-shaped particle.

The ratio L/W is called the elongation or aspect ratio. Particles with elongation greater than 3.0 are often classified as fibers. The inverse of elongation is known as the *chunkiness*, which is often used in preference to elongation, as it necessarily varies between 0 and 1.0 and so is easier to use when plotting graphs. The ratio W/t is known as the *flakiness* factor. Particles with high flakiness can be difficult to filter and flow out of hoppers due to their tendency to stack.

The diameter of a circle that has the same projected area as the particle is:

$$d_a = 2\sqrt{\frac{A_{ps}}{\pi}} \quad (18.1)$$

where:

d_a = diameter of a circle with same projected area as the particle

A_{ps} = projected area of the particle

A circle with the same perimeter as the particle would have diameter d_{pr} given by:

$$d_{pr} = \frac{P_{ps}}{\pi} \quad (18.2)$$

where:

P_{ps} = perimeter of the particle

The conventional definition of the effective hydraulic diameter, d_h , is four times the area divided by the wetted perimeter:

$$d_h = \frac{4 A_{ps}}{P_{ps}} \quad (18.3)$$

The Stokes diameter, d_{st} , is defined as the diameter of a smooth sphere of the same density as the particle that settles at the same velocity in a viscous liquid.

The *sphericity* of a particle, Ψ , can be defined as:

$$\Psi = \frac{\text{Surface area of a sphere with the same volume as the particle}}{\text{Surface area of the particle}} \quad (18.4)$$

Depending on the physical basis of the method used, different measurement techniques will measure different dimensions as the effective “diameter.” For example, for sieves, screens, and filters, the governing dimension is the width, W . Sedimentation and elutriation methods will measure the Stokes diameter, d_{st} . Optical analysis by microscopy will measure the area, A_{ps} , and hence d_a . Methods have been developed to relate d_{st} to optical or sieve measurements; see, for example, [Heywood \(1961\)](#).

Screening results are usually given in terms of screen mesh sizes. Different standard mesh sizes are used in the United States and internationally. [Table 18.1](#) gives the dimensions corresponding to different standard wire mesh sieve sizes.

Density and porosity

The density of particulate materials can also be difficult to measure due to the presence of internal pores, cracks, and imperfections in the particles.

The *true density*, also known as *framework*, *crystalline*, or *structural* density, is defined as the density of the material from which the particles are constituted. The *apparent density*, *piece density*, or *particle density* is the observed density of the particles allowing for internal porosity within a particle. The *bulk density* is the density of a large collection of particles, which is further reduced because of the presence of void space between particles.

The bulk density is related to the apparent density and bed void fraction, ϵ :

$$\rho_b = \rho_p(1 - \epsilon) \quad (18.5)$$

where:

ρ_b = bulk density

ρ_p = particle density

ϵ = bed void fraction

TABLE 18.1 Standard wire mesh sieve sizes

ASTM (E-11)	Tyler equivalent	Sieve opening (mm)	Sieve opening (in)
100 mm		100.0	~4.00
75 mm		75.0	~3.00
50 mm		50.0	~2.00
25 mm		25.0	~1.00
12.5 mm		12.5	~0.50
6.3 mm		6.3	~0.25
No. 4	4 mesh	4.75	0.187
No. 5	5 mesh	4.00	0.157
No. 6	6 mesh	3.35	0.132
No. 7	7 mesh	2.80	0.110
No. 8	8 mesh	2.36	0.0937
No. 10	9 mesh	2.00	0.0787
No. 12	10 mesh	1.70	0.0661
No. 14	12 mesh	1.40	0.0555
No. 16	14 mesh	1.18	0.0469
No. 18	16 mesh	1.00	0.0394
No. 20	20 mesh	0.85	0.0331
No. 25	24 mesh	0.71	0.0278
No. 30	28 mesh	0.60	0.0234
No. 35	32 mesh	0.500	0.0197
No. 40	35 mesh	0.425	0.0165
No. 45	42 mesh	0.355	0.0139
No. 50	48 mesh	0.300	0.0117
No. 60	60 mesh	0.250	0.0098
No. 70	65 mesh	0.212	0.0083
No. 80	80 mesh	0.180	0.0070
No. 100	100 mesh	0.150	0.0059
No. 120	115 mesh	0.125	0.0049
No. 140	150 mesh	0.106	0.0041
No. 200	200 mesh	0.075	0.0029
No. 400	400 mesh	0.038	0.0015

S.I. units are ASTM standard. 25.4 mm = 1.00 in.

Values in bold are those for which the ASTM designations and Tyler equivalent series match.

Similarly, the particle density is related to the true density and the internal porosity (internal pore volume fraction), χ :

$$\rho_p = \rho_t(1 - \chi) \quad (18.6)$$

where:

ρ_t = true density

χ = internal pore volume fraction

The density of a fully wetted particle in a fluid of density ρ_f is:

$$\rho_w = \rho_p + \rho_f \chi \quad (18.7)$$

where:

ρ_w = wetted particle density

If a container of known volume is filled with particles and weighed, then the bulk density can be calculated. Adding a liquid that fills the interparticle voids but does not fill the internal cracks and pores allows measurement of the particle external volume and hence the apparent density. Adding a liquid that fills the internal cracks and pores or measuring the gas volume added to achieve a given change in pressure similarly allows measurement of the particle framework volume and hence the true density.

Bulk density is highly sensitive to packing of the particles and can be increased by tapping or compacting the bed, particularly for fine powders or materials with high flakiness. A compacted bed will typically have 5% to 20% higher density than an uncompacted bed. Typical values of bulk density are given in [Table 18.2](#).

In most design situations, the apparent density is used for individual particles and the average bulk density is used for sizing fixed beds, bins, hoppers, and conveyors. The apparent density is valid for dry particles in gas flows and for particles that are wetted by a surrounding fluid. The wetted density should be used when a wet particle is surrounded by a gas (as in a dryer) to allow for the liquid in the pores. An initial estimate of particle density can often be made by assuming a typical value of 0.4 for the bed void fraction in Equation 18.5.

The internal porosity of particles is very important in many applications. Internal porosity greatly increases the surface area of the particles, increasing the rate of dissolution in a solvent and providing additional surface area for adsorption, reaction, or catalysis. Because the internal surface area is usually more important than the fraction of internal volume occupied by pores, porosity is often expressed as surface area per unit mass, typically given in m^2/g . Internal surface area is measured by gas adsorption studies; see [Allen \(1996\)](#).

Particle strength and hardness

The strength of individual particles of a granular material depends on the material from which the particles are made, as well as the microstructure of the particles. Many solid products are actually composites or agglomerates, formed by binding smaller particles together in a matrix of liquid or solid binder. The strength of agglomerates depends on the chemical and physical nature of the binder; see [Pietsch \(1997\)](#). The use of agglomeration to increase particle size is discussed in [Section 18.8](#).

Particle strength is an important property of formed particles such as pharmaceutical tablets, catalyst pellets, and processed foods such as candies and breakfast cereals. Formed particles must be strong enough to survive processing during manufacture as well as end-use requirements. Catalyst pellets must be strong enough to withstand the forces exerted on them when loaded into a fixed bed or the impacts encountered in fluidized-bed or moving-bed reactors. Tablets must often be designed to be strong enough to keep their shape during packaging and shipping but weak enough to be easily split in two by the patient or caregiver. Formed foods such as breakfast cereals and baked snacks must have sufficient strength to be crunchy but not be so strong that the consumer breaks their teeth.

Particle strength is often confused with hardness, although they are not strictly related except for particles that are microscopically homogeneous and free from defects. Hardness is defined as a measure of the ability of a material to scratch or wear another material. Hardness is a good predictor of how abrasive a material will be and hence how rapidly it will cause wear or erosion of contacting surfaces such as cyclones or grinding equipment. The grindability of a nonuniform solid depends more on the particle strength than the hardness. For example, small inclusions of quartz in raw minerals can make the material very abrasive without making it resistant to crushing or grinding. Hardness is typically measured on the Moh hardness scale. The Moh hardness scale is a relative ranking of hardness ranging from 1 (talc) to 10 (diamond), where materials are ranked based on their scratch test performance relative to other materials. The Moh hardness scale is given in [Fig. 18.69](#) in the discussion of grinding in [Section 18.9](#). Values of Moh hardness for some common materials are given in [Table 18.2](#).

The strength of individual particles is usually stated as the *crush strength*, which is the resistance to compression measured in a load cell. Standard methods for measuring particle crush strength are ASTM D6175 and ASTM D4179. The particle crush strength is sometimes referred to as *piece crush strength*. Values of crush strength are conventionally reported as the average force in lb_f required to break a particle. In some cases, specifications are set on the standard deviation as well as the mean crush strength of the product.

TABLE 18.2 Typical properties of particulate materials

Material	Average bulk density (kg/m ³)	Moh Hardness	Angle of repose, β (degrees)	Bond work index (kWh/2000 lb)
Alumina beads	2050	2	34	
Ash, dry fly ash	610	1	40	
Bauxite	1090	5	31	9.45
Calcium oxide	430		43	
Carbon	800	1	21	
Catalyst, 100 micron FCC	510	2	32	
Cement, Portland	1520		39	10.5
Clay, kaolin, crushed	1025		35	7.1
Coal, bituminous, dry	670	2	29	11.4
Coal, bituminous, wet	800	2	40	
Coffee beans	670		25	
Coke, petroleum	640	2	34	73.8
Corn, shelled	720	1	21	
Dolomite, crushed	740	4	41	11.3
Glass beads, 280 micron	1500	7	26	
Gypsum, ground	900	2	40	8.16
Lead shot, 0.25 in.	6600	1	33	
Lime, burned and pulverized	430	3	43	
Limestone, coarse	1570	4	25	11.6
Mica, ground	220		36	135
Phosphate, dicalcium	960		30	
Phosphate rock	960	4	40	10.1
Phthalic anhydride, flakes	670	1	24	
Rice	800		20	
Rubber, ground tire	370		35	
Salt, granulated	1300	2	31	
Sand, coarse	1500	7	30	16.5
Sand, very fine	1500	7	32	
Sawdust	320	1	45	

Soap powder	160	1	30
Soda ash, light	480	1	37
Sodium nitrate	1090		24
Soybeans	770	1	39
Sulfur powder	800	3	45
Wheat	770	1	23
Wood chips	350		36

Notes:

1 Adapted from data in Zenz and Othmer (1960) and Green and Southard (2018).

2 Average bulk densities can vary $\pm 10\%$

The strength of an aggregate of particles is not the same as the strength of an individual particle and can be measured using ASTM method D7084. The strength of bulk minerals is more often reported as a *work index*, which is a measure of the difficulty of grinding the material. The work index is defined and described in [Section 18.9](#).

An important property related to particle strength is attrition resistance, which is a measure of how easily smaller particles can be broken off from the surface of a larger particle. Particles that are easily reduced to smaller particles by friction or that crumble under low pressures are described as *friable* and tend to form dust during transport, handling, processing, and storage. Many hazards are associated with dust formation (see [Section 18.11](#)), so special care must be taken when designing plant for friable solids. The attrition resistance of formed products can be measured using the methods described in ASTM D4058.

Particle chemical properties

Chemical properties of the material of which particles are composed can have a strong influence on particle density, porosity, and strength or can introduce health and safety hazards that affect processing. Chemical reactions can also lead to changes in these properties, particularly when particles are exposed to air or atmospheric moisture.

Solid materials that are toxic, corrosive, combustible, or explosive must be handled with particular care. Hazards of processing solids are described in [Section 18.11](#). Materials that would normally be considered inert can become extremely hazardous during processing, for example, if they become saturated with a flammable solvent or if grinding or abrasion exposes an unoxidized surface that can react with air.

Many solid materials are hygroscopic and will absorb water if exposed to ambient air. This affects the particle density and can also affect particle strength. Many minerals have a different crystalline structure in the hydrated state. Moisture content can also dramatically change the bulk flow properties of granular materials; see [Armstrong et al. \(2014\)](#) for examples.

Many solid materials are susceptible to oxidation in air. This can be particularly hazardous during grinding, as fresh surface area will be exposed. Oxidation is an exothermic reaction that can run away and cause ignition of the solids.

Grinding can also cause release of volatile compounds from a solid when fresh surface area is exposed. This can affect the chemical composition of the solids. In many foods, volatile organic compounds (VOCs) play a critical role in determining flavor, and process operations must be designed to reduce loss of these compounds by evaporation. For example, this effect is important in processing of coffee and cacao beans and explains why many consumers prefer to grind coffee close to the point of consumption.

18.2.2 Bulk and flow properties of particulate materials

A bulk mass of particles has some characteristics that are aggregate properties of the individual particles, as well as properties that depend on interparticle interactions. The properties of bulk solids are strongly dependent on the distribution of particle size and can also vary significantly from sample to sample within the same material.

Particle size distribution

All solids streams encountered in the process industries have a distribution of particle sizes. Even narrow-cut screened fractions will have a range of sizes between the upper and lower screen aperture sizes.

In some processes and unit operations the use of a mean “diameter” is sufficient. The choice of dimensions to use as diameter is discussed in [Section 18.2.1](#) and in the sections on each unit operation. Several different mean diameters can also be defined, each of which is most appropriate in given situations. The simplest definitions are the arithmetic and geometric mean diameters:

$$\overline{d}_a = \sum_{i=1}^N \frac{d_i}{N} \quad (18.8)$$

$$\overline{d}_g = \left(\prod_{i=1}^N d_i \right)^{1/N} \quad (18.9)$$

where:

$\overline{d}_a$ = arithmetic mean diameter

$\overline{d}_g$ = geometric mean diameter

d_i = diameter of particle i

N = number of particles in sample

Many chemical engineering operations involve a balance between phenomena related to area and phenomena related to volume (or mass), so one of the most useful definitions is the volume–surface mean diameter:

$$\overline{d_{vs}} = \frac{\sum_{i=1}^N d_i^3}{\sum_{i=1}^N d_i^2} \quad (18.10)$$

where:

$\overline{d_{vs}}$ = volume surface mean diameter

Although a mean diameter may be appropriate in many design problems, there are several unit operations that have different outcomes depending on particle size. For these operations, the design engineer must work with a distribution of particle sizes. For example, the efficiency of a cyclone for capturing particles depends strongly on the particle size and will be substantially lower for very fine particles less than 5 μm in diameter.

Particle size distributions are usually discretized and reported as number or mass fractions of particles within a given size range. Care must be taken when reading a particle size distribution to ensure that the fraction stated is over or under the given size.

Commercial process simulation programs usually have the ability to track particle size distribution of solids streams and allow the user to manipulate the particle size distribution in operations such as crystallizers, cyclones, filters, and agglomerators.

Voidage and bulk density

The void fraction (voidage) and average bulk density of a collection of particles were introduced in [Section 18.2.1](#). The bulk density and voidage are related by Equation 18.5.

For fine particles, the void fraction can be a strong function of bed packing. Vibrating or tamping a bed of fine particles causes the particles to settle and reduces the void fraction. For regular-shaped, cohesionless materials with narrow size distribution, the change in bed density is usually in the range of 1% to 10%, but for materials with a wide range of particle size, the density can be increased by up to 40% (Seville et al., 1997). When there is a wide range of particle size, the smaller particles can fit into the spaces between the larger particles, as illustrated schematically in [Fig. 18.2](#), and hence lower voidage is possible.

Compaction or compression of a bed of particles is sometimes desired, for example, when the design calls for maximizing the amount of a solid such as an adsorbent in a given volume. In such cases, the bed is dense loaded and compacted during loading. More often, compaction is not desired but occurs naturally when solids are stored or transported. It can be difficult to initiate free flow from a compacted bed of solids, and the pressure drop is higher when flowing a fluid through a compacted bed because of the lower voidage.

Cohesion

Cohesion is a measure of the tendency of particles to stick together. Cohesion arises from several different particle-to-particle interactions:

1. Van der Waals forces between particles
2. Electrostatic forces between particles
3. Interlocking of irregularly shaped particles
4. Capillary action and surface tension forces due to liquid bridges between particles
5. Forces between adsorbed layers or coatings on particles

Cohesion should not be confused with friction forces between particles that are moving and can be thought of as the force that must be overcome to initiate movement of particles rather than the force to maintain movement.

Cohesive forces tend to be stronger for small particles and particles that are moist or partially wetted, particularly if the liquid has a high surface tension. For example, sand, grain, and sugar have low cohesion when dry, but can be very cohesive when moist. It is the cohesion of wet sand that allows the construction of sand castles with near-vertical walls.

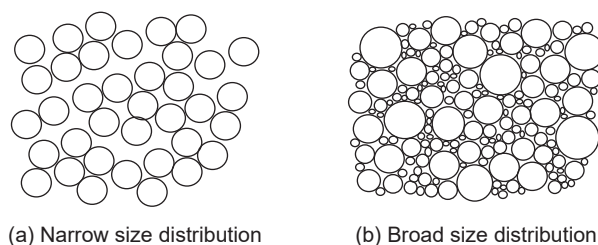


FIG. 18.2 Packing of beds of particles with different size distributions.

Flow properties

Powders and particulate materials can be treated as rigid, semi-plastic, continuous materials for the purposes of modeling flow in bins and hoppers. A rigid, semi-plastic Coulomb solid has a linear yield locus, as shown in Fig. 18.3. If the applied shear stress is greater than the yield locus, slip will occur and the material will flow.

For a Coulomb solid:

$$\tau = c + \sigma \tan \alpha \quad (18.11)$$

where:

τ = shear stress

c = cohesion

σ = normal stress (applied load)

α = angle of internal friction

The angle of internal friction should not be confused with the angle of repose of the solids. The angle of repose, β , is the angle formed by a pile of solids loosely tipped onto a surface (Fig. 18.4). The angle of internal friction can be measured as the angle formed between stationary and flowing material when solids are flowed out from a flat-bottomed slotted vessel (Fig. 18.5). The cohesion and angle of internal friction can also be measured in appropriately designed shear cells; see, for example, Thomson (1997). Some examples of angle of repose are given in Table 18.2.

The angle of internal friction can be affected by packing density and consolidation. The angle of internal friction tends to be lower for materials that are dry and have small particles and higher for materials that are cohesive, are moist, or contain large particles.

When a Coulomb material is placed on an inclined surface, the friction between the solid and the wall can be less than the internal friction, causing the material to slide down the wall. If there are no cohesive forces between the material and the wall, then a wall yield locus can be defined:

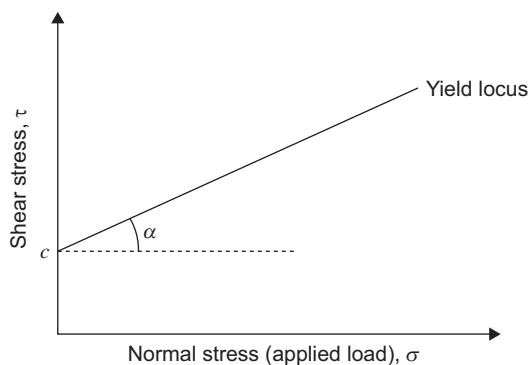


FIG. 18.3 Yield locus of a bed of solid particles.

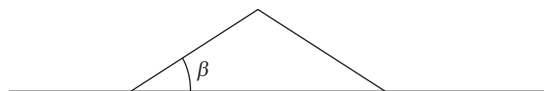


FIG. 18.4 Angle of repose of a pile of solid particles.

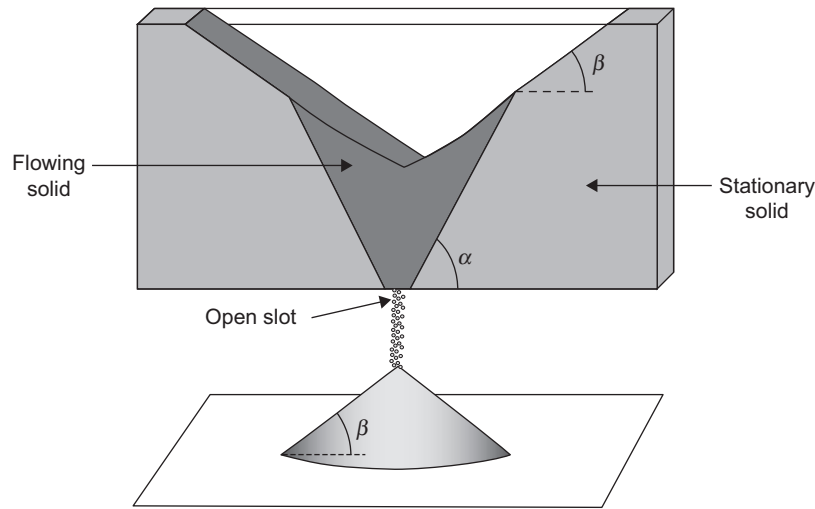


FIG. 18.5 Measuring the angle of internal friction and angle of repose.

$$\tau_w = \sigma_w \tan \alpha_w \quad (18.12)$$

where:

τ_w = shear stress at the wall

σ_w = normal stress at the wall

α_w = angle of wall friction

The angle of wall friction can be measured as the slope of a surface along which particles will slide (Fig. 18.6).

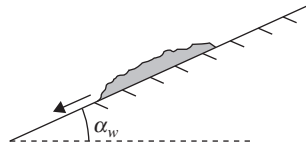


FIG. 18.6 Angle of wall friction.

Fluidization

The pressure drop of a fluid flowing through a bed of granular solids can be estimated using the Ergun equation (Ergun, 1952):

$$\frac{\Delta P}{L_b} = 150 \frac{(1 - \epsilon)^2}{\epsilon^3} \frac{\mu U}{d_p^2} + 1.75 \frac{(1 - \epsilon)}{\epsilon^3} \frac{\rho_f U^2}{d_p} \quad (18.13)$$

where:

ΔP = pressure drop, N/m²

L_b = length of the bed of solids, m

ϵ = void fraction

μ = fluid viscosity, Ns/m²

U = fluid superficial velocity, m/s

d_p = particle effective diameter, m

ρ_f = fluid density, kg/m³

If the fluid flows upwards, then at some point the force due to pressure drop balances the weight of the particles and the bed begins to expand and become fluidized (Fig. 18.7b). As the fluid flow rate increases, the bed expands further and can enter bubbling, slugging, or spouting regimes, illustrated in Figs. 18.7(c) and (d). If the fluid flow is increased to the point where the superficial velocity is equal to the terminal velocity of the smallest particles, these particles are elutriated from the bed and carried out with the fluid. Ultimately, at high enough velocities, the entire bed can be entrained into the fluid flow, and pneumatic or hydraulic conveying of the particles occurs.

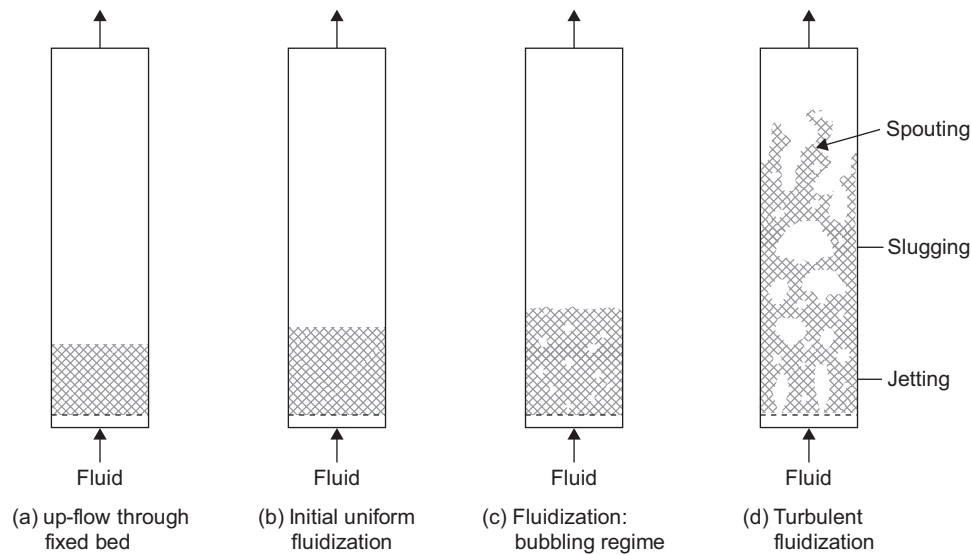


FIG. 18.7 Fluidization of a bed of particles.

The onset of fluidization can be predicted by balancing the pressure drop against the buoyant weight of the bed:

$$\frac{\Delta P}{L_b} = (1 - \epsilon_{mf})(\rho_p - \rho_f)g \quad (18.14)$$

where:

ϵ_{mf} = void fraction at minimum fluidization

g = gravitational acceleration, m/s²

ρ_p = particle density, kg/m³

Substituting the Ergun equation for pressure drop:

$$(1 - \epsilon_{mf})(\rho_p - \rho_f)g = 150 \frac{(1 - \epsilon_{mf})^2}{\epsilon_{mf}^3} \frac{\mu U_{mf}}{d_p^2} + 1.75 \frac{(1 - \epsilon_{mf})}{\epsilon_{mf}^3} \frac{\rho_f U_{mf}^2}{d_p}$$

where U_{mf} is the superficial fluid velocity at minimum fluidization.

Rearranging:

$$\frac{\rho_f(\rho_p - \rho_f)g}{\mu^2} d_p^3 = 150 \frac{(1 - \epsilon_{mf})}{\epsilon_{mf}^3} \frac{\rho_f d_p U_{mf}}{\mu} + \frac{1.75}{\epsilon_{mf}^3} \frac{\rho_f^2 d_p^2 U_{mf}^2}{\mu^2} \quad (18.15)$$

The Reynolds number at minimum fluidization, Re_{mf} , is:

$$\text{Re}_{mf} = \frac{\rho_f d_p U_{mf}}{\mu} \quad (18.16)$$

We can also define the dimensionless Archimedes number, Ar :

$$\text{Ar} = \frac{\rho_f(\rho_p - \rho_f)g}{\mu^2} d_p^3 \quad (18.17)$$

The Archimedes number is a dimensionless volume, and $\text{Ar}^{1/3}$ is a dimensionless diameter. We can then rewrite Equation 18.15 as:

$$\text{Ar} = 150 \frac{(1 - \epsilon_{mf})}{\epsilon_{mf}^3} \text{Re}_{mf} + \frac{1.75}{\epsilon_{mf}^3} \text{Re}_{mf}^2 \quad (18.18)$$

This is not particularly useful if we have no knowledge or measurement of ε_{mf} , but Wen and Yu (1966) correlated a large set of data to obtain:

$$Ar = 1650 Re_{mf} + 24.5 Re_{mf}^2 \quad (18.19)$$

Note that although the Wen and Yu correlation has the same form as Equation 18.18, it does not yield a unique value for ε_{mf} .

The different types of fluidization behaviors can be plotted on a graph of dimensionless superficial velocity versus dimensionless diameter ($Ar^{1/3}$), as shown in Fig. 18.8, which is taken with permission from Grace (1986). It can be seen that conventional fluidized beds are usually stable over about an order of magnitude of superficial velocity before turbulence and elutriation begin, although this is obviously somewhat sensitive to the particle size distribution. The classification of particles as Groups A, B, C, and D refers to the system developed by Geldart (1973), summarized in Table 18.3. For more detail on prediction of different fluidization regimes, see Seville et al. (1997), Zenz and Othmer (1960), or Zenz (1997).

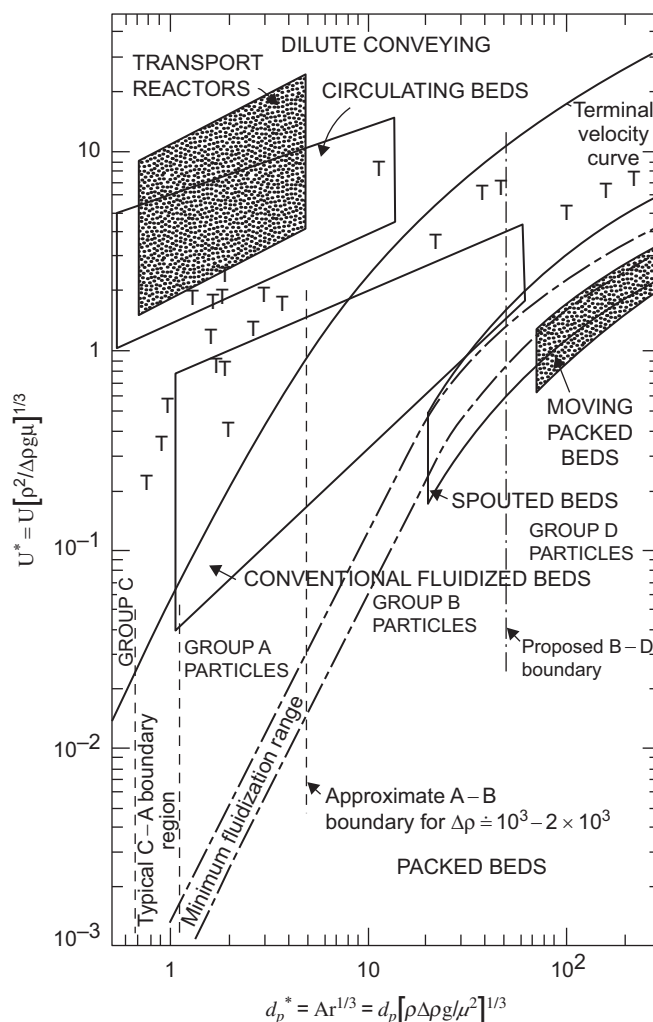


FIG. 18.8 Fluidization regimes. Taken, with permission, from Grace, J. R. (1986). *Contacting modes and behavior classification of gas-solid and other two-phase suspensions*, Canadian Journal of Chemical Engineering, 64(3), 353. "T" indicates experimentally observed onset of turbulent fluidization.

Most fluidized beds are operated under conditions where they are well-mixed in the solids phase and have very uniform temperature. They have high rates of heat transfer to internal and external surfaces, as heat is effectively carried and transferred by the solid particles. The uniform temperature and high heat transfer experienced in fluidized beds can be advantageous for carrying out reactions. Fluidized beds can also be used to transport heat from a hot zone such as a regeneration reactor to an endothermic reactor, as occurs in catalytic cracking; see Section 15.8.4.

TABLE 18.3 Geldart's classification of fluidization behavior

Group	C	A	B	D
Major characteristic	Cohesive Hard to fluidize	Bubble-free over some range	Bubbles at U_{mf}	Coarse solids
Bed expansion	Low, prone to Channeling	High	Moderate	Low
Bubble behavior	Channels	Bubbles coalesce to limiting size	Large stable bubbles	Large slugs
Solids mixing	Low	High	Moderate	Low
Gas back-mixing	Low	High	Moderate	Low
Spouting	No	Shallow beds only	Shallow beds only	Yes
Examples	Flour, cement	Cracking catalyst	Sand, salt	Crushed rock, beans

From Geldart (1973).

Fluidized beds also allow easy continuous addition and removal of solids, which can be important in catalytic reactions if the catalyst deactivates.

The main disadvantage of fluidized beds is that attrition can cause erosion of the vessel internals and formation of fine solid particles. Fluidized-bed equipment is often lined with refractory material to prevent erosion of the structural metal. Fluidized beds are usually fitted with cyclones to capture and return larger particles, but secondary or tertiary capture systems may be needed for environmental reasons or to prevent loss of expensive catalyst. Caution is also needed when designing a fluidized bed at higher velocities, particularly for gases, as bubbling or slugging can cause part of the fluid flow to bypass the bed, reducing reaction or heat-transfer efficiency. See Cocco et al. (2014a) for design approaches to address these problems. Solids that have high cohesion or high tendency to agglomerate are usually not suitable for fluidization.

Fluidization is important in several solids-handling and processing operations. Some catalytic reactors are designed to operate as fluidized beds; see Sections 15.7 and 15.8, whereas packed beds that are used in up-flow in reactors or adsorption columns must be designed to avoid fluidization. Fluidization is also used in some designs of solids heaters and dryers, in some agglomerators, and in pneumatic and hydraulic conveying. Fluidized beds are widely used in minerals processing and in the combustion of coal. For more information on fluidization and its applications, see the introductory article by Cocco et al. (2014b) and the books by Zenz and Othmer (1960), Seville et al. (1997), Kunii and Levenspiel (1991), and Yang (1999, 2003).

18.3 Storage and transport of solids

18.3.1 Storage of bulk solids

The simplest way to store raw bulk solids is to pile them on the ground in the open air. This is satisfactory for the long-term storage of materials that do not deteriorate on exposure to the elements, for example, the seasonal stockpiling of coal at mines and power stations. For large stockpiles, permanent facilities are usually installed for distributing and reclaiming the material; traveling gantry cranes, grabs, and drag scrapers feeding belt conveyors are used. For small, temporary storage, mechanical shovels, front-end loaders, and trucks can be used. When the cost of recovery from the stockpile is large compared with the value of the stock held, storage in silos or bunkers should be considered. There is no upper limit to the amount of material that can be stored in open piles, but the design engineer should be concerned about visual impact and the possibility of water run-off contaminating local surface and ground water.

Solids can also be piled inside storage sheds or warehouses. This affords protection from the weather and can be cheaper than using silos for large quantities. Indoor storage is used for stockpiling seasonal products such as fertilizers that are water-soluble and so cannot be stored outside.

Storage bins are used for solids that must be protected from the elements and for storing smaller quantities of solids for process use. A *bin* is any vertical walled vessel used for storing solids. Most bins have a conical or wedge-shaped *hopper* at the base, though some bins are flat-bottomed. Large bins with a height-to-diameter ratio greater than 1.5 are also called *silos*, whereas bins with a height-to-diameter ratio less than 1.5 are termed *bunkers*.

Large bunkers and silos used for agricultural products, cement, and bulk minerals typically hold 50 to 1000 metric tons of solids and are usually constructed from reinforced concrete. The design of large bins requires analysis of the wall stresses caused during filling and discharge of the solids and is beyond the scope of this book; see [Fayed and Otten \(1997\)](#) and [Seville et al. \(1997\)](#). Large-scale concrete silos are often used to store material for days, weeks, or even months.

Smaller bins and silos for handling the feeds to chemical and pharmaceutical processes typically contain less than 10 tons of material and are constructed from steel or glass-reinforced polymer (GRP). Bins and silos for process use are usually sized to contain enough material to ensure correct operation of the feeder device at the base of the hopper (see [Section 18.3.2](#)) and to provide adequate residence time to compensate for short interruptions in any upstream system that is feeding the bin. A residence time of 1 to 3 hours is typical, but longer residence times are used when the flow rate is small or when the upstream operation is highly discontinuous. The sizing of a process feed bin is illustrated in Example 18.1.

18.3.2 Discharge from bins and hoppers

Process bins and silos are usually designed to give a uniform flow rate of solids to downstream operations. The flow of solids from a bin or hopper is influenced by the shape of the hopper, the size of the discharge opening, and the use of devices known as *feeders* to regulate the flow.

Flow patterns in bins and hoppers

There are two main types of flow from a bin or hopper, as illustrated schematically in [Fig. 18.9](#). When the wall angle of the hopper is steep enough, mass flow occurs, as illustrated in [Fig. 18.9\(a\)](#). In mass flow, slip occurs between the solids and the wall and all of the particulate material is moving downwards. There are no stagnant regions in the bin or hopper, and there is a reduced tendency for the particles to segregate by particle size during flow. This flow pattern is also known as “first-in-first-out,” as material from the top of the bin does not channel to the exit.

If the wall angle of the hopper is not sufficiently steep, or if discharge occurs from an opening at the base of a flat-bottomed bin, then funnel flow occurs, as illustrated in [Fig. 18.9\(b\)](#). In funnel flow, slip occurs within the solid at an angle roughly equal to the angle of internal friction if the solids have low cohesion. In extreme cases, particularly when the solids are cohesive, rat-holes, bridges, or arches can occur ([Fig. 18.10](#)). The formation of rat-holes and arches causes numerous problems. Flow is interrupted while the hopper is still largely full, and level detection devices may no longer function correctly. Also, if a bridge or rat-hole in a fine powder collapses suddenly, the void fraction increases rapidly, and uncontrollable flow through the discharge known as *flushing* occurs. Volumetric feeders are neither accurate nor effective if the void fraction of solids varies widely, so hoppers in process service are usually designed to avoid funnel flow.

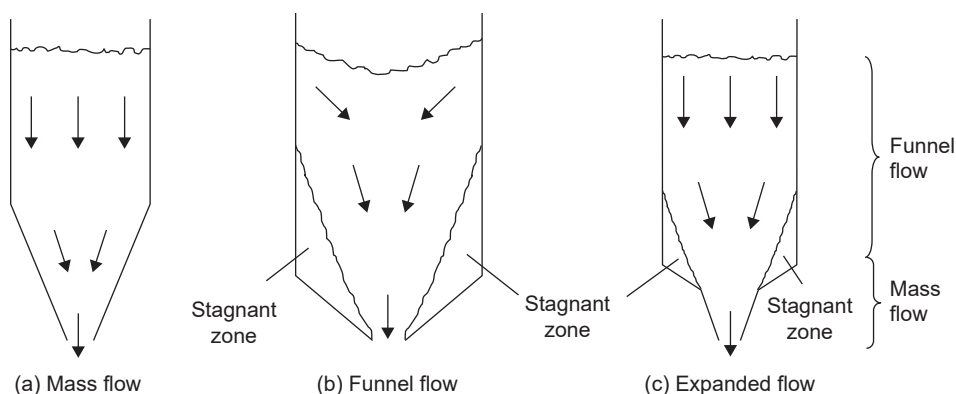


FIG. 18.9 Flow of solid particles from a storage bin.

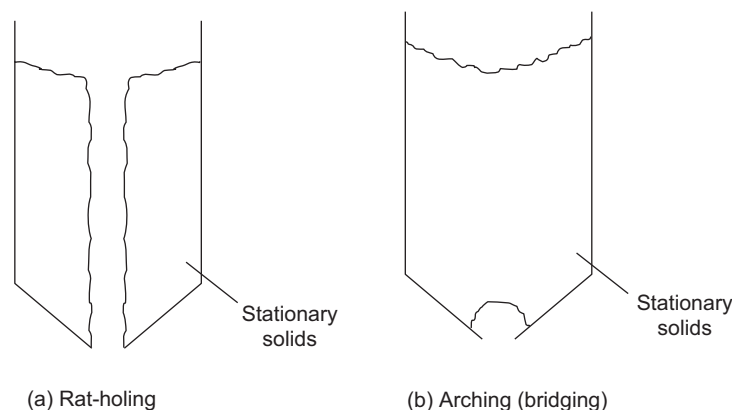


FIG. 18.10 Problems of funnel flow.

One method of avoiding the problems of funnel flow is to use a steep-walled hopper at the base of a funnel flow bin so as to create a mass flow zone at the base with a funnel flow zone above. The entry of the mass flow section is sized to be large enough that a stable arch cannot form. The resulting combination of funnel flow and mass flow is known as expanded flow and is illustrated in Fig. 18.9(c).

Jenike (1967) and Jenike and Johnson (1970) have studied the flow of solids in containers and developed design methods. All aspects of the design of bins and hoppers, including feeding and discharge systems, are covered in books by Reisner (1971) and Brown and Nielsen (1998); see also Rhodes (2008) and the journal articles by Mehos and Maynard (2009), Mehos and Morgan (2016), and Mehos et al. (2018). The use of inserts to correct solids flow problems with existing hoppers is addressed in an article by Bates et al. (2010). The design of large concrete bins and silos is covered by the American Concrete Institute standard ACI-313. See also the British Material Handling Board's code of practice on the design of silos and bunkers: BMHB (1992), Seville et al. (1997), and Thomson (1997). There is no international standard on hopper or silo design at the time of writing.

Flow of solids from an unregulated orifice

The mass flow rate of solids from an open conical hopper or flat-bottomed bin has been shown to be constant and independent of the vessel diameter, D , or depth of solids, H , as long as $H > D$ (Brown & Richards, 1959) and $D > 2.5D_o$, where D_o is the diameter of the opening at the base of the vessel (Fig. 18.11).

Most correlations for solids flow rate are derived from that of Beverloo et al. (1961):

$$m = C \rho_b g^{1/2} (D_o - k d_p)^{5/2} \quad (18.20)$$

where:

m = mass flow rate of solids, kg/s

C = a constant

ρ_b = bulk density, kg/m³

g = gravitational acceleration, m/s²

k = a shape parameter

d_p = particle "diameter"

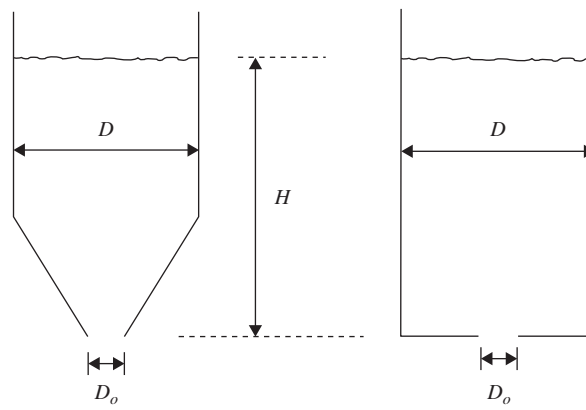


FIG. 18.11 Bin dimensions.

The choice of constant k depends on the measure used for d_p . For uniform spheres, $k = 1.5$, whereas more angular particles have higher values. Beverloo et al. and subsequent workers found quite a large variation in the constant C . It was surmised that this could be because of differences between the bulk density and flowing density of the solids. Nedderman (1992) proposed:

$$m = C' \rho_{bf} g^{1/2} (D_o - k d_p)^{5/2} \quad (18.21)$$

where:

C' = a constant, typically about 0.585

ρ_{bf} = the flowing bulk density of the solids, kg/m³

The flowing density can be measured at the discharge using γ -ray tomography and is typically 10% to 20% less than the static bulk density (Hosseini-Ashrafi & Tüzün, 1993).

The effects of orifice shape and discharge through side-wall ports are discussed by Dhodapkar et al. (2016).

Note that applying a force to the top of a bed of solids does not improve the flow rate. In fact, conversely, a force on the top of the bed will cause densification of the bed, which increases cohesion and increases the probability of arching or bridging.

Most hoppers are not designed for unregulated flow, as the flow is usually metered into downstream operations. It is, however, important to understand the unregulated flow rate so that the opening is not made too small for the desired design flow rate. To ensure good controllability of the solids flow, the opening is usually sized to allow a much greater flow rate of solids than is needed by the process. The outlet opening must also be large enough to prevent the solids from forming a bridge or arch; see Rhodes (2008).

Volumetric and gravimetric feeders

In most process applications, the flow rate of solids leaving a bin or hopper is governed by a device that ensures a constant flow rate on a volume flow or mass flow basis and that permits the flow to be turned off when necessary. The design of the feeder is an integral part of the design of the bin and hopper. For mass flow to occur in the bin, the feeder shape and opening must allow material to discharge uniformly across the open area at the base of the hopper. The feeder must also allow flow of solids to be interrupted and resumed and must prevent the stationary solids from wicking moisture from downstream fluid operations.

In a *belt feeder* (Fig. 18.12), the base of the hopper section discharges slightly above a conveyor belt. If the belt is stopped, the solids cannot spill out and flow ceases. The volume flow of solids can be varied by adjusting belt speed. Belt feeders are also known as *apron feeders*. A similar principle is used in a *screw feeder* (Fig. 18.13), in which a

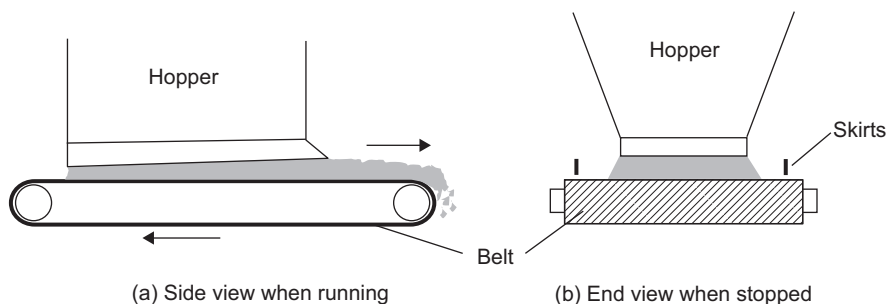


FIG. 18.12 Belt feeder.

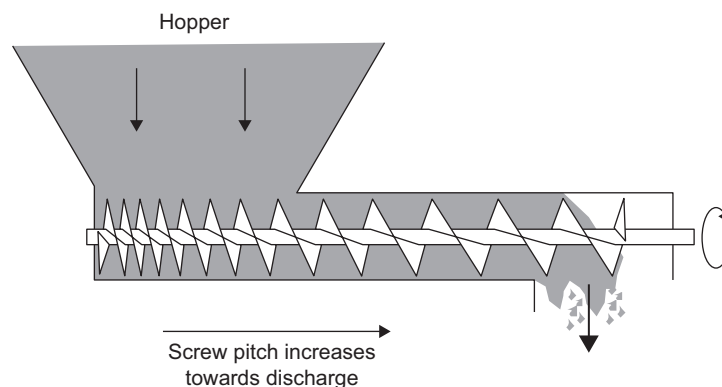


FIG. 18.13 Screw feeder.

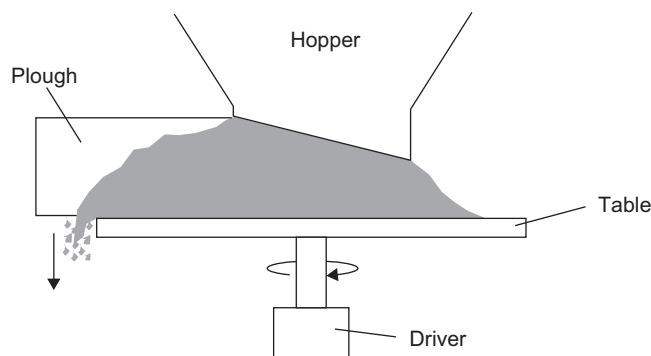


FIG. 18.14 Table feeder.

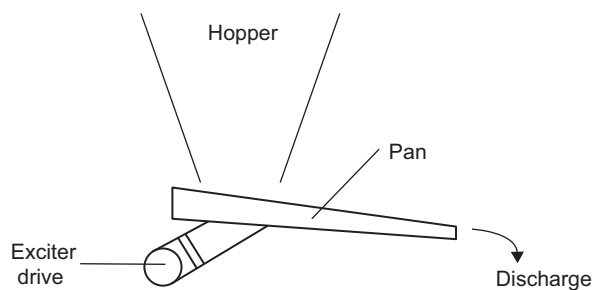


FIG. 18.15 Vibratory feeder.

variable-pitch screw is used to move the solids, and in a *table feeder* (Fig. 18.14), in which the solids discharge onto a rotating table and are scraped off the table by a stationary plow.

For lower solids feed rates that require precise control, a *vibratory feeder*, shown in Fig. 18.15, can be used. With a vibratory feeder, the hopper discharges onto a pan that is vibrated at a controlled speed to give a constant volume flow of solids. Vibratory feeders have a high range of turndown, from 5% to 100% of capacity, but they can be difficult to operate with fine powders that settle or deaerate on the pan.

Rotary feeders (Fig. 18.16) are used for precise control of volume feed rate from larger openings. Rotary feeders are also known as rotary vane feeders or star feeders. The feed rate can be controlled by adjusting the speed of the rotary valve (star valve). The rotary valve can be designed to provide a positive seal between a gas atmosphere in the process and air in the hopper, and the feeder and hopper can be purged with nitrogen if the designer needs to prevent entry of air into the process. Rotary vane feeders are particularly useful when feeding into contained systems such as pneumatic conveyors or fluidized beds. Rotary vane feeders can be prone to jamming when the tolerances between the vanes and casing are small.

Volumetric feeders can only provide a constant mass flow rate to a process if the flowing density of the solids at the hopper discharge remains constant. When more precise control of mass flow rate is needed, a gravimetric feeder must be used. Gravimetric feeders usually consist of a weighing device in line with a volumetric feeder so that the volumetric feeder rate can be adjusted using feedback from the weighing device. Batch and continuous weighing devices are used; see Green and Southard (2018) and Liptak (1993).

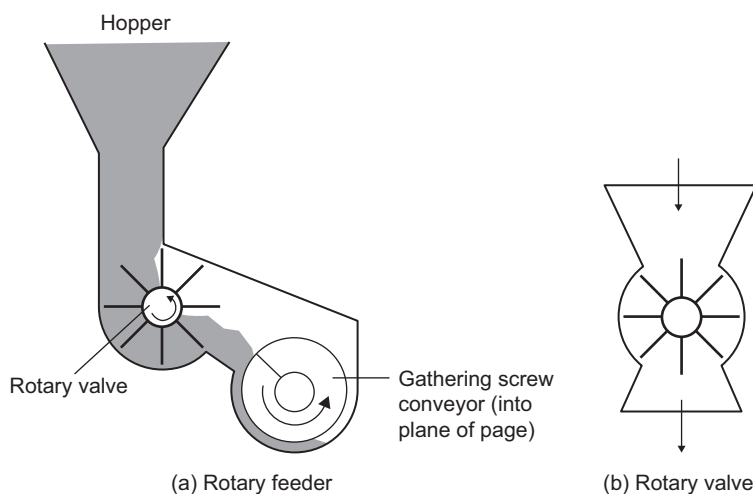


FIG. 18.16 Rotary feeders.

When small bins and hoppers are used, it is sometimes sufficient to place the bin on a load cell and record the loss in mass of the contents against time.

There are many suppliers of bins, hoppers, and feeder systems. Most of them have catalogs or selection tools on their websites that allow preliminary identification of a suitable device for a given application. However, it is always necessary to consult a vendor for a firm warranty that their device is suitable before a final selection is made. It may also be worthwhile to make test runs using the vendor's test facilities, particularly when dealing with cohesive materials or fine powders. Guidelines for feeder selection are given in an article by [Marinelli and Miller \(2017\)](#).

Example 18.1

A tablet-forming process is to be continuously fed with 10 kg/h or USP-grade sorbitol powder when in operation. The sorbitol is supplied in 35-kg polypropylene bags with polyethylene liners, stacked 24 bags to a pallet, minimum order 20 pallets, with 45-day lead time for orders.

Design a storage and feed system for the sorbitol powder. Sorbitol has average bulk density 1.48 kg/liter and can be assumed to have an average particle size of 200 μm .

Solution

Feed rate of 10 kg/h = 240 kg/day, hence $240/35 = 6.9$ bags per day, or exactly 2.0 pallets per week.

A minimum order of 20 pallets should therefore last for 10 weeks, which is longer than the 45-day lead time for orders. The plant can therefore order the minimum quantity, placing an order when there are ~ 14 pallets remaining. This would be sufficient for 49 days operation, thereby allowing 4 days margin above the 45-day delivery lead time quoted.

Because the feed arrives packed in dry bags, there is no need to unpack it immediately. Sufficient warehouse space should be allowed for storage of an incoming order of 20 pallets, plus a few pallets extra to allow for potential interruption of supply. If we allow sufficient storage for an additional 2 weeks of operation (4 pallets), then we would need warehouse space for 24 pallets. A standard pallet is 1.065×1.22 m, that is, 1.3 m^2 area, so we need to set aside at least $24 \times 1.3 = 31 \text{ m}^2$ of warehouse space, plus allowances for forklift truck movement and access.

The sorbitol must then be transferred from the warehouse to a bin that feeds the process. This will require an operator to move a pallet to the bin twice per week. A suitable elevator, lift, or hoist system must be designed to enable the operators to move the sorbitol bags to the top of the feed bin. In a region with low labor costs, this could be as simple as operators carrying the bags one at a time, although the safety risks of operators carrying 35-kg bags over any significant distance would be deemed unacceptable by most companies. A better design would be to use a hoist or lift to raise the pallet to a platform above the feed bin. Alternatively, a pallet could be loaded into a freight elevator and then the bags could be unloaded manually onto a platform above the feed bin. The hoist can be designed and costed knowing that the mass of the pallet is about 30 kg and the mass of the contents is $24 \times 35 = 840$ kg, so the hoist should be rated for at least 1000 kg to allow some design margin.

It is not necessary to design the feed bin to hold the entire contents of a pallet of bags, as provision can be made to store a few bags on the platform above the feed bin. In fact, there may be advantages to designing a smaller bin, as it will have a lower residence time and reduce the chance of compaction or moisture adsorption by the sorbitol, which is hygroscopic.

The smallest practical size of feed bin would be sized to hold a single 35-kg bag of feed and would require refilling every $35/10 = 3.5$ hours. This would not be convenient with the usual 8-hour shift cycle, so if we design a feed bin that can hold four bags of sorbitol, then the operators can load two bags at some point during the shift when a low level is indicated by the control system.

$$\text{Bin volume} = \frac{4 \times 35}{1.48} = 95 \text{ liters}$$

Assuming a 2:1 cylindrical bin, volume = $\pi D^3/2$:

$$\text{Bin diameter} = \left(\frac{0.095 \times 2}{\pi} \right)^{1/3} = 0.4 \text{ m}$$

$$\text{Bin height} = 0.8 \text{ m}$$

Using Nedderman's version of the Beverloo equation (Equation 18.21):

$$\frac{10}{3600} = 0.585 \times \left(\frac{1480}{1.2} \right) \times (9.81)^{1/2} (D_o - 1.5 \times 200 \times 10^{-6})^{5/2} \quad (18.21)$$

$$D_o = 300 \times 10^{-6} + (1.229 \times 10^{-6})^{2/5} = 4.62 \times 10^{-3} \text{ m}$$

A 5-mm opening would obviously be too small for a fine powder such as sorbitol, so a much larger opening would be used in combination with a precise gravimetric feeder. Details of the hopper design and feeder selection can now be worked out in consultation with a vendor. Other factors may also need to be considered, for example, ensuring that the feed system is in accordance with Current Good Manufacturing Practices (CGMP) if the tablet-making process is for food or drug manufacture.

Another alternative that could be considered would be to use a small feeder bin placed at ground level with a screw or pneumatic conveyor to lift the solids into the feed storage bin. The operators would then empty bags into the feeder bin without requiring a hoist or elevator. Several manufacturers supply proprietary designs of such feeder systems. The cost of these proprietary systems could be compared with the cost of the design proposed earlier.

18.3.3 Packaging and storage of solid products

Some solid products such as polymer chips, petroleum coke, and foods such as flour and sugar are shipped in bulk in railroad hopper cars and bulk ship containers. Standard railroad hopper cars in the United States are designed to hold 55 tons or 100 tons. Dimensions of hopper cars are given in [Green and Southard \(2018\)](#).

Most solid products are not suitable for bulk shipment and are packaged in bags or polyethylene-lined 55-gallon drums. A wide variety of bag and drum filling devices exists, and details of specific designs can be obtained from manufacturer's websites. Small sacks are usually designed to be easily lifted by one or two workers and do not hold more than 80 lb (36 kg). Larger sewn bags and flexible containers can hold as much as a metric ton. The filled bags or drums are stacked on standard 1.065 m by 1.22 m pallets and are usually wrapped with stretch-wrap or shrink-wrap or taped to prevent the contents from slipping off the pallet during transportation.

Consumer products, foods, and pharmaceutical products are usually packed directly into the final packaging for quality control purposes and are then boxed for delivery to wholesalers. Because the final packaging contains a relatively small quantity of material (typically less than 1 kg), the packaging operations are highly automated. The food, beverage, and consumer products industries have developed highly integrated systems for weighing (or metering), filling, sealing, inspecting, stacking, and boxing small containers at rates up to the order of ten units per second. Details of packaging and filling equipment must be obtained in consultation with the equipment manufacturer.

18.3.4 Conveying of solids

The movement and storage of solids are usually more expensive than the movement of liquids and gases, which can be easily pumped down a pipeline. The best equipment to use to move solids will depend on a number of factors:

1. The throughput
2. Length of travel

3. Change in elevation

4. Nature of the solids: size, bulk density, angle of repose, angle of wall friction, abrasiveness, corrosiveness, wet or dry

Belt conveyors

Belt conveyors are the most commonly used type of equipment for the continuous transport of solids. They can carry a wide range of materials economically over long and short distances, either horizontally or at an appreciable angle, depending on the angle of repose and angle of wall (belt) friction of the solids. A belt conveyor consists of an endless belt of a flexible material, supported on rollers (idlers), and passing over larger rollers at each end, one of which is driven. The belt material is usually fabric-reinforced rubber or plastic; segmental metal belts are also used. Belts can be specified to withstand abrasive and corrosive materials. The design, installation, operation, and maintenance of conveyor belts are covered in the Conveyor Equipment Manufacturers Association standard (CEMA, 2014) and are also addressed by several international (ISO) standards; see BS EN ISO 21183, BS EN ISO 14890, and BS EN ISO 15236.

Polymer or rubber belt conveyors are restricted to running at ambient conditions and cannot handle hot solids, though metal belts can run hot and are used in belt dryers. The belt width and speed depend on the particle size and bulk density of the material. The lowest cost is obtained by using a narrow belt running at high speed, but dust formation can limit the obtainable speed, particularly if the belt is not covered. Speeds up to 5 ms^{-1} are possible, although 2 to 3 ms^{-1} is most typical. The carrying capacity can be determined from the belt speed, load cross-sectional area, and material bulk density. The load cross-sectional area depends on the angle of repose of the solids, the depth of filling, and whether the conveyor has side idlers to prevent spillage at the sides. Table 18.4, which is adapted from data of Raymus (1984), gives the relationships between belt width, maximum lump size, and load cross-sectional area for a troughed belt with idlers. Load cross-sectional areas for other designs and belt sizes are given in CEMA (2014). Belt conveyors used in mining can be several kilometers long, but most belt conveyors in the process industries are shorter than 100 m.

TABLE 18.4 Belt conveyor capacity

Belt width		Max lump size		Load cross-sectional area	
(in)	(m)	(in)	(mm)	(ft ²)	(m ²)
14	0.35	2.0	51	0.11	0.01
16	0.4	2.5	64	0.14	0.013
20	0.5	3.5	89	0.22	0.02
24	0.6	4.5	114	0.33	0.03
30	0.75	7.0	178	0.53	0.05
36	0.9	8.0	203	0.78	0.072
48	1.2	12.0	305	1.46	0.136
60	1.5	16.0	406	2.40	0.223

Adapted from Raymus (1984).

The power required to operate a belt conveyor is made up of the power required to run the empty belt, the work required to raise or lower the load, and the work required to overcome friction caused by the load moving over the rotating parts and causing flexure or indentation of the belt. The magnitude of each contribution can vary depending on the application (Alspaugh, 2004). For large conveyors carrying high loads of bulk minerals, the power is closely proportional to the mass, as both the potential energy required and the friction at the belt are proportional to the mass conveyed; however, for small conveyors carrying low loads over short distances (as often encountered in the process industries), the power consumption can be a weak function of the mass transported if there is no change in elevation, as the belt itself may make up a significant fraction of the mass moved. Power consumption is therefore best determined in consultation with an equipment vendor. Preliminary estimates of power consumption can be made using process simulation programs that include conveying as a unit operation; see Chapter 4.

The mechanical conveying of solids is the subject of books by Colijn (1985), Fayed and Skocir (1996), Levy and Kalman (2001), and McGlinchey (2008). The conveying of wet solids is addressed by Heywood (1991).

Screw conveyors

Screw conveyors, also called worm conveyors, are used for materials that are free flowing. The basic principle of the screw conveyor has been known since the time of Archimedes. The modern conveyor consists of a helical screw rotating in a U-shaped trough or enclosed pipe. They can be used horizontally or, with some loss of capacity, at an incline to lift materials. Screw conveyors are less efficient than belt conveyors due to the friction between the solids and the flights of the screw and the trough, but are cheaper and easier to maintain. They are used to convey solids over short distances and when some elevation (lift) is required. They can also be used for delivering a metered flow of solids.

Green and Southard (2018) give data on screw conveyor power consumption that can be correlated with the following equation:

$$p = 0.038 + 0.0072mL_c^{0.8} \quad (18.22)$$

where:

p = power consumption (hp)

m = mass flow (tons/h)

L_c = conveying distance (ft)

Equation 18.22 can be used to obtain a preliminary estimate of the work required for a screw conveyor; however, an accurate estimate will require consultation with a vendor. An excellent overview of screw compressor design is given by Roberts (2015). Screw conveyors are described in detail in the book by Forcade (1999).

Pneumatic and hydraulic conveying

Pneumatic conveying is used for movement of solids over relatively short distances. It is generally suitable only for free-flowing particles in the range of 20 μm to 50 mm, as finer dusts tend to stick to the pipes, while larger particles are hard to entrain. In pneumatic conveying, the solids are transported in suspension in a gas. The solids may be either dilute phase, with void fraction typically greater than 95%, or dense phase, with void fraction as low as 50%. The velocity of carrier fluid must be large enough to keep the particles suspended; see Chhabra and Gurappa (2019). Pneumatic conveying can be used for both horizontal and vertical transport of solids, including making pipe turns. Sharp turns are generally avoided, as these cause solids attrition and pipe abrasion. In a pneumatic conveying system, the solids are usually fed from a hopper via a rotary vane feeder that prevents backflow and are entrained into the gas flow, as shown in Fig. 18.17(b). Pneumatic conveying can also be used under slight vacuum as a means of

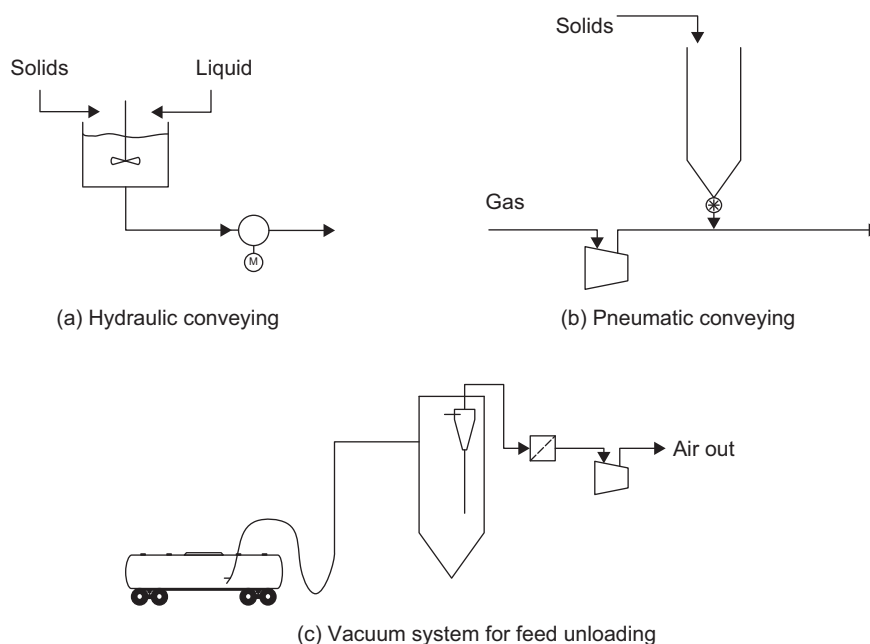


FIG. 18.17 Hydraulic and pneumatic conveying.

unloading feed tankers or lifting solids into silos. A flexible feed hose attached to the silo is used to suck up the solids, rather like a domestic vacuum cleaner. A vacuum compressor is used to pull air out of the silo. A combination of a cyclone and filter will usually be used upstream of the vacuum compressor to prevent damage from particulates, as illustrated in Fig. 18.17(c).

Small, portable pneumatic conveyors with flexible hoses instead of piping are often used for conveying solids over small distances in the manufacture of specialty chemicals and pharmaceuticals.

Hydraulic conveying is similar to pneumatic conveying, but uses a liquid rather than a gas as the carrier fluid. The smaller difference in specific gravity between the particles and carrier fluid makes it easier to entrain solids, but it is much harder to separate the solids from the carrier. Because pumps are less susceptible to damage by solids than compressors, hydraulic conveying systems usually mix the solid and liquid upstream of the pump and then pump the resulting slurry, as shown in Fig. 18.17(a). Hydraulic conveying is widely used in mining, coal processing, and oil drilling, but is only used in the process industries when contamination of the solids with the carrier fluid is acceptable, for example, when processing solids in water or conveying a catalyst in process feed.

The work required in hydraulic conveying can be estimated from the pipe pressure drop by treating the slurry as a fluid, as described in Section 20.7.2. For pneumatic conveying, the calculation is more complex, as allowance must be made for the compressible nature of the gas and the acceleration that occurs when the solids enter the gas flow. Methods for pressure drop in horizontal and vertical pneumatic conveying are given in Seville et al. (1997) and Jones (1997). Pneumatic and hydraulic conveying are discussed in a book by Mills (2015); see also Mills et al. (2004), Rhodes (2008), and Wilms and Dhodapkar (2014).

Pipe conveyors

Pipe conveying is a relatively new technology that is becoming widely used in minerals handling. A pipe conveyor is similar to a belt conveyor in that solids are dropped onto a flexible belt. After loading the solids, the belt passes through rollers that fold the sides over to cover the solid (Fig. 18.18). The rolled-up tube containing the solids then passes through idlers that keep the tube rolled until the destination, where the tube is unrolled and the solids are discharged. Pipe conveyors cost only slightly more than belt conveyors and have a number of advantages. They cause less dust formation, there is no need to cover the conveyor to keep material dry, and there is no need for the pipes to follow straight flights, so a pipe conveyor can more easily traverse difficult terrain.

Bucket elevators

The most widely used equipment where a vertical lift is required is the bucket elevator. This consists of buckets fitted to an endless chain or belt, which passes over a driven roller or sprocket at the top end. Bucket elevators can

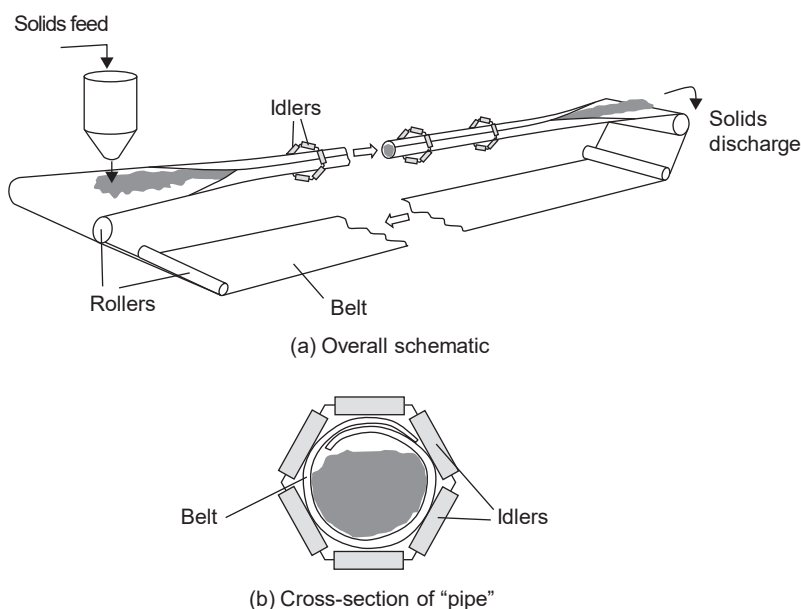


FIG. 18.18 Pipe conveyor.

handle a wide range of solids, from heavy lumps to fine powders, and are suitable for use with wet solids and slurries.

18.3.5 Pressurization of solid feeds

Bringing a solid feed into a process that operates under pressure can be challenging, as there will always be a tendency for high-pressure fluids to flow back through the interstitial spaces between solid particles if a positive seal is not provided.

When the process fluid is a nonflammable liquid, the simplest means of pressurizing the solid is to slurry it into the process liquid and then pump the resulting slurry to the desired pressure. This is the same concept as the hydraulic conveying system shown in Fig. 18.17(a). If the process fluid is flammable, it can still be used to slurry the solids, but a closed atmospheric-pressure slurring tank must be used, and the solids must be fed through a rotary vane feeder or rotary valve that minimizes back-flow of vapor. The head space in the mixing tank should also be vented with inert gas, as shown in Fig. 18.19. Additional treatment of the vent gas may be needed to control particulate (dust) and VOC emissions.

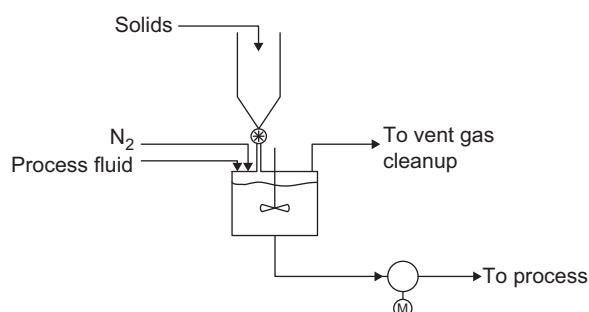


FIG. 18.19 Vented and purged system for feeding solids into hydrocarbon.

For a slurry to be pumpable, the void fraction must be greater than 0.7, and greater than 0.8 is usually preferred. It is therefore necessary to feed roughly four volumes of liquid per volume of solid with this approach. When such high volumes of liquid are unacceptable, a dilute slurry can be prepared, pumped to pressure, and then fed to a hydrocyclone to form a more concentrated slurry. The dilute stream from the hydrocyclone can be recycled to the slurring tank.

Feeding a solid into a high-pressure gas-phase process is substantially more difficult. Rotary valves and screw conveyors can be designed with tight enough tolerances to prevent gas back-flow for relatively small pressure differences. If the distance between flights on a screw conveyor is tapered, then compression will occur, so a screw conveyor can be designed to act as a solids compressor over a small range of pressure increase.

For larger pressure increases, a *dual hopper* system can be used, as illustrated in Fig. 18.20. Dual hoppers are also known as *charge hoppers* or *lock hoppers*. In a dual hopper arrangement, the solids flow is metered from the upper hopper into the lower hopper while the lower hopper is isolated from the process. The upper hopper is then isolated from the lower hopper and the lower hopper is brought to process pressure, usually by up-flow of gas to avoid compressing the bed of solids. The isolation valve at the lower hopper outlet is then opened to discharge solids into the process. Once the solids are discharged, the lower hopper is isolated, is vented to low pressure, and is

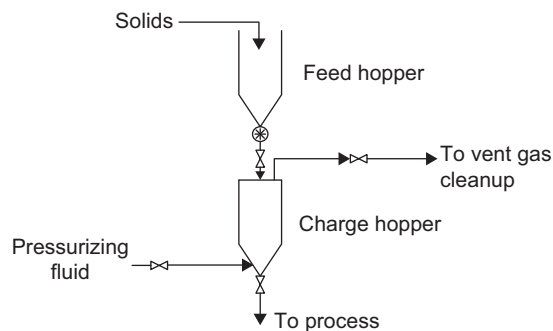


FIG. 18.20 Dual hopper feed system.

then ready to be charged again. The timing of the cycle is set to ensure the desired volumetric or mass flow rate of solids to the process. Specially designed valves are used that can isolate gas flow in the presence of solids. In some designs a double valve arrangement is used, where the upper valve can close on a flowing solids stream, and the lower valve subsequently closes to provide a gas-tight seal.

A dual hopper flow scheme always operates in a semi-batch mode. The design engineer must ensure that the equipment is sized for the actual instantaneous flowing rate of solids or gas and not for the time-averaged flow rate.

Dual hopper schemes can also be used when a solid is moved from an air- or oxygen-containing environment to a hydrogen- or hydrocarbon-containing environment or vice versa. Purging the charge hopper with an inert gas such as steam or nitrogen prevents either flammable gas or oxidant from being entrained with the solids flow.

18.4 Separation and mixing of solids

Processes and equipment are required to separate valuable solids from unwanted material and for size grading (classifying) solid raw materials and products.

The equipment used for solid–solid separation processes was developed primarily for the minerals processing and metallurgical industries for the beneficiation (upgrading) of ores. The techniques used depend on differences in physical, rather than chemical, properties, though chemical additives may be used to enhance separation. The principal techniques used are shown in Fig. 18.21, which can be used to select the type of processes likely to be suitable for a particular material and size range. Sorting material by appearance, by hand, is now rarely used due to the high cost of labor.

Blending and mixing devices are used when intermingling of particles is desired. Solid particles may require mixing to achieve a desired blend for formulation purposes or to prevent segregation by particle size or density during processing.

18.4.1 Screening (sieving)

Screens separate particles on the basis of size. Their main application is in grading raw materials and products into size ranges, but they are also used for the removal of trash (oversized and undersized contaminants) and for dewatering. Industrial screening equipment is used over a wide range of particle sizes, from fine powders to large rocks. For small particles, woven cloth or wire screens are used, and for larger sizes, perforated metal plates or grids.

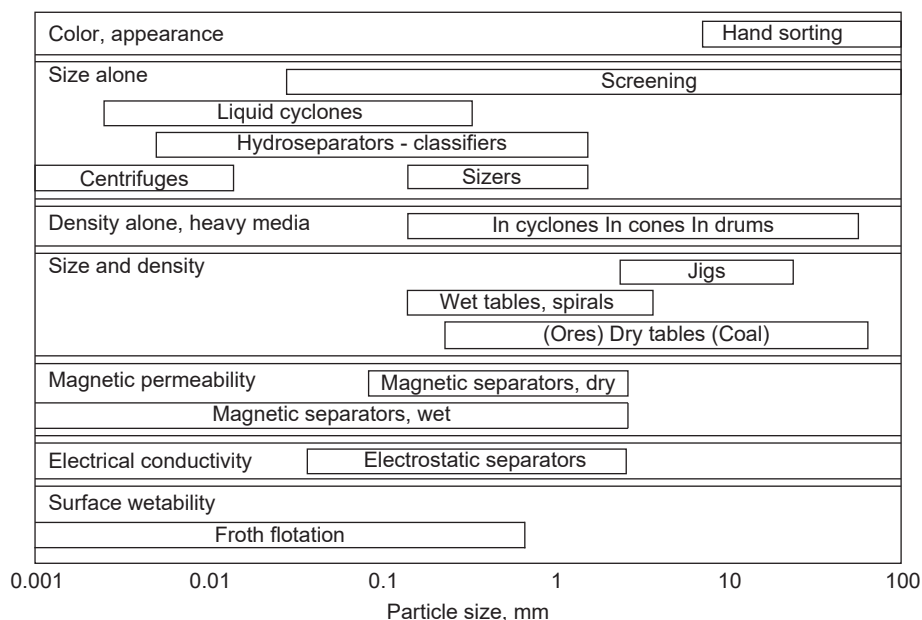


FIG. 18.21 A particle size selection guide to solid–solid separation techniques and equipment. (From Roberts, E. J., Stavenger, P., Bowersox, J. P., Walton, A. K., & Mehta, M. (1971). *Solid/solid separation*. Chemical Engineering, New York, 78(Feb. 15), 89.)

Screen sizes are defined in two ways: by a mesh size number for small sizes and by the actual size of opening in the screen for the larger sizes. There are several different standards in use for mesh size, and it is important to quote the particular standard used when specifying particle size ranges by mesh size. In the United States, the appropriate ASTM standards should be used, ASTM E11, but Tyler mesh sizes will often be encountered. For overseas designs the International Organization for Standardization (ISO) standards will usually be used. These are available in English as BS 7792 (ISO 10630) and BS ISO 14315. The relationship between particle size and standard mesh sizes is given in Table 18.1 in Section 18.2.1.

The simplest industrial screening devices are stationary screens, over which the material to be screened flows. Typical of this type are “Grizzly” screens, which consist of rows of equally spaced parallel bars and which are used to “scalp” off oversized rocks in the feed to crushers.

Dynamic screening equipment can be categorized according to the type of motion used to shake up and transport the material on the screen. The principal types used in the chemical process industries are described briefly next.

Vibrating screens: Horizontal and inclined screening surfaces vibrated at high frequencies (1000 to 7000 Hz). These are high-capacity units, with good separating efficiency, and are used for a wide range of particle sizes.

Oscillating screens: Operated at lower frequencies than vibrating screens (100 to 400 Hz) with a longer, more linear stroke.

Reciprocating screens: Operated with a shaking motion, a long stroke at low frequency (20 to 200 Hz). Used for conveying with size separation.

Sifting screens: Operated with a circular motion in the plane of the screening surface. The actual motion may be circular, gyratory, or circularly vibrated. Used for the wet and dry screening of fine powders.

Revolving screens: Inclined, cylindrical screens rotated at low speeds (10 to 20 rpm). Used for the wet screening of relatively coarse material, but have now been largely replaced by vibrating screens.

Fig. 18.22, which is based on a similar chart given by Matthews (1971), can be used to select the type of screening equipment likely to be suitable for a particular size range. Equipment selection will normally be based on laboratory and pilot-scale screening tests conducted by the equipment vendors or with their cooperation. The main factors to be considered, and the information that would be required by the firms supplying proprietary screening equipment, are listed here:

1. Rate, throughput required
2. Size range (test screen analysis)
3. Characteristics of the material: free-flowing or sticky, bulk density, abrasiveness
4. Hazards: flammability, toxicity, dust explosion
5. Wet or dry screening to be used

18.4.2 Liquid–solid cyclones

Cyclones can be used for the classification of solids, as well as for liquid–solid and liquid–liquid separations. The design and application of liquid cyclones (hydrocyclones) is discussed in Section 18.6.4. A typical unit is shown in Fig. 18.23.

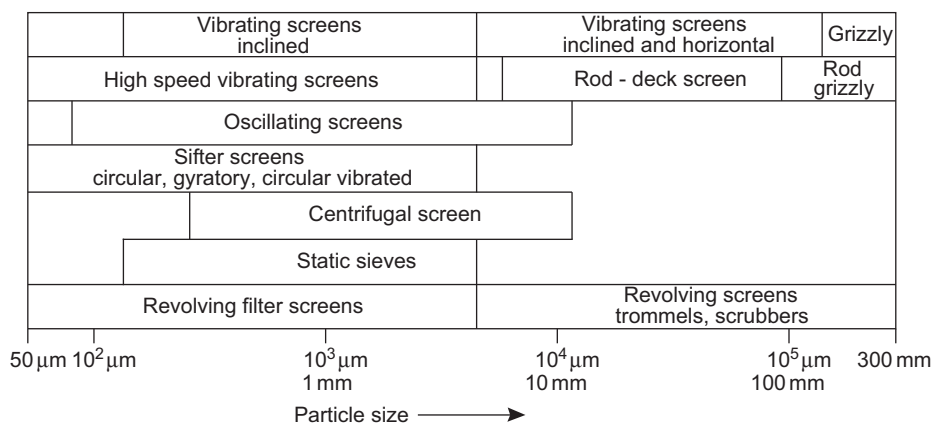


FIG. 18.22 Screen selection by particle size range.

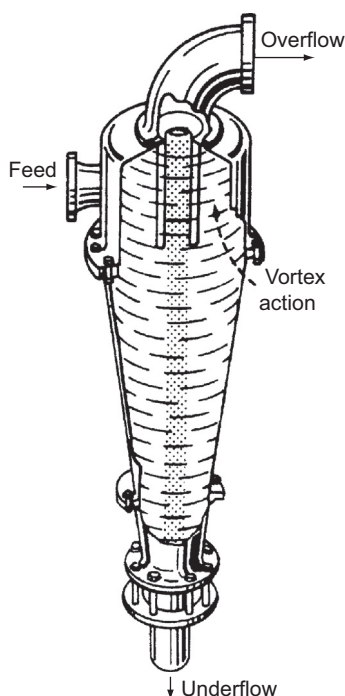


FIG. 18.23 Liquid–solid cyclone (hydrocyclone).

Liquid cyclones can be used for the classification of solid particles over a size range from 5 to 100 μm . Commercial units are available in a wide range of materials of construction and sizes, from as small as 10 mm to up to 30 m in diameter. The separating efficiency of liquid cyclones depends on the particle size and density and the density and viscosity of the liquid medium.

18.4.3 Hydroseparators and sizers (classifiers)

Classifiers that depend on the difference in the settling rates of different size particles in water are frequently used for separating fine particles in the 50 to 300 μm range. Various designs are used. The principal ones used in the chemical process industries are described next.

Thickeners: Thickeners are primarily used for liquid–solid separation (see Section 18.6). When used for classification, the feed rate is such that the overflow rate is greater than the settling rate of the slurry, and the finer particles remain in the overflow stream.

Rake classifiers: These are inclined, shallow, rectangular troughs fitted with mechanical rakes at the bottom to rake the deposited solids to the top of the incline (Fig. 18.24). Several rake classifiers can be used in series to separate the feed into different size ranges.

Bowl classifiers: These are shallow bowls with concave bottoms fitted with rakes. Their operation is similar to that of thickeners.

18.4.4 Hydraulic jigs

Jigs separate solids by difference in density and size. The material is immersed in water, supported on a screen (Fig. 18.25). Pulses of water are forced through the bed of material either by moving the screen or by pulsating the water level. The flow of water fluidizes the bed and causes the solids to stratify with the lighter material at the top and the heavier at the bottom.

18.4.5 Tables

Tables are used wet and dry. The separating action of a wet table resembles that of the traditional miner's pan. Riffled tables (Fig. 18.26) are basically rectangular decks, inclined at a shallow angle to the horizontal (2 to 5 degrees),

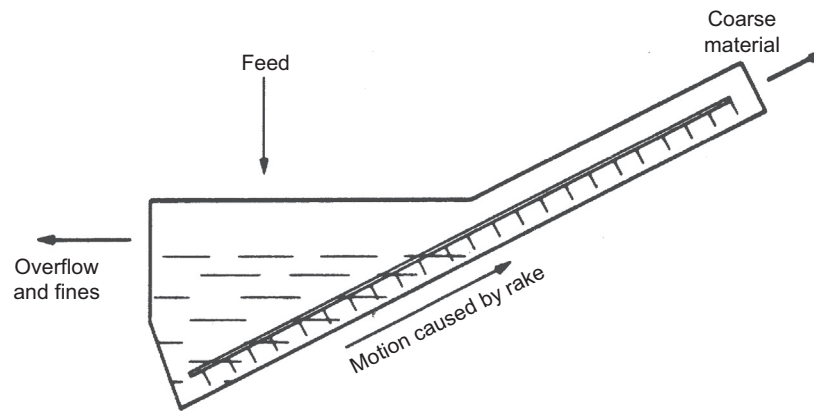


FIG. 18.24 Rake classifier.

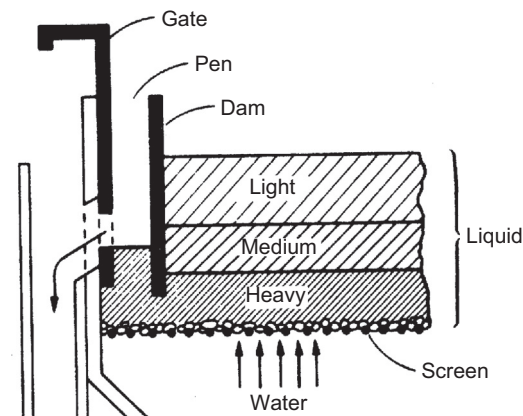


FIG. 18.25 A hydraulic jig.

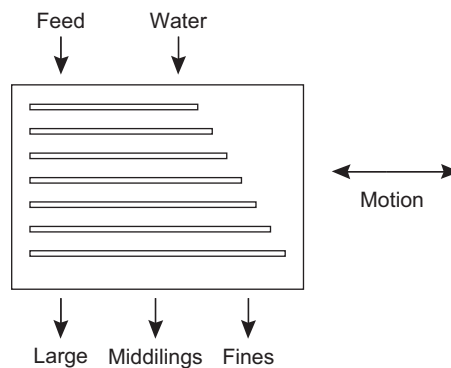


FIG. 18.26 Wilfley riffled table.

with shallow slats (riffles) fitted to the surface. The table is mechanically shaken, with a slow stroke in the forward direction and a faster backward stroke. The particles are separated into different size ranges under the combined action of the vibration, water flow, and resistance to flow over the riffles.

18.4.6 Classifying centrifuges

Centrifuges are used for the classification of particles in size ranges below $10\ \mu\text{m}$. Two types are used: solid bowl centrifuges, usually with a cylindrical, conical bowl, rotated about a horizontal axis and “nozzle” bowl machines, fitted with discs. These types are described in [Section 18.6.3](#).

18.4.7 Dense-medium separators (sink and float processes)

Solids of different densities can be separated by immersing them in a fluid of intermediate density. The heavier solids sink to the bottom and the lighter float to the surface. Water suspensions of fine particles are often used as the dense liquid (heavy-medium). The technique is widely used in the beneficiation (concentration) of mineral ores.

18.4.8 Flotation separators (froth-flotation)

Froth-flotation processes are used extensively for the separation of finely divided solids. Separation depends on differences in the surface properties of the materials. The particles are suspended in an aerated liquid (usually water), and air bubbles adhere preferentially to the particles of one component and bring them to the surface. Frothing agents are used so that the separated material is held on the surface as froth and can be removed.

Froth-flotation is an extensively used separation technique, having a wide range of applications in the minerals processing industries and other industries. It can be used for particles in the size range from 50 to 400 μm .

18.4.9 Magnetic separators

Magnetic separators can be used for materials that are affected by magnetic fields; the principle is illustrated in Fig. 18.27. Rotating-drum magnetic separators are used in the minerals processing industries. They can be designed to handle relatively high throughputs, up to 3000 kg/h per meter length of drum. Simple magnetic separators are often used for the removal of iron from the feed to a crusher.

The various types of magnetic separators used and their applications are described by Bronkala (1988).

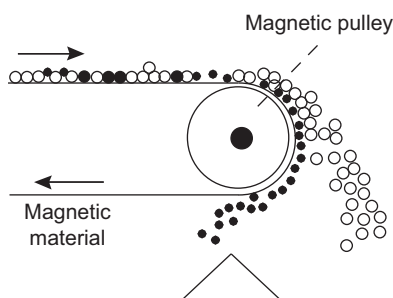


FIG. 18.27 Magnetic separator.

18.4.10 Electrostatic separators

Electrostatic separation depends on differences in the electrical properties (conductivity) of the materials to be treated. In a typical process the particles pass through a high-voltage electric field as they are fed on to a revolving drum, which is at earth potential (Fig. 18.28). Those particles that acquire a charge adhere to the drum surface and are carried further around the drum before being discharged.

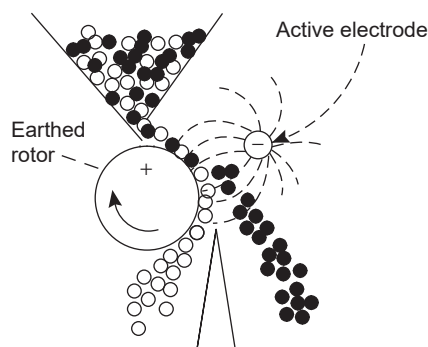


FIG. 18.28 Electrostatic separator.

18.4.11 Solids blending and mixing

Many solid products are blends of different solid materials. Familiar examples include food mixes, soap powders, candy mixes, and lawn fertilizer. Much of the equipment used for mixing solid powders and blending solids into liquids was initially developed for use in the food processing industry. Blends of powders or particulates are also used in the preparation of formed solids such as tablets, catalyst pellets, and many polymer products. Some processes also require solids mixing to prevent segregation of particles by size during processing.

Mixing of solids can be accomplished with or without changes in size distribution. Substantial mixing occurs in size reduction devices; see [Section 18.9](#).

A great variety of specialized equipment has been developed for mixing dry solids and pastes (wet solids). The principal types of equipment and their fields of application are given in [Table 18.5](#). Descriptions of the equipment can be found in the literature: [Green and Southard \(2018\)](#), [Harnby et al. \(1997\)](#), and [Reid \(1979\)](#). A selection chart for solids mixing equipment is given by [Jones \(1985\)](#).

Most solids mixing devices are operated in batch mode. A charge of solids is loaded into the mixer, which is then run for the desired time to achieve mixing and unloaded. Cone blenders are used for free-flowing solids. Ribbon blenders can be used for dry solids and for blending liquids with solids. Z-blade mixers and pan mixers are used for kneading heavy pastes and doughs. Ribbon blenders, fluidized mixers, and paddle mixers can be operated in continuous mode. Static mixers are always operated continuously in a flowing fluid. The power requirement for mixing solids depends on the device used and should be determined in consultation with the device manufacturer or through plant trials. Most manufacturers state the power consumption in their brochures or on their websites. More information on solids mixing is given in [Weinekötter and Gericke \(2000\)](#), [Kaye \(1996\)](#), [Green and Southard \(2018\)](#), [Fayed and Otten \(1997\)](#), and [Paul et al. \(2003\)](#).

TABLE 18.5 Solids and paste mixers

Type of equipment	Mixing action	Applications	Examples
Rotating: cone, double cone, drum	Tumbling action	Blending dry, free-flowing powders, granules, crystals	Pharmaceuticals, food, chemicals
Air blast fluidization	Air blast lifts and mixes particles	Dry powders and granules	Milk powder; detergents, chemicals
Horizontal trough mixer, with ribbon blades, paddles, or beaters	Rotating element produces contra-flow movement of materials	Dry and moist powders	Chemicals, food, pigments, tablet granulation
Z-blade mixers	Shearing and kneading by the specially shaped blades	Mixing heavy pastes, creams, and doughs	Bakery industry, rubber doughs, plastic dispersions
Pan mixers	Vertical, rotating paddles, often with planetary motion	Mixing, whipping, and kneading of materials ranging from low-viscosity pastes to stiff doughs	Food, pharmaceuticals and chemicals, printing inks, and ceramics
Cylinder mixers, single and double	Shearing and kneading action	Compounding of rubbers and plastics	Rubbers, plastics, and pigment dispersion

Tumbling drums

The simplest solids mixers are rotating drums known as tumblers. These are usually double-cone shaped, as shown in [Fig. 18.29\(a\)](#), but twin-shell, V-shaped drums are also used. Tumblers can contain baffles or internals to break up agglomerates and promote mixing. They are generally operated at low speed and are suitable for gentle blending without changes in particle size distribution.

Internally agitated mixers

Several mixer designs use a static drum with moving internals to accomplish mixing of the solids. In a ribbon blender, illustrated schematically in [Fig. 18.29\(b\)](#), a helical ribbon rotates horizontally inside a drum to blend the solids. Vertical screw mixers, shown in [Fig. 18.29\(c\)](#), also use a screw, which lifts the solids along the wall of the vessel and orbits around a central axis. This type is often referred to as a Nauta mixer. In a paddle mixer (see [Fig. 18.29d](#)), a horizontal drum containing the solids is agitated by pins or paddles located along a central shaft. Twin-screw paddle mixers are also used, in which two shafts are located in the drum or tank.

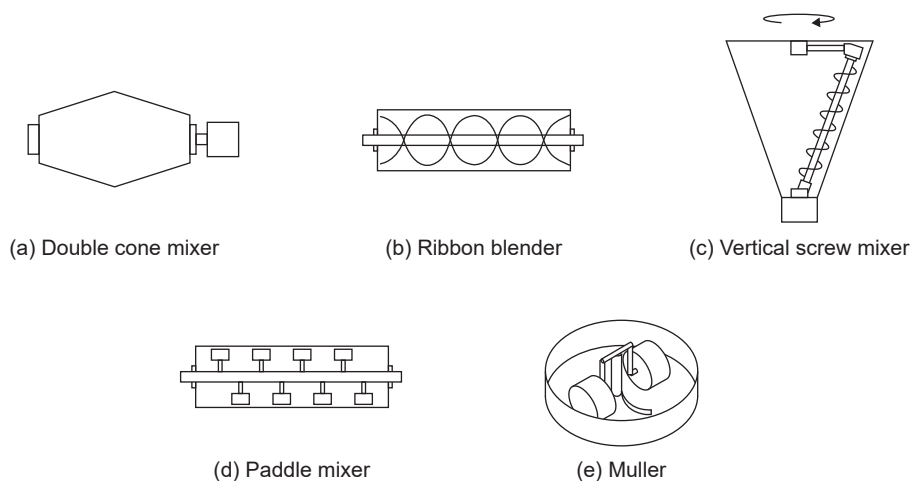


FIG. 18.29 Solids mixers.

A muller, illustrated in Fig. 18.29(e), is an open-pan mixer in which heavy rollers are driven over the material. A plow or scrapers is also incorporated to lift and turn the solids. In some mullers the pans also rotate in the opposite direction to the rollers. Mullers are suitable for materials that agglomerate but are not sticky.

Fluidized mixers

A fluidized bed can be operated to give rapid mixing of solid particles. Fluidized bed mixers can be operated continuously or in batch mode. The mixed particles can be removed continuously from the bed through a stand pipe during operation. Fluidization is discussed in more detail in Section 18.2.2. Fluidized mixers are easily integrated with pneumatic or hydraulic conveying systems.

Static mixers

Static mixers, also known as in-line mixers, can be used to prevent settling of particles or segregation of particles by size during pneumatic or hydraulic conveying. Several different proprietary designs are used, all based on the principle of using shaped internals to break up the flow pattern and remix the solid particles. Static mixers are also used for mixing liquid streams; a typical design is illustrated in Fig. 15.10.

18.5 Gas–solids separations (gas cleaning)

The primary need for gas–solid separation processes is for gas cleaning: the removal of dispersed, finely divided solids (dust) and liquid mists from gas streams. Process gas streams must often be cleaned up to prevent contamination of catalysts or products and to avoid damage to equipment, such as compressors. Also, effluent gas streams must be cleaned to comply with air pollution regulations and for reasons of hygiene, to remove toxic and other hazardous materials; see Muir (1992) and Section 11.5.1.

There is often a need for clean, filtered air for processes that use air as a raw material and where clean working atmospheres are needed: for instance, in the pharmaceutical and electronics industries. Air for process instruments must be clean, dry, and dust-free.

The particles to be removed may range in size from large molecules, measuring a few hundredths of a micrometer, to the coarse dusts arising from the attrition of catalysts or fly ash from the combustion of pulverized fuels.

A variety of equipment has been developed for gas cleaning. The principal types used in the process industries are listed in Table 18.6, which is adapted from a selection guide given by Sargent (1971). Table 18.6 shows the general field of application of each type in terms of the particle size separated, the expected separation efficiency, and the throughput. It can be used to make a preliminary selection of the type of equipment likely to be suitable for a particular application. Descriptions of the equipment shown in Table 18.6 can be found in various handbooks: Green and Southard (2018) and Schweitzer (1997) and in specialist texts: Strauss (1975). Gas cleaning is also covered in Chhabra and Gurappa (2019).

TABLE 18.6 Gas-cleaning equipment

Type of equipment	Minimum particle size (μm)	Minimum loading (mg/m^3)	Approx. efficiency (%)	Typical gas velocity (m/s)	Maximum capacity (m^3/s)	Gas pressure drop ($\text{mm H}_2\text{O}$)	Liquid rate ($\text{m}^3/10^3 \text{ m}^3 \text{ gas}$)	Space required (relative)
<i>Dry collectors</i>								
Settling chamber	50	12,000	50	1.5–3	none	5	—	Large
Baffle chamber	50	12,000	50	5–10	none	3–12	—	Medium
Louver	20	2500	80	10–20	15	10–50	—	Small
Cyclone	10	2500	85	10–20	25	10–70	—	Medium
Multiple cyclone	5	2500	95	10–20	100	50–150	—	Small
Impingement	10	2500	90	15–30	none	25–50	—	Small
<i>Wet scrubbers</i>								
Gravity spray	10	2500	70	0.5–1	50	25	0.05–0.3	Medium
Centrifugal	5	2500	90	10–20	50	50–150	0.1–1.0	Medium
Impingement	5	2500	95	15–30	50	50–200	0.1–0.7	Medium
Packed	5	250	90	0.5–1	25	25–250	0.7–2.0	Medium
Jet	0.5 to 5 (range)	250	90	10–100	50	none	7–14	Small
Venturi	0.5	250	99	50–200	50	250–750	0.4–1.4	Small
<i>Others</i>								
Fabric filters	0.2	250	99	0.01–0.1	100	50–150	—	Large
<i>Electrostatic</i>								
Precipitators	2	250	99	5–30	1000	5–25	—	Large

Gas-cleaning equipment can be classified according to the mechanism employed to separate the particles: gravity settling, impingement, centrifugal force, filtering, washing, and electrostatic precipitation.

18.5.1 Gravity settlers (settling chambers)

Settling chambers are the simplest form of industrial gas-cleaning equipment but have only a limited use; they are suitable for coarse dusts, particles larger than $50\text{ }\mu\text{m}$. They are essentially long, horizontal, rectangular chambers through which the gas flows. The solids settle under gravity and are removed from the bottom of the chamber. Horizontal plates or vertical baffles are used in some designs to improve the separation. Settling chambers offer little resistance to the gas flow and can be designed for operation at high temperature and high pressure and for use in corrosive atmospheres.

The length of chamber required to settle a given particle size can be estimated from the settling velocity (calculated using Stokes law) and the gas velocity. A design procedure is given by [Jacob and Dhodapkar \(1997\)](#).

18.5.2 Impingement separators

Impingement separators employ baffles to achieve the separation. The gas stream flows easily around the baffles, whereas the solid particles, due to their higher momentum, tend to continue in their line of flight, strike the baffles, and are collected. Many different baffle designs are used in commercial equipment; a typical example is shown in [Fig. 18.30](#). Impingement separators cause a higher pressure drop than settling chambers but are capable of separating smaller particle sizes: 10 to $20\text{ }\mu\text{m}$.

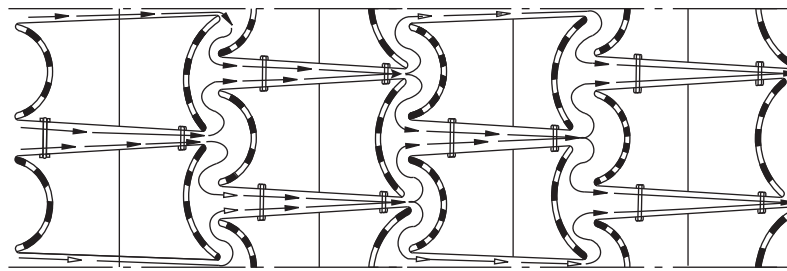


FIG. 18.30 Impingement separator (section showing gas flow).

18.5.3 Centrifugal separators (cyclones)

Cyclones are the principal type of gas–solids separator employing centrifugal force and are widely used. They are simple constructions, can be made from a wide range of materials, and can be designed for high-temperature and high-pressure operation.

Cyclones are suitable for separating particles above about $5\text{ }\mu\text{m}$ in diameter; smaller particles, down to about $0.5\text{ }\mu\text{m}$, can be separated when agglomeration occurs.

The most commonly used design is the reverse-flow cyclone ([Fig. 18.31](#)); other configurations are used for special purposes. In a reverse-flow cyclone the gas enters the top chamber tangentially and spirals down to the apex of the conical section; it then moves upward in a second smaller-diameter spiral and exits at the top through a central vertical pipe. The solids move radially to the walls, slide down the walls, and are collected at the bottom. Design procedures for cyclones are given by [Constantinescu \(1984\)](#), [Strauss \(1975\)](#), [Koch and Licht \(1977\)](#), and [Stairmand \(1951\)](#). The theoretical concepts and experimental work on which the design methods are based are discussed in [Chhabra and Gurappa \(2019\)](#). Stairmand's method is outlined here and illustrated in Example 18.2.

Cyclone design

Stairmand developed two standard designs for gas–solid cyclones: a high-efficiency cyclone ([Fig. 18.32a](#)) and a high-throughput design ([Fig. 18.32b](#)). The performance curves for these designs, obtained experimentally under standard test conditions, are shown in [Figs 18.33\(a\) and 18.33\(b\)](#). These curves can be transformed to other cyclone sizes and operating conditions by use of the following scaling equation for a given separating efficiency:

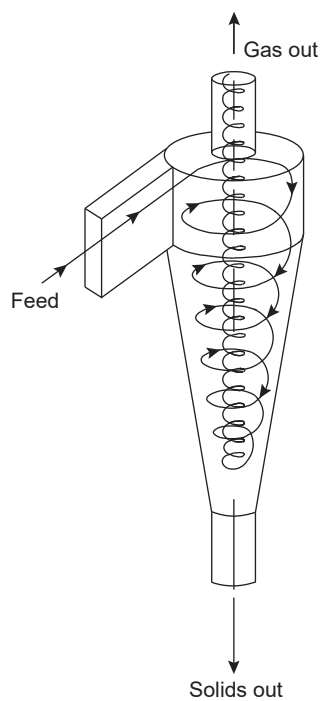


FIG. 18.31 Reverse-flow cyclone.

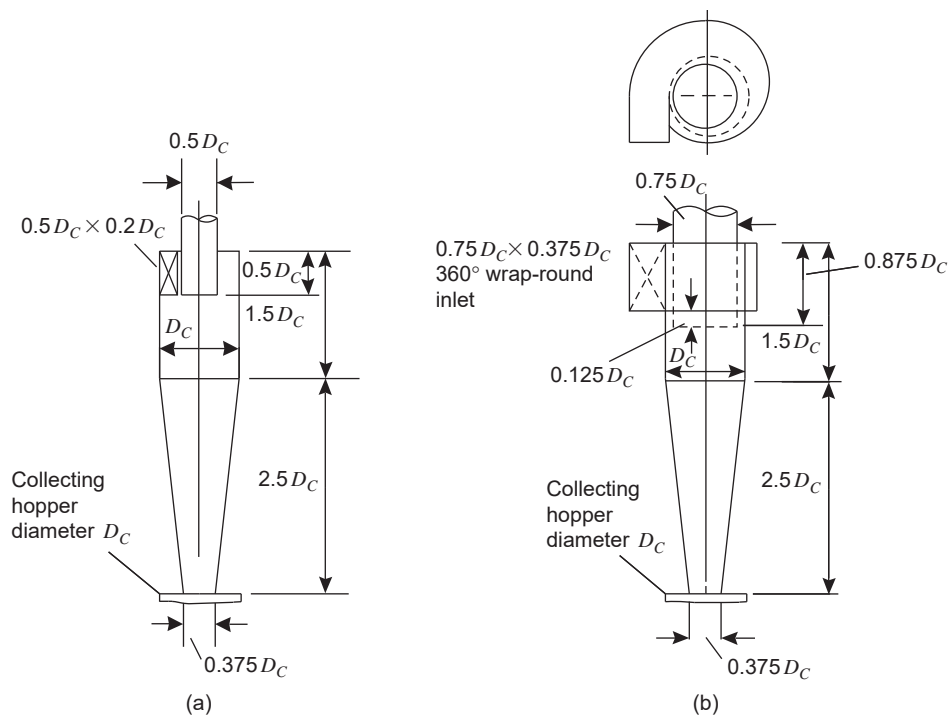


FIG. 18.32 Standard cyclone dimension. (a) High-efficiency cyclone. (b) High-gas-rate cyclone.

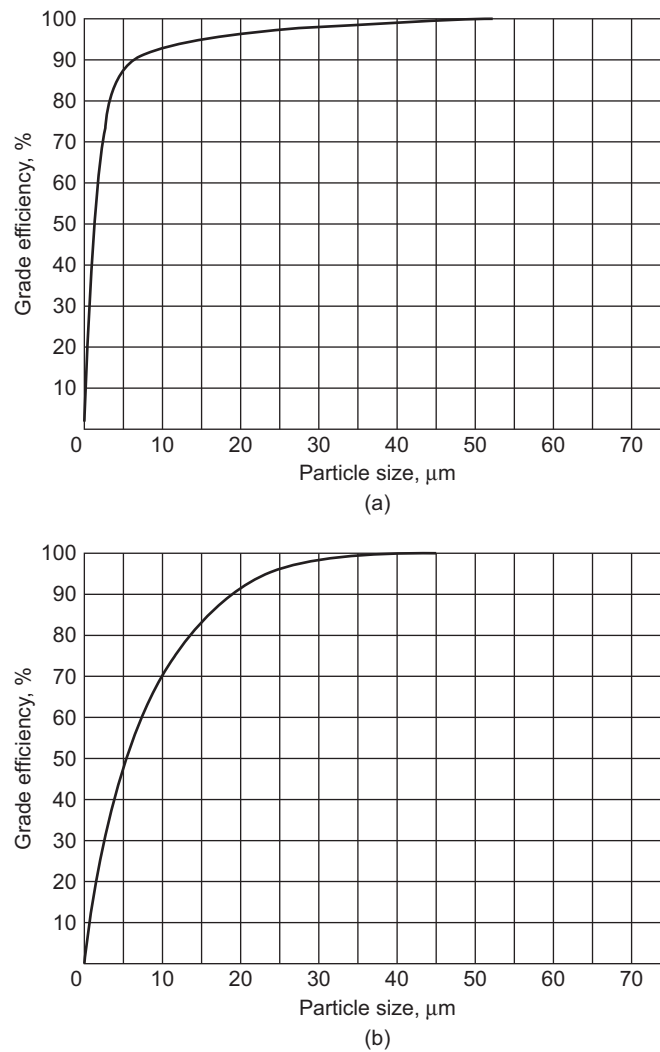


FIG. 18.33 Performance curves, standard conditions. (a) High-efficiency cyclone performance curves, standard conditions. (b) High-gas-rate cyclone.

$$d_2 = d_1 \left[\left(\frac{D_{c2}}{D_{c1}} \right)^3 \times \frac{Q_1}{Q_2} \times \frac{\Delta\rho_1}{\Delta\rho_2} \times \frac{\mu_2}{\mu_1} \right]^{1/2} \quad (18.23)$$

where d_1 = mean diameter of particle separated at the standard conditions, at the chosen separating efficiency (see Figs. 18.33a or 18.33b)

d_2 = mean diameter of the particle separated in the proposed design at the same separating efficiency

D_{c1} = diameter of the standard cyclone = 8 in (203 mm)

D_{c2} = diameter of proposed cyclone, mm

Q_1 = standard flow rate:

for high efficiency design = 223 m³/h

for high throughput design = 669 m³/h

Q_2 = proposed flow rate, m³/h

$\Delta\rho_1$ = solid–fluid density difference in standard conditions = 2000 kg/m³

$\Delta\rho_2$ = density difference, proposed design

μ_1 = test fluid viscosity (air at 1 atm, 20 °C) = 0.018 mN s/m²

μ_2 = viscosity, proposed fluid

A performance curve for the proposed design can be drawn up from Fig. 18.33(a) or 18.33(b) by multiplying the grade diameter at, say, each 10% increment of efficiency by the scaling factor given by Equation 18.23, as shown in Fig. 18.34.

An alternative method of using the scaling factor that does not require redrawing the performance curve is used in Example 18.2. The cyclone should be designed to give an inlet velocity of between 9 and 27 m/s (30 to 90 ft/s); the optimum inlet velocity has been found to be 15 m/s (50 ft/s).

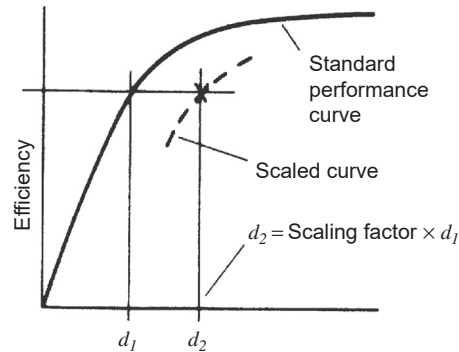


FIG. 18.34 Scaled performance curve.

Pressure drop

The pressure drop in a cyclone will be due to the entry and exit losses and friction and kinetic energy losses in the cyclone. The empirical equation given by Stairmand (1949) can be used to estimate the pressure drop:

$$\Delta P = \frac{\rho_f}{203} \left\{ u_1^2 \left[1 + 2\phi^2 \left(\frac{2r_t}{r_e} - 1 \right) \right] + 2u_2^2 \right\} \quad (18.24)$$

where ΔP = cyclone pressure drop, millibars

ρ_f = fluid (gas) density, kg/m³

u_1 = inlet duct velocity, m/s

u_2 = exit duct velocity, m/s

r_t = radius of circle to which the center line of the inlet is tangential, m

r_e = radius of exit pipe, m

ϕ = factor from Fig. 18.35

Ψ = parameter in Fig. 18.35, given by:

$$\Psi = f_c \frac{A_s}{A_1}$$

f_c = friction factor, taken as 0.005 for gases

A_s = surface area of cyclone exposed to the spinning fluid, m²

For design purposes, this can be taken as equal to the surface area of a cylinder with the same diameter as the cyclone and length equal to the total height of the cyclone (barrel plus cone).

A_1 = area of inlet duct, m²

Stairmand's equation is for the gas flowing alone, containing no solids. The presence of solids will normally increase the pressure drop over that calculated using Equation 18.24, depending on the solids loading. Alternative design methods for cyclones, which include procedures for estimating the true pressure drop, are given by Green and Southard (2018) and Yang (1999); see also Zenz (2001).

General design procedure

1. Select either the high-efficiency or high-throughput design, depending on the performance required.
2. Obtain an estimate of the particle size distribution of the solids in the stream to be treated.
3. Estimate the number of cyclones needed in parallel.

4. Calculate the cyclone diameter for an inlet velocity of 15 m/s (50 ft/s). Scale the other cyclone dimensions from Fig. 18.32(a) or 18.32(b).
5. Calculate the scale-up factor for the transposition of Fig. 18.33(a) or 18.33(b).
6. Calculate the cyclone performance and overall efficiency (recovery of solids). If unsatisfactory, try a smaller diameter.
7. Calculate the cyclone pressure drop and, if required, select a suitable blower.
8. Cost the system and optimize to make the best use of the pressure drop available or, if a blower is required, to give the lowest operating cost.

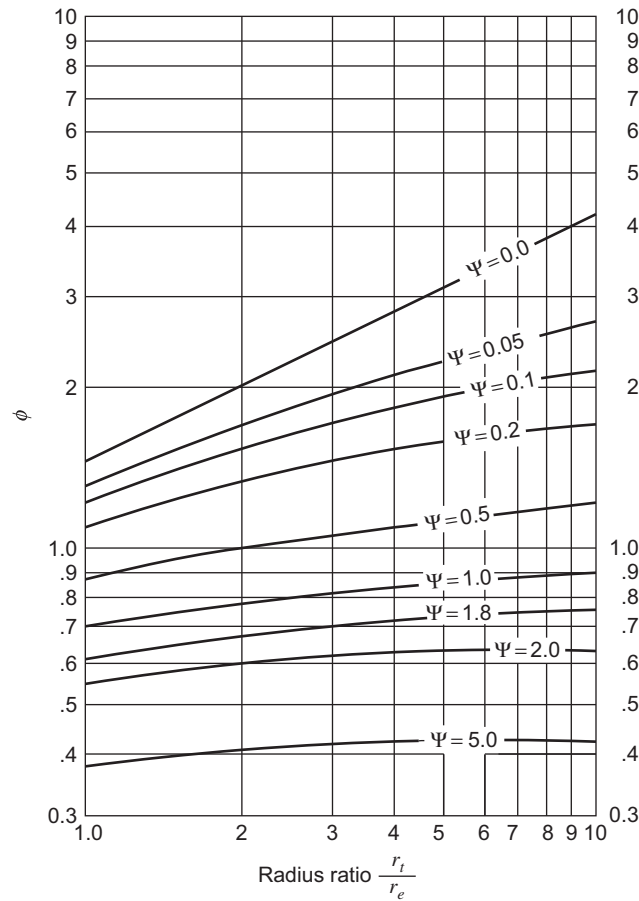


FIG. 18.35 Cyclone pressure drop factor.

Example 18.2

Design a cyclone to recover solids from a process gas stream. The anticipated particle size distribution in the inlet gas is given here. The density of the particles is 2500 kg/m^3 , and the gas is essentially nitrogen at 150°C . The stream volumetric flow rate is $4000 \text{ m}^3/\text{h}$, and the operation is at atmospheric pressure. An 80% recovery of the solids is required.

Particle size, μm	50	40	30	20	10	5	2
Percentage by weight less than	90	75	65	55	30	10	4

Solution

As 30% of the particles are below 10 μm , the high-efficiency design will be required to give the specified recovery.

$$\text{Flow rate} = \frac{4000}{3600} = 1.11 \text{ m}^3/\text{s}$$

$$\text{Area of inlet duct, at } 15 \text{ m/s} = \frac{1.11}{15} = 0.07 \text{ m}^2$$

From Fig. 18.32(a), duct area = $0.5 D_c \times 0.2 D_c$,

$$\text{so, } D_c = 0.84 \text{ m}$$

This is clearly too large compared with the standard design diameter of 0.203 m.

Try four cyclones in parallel, $D_c = 0.42 \text{ m}$.

$$\text{Flow rate per cyclone} = 1000 \text{ m}^3/\text{h}$$

$$\text{Density of gas at } 150^\circ\text{C} = \frac{28}{22.4} \times \frac{273}{423} = 0.81 \text{ kg/m}^3,$$

This is negligible compared with the solids density:

$$\text{Viscosity of } \text{N}_2 \text{ at } 150^\circ\text{C} = 0.023 \text{ cp (mN s/m)}$$

From Equation 18.23,

$$\text{scaling factor} = \left[\left(\frac{0.42}{0.203} \right)^3 \times \frac{223}{1000} \times \frac{2000}{2500} \times \frac{0.023}{0.018} \right]^{1/2} = \underline{\underline{1.42}}$$

The performance calculations, using this scaling factor and Fig. 18.33(a), are set out in the following table:

Calculated performance of cyclone design, Example 18.2

1 Particle size (μm)	2 Percent in range	3 Mean particle size $\div$ scaling	4 Efficiency at scaled size % Factor	5 Collected $\frac{(2) \times (4)}{100}$ (Fig. 18.33a)	6 Grading at exit (2) – (5)	7 Percent at exit
>50	10	35	98	9.8	0.2	1.8
50–40	15	32	97	14.6	0.4	3.5
40–30	10	25	96	9.6	0.4	3.5
30–20	10	18	95	9.5	0.5	4.4
20–10	25	11	93	23.3	1.7	15.1
10–5	20	5	86	17.2	2.8	24.8
5–2	6	3	72	4.3	1.7	15.1
2–0	4	1	10	0.4	3.6	31.8
		100	Overall collection Efficiency	88.7	11.3	100.0

The collection efficiencies shown in column 4 of the table were read from Fig. 18.33(a) at the scaled particle size, column 3. The overall collection efficiency satisfies the specified solids recovery. The proposed design with dimensions in the proportions given in Fig. 18.32(a) is shown in Fig. 18.36.

Pressure-drop calculation

$$\text{Area of inlet duct, } A_1, = 210 \times 80 = 16,800 \text{ mm}^2$$

$$\begin{aligned} \text{Cyclone surface area, } A_s &= \pi \times 420 \times (630 + 1050) \\ &= 2.218 \times 10^6 \text{ mm}^2 \end{aligned}$$

f_c taken as 0.005:

$$\psi = \frac{f_c A_s}{A_1} = \frac{0.005 \times 2.218 \times 10^6}{16,800} = 0.66$$

$$\frac{r_t}{r_e} = \frac{(420 - (80/2))}{210} = 1.81$$

From Fig. 18.35, $\phi = 0.9$.

$$u_1 = \frac{1000}{3600} \times \frac{10^6}{16,800} = 16.5 \text{ m/s}$$

$$\text{Area of exit pipe} = \frac{\pi \times 210^2}{4} = 34,636 \text{ mm}^2$$

$$u_2 = \frac{1000}{3600} \times \frac{10^6}{34,636} = 8.0 \text{ m/s}$$

From Equation 18.24:

$$\begin{aligned} \Delta P &= \frac{0.81}{203} [16.5^2 [1 + 2 \times 0.9^2 (2 \times 1.81 - 1)] + 2 \times 8.0^2] \\ &= \underline{\underline{6.4 \text{ millibar}}} \quad (67 \text{ mm H}_2\text{O}) \end{aligned}$$

This pressure drop looks reasonable.

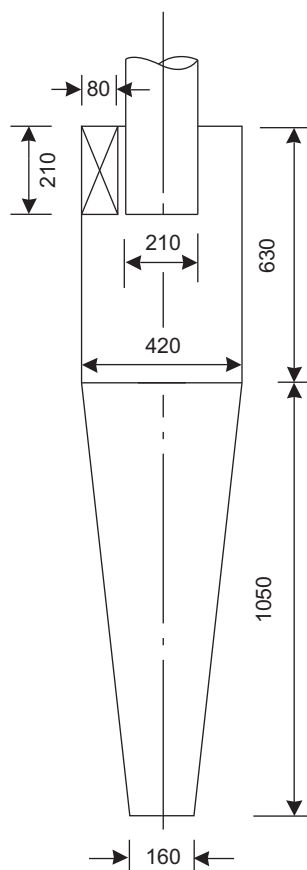


FIG. 18.36 Proposed cyclone design, all dimensions mm (Example 10.6).

18.5.4 Filters

The filters used for gas cleaning separate the solid particles by a combination of impingement and filtration; the pore sizes in the filter media used are too large simply to filter out the particles. The separating action relies on the precoating of the filter medium by the first particles separated, which are separated by impingement on the filter medium fibers. Woven or felted cloths of cotton and various synthetic fibers are commonly used as the filter media. Glass-fiber mats and paper filter elements are also used.

A typical example of this type of separator is the bag filter, which consists of a number of bags supported on a frame and housed in a large rectangular chamber (Fig. 18.37). The deposited solids are removed by mechanically vibrating the bag or by periodically reversing the gas flow. Bag filters can be used to separate small particles, down to around $1\ \mu\text{m}$, with a high separating efficiency. Commercial units are available to suit most applications and should be selected in consultation with a vendor.

The design and specification of bag filters (baghouses) is covered by Kraus (1979).

Air filters

Dust-free air is required for many process applications. The requirements of air filtration differ from those of process gas filtration mainly in that the quantity of dust to be removed will be lower, typically less than $10\ \text{mg}/\text{m}^3$ (~ 5 grains per $1000\ \text{ft}^3$), and also in that there is no requirement to recover the material collected.

Three basic types of air filters are used: viscous, dry, and continuous. Viscous and dry units are similar in construction, but the filter medium of the viscous type is coated with a viscous material, such as a mineral oil, to retain the dust. The filters are made up from standard, preformed, sections, supported on a frame in a filter housing. The sections are removed periodically for cleaning or replacement. Various designs of continuous filtration equipment are also available, employing either viscous or dry filter elements, but in which the filter is cleaned continuously. A comprehensive description of air filtration equipment is given by Strauss (1975).

18.5.5 Wet scrubbers (washing)

In wet scrubbing, the dust is removed by countercurrent washing with a liquid, usually water, and the solids are removed as a slurry. The principal mechanism involved is the impact (impingement) of the dust particles and the water droplets. Particle sizes down to $0.5\ \mu\text{m}$ can be removed in suitably designed scrubbers. In addition to

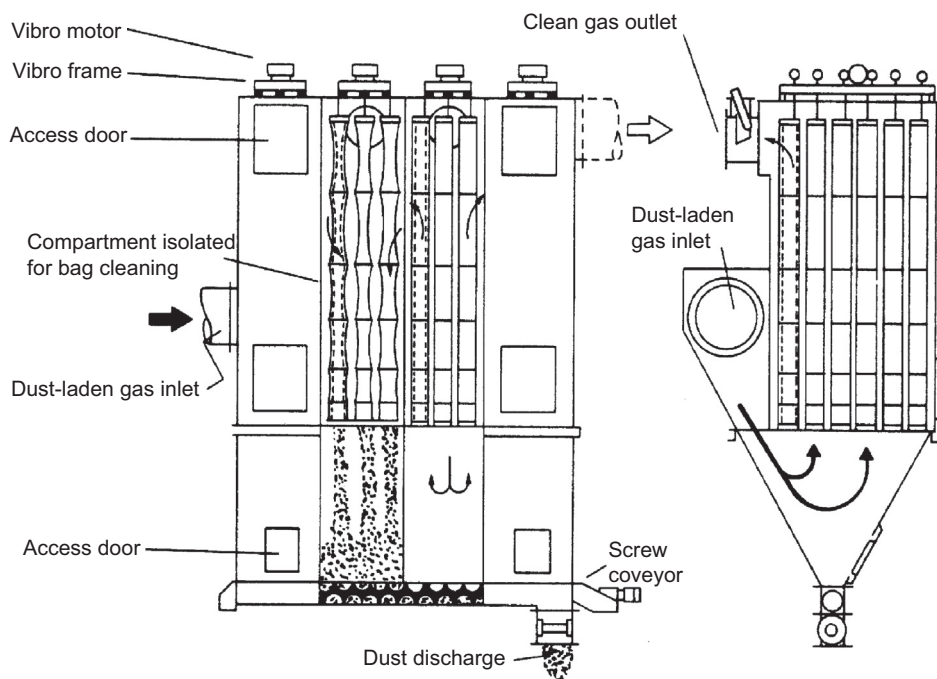


FIG. 18.37 Multicompartiment vibro bag filter.

removing solids, wet scrubbers can be used to simultaneously cool the gas and neutralize any corrosive constituents.

Spray towers, plate, and packed columns are used, as well as a variety of proprietary designs. Spray towers have a low pressure drop but are not suitable for removing very fine particles, below $10\text{ }\mu\text{m}$. The collecting efficiency can be improved by the use of plates or packing but at the expense of a higher pressure drop. A packed or plate column for wet scrubbing would typically be designed with three to five plates or height equivalent to three theoretical stages of packing.

Venturi and orifice scrubbers are simple forms of wet scrubbers. The turbulence created by the venturi or orifice is used to atomize water sprays and promote contact between the liquid droplets and dust particles. The agglomerated particles of dust and liquid are then collected in a centrifugal separator, usually a cyclone.

18.5.6 Electrostatic precipitators

Electrostatic precipitators are capable of collecting very fine particles, $<2\text{ }\mu\text{m}$, at high efficiencies; however, their capital and operating costs are high. Electrostatic precipitation should only be considered in place of alternative processes, such as filtration, where the gases are hot or corrosive.

Electrostatic precipitators are used extensively in the metallurgical, cement, and electrical power industries. Their main application is probably in the removal of the fine fly ash formed in the combustion of pulverized coal in power station boilers. The basic principle of operation is simple. The gas is ionized in passing between a high-voltage electrode and an earthed (grounded) electrode; the dust particles become charged and are attracted to the earthed electrode. The precipitated dust is removed from the electrodes mechanically, usually by vibration or by washing. Wires are normally used for the high-voltage electrode and plates or tubes for the earthed electrode. A typical design is shown in Fig. 18.38. A full description of the construction, design, and application of electrostatic precipitators is given by Schneider et al. (1975) and Parker (2002).

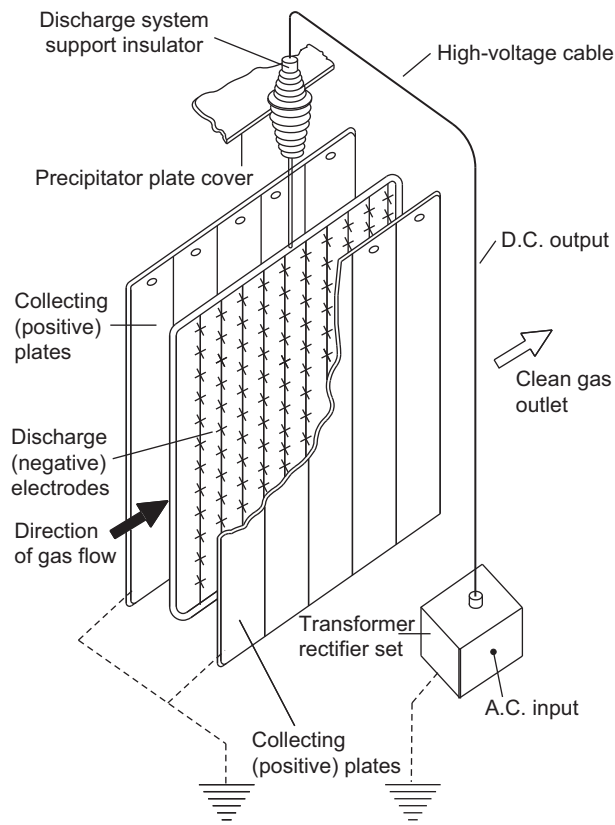


FIG. 18.38 Electrostatic precipitator.

18.6 Separation of solids from liquids

The need to separate solid and liquid phases is probably the most common phase separation requirement in the process industries, and many techniques are used (Fig. 18.39). The most suitable technique will depend on the solids concentration and feed rate, as well as the size and nature of the solid particles. The range of application of various techniques and equipment, as a function of slurry concentration and particle size, is shown in Fig. 18.40.

The choice of equipment also depends on whether the prime objective is to obtain a clear liquid or a solid product and on the degree of dryness of the solid required.

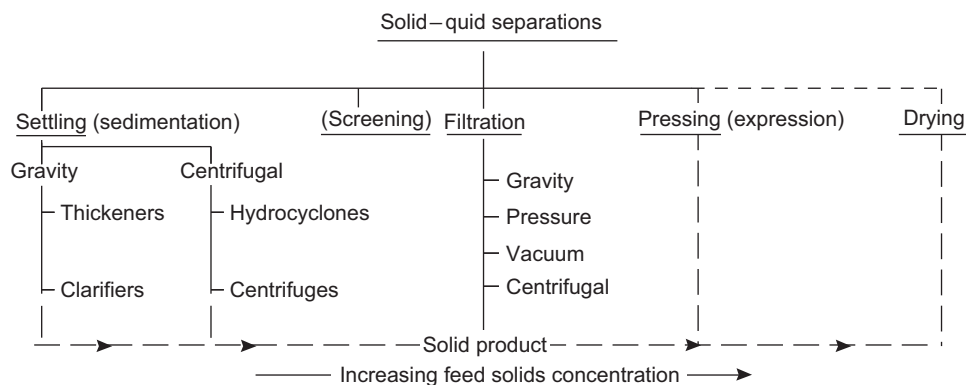


FIG. 18.39 Solid-liquid separation techniques.

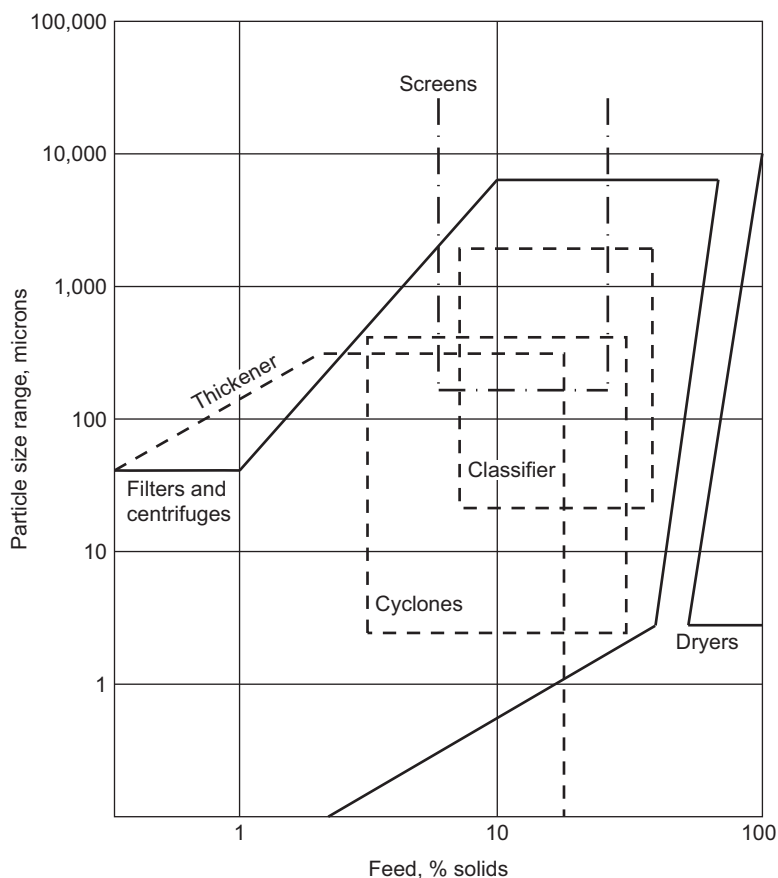


FIG. 18.40 Solid-liquid separation techniques. (From Dahlstrom & Cornell, 1971.)

The design, construction, and application of thickeners, centrifuges, and filters is a specialized subject, and firms that have expertise in these fields should be consulted when selecting and specifying equipment for new applications. Several specialist texts on the subject are available: [Svarovsky \(2001\)](#), [Rushton et al. \(2000\)](#), and [Wakeman and Tarleton \(1998\)](#). The theory of sedimentation processes and filtration is covered in [Chhabra and Gurappa \(2019\)](#).

18.6.1 Thickeners and clarifiers

Thickening and clarification are sedimentation processes, and the types of equipment used for the two techniques are similar. The primary purpose of thickening is to increase the concentration of a relatively large quantity of suspended solids, whereas that of clarifying, as the name implies, is to remove a small quantity of fine solids to produce a clear liquid effluent. Thickening and clarification are relatively cheap processes when used for the treatment of large volumes of liquid.

A thickener, or clarifier, consists essentially of a large circular tank with a rotating rake at the base. Rectangular tanks are also used, but the circular design is preferred. They can be classified according to the way the rake is supported and driven. The three basic designs are shown in [Fig. 18.41](#). Various designs of rake are used, depending on the nature of the solids.

The design and construction of thickeners and clarifiers are described by [Dahlstrom and Cornell \(1971\)](#).

Flocculating agents are often added to promote the separating performance of thickeners.

18.6.2 Filtration

In filtration processes, the solids are separated from the liquid by passing (filtering) the slurry through some form of porous filter medium. Filtration is a widely used separation process in the chemical and other process industries. Many types of equipment and filter media are used, designed to meet the needs of particular applications.

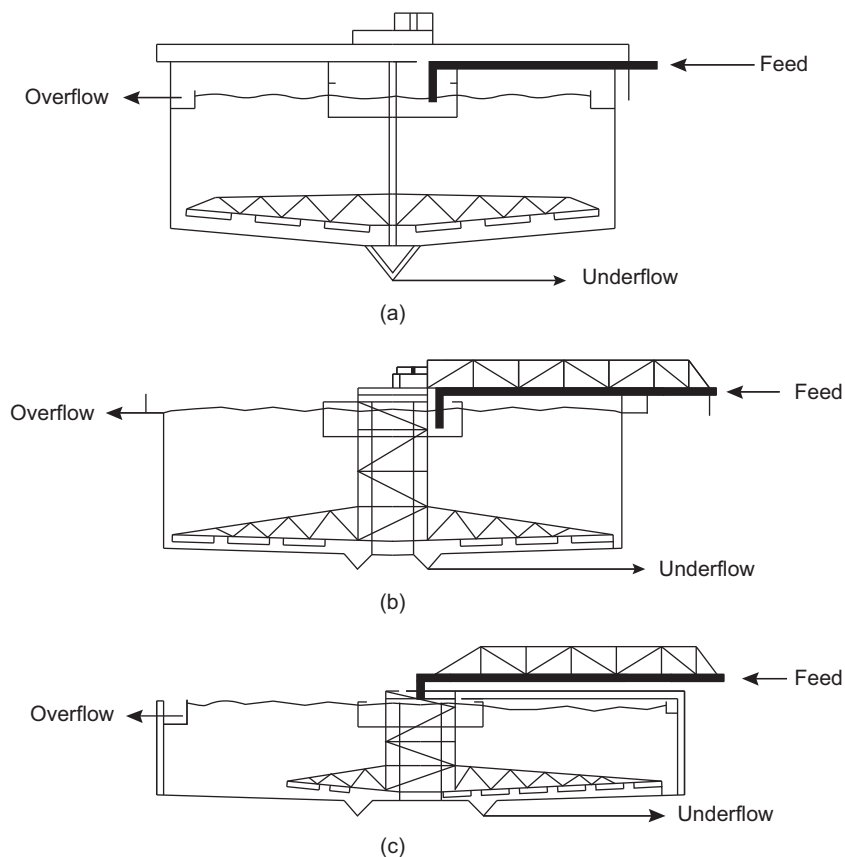


FIG. 18.41 Types of thickeners and clarifiers. (a) Bridge supported (up to <40 m dia.). (b) Center column supported (<30 m dia.). (c) Traction driven (<60 m dia.).

Descriptions of the filtration equipment used in the process industries and their fields of application can be found in various handbooks: [Green and Southard \(2018\)](#), [Sparks and Chase \(2015\)](#), and [Schweitzer \(1997\)](#) and in specialist texts on the subject: [Cheremisinoff \(1998\)](#), [Orr \(1977\)](#), [Cheryan \(2016\)](#), and [Wakeman and Tarleton \(2005\)](#). Biopharmaceutical applications are discussed in [Jornitz \(2019\)](#). A short discussion of filtration theory and descriptions of the principal types of equipment is given in [Chhabra and Gurappa \(2019\)](#).

The most commonly used filter medium is woven cloth, but many other media are also used. The main types are listed in [Table 18.7](#). A comprehensive discussion of the factors to be considered when selecting filter media is given by [Purchas \(1971\)](#) and [Mais \(1971\)](#); see also [Purchas and Sutherland \(2001\)](#). Filter aids are often used to increase the rate of filtration of difficult slurries. They are either applied as a precoat to the filter cloth or added to the slurry and deposited with the solids, assisting in the formation of a porous cake.

Industrial filters use vacuum, pressure, or centrifugal force to drive the liquid (filtrate) through the deposited cake of solids. Filtration is essentially a discontinuous process. With batch filters, such as plate and frame presses, the equipment has to be shut down to discharge the cake, and even with those filters designed for continuous operation, such as rotating drum filters and cross-flow filters, periodic stoppages are necessary to change the filter cloths. Batch filters can be coupled to continuous plant by using several units in parallel or by providing buffer storage capacity for the feed and product.

The principal factors to be considered when selecting filtration equipment are:

1. The nature of the slurry and the cake formed
2. The solids concentration in the feed
3. The throughput required
4. The nature and physical properties of the liquid: viscosity, flammability, toxicity, corrosiveness
5. Whether cake washing is required
6. The cake dryness required
7. Whether contamination of the solid by a filter aid is acceptable
8. Whether the valuable product is the solid or the liquid, or both

The overriding factor will be the filtration characteristics of the slurry: whether it is fast filtering (low specific cake resistance) or slow filtering (high specific cake resistance). The filtration characteristics can be determined by laboratory or pilot plant tests. A guide to filter selection by the slurry characteristics is given in [Table 18.8](#), which is based on a similar selection chart given by [Porter et al. \(1971\)](#).

The principal types of industrial-scale filters used are described briefly next.

Nutsche (gravity and vacuum operation)

This is the simplest type of batch filter. It consists of a tank with a perforated base, which supports the filter medium.

TABLE 18.7 Filter media

Type	Example	Minimum size particle trapped (μm)
1. Solid fabrication	Scalloped washers, wire-wound tubes	5
2. Rigid porous media	Ceramics, stoneware	1
	Sintered metal	3
3. Metal	Perforated sheets	100
	Woven wire	5
4. Porous plastics	Pads, sheets	3
	Membranes	0.005
5. Woven fabrics	Natural and synthetic fiber cloths	10
6. Nonwoven fabrics	Felts, lap	10
	Paper, cellulose	5
7. Cartridges	Yarn-wound spools, graded fibers	2
8. Loose solids	Fibers, asbestos, cellulose	Submicron

TABLE 18.8 Guide to filter selection

Slurry characteristics	Fast filtering	Medium filtering	Slow filtering	Dilute	Very dilute
Cake formation rate	cm/s	mm/s	0.02–0.12 mm/s	0.02 mm/s	No cake
Normal concentration	>20%	10–20%	1–10%	<5%	<0.1%
Settling rate	Very fast	Fast	Slow	Slow	—
Leaf test rate, kg/h m ²	>2500	250–2500	25–250	<25	—
Filtrate rate, m ³ /h m ²	>10	5–10	0.02–0.05	0.02–5	0.02–5
<i>Filter application</i>					
Continuous vacuum filters					
Multicompartiment drum					
Single compartment drum					
Top feed drum					
Scroll discharge drum					
Tilting pan					
Belt					
Disc					
Batch vacuum leaf					
Batch nutsche					
Batch pressure filters					
Plate and frame					
Vertical leaf					
Horizontal plate					
Cartridge edge					

Plate and frame press (pressure operation) (Fig. 18.42)

These are the oldest and most commonly used batch filters. The equipment is versatile, made in a variety of materials, and capable of handling viscous liquids and cakes with a high specific resistance.

Leaf filters (pressure and vacuum operation)

Various types of leaf filters are used, with the leaves arranged in horizontal or vertical rows. The leaves consist of metal frames over which filter cloths are draped. The cake is removed either mechanically or by sluicing it off with jets of water. Leaf filters are used for similar applications as plate and frame presses, but generally have lower operating costs.

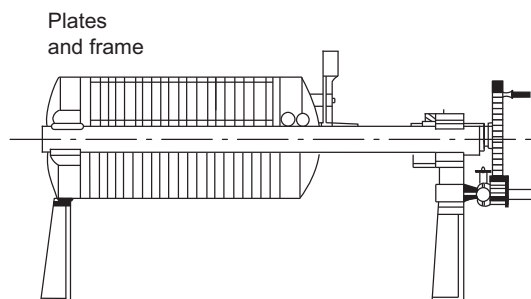


FIG. 18.42 Plate and frame filter press.

Rotary drum filters (usually vacuum operation) (Fig. 18.43)

A drum filter consists essentially of a large hollow drum around which the filter medium is fitted. The drum is partially submerged in a trough of slurry, and the filtrate is sucked through the filter medium by vacuum inside the drum. Wash water can be sprayed on to the drum surface, and multicompartment drums are used so that the wash water can be kept separate from the filtrate. A variety of methods are used to remove the cake from the drum: knives, strings, air jets, and wires. Rotating drum filters are essentially continuous in operation. They can handle large throughputs and are widely used for free-filtering slurries.

Disc filters (pressure and vacuum operation)

Disc filters are similar in principle to rotary filters, but consist of several thin discs mounted on a shaft, in place of the drum. This gives a larger effective filtering area on a given floor area, and vacuum disc filters are used in preference to drum filters where space is restricted. At sizes above approximately 25 m² filtration area, disc filters are cheaper, but their applications are more restricted, as they are not as suitable for the application of wash water or precoating.

Belt filters (vacuum operation) (Fig. 18.44)

A belt filter consists of an endless reinforced rubber belt with drainage holes along its center, which supports the filter medium. The belt passes over a stationary suction box into which the filtrate is sucked. Slurry and wash water are sprayed on to the top of the belt.

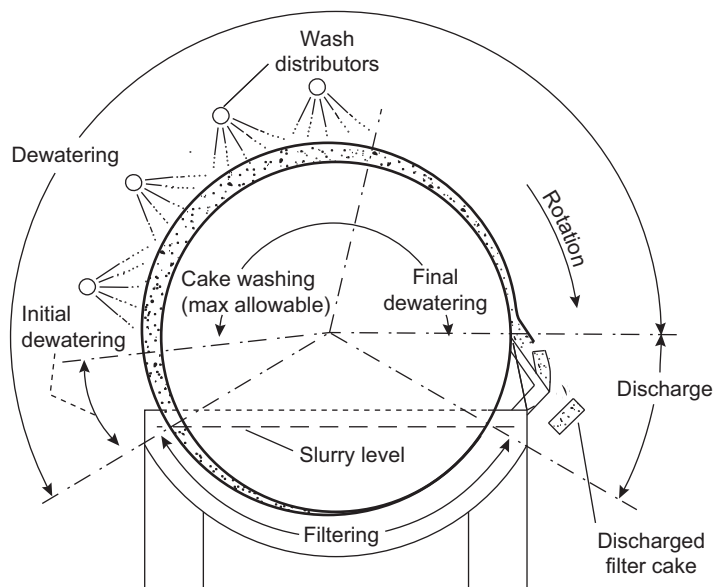


FIG. 18.43 Drum filter.

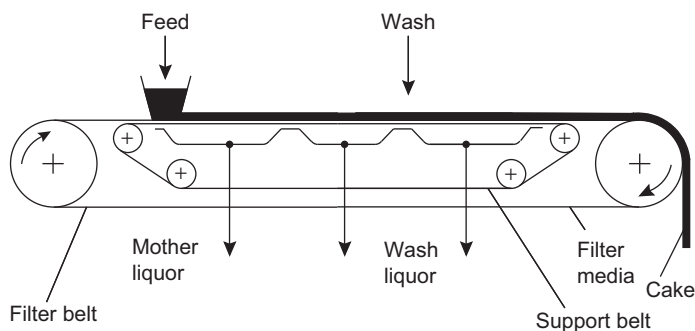


FIG. 18.44 Belt filter.

Horizontal pan filters (vacuum operation) (Fig. 18.45)

This type is similar in operation to a vacuum Nutsche filter. It consists of shallow pans with perforated bases, which support the filter medium. By arranging a series of pans around the circumference of a rotating wheel, the operation of filtering, washing, drying, and discharging can be made automatic.

Centrifugal filters

Centrifugal filters use centrifugal force to drive the filtrate through the filter cake. The equipment used is described in the next section.

Cross-flow filters

Cross-flow filters are used to reject liquid from a slurry, similar to thickeners. The filter is arranged as tubular modules and is often a porous membrane; see Section 16.5.4. The slurry is usually fed through the tubes and clear liquid is withdrawn through the tube wall as filtrate, as illustrated in Fig. 18.46. The flow on the inside of the tube prevents the accumulation of solids, and the concentrated slurry can then be sent for further processing.

Cross-flow filtration for solvent rejection is often carried out in a “feed and bleed” flow scheme, shown in Fig. 18.47. A portion of the retentate from the filtration unit is recycled to the feed. This flow scheme allows higher velocities to be used on the retentate side, and hence higher concentrations can be reached without fouling the filter surface.

Cross-flow filtration processes such as microfiltration and ultrafiltration are widely used in food processing, for example, in concentrating orange juice, and also in the recovery of products from fermentation processes. See Harrison (2014) and Gabelman (2017) for examples.

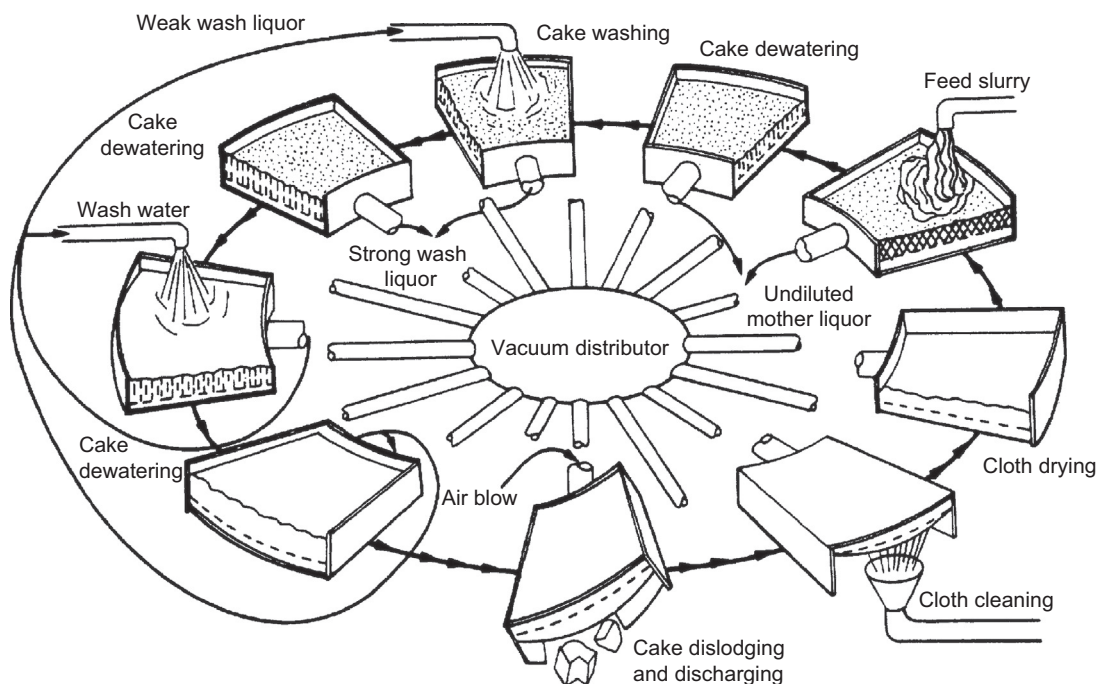


FIG. 18.45 Pan filters.

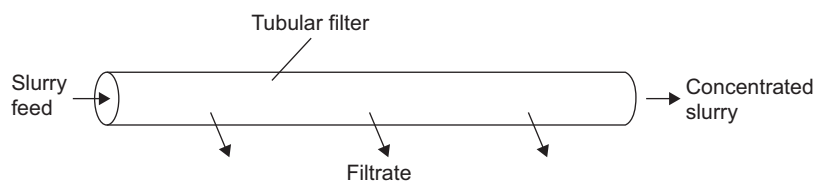


FIG. 18.46 Cross-flow filtration.

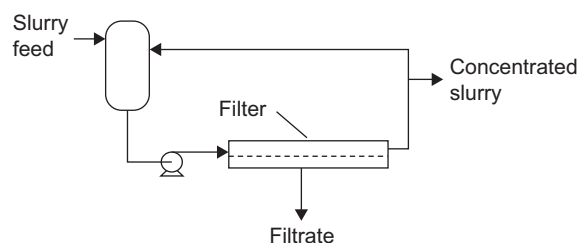


FIG. 18.47 Feed and bleed filtration.

18.6.3 Centrifuges

Centrifuges are classified according to the mechanism used for solids separation:

1. Sedimentation centrifuges, in which the separation is dependent on a difference in density between the solid and liquid phases (solid heavier).
2. Filtration centrifuges, which separate the phases by filtration. The walls of the centrifuge basket are porous, and the liquid filters through the deposited cake of solids and is removed.

The choice between a sedimentation or filtration centrifuge for a particular application will depend on the nature of the feed and the product requirements. The main factors to be considered are summarized in Table 18.9. As a general rule, sedimentation centrifuges are used when a clarified liquid product is required, and filtration centrifuges to produce a pure, dry, solid product.

A variety of centrifugal filter and sedimenter designs are used. The main types are listed in Table 18.10. They can be classified by a number of design and operating features, such as:

1. Mode of operation — batch or continuous
2. Orientation of the bowl/basket — horizontal or vertical
3. Position of the suspension and drive — overhung or underhung
4. Type of bowl — solid, perforated basket, disc bowl
5. Method of solids cake removal
6. Method of liquid removal

Descriptions of the various types of centrifuges and their fields of application can be found in various handbooks; in a book by Leung (1998); and in articles by Ambler (1971), Linley (1984), and Patnaik (2012).

The fields of application of each type, classified by the size range of the solid particles separated, are given in Fig. 18.48. A similar selection chart is given by Schroeder (1998).

Sedimentation centrifuges

There are four main types of sedimentation centrifuges:

TABLE 18.9 Selection of sedimentation or filter centrifuge

Factor	Sedimentation	Filtration
Solids size, fine		x
Solids size, >150 μm	x	
Compressible cakes	x	
Open cakes		x
Dry cake required		x
High filtrate clarity	x	
Crystal breakage problems		x
Pressure operation		
High-temperature	will depend on the type of operation centrifuge used	

TABLE 18.10 Centrifuge types

Sedimentation	Filtration-fixed bed
Laboratory	Vertical basket
Bottle	Manual discharge
Ultra	Bag discharge
	Knife discharge
Tubular bowl	Horizontal basket
	Inclined basket
Disc	
Batch bowl	
Nozzle discharge	
Valve discharge	Filtration-moving bed
Opening bowl	
Imperforate basket	Conical bowl
Manual discharge	Wide angle
Skimmer discharge	Vibrating
	Torsional
	Tumbling
Scroll discharge	Scroll discharge
Horizontal	
Cantilevered	Cylindrical bowl
Vertical	Scroll discharge
Screen bowl	Pusher

From Sutherland (1970)

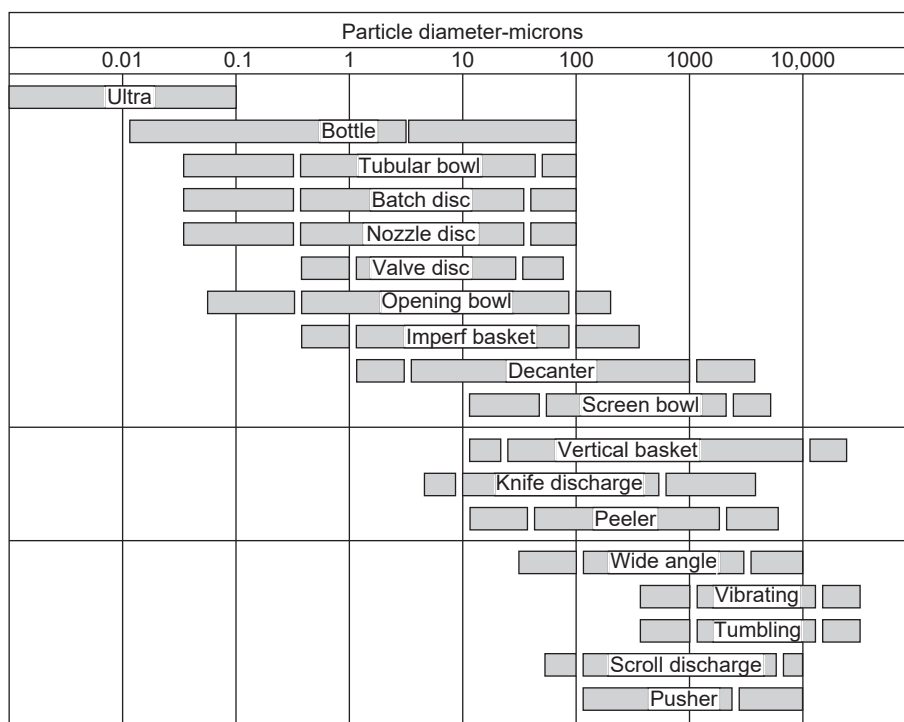


FIG. 18.48 Classification of centrifuges by particle size. (From Sutherland, K. S. [1970]. *How to specify a centrifuge*. Chemical Processing, 16[May], 10.)

1. Tubular bowl (Fig. 18.49)

High-speed, vertical-axis, tubular bowl centrifuges are used for the separation of immiscible liquids, such as water and oil, and for the separation of fine solids. The bowl is driven at speeds of around 15,000 rpm (250 Hz), and the centrifugal force generated exceeds 130,000 N.

2. Disc bowl (Fig. 18.50)

The conical discs in a disc bowl centrifuge split the liquid flow into a number of very thin layers, which greatly increases the separating efficiency. Disc bowl centrifuges are used for separating liquids and fine solids and for solids classification.

3. Scroll discharge

In this type of machine, the solids deposited on the wall of the bowl are removed by a scroll (a helical screw conveyer), which revolves at a slightly different speed from the bowl. Scroll discharge centrifuges can be designed so that solids can be washed and relatively dry solids can be discharged.

4. Solid bowl batch centrifuge

The simplest type; similar to the tubular bowl machine, but with a smaller bowl length-to-diameter ratio (less than 0.75).

The tubular bowl type is rarely used for solids concentrations above 1% by volume. For concentrations between 1% and 15%, any of the other three types can be used. Above 15%, either the scroll discharge type or the batch type may be used, depending on whether continuous or intermittent operation is required.

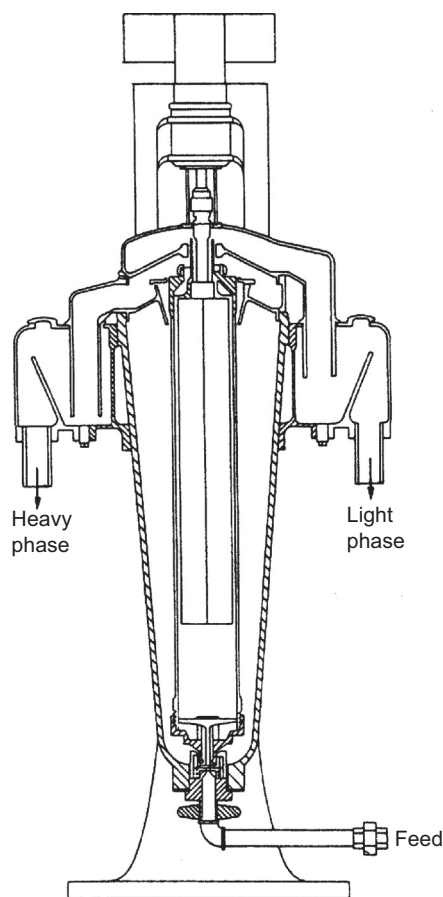


FIG. 18.49 Tubular bowl centrifuge.

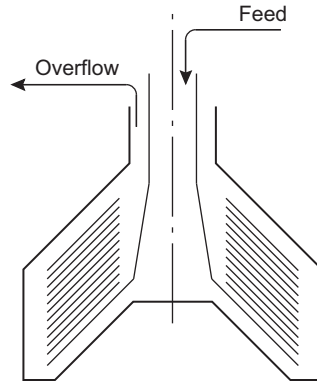


FIG. 18.50 Disc bowl centrifuge.

Sigma theory for sedimentation centrifuges

The basic equations describing sedimentation in a centrifugal field are given in Chhabra and Gurappa (2019). In that discussion the term *sigma* (Σ) is introduced, which can be used to define the performance of a centrifuge independently of the physical properties of the solid–fluid system. The sigma value of a centrifuge, normally expressed in cm^2 , is equal to the cross-sectional area of a gravity settling tank having the same clarifying capacity.

This approach to describing centrifuge performance has become known as the “sigma theory.” It provides a means for comparing the performance of sedimentation centrifuges and for scaling up from laboratory and pilot-scale tests; see Ambler (1952) and Trowbridge (1962).

In the general case, it can be shown that:

$$Q = 2u_g\Sigma \quad (18.25)$$

and (where Stokes law applies)

$$u_g = \frac{\Delta\rho d_s^2 g}{18\mu} \quad (18.26)$$

Note: The factor of 2 is included in Equation 18.25, as d_s is the *cut-off* size; 50% of particles of this size will be removed in passage through the centrifuge.

where Q = volumetric flow of liquid through the centrifuge, m^3/s

u_g = terminal velocity of the solid particle settling under gravity through the liquid, m/s

Σ = sigma value of the centrifuge, m^2

$\Delta\rho$ = density difference between solid and liquid, kg/m^3

d_s = the diameter of the solid particle, the *cut-off* size, m ($\mu\text{m} \times 10^{-6}$)

μ = viscosity of the liquid, Nm^{-2}s

g = gravitational acceleration, $9.81 \text{ m}/\text{s}^2$

Morris (1966) gives a method for the selection of the appropriate type of sedimentation centrifuge for a particular application based on the ratio of the liquid overflow to sigma value (Q/Σ). His values for the operating range of each type, and their approximate efficiency rating, are given in Table 18.11. The efficiency term is used to account for the different amounts by which the various designs differ from the theoretical sigma values given by Equation 18.25. Sigma values depend solely on the geometrical configuration and speed of the centrifuge. Details of the calculation for various types are given by Ambler (1952). To use Table 18.11, it is necessary to know the feed rate of slurry (and hence the liquid overflow Q), the density of the liquid and solid, the liquid viscosity, and the diameter of the particle for, say, a 98% size removal. The use of Table 18.11 is illustrated in Example 18.3.

TABLE 18.11 Selection of sedimentation centrifuges

Type	Approximate efficiency (%)	Normal operating range Q , m^3/h at Q/Σ m/s
Tubular bowl	90	0.4 at 5×10^{-8} to 4 at 3.5×10^{-7}
Disc	45	0.1 at 7×10^{-8} to 110 at 4.5×10^{-7}
Solid bowl (scroll discharge)	60	0.7 at 1.5×10^{-6} to 15 at 1.5×10^{-5}
Solid bowl (basket)	75	0.4 at 5×10^{-6} to 4 at 1.5×10^{-4}

A selection guide for sedimentation centrifuges by Lavanchy et al. (1964), which includes other types of solid–liquid separators, is shown in Fig. 18.51, adapted to SI units.

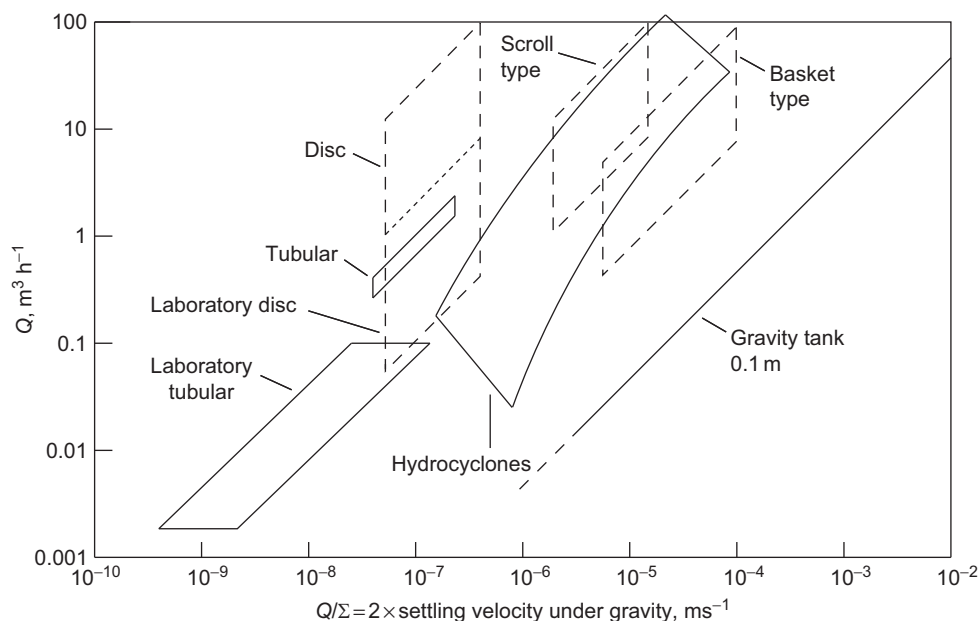


FIG. 18.51 Performance of sedimentation equipment. (From Lavanchy, A. C., Keith, F. W., & Beams, J. W. [1964]. *Centrifugal separation*. In Kirk-Othmer encyclopedia of chemical technology, 2nd ed. Interscience.)

Example 18.3

A precipitate is to be continuously separated from a slurry. The solids concentration is 5% and the slurry feed rate $5.5 \text{ m}^3/\text{h}$. The relevant physical properties at the system operating temperature are:

liquid density 1050 kg/m^3 , viscosity 4 cP (mNm^{-2}s),

solid density 2300 kg/m^3 , "cut-off" particle size $10 \mu\text{m} = 10 \times 10^{-6} \text{ m}$.

Solution

$$\text{Overflow rate, } Q = 0.95 \times 5.5 = 5.23 \text{ m}^3/\text{h}$$

$$= \frac{5.13}{3600} = 1.45 \times 10^{-3} \text{ m}^3/\text{s}$$

$$\Delta\rho = 2300 - 1050 = 1250 \text{ kg/m}^3$$

From Equations 18.25 and 18.26:

$$\frac{Q}{\Sigma} = 2 \times \frac{1250(10 \times 10^{-6})^2}{18 \times 4 \times 10^{-3}} \times 9.81 = 3.4 \times 10^{-5}$$

From Table 18.11, for a Q of $5.23 \text{ m}^3/\text{h}$ at a Q/Σ of 3.4×10^{-5} , a solid bowl basket type should be used.

To obtain an idea of the size of the machine needed, the sigma value can be calculated using the efficiency value from Table 18.11.

From Equation 18.25:

$$\Sigma = \frac{Q}{\text{eff.} \times 2u_g} = \frac{1.45 \times 10^{-3}}{0.75 \times 3.4 \times 10^{-5}} = \underline{56.9 \text{ m}^2}$$

The sigma value is the equivalent area of a gravity settler that would perform the same separation as the centrifuge. Once the sigma value is known, a suitable machine can be selected by reviewing performance data on manufacturers' websites.

Filtration centrifuges (centrifugal filters)

It is convenient to classify centrifugal filters into two broad classes, depending on how the solids are removed: fixed bed or moving bed.

In the fixed-bed type, the cake of solids remains on the walls of the bowl until removed either manually or automatically by means of a knife mechanism. It is essentially cyclic in operation. In the moving-bed type, the mass of solids is moved along the bowl by the action of a scroll (similar to the solid-bowl sedimentation type), or by a ram (pusher type), or by a vibration mechanism, or by the bowl angle. Washing and drying zones can be incorporated into the moving-bed type.

Bradley (1965) has grouped the various types into the family tree shown in Fig. 18.52.

Schematic diagrams of the various types are shown in Fig. 18.53. The simplest machines are the basket types (Fig. 18.53a, b, and c), and these form the basic design from which the other types have been developed (Fig. 18.53d to o).

The various arrangements of knife mechanisms used for automatic removal of the cake are shown in Fig. 18.53(d) to (h). The bottom discharge-type machines (Fig. 18.53d and e) can be designed for variable-speed automatic

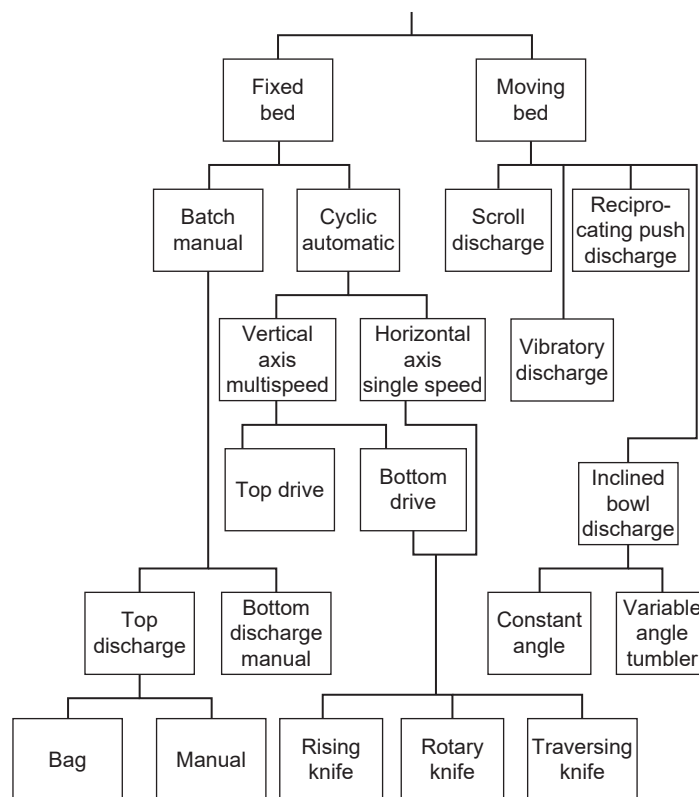


FIG. 18.52 Filtration centrifuge family tree. (From Bradley, D. [1965]. *Medium-speed centrifuges*. Chemical and Process Engineering, 595.)

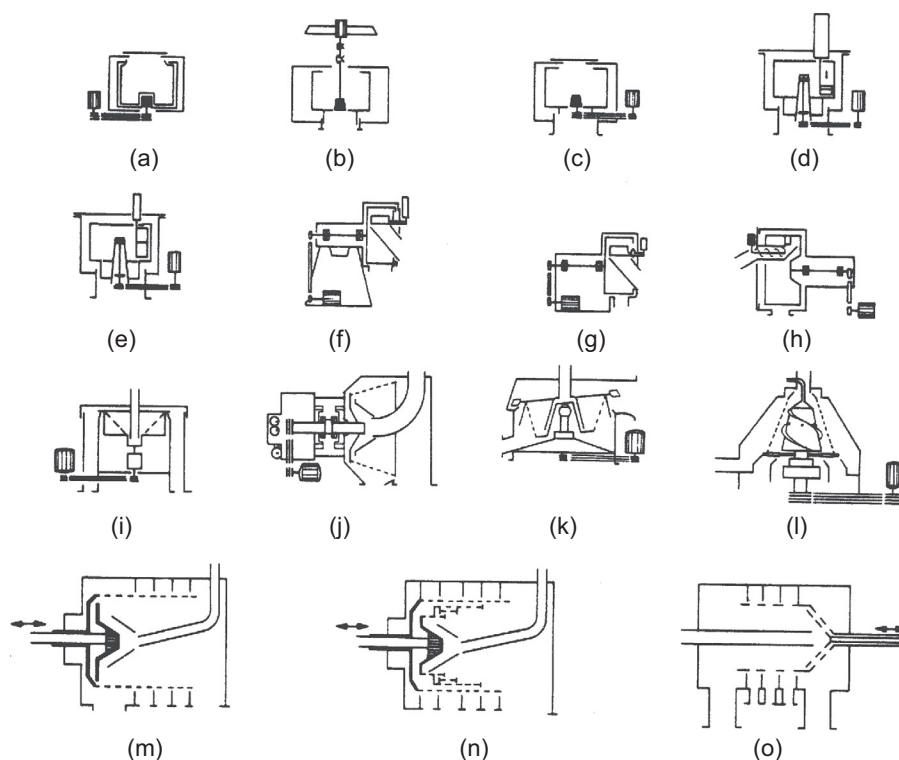


FIG. 18.53 Schematic diagrams of filtration centrifuge types. (a) Bottom drive batch basket with bag. (b) Top drive bottom discharge batch basket. (c) Bottom drive bottom discharge batch basket. (d) Bottom drive automatic basket, rising knife. (e) Bottom drive automatic basket, rotary knife. (f) Single reversing knife rising knife. (g) Single-speed automatic rotary knife. (h) Single-speed automatic traversing knife. (i) Inclined wall self-discharge. (j) Inclined vibrating wall self-discharge. (k) Inclined "tumbling" wall self-discharge. (l) Inclined wall scroll discharge. (m) Traditional single-stage pusher. (n) Traditional multistage pusher. (o) Conical pusher with de-watering cone. (From Bradley, 1965.)

discharge and are suitable for use with fragile or plate or needle-shaped crystals, where it is desirable to avoid breakage or compaction of the bed. They can be loaded and discharged at low speeds, which reduces breakage and compaction of the cake. The single-speed machines (Fig. 18.53f, g, and h) are used where cakes are thin and short cycle times are required. They can be designed for high-temperature and high-pressure operation. When continuous operation is required, the scroll, pusher, or other self-discharge types are used (Fig. 18.53i to o). The scroll discharge centrifuge is a low-cost, flexible machine, capable of a wide range of applications but is not suitable for handling fragile materials. It is normally used for coarse particles, where some contamination of the filtrate with fines can be tolerated.

The capacity of filtration centrifuges is highly dependent on the solids concentration in the feed. For example, at 10% feed slurry concentration, 9 kg of liquid will be centrifuged for every 1 kg of solids separated, whereas with a 50% solids concentration, the quantity will be less than 1 kg. For dilute slurries it is well worth considering using some form of preconcentration, such as gravity sedimentation, hydrocyclones, or cross-flow filtration.

18.6.4 Hydrocyclones (liquid cyclones)

Hydrocyclones are used for solid–liquid separations as well as for solids classification and liquid–liquid separation. A hydrocyclone is a centrifugal device with a stationary wall, with the centrifugal force being generated by the liquid motion. The operating principle is basically the same as that of the gas cyclone described in Section 18.5.3. Hydrocyclones are simple, robust separating devices that can be used over the particle size range from 4 to 500 μm . They are often used in groups, as illustrated in Fig. 18.54. The design and application of hydrocyclones is discussed fully in books by Abulnaga (2002) and Svarovsky and Thew (1992). Design methods and charts are also given by Zanker (1977), Day et al. (1997), and Moir (1985).

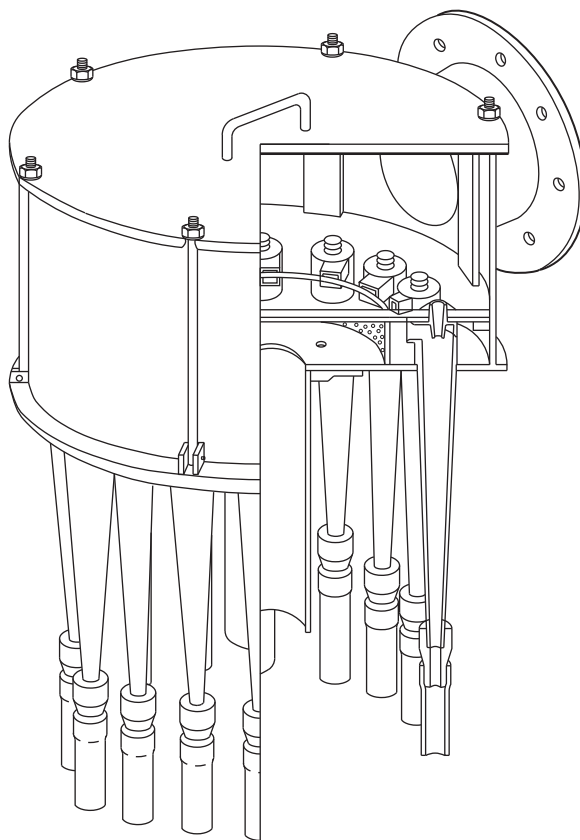


FIG. 18.54 A "clog" assembly of 16×2 in (50 mm) diameter hydrocyclone. (Courtesy of Richard Mozley Ltd.)

The nomographs by Zanker can be used to make a preliminary estimate of the size of cyclone needed. The specialist manufacturers of hydrocyclone equipment should be consulted to determine the best arrangements and design for a particular application.

Zanker's method is outlined here and illustrated in Example 18.4. Fig. 18.56 is based on an empirical equation by Bradley (1960):

$$d_{50} = 4.5 \left[\frac{D_c^3 \mu}{L^{1.2} (\rho_s - \rho_L)} \right] \quad (18.27)$$

where d_{50} = the particle diameter for which the cyclone is 50% efficient, μm

D_c = diameter of the cyclone chamber, cm

μ = liquid viscosity, centipoise (mN s/m^2)

L = feed flow rate, l/min

ρ_L = density of the liquid, g/cm^3

ρ_s = density of the solid, g/cm^3

The equation gives the chamber diameter required to separate the so-called d_{50} particle diameter as a function of the slurry flow rate and the liquid and solid physical properties. The d_{50} particle diameter is the diameter of the particle, 50% of which will appear in the overflow and 50% in the underflow. The separating efficiency for other particles is related to the d_{50} diameter by Fig. 18.55, which is based on a formula by Bennett (1936).

$$\eta = 100 \left[1 - e^{-(d/d_{50} - 0.115)^3} \right] \quad (18.28)$$

where η = the efficiency of the cyclone in separating any particle of diameter d , percent

d = the selected particle diameter, μm

The method applies to hydrocyclones with the proportions shown in Fig. 18.57.

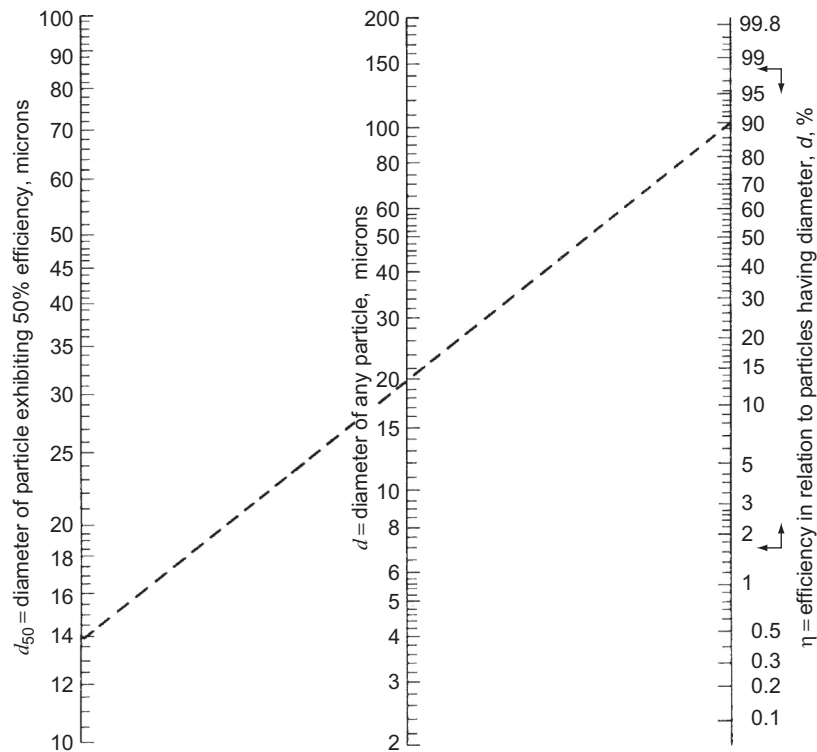


FIG. 18.55 Determination of d_{50} from the desired particle separation (Equation 18.27, Example 18.4). (From Zanker, 1977.)

Example 18.4

Estimate the size of hydrocyclone needed to separate 90% of particles with a diameter greater than $20\text{ }\mu\text{m}$ from $10\text{ m}^3/\text{h}$ of a dilute slurry.

Physical properties: solid density 2000 kg/m^3 , liquid density 1000 kg/m^3 , viscosity 1 mN s/m^2 .

Solution

$$\text{Flow rate} = \frac{10 \times 10^3}{60} = 1.6671/\text{min}$$

$$(\rho_s - \rho_L) = 2.0 - 1.0 = 1.0\text{ g/cm}^3$$

From Fig. 18.55, for 90% removal of particles above $20\text{ }\mu\text{m}$:

$$d_{50} = 14\text{ }\mu\text{m}$$

From Fig. 18.56, for $\mu = 1\text{ mN s/m}^2$, $(\rho_s - \rho_L) = 1.0\text{ g/cm}^3$, $L = 167/\text{min}$

$$D_c = \underline{\underline{16\text{ cm}}}$$

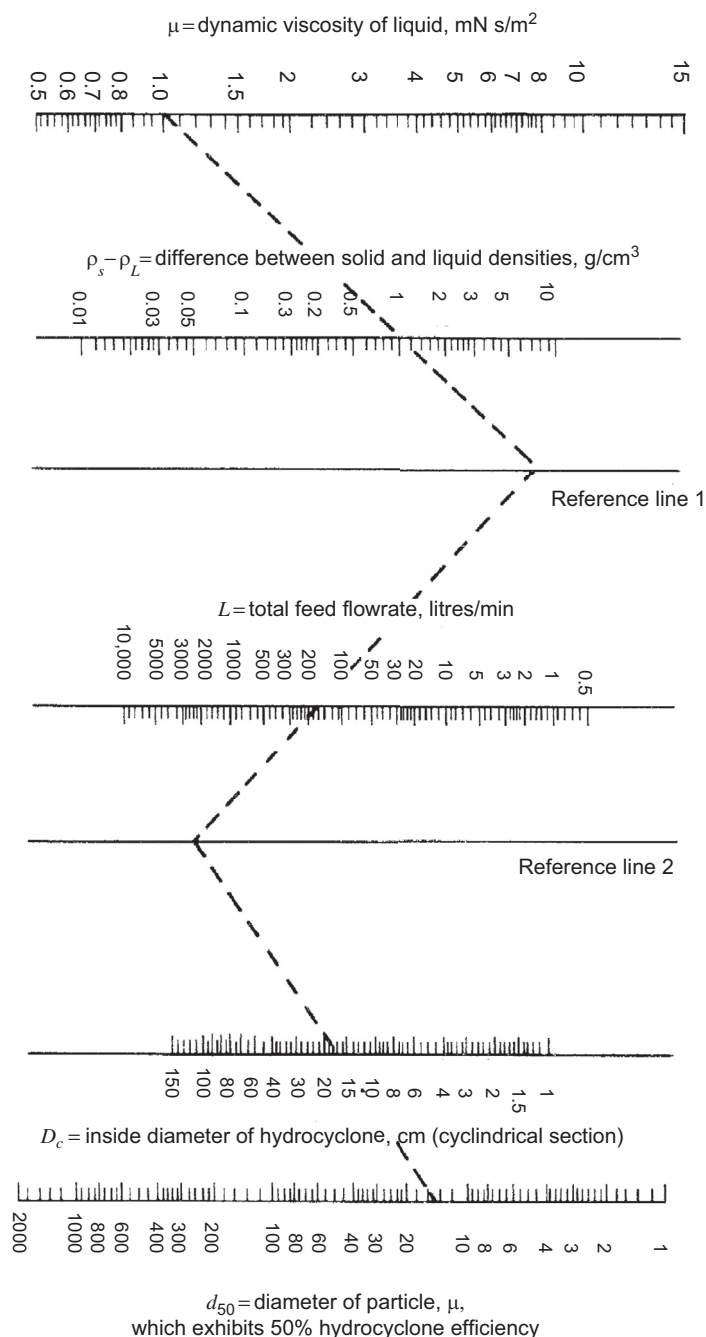


FIG. 18.56 Chamber dia. D_c from flow rate, physical properties, and d_{50} particle size (Equation 18.28, Example 18.4). (From Zanker, 1977.)

8.6.5 Pressing (expression)

Pressing, in which the liquid is squeezed (expressed) from a mass of solids by compression, is used for certain specialized applications. Pressing consumes a great deal of energy and should not be used unless no other separating technique is suitable; however, in some applications dewatering by pressing can be competitive with drying.

Presses are of two basic types: hydraulic batch presses and screw presses. Hydraulic presses are used for extracting fruit juices, and screw presses for dewatering materials such as paper pulp, garbage, and manure. The equipment used is described in [Green and Southard \(2018\)](#).

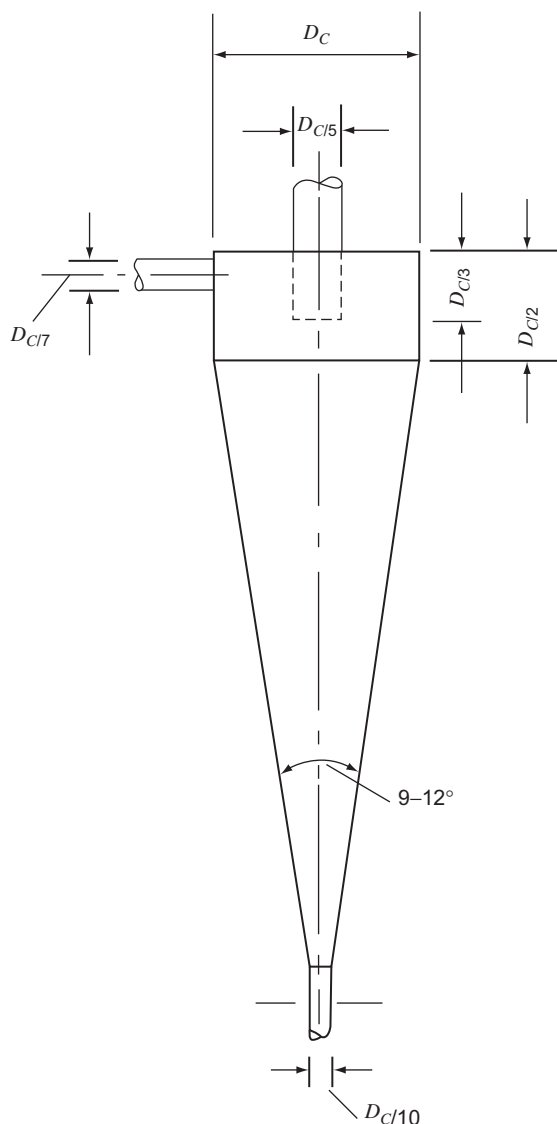


FIG. 18.57 Hydrocyclone-typical proportions.

18.7 Separation of liquids from solids (drying)

Drying is the removal of water, or other volatile liquids, by evaporation. Most solid materials require drying at some stage in their production. The choice of suitable drying equipment often cannot be separated from the selection of the upstream equipment feeding the drying stage.

The overriding consideration in the selection of drying equipment is the nature and concentration of the feed. Drying is an energy-intensive process, and the removal of liquid by thermal drying will be more costly than by mechanical separation techniques.

Drying equipment can be classified according to the following design and operating features:

1. Batch or continuous
2. Physical state of the feed: liquid, slurry, wet solid
3. Method of conveyance of the solid: belt, rotary, fluidized
4. Heating system: conduction, convection, radiation

Hot air is usually used as the heating and mass transfer medium in industrial dryers unless there are concerns about solvent flammability, in which case nitrogen, depleted air, or recirculating flue gas are used. The air may be directly

heated by firing a burner into the air stream (using oil, gas, or coal as fuel) or indirectly heated, usually by banks of steam-heated finned tubes. The heated air is usually propelled through the dryer by electrically driven fans.

Table 18.12, adapted from a similar selection guide by Parker (1963), shows the basic features of the various types of solids dryers used in the process industries; Table 18.13, by Williams-Gardner (1965), shows typical applications.

Batch dryers are normally used for small-scale production and when the drying cycle is likely to be long. Continuous dryers require less labor and less floor space and produce a more uniform quality product.

It is important to present the material to the dryer in a form that will produce a bed of solids with an open, porous structure. For pastes and slurries, some form of pretreatment equipment will normally be needed, such as extrusion or granulation.

Liquids can also be extracted from solids by leaching; see Section 16.5.6.

TABLE 18.12 Dryer selection

Mode of operation	Generic type	Feed condition			Specific dryer types	Jacketed	Suitable for heat-sensitive materials	Suitable for vacuum service	Retention or cycle time	Heat transfer method	Capacity	Typical evaporation capacity
		1	2	3								
Batch	Stationary		↔		1. Shelf							
			↔		2. Cabinet	Yes	Yes	Yes	6.48 h	Radiant and conduction	Limited	0.15–1.0
			↔		3. Compartment							
		↔			Truck	No	Yes	No	6.48 h	Convection	Limited	0.15–1.0
		↔			1. Kettle	Yes	No	Yes	3.12 h	Conduction	Limited	1.5–15
		↔			2. Pan							
Continuous	Drum		↔		Rotary shell	Yes	Yes	Yes	4.48 h	Conduction	Limited	0.5–12
			↔		Rotary internal	Yes	Yes	Yes	4.48 h	Conduction	Limited	0.5–12
			↔		Double cone	Yes	Yes	Yes	3.12 h	Conduction	Limited	0.5–12
		↔			1. Single drum							
		↔			2. Double drum	No	Yes	Yes	Very short	Conduction	Medium	5–50
		↔			3. Twin drum							
	Rotary		↔		Rotary direct heat	No	No	No	Long	Convection	High	3–110
			↔		Rotary, indirect heat	No	No	No	Long	Conduction	Medium	15–200
			↔		Rotary, steam tube	No	Depends on material	No	Long	Conduction	High	15–200
			↔		Rotary, direct-indirect heat	No	No	No	Long	Conduction	High	50–150
			↔		Louver	No	Depends on material	No	Long	Convection	High	5–240
			↔		Tunnel belt, screen	No	Yes	No	Long	Convection	Medium	1.5–35
	Conveyor		↔		Rotary shelf	Yes	Depends on material	No	Medium	Conduction	Medium	0.5–10
			↔		Trough	Yes	Depends on material	Yes	Varies	Conduction	Medium	0.5–15
			↔		Vibrating	Yes	Depends on material	No	Medium	Convection	Medium	0.5–100
			↔		Turbo	No	Depends on material	No	Medium	Conduction	Medium	1–10
	Suspended particle	↔			Spray	No	Yes	No	Short	Convection	High	1.5–50
			↔		Flash	No	Yes	No	Short	Convection	High	–
			↔		Fluid bed	No	Yes	No	Short	Convection	Medium	–

↔ = applicable to feed conditions noted

Key to feed conditions:

1. Solutions, colloidal suspensions and emulsions, pumpable solids suspensions, pastes and sludges.
2. Free-flowing powders, granular, crystalline or fibrous solids that can withstand mechanical handling.
3. Solids incapable of withstanding mechanical handling.

TABLE 18.13 Dryer applications

Dryer type	System	Feed form	Typical products
Batch ovens	Forced convection	Paste, granules, extrude cake	Pigment dyestuffs, pharmaceuticals, fibers
	Vacuum	Extrude cake	Pharmaceuticals
" pan (agitated)	Atmospheric and vacuum	Crystals, granules, powders	Fine chemicals, food products
" rotary	Vacuum	Crystals, granules solvent recovery	Pharmaceuticals
" fluid bed	Forced convection	Granular, crystals	Fine chemicals, pharmaceuticals, plastics
" infrared	Radiant	Components sheets	Metal products, plastics
Continuous rotary	Convection Direct/indirect Direct Indirect Conduction	Crystals, coarse powders, extrudes preformed cake lumps, granular paste and fillers cakes back-mixed with dry product	Chemical ores, food products, clays, pigments, chemicals Carbon black
" film drum	Conduction	Liquids, suspensions	Foodstuffs, pigment
" trough	Conduction		Ceramics, adhesives
" spray	Convection	Liquids, suspensions	Foodstuffs, pharmaceuticals, ceramics, fine chemicals, detergents, organic extracts
" band	Convection	Preformed solids	Foodstuffs, pigments, chemicals, rubber, clays, ores, textiles
" fluid bed	Convection	Preformed solids granules, crystals	Ores, coal, clays, chemicals
" pneumatic	Convection	Preformed pastes, granules, crystals, coarse products	Chemicals, starch, flour, resins, wood-products, food products
" infra-red	Convection	Components sheets	Metal products, moulded fiber articles, painted surfaces

18.7.1 Theory of drying

Drying involves a combination of heat transfer and mass transfer. Heat is transferred to the surface of the solid to provide the heat of vaporization. The liquid evaporates and diffuses away from the surface into the bulk of the drying gas. Most dryers use a flowing gas to ensure that there is always an adequate partial pressure driving force for evaporation. Convective heat transfer is important in all dryers, although some designs also use conduction and radiation to increase the heating rate.

Much of the terminology used in drying refers to air–water systems, as these are widely encountered in the food, paper, textile, and minerals-processing industries. The same concepts can be extended to other liquids or solvents and to other drying gases.

The amount of drying that can be achieved depends on the nature of the solid and the moisture (or solvent) content of the inlet air (or gas). Nonporous solids such as sand can be dried to near-zero moisture regardless of the inlet gas condition. Porous solids, cellular and fibrous materials, and hygroscopic solids will reach a moisture content that is in equilibrium with the humidity of the entering gas. Consequently, to reach the desired moisture content, the designer will have to select the correct combination of inlet gas humidity, inlet gas temperature, and dryer residence time. In the case of hydrated salts that carry water of crystallization, high temperatures may be needed to drive off the final moisture.

The moisture content of the solid phase is usually expressed as mass of moisture per mass of bone-dry solids. The residual moisture remaining when the solids reach equilibrium with the entering gas is referred to as the equilibrium moisture content. When a wet solid is dried experimentally, a curve like that shown in Fig. 18.58 is obtained. The rate of drying initially increases during a warm-up period, marked A to B in the figure. A constant-rate period is then observed, shown as B to C, during which free moisture is evaporated from the solid surface. During this period there

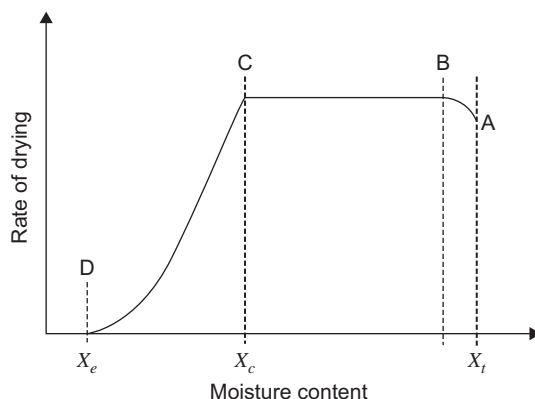


FIG. 18.58 Typical drying curve.

is no internal resistance to movement of moisture and the rate of evaporation is the same as that of a liquid of the same surface area. Once the free moisture is evaporated, the rate of drying starts to fall as internal mechanisms such as capillary drying and pore diffusion become more important, C to D. The falling-rate period is sometimes subdivided into regions dominated by different mechanisms.

The moisture content at the end of the constant-rate period is known as the critical moisture content, X_c . Moisture in excess of X_c is described as *free moisture*, and moisture removed between X_c and the equilibrium moisture content, X_e , is described as *bound moisture*. The critical moisture content can depend on the type of material, the particle size distribution, the type of dryer, and the thickness of solid layer (for tray and belt dryers). The critical moisture content increases with particle size and thickness of solids layer. The critical moisture content is easily determined experimentally in a drying test, but cannot reliably be predicted from theory. Some typical values of X_c are given in [Green and Southard \(2018\)](#) and range from 300% w/w for gelatin beads to 3% w/w for a 0.25-inch layer of sand on trays. Because the critical moisture content cannot be determined from first principles, estimation of dryer residence time must always be made experimentally, usually in plant trials carried out on an existing production line or in consultation with a dryer vendor.

Expressions for the rate of drying in the constant-rate period and falling-rate period are given in most textbooks on separations or mass transfer; see [Chhabra and Gurappa \(2019\)](#), [McCabe et al. \(2004\)](#), and [Green and Southard \(2018\)](#). During the constant-rate period the rate of drying is independent of the moisture content and depends only on the rates of heat and mass transfer. For a system that has only convective heat transfer:

$$-\frac{dX}{dt} = \frac{h A \Delta T}{\lambda} = K_G A (p_s - p_b) \quad (18.29)$$

where:

X = moisture content

t = time

h = heat transfer coefficient

A = gas–solids contact area

ΔT = temperature difference between the gas and solids, $= T_g - T_s$, where T_g is the temperature of the bulk gas and T_s is the temperature at the gas–liquid interface

λ = latent heat of vaporization

K_G = mass transfer coefficient

p_s = vapor pressure of liquid at the surface

p_b = vapor pressure of liquid in the bulk gas

By writing the vapor pressure of the liquid at the surface, p_s , as a function of the surface temperature, T_s , using an equation such as the Antoine equation, Equation 18.29 can be solved using the j-factor analogy to relate the heat and mass transfer coefficients, and hence the surface temperature during the constant-rate period can be found. This temperature will always be less than the boiling point of the liquid and corresponds to the wet bulb temperature of the gas. This does not give a good estimate of the solids outlet temperature, though, as the solids will heat up further during the falling-rate period and can approach the inlet gas temperature if the flow in the dryer is countercurrent.

Unfortunately, integration of Equation 18.29 to predict the residence time requires knowledge of the critical moisture content, X_c . The critical moisture content is also required to solve the equations for residence time in the falling-rate section. Because the experiments that measure X_c also yield the residence time, the theoretical analysis is somewhat superfluous and is generally only used when making modifications to an existing dryer.

An important consideration in dryer design is to properly match the solids and gas flows. The gas flow rate must be large enough to take up the moisture removed and to provide the necessary heat without requiring an excessively high inlet gas temperature. Depending on the dryer design, the gas velocity must also be chosen so as to prevent (or, if required, cause) fluidization or entrainment of the solids. The gas outlet condition depends on whether the gas flow is co-current, countercurrent, or cross-flow to the solids, but will generally be approaching equilibrium with the solids at the point where the gas exits the dryer.

The gas inlet temperature is often constrained by the maximum temperature to which the solids can be exposed. Thermally sensitive materials such as foods, pharmaceutical products, polymers, paper, and textiles can become charred or degraded if exposed to high temperatures. Operation with a lower inlet temperature will lead to a higher gas flow rate and may require a larger dryer.

The gas inlet moisture (or solvent) content depends on whether the gas is recirculated or predried; see [Section 18.7.3](#). An initial estimate of the gas rate can be made from a simple heat balance:

$$m_s \lambda (X_i - X_o) + m_s C_{ps} (T_{si} - T_{so}) = (m_g + m_e) C_{pg} (T_i - T_o) \quad (18.30)$$

where:

m_s = mass flow of bone-dry solids

m_g = mass flow of dry gas

m_e = mass flow of solvent or water in entering gas

X_i = inlet moisture content

X_o = outlet moisture content

C_{ps} = specific heat capacity of solids

C_{pg} = specific heat capacity of gas

T_{si} = solids inlet temperature

T_{so} = solids outlet temperature

T_i = gas inlet temperature

T_o = gas outlet temperature

λ = latent heat of vaporization

The sensible heat change of the evaporated moisture is neglected in Equation 18.30, and the sensible heat change of the solids can also be neglected for an initial estimate. The designer should then confirm that the gas flow rate calculated is sufficient to provide a driving force for mass transfer at the outlet conditions:

$$\frac{(m_o/M_{wL})}{((m_g/M_{wG}) + (m_o/M_{wL}))} < \frac{p_s}{P} \quad (18.31)$$

where:

m_o = mass flow of moisture at the outlet = $m_e + m_s(X_i - X_o)$

M_{wG} = average molecular weight of the gas

M_{wL} = molecular weight of the liquid

P = pressure

Typically, a dryer will be designed for about 80% or 90% approach to equilibrium at the gas exit:

$$\frac{(m_o/M_{wL})}{((m_g/M_{wG}) + (m_o/M_{wL}))} \approx \frac{0.8p_s}{P} \quad (18.32)$$

Equations 18.30 and 18.32 can be solved together until satisfactory values of m_g , m_e , and T_i are obtained. These calculations can also be made using process simulation programs.

18.7.2 Selection and design of dryers

The main factors to be considered when selecting a dryer are:

1. Feed condition: solid, liquid, paste, powder, crystals
2. Feed concentration, the initial liquid content
3. Product specification: dryness required, physical form
4. Throughput required
5. Heat sensitivity of the product
6. Nature of the vapor: toxicity, flammability
7. Nature of the solid: flammability (dust explosion hazard), toxicity

The drying characteristics of the material can be investigated by laboratory and pilot plant tests, which are best carried out in consultation with the equipment vendors. Full descriptions of the various types of dryers and their applications are given in [Green and Southard \(2018\)](#), [Mujumdar \(2014\)](#), and [Couper et al. \(2012\)](#). Only brief descriptions of the principal types will be given in this section.

The basic types used in the chemical process industries are tray, band, rotary, fluidized, pneumatic, drum, and spray dryers. For removal of low amounts of moisture under diffusion-limited conditions, gravity dryers can also be considered. These consist of a silo with the solids in mass flow and a countercurrent flow of sweep gas; see [Mehos \(2009\)](#).

Tray dryers ([Fig. 18.59](#))

Batch tray dryers are used for drying small quantities of solids and are used for a wide range of materials.

The material to be dried is placed in either solid bottomed trays over which hot air is blown or perforated bottom trays through which the air passes. If air passes through the trays, then the designer must ensure that the air velocity is not high enough to cause fluidization; see [Section 18.2.2](#).

Batch dryers have high labor requirements, but close control can be maintained over the drying conditions and the product inventory, and they are suitable for drying fine powders and valuable products.

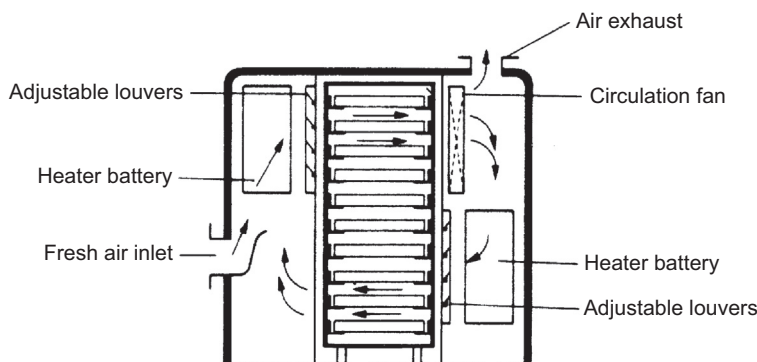


FIG. 18.59 Tray dryer.

Conveyor dryers (continuous circulation band dryers) ([Fig. 18.60](#))

In this type, the solids are fed on to an endless, perforated conveyor belt. Hot air or other drying gas is either passed over the solids on the belt or forced through the belt. The belt is housed in a long rectangular cabinet, which is divided up into zones, so that the flow pattern and temperature of the drying air can be controlled. The relative movement through the dryer of the solids and drying air can be co-current, cross-flow, or, more usually, countercurrent.

This type of dryer is clearly only suitable for materials that form a bed with an open structure. High drying rates can be achieved with good product quality control. Thermal efficiencies are high, and with steam heating, steam usage can be as low as 1.5 kg per kg of water evaporated. The disadvantages of this type of dryer are high initial cost and, due to the mechanical belt, high maintenance costs. Care must be taken to ensure that the air velocity at dryer conditions is not high enough to cause fluidization of the bed if the flow is up through the belt.

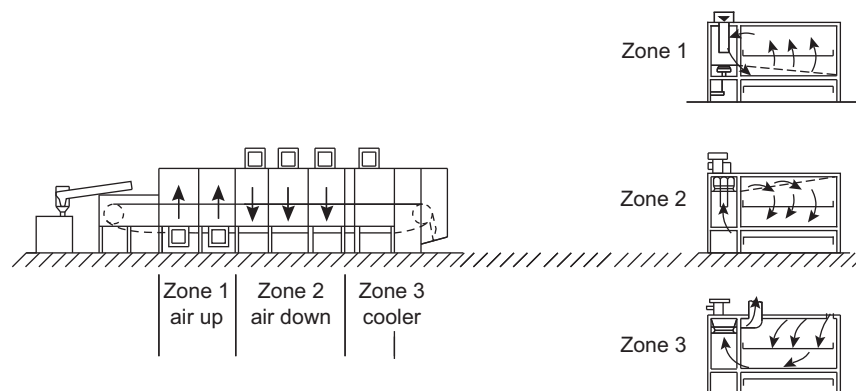


FIG. 18.60 Conveyor dryer.

Trays can also be used with belt dryers to allow continuous drying of fine powders. The trays are placed on the belt, filled with wet solids, passed through the dryer, emptied, and returned to the start. The trays can be fastened to the belt to eliminate manual retrieval.

Rotary dryer (Fig. 18.61)

In rotary dryers, the solids are conveyed along the inside of a rotating, inclined cylinder and are heated and dried by direct contact with hot air or gases flowing through the cylinder. In some, the cylinders are indirectly heated. Most commonly, the drying gas is heated by a steam heater at the dryer inlet or is direct-fired with a fuel.

Rotating dryers are suitable for drying free-flowing granular materials. They are suitable for continuous operation at high throughputs, have a high thermal efficiency, and have relatively low capital cost and labor costs. Some disadvantages of this type are nonuniform residence time, dust generation, and high noise levels. The superficial velocity of the gas in a rotary dryer should be low enough that the particles are not entrained in the gas flow; this sets the cylinder diameter if the gas rate is known. Rotary dryers can be quite large, and diameters over 15 ft are common, although 3 to 8 ft is the most typical size range. The main cost of a rotary dryer is usually the heater for the air, as the shell is not usually pressure-retaining and so is not a high-cost item. A cost correlation is given in Table 7.2 in terms of the heater surface area.

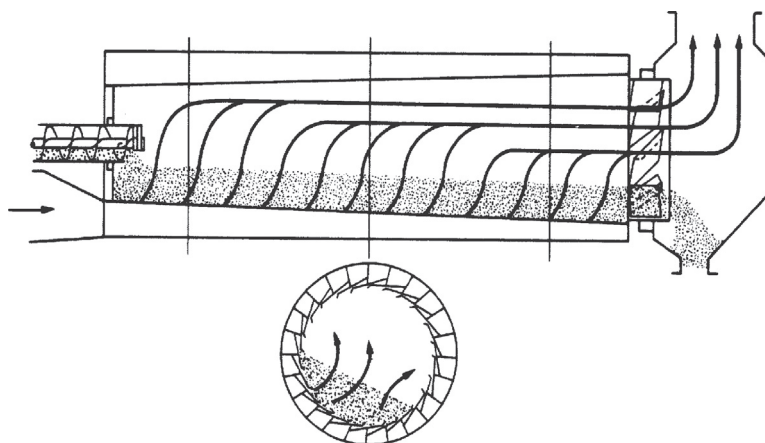


FIG. 18.61 Rotary dryer.

Fluidized-bed dryers (Fig. 18.62)

In this type of dryer, the drying gas is passed through the bed of solids at a velocity sufficient to keep the bed in a fluidized state, which promotes high heat transfer and drying rates.

Fluidized-bed dryers are suitable for granular and crystalline materials within the particle size range 0.5 to 3 mm. They are designed for continuous and batch operation.

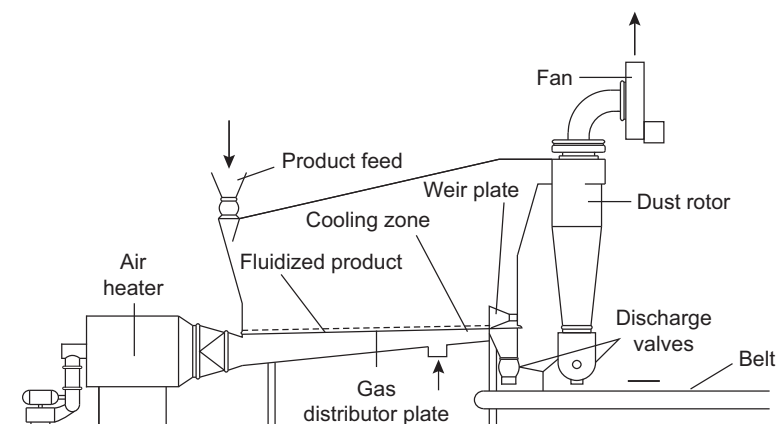


FIG. 18.62 Fluidized bed dryer.

The main advantages of fluidized dryers are rapid and uniform heat transfer; short drying times, with good control of the drying conditions; and low floor area requirements. The power requirements are high compared with other types.

A fluidized-bed dryer can be sized knowing the drying gas velocity and the superficial velocity required to cause fluidization; see [Section 18.2.2](#). Cyclones are usually incorporated on the exit gas to prevent formation of dust from fine particles that are elutriated by the drying gas.

Pneumatic dryers (Fig. 18.63)

Pneumatic dryers, also called flash dryers, are similar in their operating principle to spray dryers. The product to be dried is dispersed into an upward-flowing stream of hot gas by a suitable feeder. The equipment acts as a

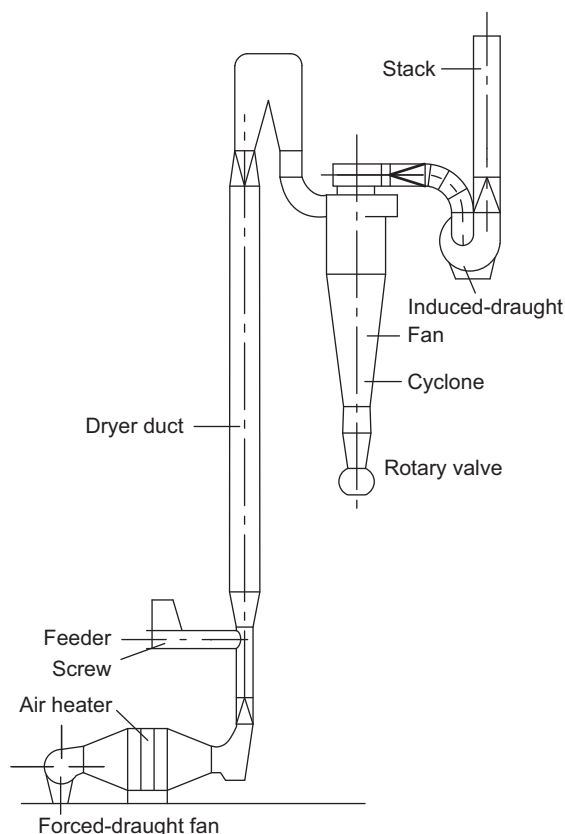


FIG. 18.63 Pneumatic dryer.

pneumatic conveyor and dryer. Contact times are short, and this limits the size of particle that can be dried. Pneumatic dryers are suitable for materials that are too fine to be dried in a fluidized-bed dryer but which are heat sensitive and must be dried rapidly. The thermal efficiency of this type is generally low.

Spray dryers (Fig. 18.64)

Spray dryers are normally used for liquid and dilute slurry feeds, but can be designed to handle any material that can be pumped. The material to be dried is atomized in a nozzle, or by a disc-type atomizer, positioned at the top of a vertical cylindrical vessel. Hot air flows up the vessel (in some designs downward) and conveys and dries the droplets. The liquid vaporizes rapidly from the droplet surface, and open, porous particles are formed. The dried particles are removed in a cyclone separator or bag filter.

The main advantages of spray drying are the short contact time, making it suitable for drying heat-sensitive materials, and good control of the product particle size, bulk density, and form. Because the solids concentration in the feed is low, the heating requirements will be high. Spray drying is discussed in the article by Møller and Fredsted (2009) and in a book by Masters (1991).

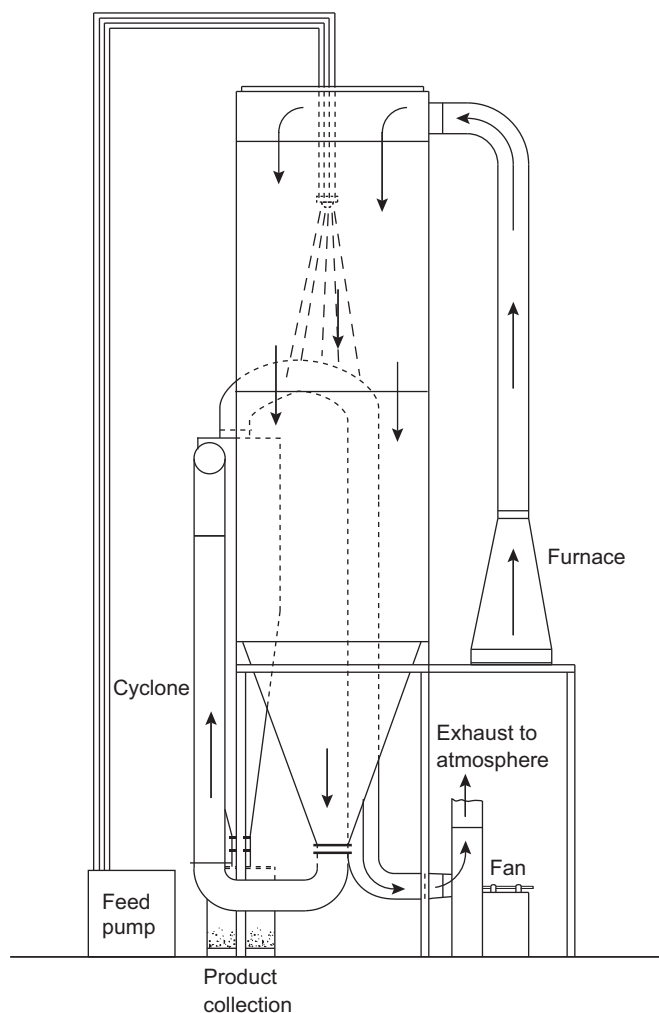


FIG. 18.64 Spray dryer.

Rotary drum dryers (Fig. 18.65)

Drum dryers are used for liquid and dilute slurry feeds. They are an alternative choice to spray dryers when the material to be dried will form a film on a heated surface and is not heat sensitive.

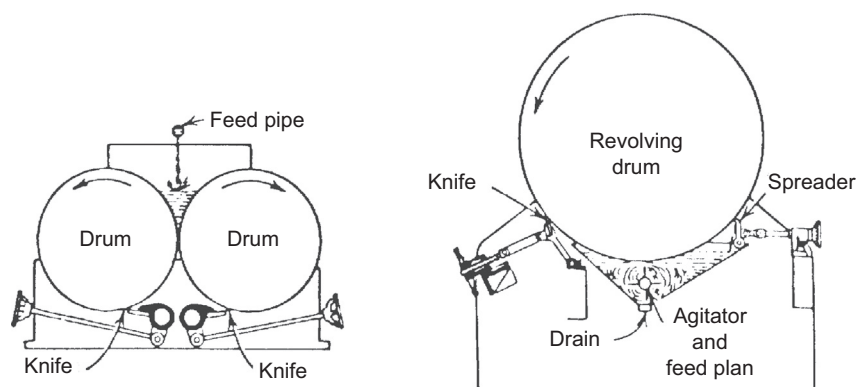


FIG. 18.65 Rotary drum dryers.

A drum dryer consists essentially of a revolving, internally heated drum, on which a film of the solids is deposited and dried. The film is formed either by immersing part of the drum in a trough of the liquid or by spraying, or splashing, the feed on to the drum surface; double drums are also used in which the feed is fed into the “nip” formed between the drums.

The drums are usually heated with steam, and steam economies of 1.3 kg steam per kg of water evaporated are typically achieved.

18.7.3 Process design and safety considerations

Dryers that remove water from solids usually use ambient air as the drying gas. The air can be heated in the dryer or pre-heated by indirect heat transfer from steam tubes at the dryer inlet. When high inlet temperatures are needed, direct heat transfer can be used by firing a burner in the inlet air. Such burners are typically fueled with natural gas or a process waste stream. The increase in inlet humidity due to the water vapor formed during combustion is not problematic if the inlet temperature is high.

For thermally sensitive products that must be dried with a low gas inlet temperature, the inlet air is sometimes preconditioned by passing it over a bed of molecular sieve adsorbent to ensure constant low inlet humidity. The adsorbent can then be regenerated in a temperature-swing cycle; see [Section 16.2.1](#). Chilling the inlet air to remove moisture by condensation is also possible but is generally avoided, as chilling increases the heat load on the heater of the dryer.

The air or flue-gas flow exiting a once-through air–water dryer is usually ducted away from the dryer and discharged away from the plant. The exiting air will be hot and moist and may contain particulate material if the solids are prone to dust formation. Gas-cleaning systems will be specified if the dust load is high or if there are environmental or safety concerns with respect to the dust; see [Section 18.5](#). The gas-cleaning equipment is usually located close to the dryer to prevent deposition of the dust or condensation in the ducting.

When a flammable solvent is removed from a solid or when a combustible dust could be formed, air should not be used as the drying gas. Although it is possible to design the dryer to operate outside the flammability envelope (see [Section 10.2.2](#)), there is nevertheless a risk that flammable conditions may occur during a process upset, with the potential to cause a fire or explosion. Instead, a closed-loop, recirculating gas system using an inert gas such as nitrogen can be used, as shown in [Fig. 18.66](#).

In a closed-loop system, the gas leaving the dryer is sent to cyclones, filters, or other gas-cleaning equipment to remove dust. The cleaned gas is cooled to allow the solvent to be condensed and recovered. The gas is then compressed by a fan or blower and returned to the heater at the dryer inlet. The dryer in a closed-loop system is designed to be gas-tight, and a small amount of make-up gas may be added to allow for the gas that flows out in the void spaces of the solid product.

Recirculating gas dryers are also used when the solid has the potential to form a toxic or corrosive dust that would be harmful if discharged to the atmosphere. Note that not all dryers are suitable for operation in recirculating-gas mode. The dryer design must enable airtight operation.

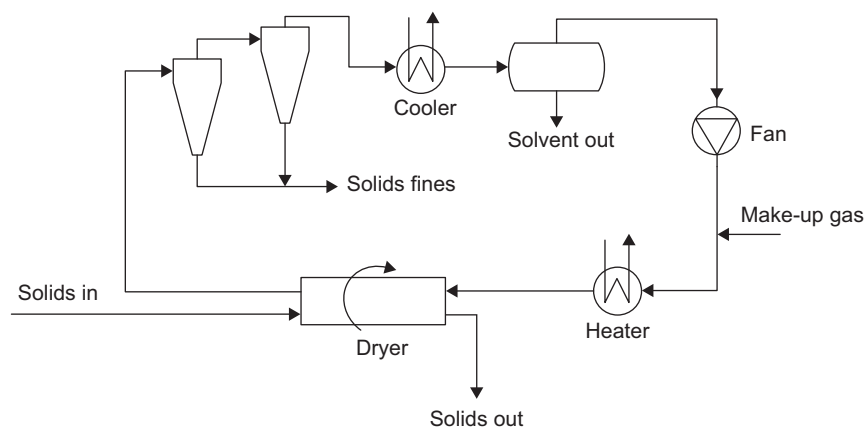


FIG. 18.66 Closed-loop drying process.

18.8 Solids formation, shaping, and size enlargement processes

Size enlargement processes are used to form larger agglomerates out of small particles. Size enlargement is a common step in processes that produce solid products, for example, in the manufacture of foods, tablets, fertilizers, catalysts, adsorbents, soap powders, solid fuels, and ceramics.

The equipment used for size enlargement is usually of proprietary design and must be specified in consultation with the vendor. The equipment vendors are easy to locate on the Web and usually have technically proficient sales engineers who can provide advice on device selection, dust handling, etc. In many cases, the vendor will be prepared to run demonstration trials to prove that their device forms an acceptable product.

There are several excellent specialist books on granulation and size enlargement: Pietsch (1991, 2005), Salman et al. (2006), and Lister and Ennis (2004). Useful introductions to the subject are also given in Green and Southard (2018), Rhodes (2008), Pietsch (1997), Higashitani et al. (2019), and Chhabra and Gurappa (2019).

18.8.1 Mechanisms of agglomeration

Agglomerates of smaller particles may be held together by natural cohesive forces between the particles or by partially melting the solids and sintering the particles together; however, in most cases a *binder* is added to the particles to stick them together. The choice of binder depends on the process temperature and product requirements. After the right size particles have been formed, the binder is sometimes driven off by drying or calcining the particles. Common binders for foods and pharmaceutical products include dextrose, starch, glucose, gelatine, and gums, whereas for catalysts or fertilizers, resins, clays, or water are usually used.

The theory of agglomeration processes is not well developed, and most size enlargement methods are difficult to model quantitatively. Instead, practitioners rely on experimentation and trial manufacturing runs to establish that the product will be formed with the desired density, strength, and other required properties. Scale-up experiments typically investigate varying the concentration and type of binder, as well as the process conditions, as it is often easier to achieve specifications by formulation changes than by changing process conditions, particularly if the product is to be made using existing equipment.

Four rate processes are important in forming an agglomerate out of smaller particles:

1. **Wetting:** in which the binder contacts the surface of the powder.
2. **Coalescence:** in which small particles adhere to one another to form a larger cluster.
3. **Consolidation:** in which the clusters are compacted by forces transmitted to the cluster as a result of processing, for example, by collisions with other particles or with the vessel walls or agitator.
4. **Breakage:** in which a weaker particle or shape is broken up due to collisions with other particles or the vessel.

Each of these processes can occur through several mechanisms. For example, breakage can occur by shattering of an agglomerate or abrasion of smaller particles from the surface.

The particle strength and shape depend on a trade-off between consolidation and breakage. High-shear devices that involve more particle–vessel and particle–particle collisions will tend to produce rounder, stronger pellets.

Particle density is obviously most affected by consolidation, which requires a high collision rate, but this must be traded off against breakage that would limit particle size. The different types of equipment used in forming solid particles are often designed to maximize the effectiveness of just one of these mechanisms. For example, tablet presses and roll presses strongly favor consolidation and seek to minimize breakage.

An overview of agglomeration mechanisms is given by [Ennis \(2010\)](#). The kinetics of particle formation and growth are discussed in more depth in [Pietsch \(1991, 1997\)](#), [Salman et al. \(2006\)](#), [Lister and Ennis \(2004\)](#), and [Green and Southard \(2018\)](#).

18.8.2 Shaping, forming, and size enlargement processes

The selection of agglomeration process depends on the size, shape, and uniformity of particles required. Some applications require a very uniform product. For example, tablets must be a standard size and shape to ensure that each contains the correct dose of active pharmaceutical ingredient. [Fig. 18.67](#) gives a guide to selection of forming process based on desired particle size, but other factors should also be taken into consideration, as described next. Details of the design, construction, and operation of these processes are given in specialized books referenced in each section and in the handbooks listed in the introduction to [Section 18.8](#).

Tablet presses and roll presses

When a uniform size product is required, a pressure compaction method is used. Rotary tablet presses or roll presses allow uniform shape and size solids to be produced at high throughputs, up to roughly 10,000 tablets per minute for a tablet press and 100 metric tons per hour for a roll press. Roll presses give less uniform product than tablet presses but allow larger briquettes to be formed.

The particles that are formed in a tablet press or roll press are usually of high density. The particle size distribution is very narrow, and particles up to the order of a few centimeters dimension can be formed. A wide range of shapes can be formed, including classic pill-shaped, lozenge-shaped, and briquettes. Tablet presses can also form intricate shapes and can imprint text or diameter indentations that make it easier for patients to split pills.

Tablet presses are widely used in the pharmaceuticals industry and are also used for food supplements and manufacture of certain types of candy. Roll presses are used in many other food processing applications, as well as in pelletizing of animal feeds and formation of briquettes.

The design and operation of tablet presses in the pharmaceutical industry are described in books by [Carstensen \(1977, 1993\)](#), [Hickey and Ganderton \(2009\)](#), and [Parikh \(2009\)](#).

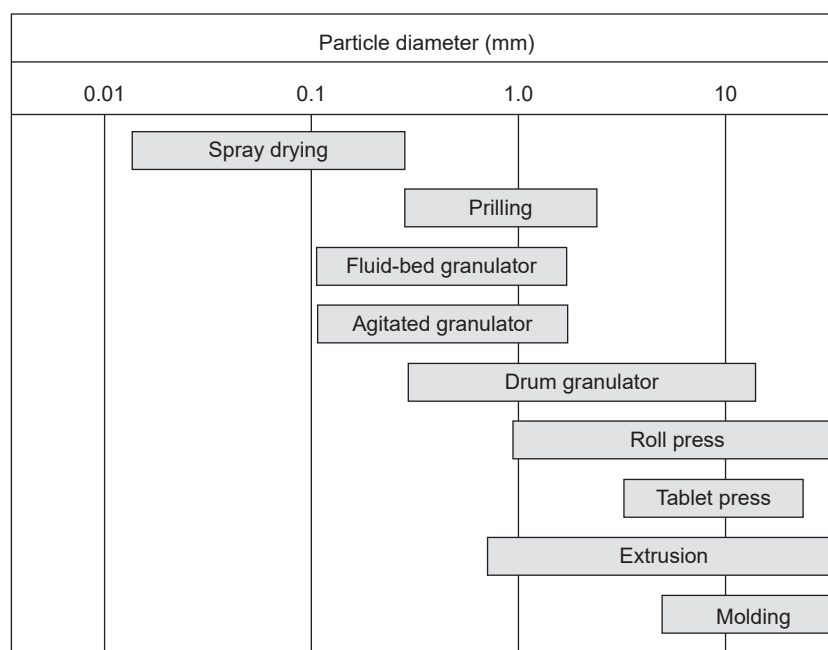


FIG. 18.67 Selection of particle-forming equipment.

Extrusion

An alternative form of pressure compaction is to form the solid and binder into a paste that can be extruded through a die. The resulting extrudate may either break up naturally, giving a product of uniform cross-section but varying length, or may be periodically cut with a knife edge or wire.

Extrusion can be used to form many different shapes of products, as evidenced by the many varieties of pasta, all of which are formed by this method. Extrudate particle cross-sectional diameter can range from less than a millimeter to a few centimeters. Extrusion processes are usually closely followed by dryers to ensure that the solids harden before they can stick together or lose their extruded shape. In addition to making pasta, some types of candy, and other formed food products, extrusion is often used for forming nonspherical catalysts. Extruded catalysts are usually formed as cylinders, but trilobes, tubes, and wagon wheel shapes are also used.

Spinning is a variant of extrusion that is used to form fibers. In polymer spinning, a melt, slurry, solution, or gel is extruded through a die known as a spinneret to form a fiber that can then be wound onto a bobbin. The spinneret usually contains several holes, allowing multiple filaments to be formed simultaneously. The fibers are often further processed by drawing (stretching) to achieve the desired fiber thickness. Spinning is used throughout the textile industry.

Molding

Mold presses can be used to accurately form large, complex shapes. Molding is used for making large, shaped ceramics and for forming shaped polymer objects. Molding processes usually incorporate a curing step, either by chemical or thermal treatment of the material inside the mold. For ceramics, this can be a high-temperature sintering, whereas for polymers, curing or cooling are used, depending on whether the polymer is thermosetting or thermoplastic. Polymers are usually melted and injected into the mold as a liquid (injection molding) to ensure full mold penetration.

Molding can allow highly accurate control of particle shape and dimensions. For large parts, additional machining may be used to remove burrs (flash) and add details. Molding is widely used in the manufacturing industries to make everything from automotive parts to toys and dinnerware.

Granulation

If a broader particle size distribution is acceptable, then particles can be agglomerated cheaply using a rotary agglomerator, also sometimes known as a granulator, pelletizer, tumbling drum, or balling drum. An agglomerator consists of an inclined drum into which the solids and binder are fed. Material leaving the drum is sent to a screen and undersize particles are recycled to the feed to act as seed particles (Fig. 18.68). Occasionally, oversize material is also separated, ground up, and returned to the feed. Inclined open pans can also be used as rotary agglomerators; these are cheaper, but produce more dust and require more recycle.

Larger size particles can also be formed in a granulation drum by spraying a slurry of small particles into the drum. The small particles coat the surface of the existing particles in the drum, causing larger particles to be formed. For crystalline materials, this method can be used to make large agglomerates containing many smaller crystalline particles.

The internally agitated mixers described in Section 18.4.11 can all be operated as granulators, with or without recycle of fines. If a binder is fed along with the solids, the solids will tend to agglomerate and form larger particles. Agitated granulators tend to form higher-density particles than drum granulators but operate at lower throughput.

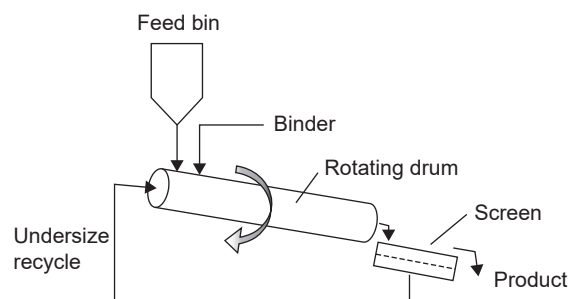


FIG. 18.68 Rotary agglomerator.

Large drum granulators are widely used in the manufacture of fertilizer pellets and can be designed for plant capacities up to the order of a thousand metric tons per hour. Granulation drums typically have 1- to 5-minute residence time, depending on the ease of agglomeration, and the solids holdup is usually between 10% and 20% of the internal volume. Smaller agitated granulators are used for making various bead products that must be approximately spherical and typically operate at a few metric tons per hour. These devices are often operated in batch mode. Some powdered foods are also made by granulation, for example, granulated sugar.

Fluidized beds can be used for granulation and also permit addition of a slurry of smaller particles. Care must be taken in a fluidized-bed granulator to ensure that the secondary feed is not elutriated before it can coat the particles of the bed. The design of fluidized beds is discussed in [Section 18.2.2](#).

Spray drying and prilling

Spray drying was described in [Section 18.7.2](#) and illustrated in [Fig. 18.64](#). Prilling is essentially the same operation as spray drying but forms a larger particle, usually by crystallization combined with drying.

Spray drying and flash drying tend to form small, low-density particles. Particle sizes between ~ 10 and ~ 500 microns are typical. The particle shape can be quite variable and can range from spheres to flakes. Prilling forms larger particles, up to a few millimeters in diameter, and tends to form spheroids. The size of particles formed is strongly dependent on the spray nozzles used, and a wide range of proprietary designs are available from nozzle vendors. If a drying gas is used in countercurrent flow to the falling solids, then fines will be elutriated and must be collected from the effluent gas and recycled to the process.

Prilling is widely used for making fertilizers such as urea and ammonium nitrate. Spray drying is used to make catalysts, powdered foods, and detergent powders.

Crystallization

Crystallization can be used to form free-flowing solid particles from materials that will crystallize. Crystallization and precipitation are described in detail in [Section 16.5](#), together with other processes for removing dissolved solids from a liquid.

18.8.3 Postforming processes

The first step after particle forming is usually to screen the particles to reject those that are undersized or oversized. Undersized material can usually be recycled to the forming step. Oversized material may need to be reground before it can be recycled.

Once the desired size and shape particles have been formed, they will often be dried or heat treated to reduce their plasticity and ensure that the shape is retained. In some cases, extreme heat treatment known as calcining will be carried out to drive off or burn off the binder.

Many solids products will subsequently be coated. Various types of coatings are applied; for example, pharmaceutical tablets may be coated with sugar or gels to make them more palatable or easier to swallow, while many solids are coated with waxes to suppress dust formation during handling. Coatings are usually applied in a tumbler or spray coating drum that is essentially the same as a rotary agglomerator. The particles must be kept moving during coating to ensure a uniform coating and prevent unwanted agglomeration. If the coating is applied in a solvent, then a rotary dryer is sometimes used as the coating drum. The selection of a coating material must consider the impact of the coating on the end-use application of the product.

18.9 Particle size reduction (comminution)

Crushing is the first step in the process of size reduction, reducing large lumps to manageable sized pieces. For some processes crushing is sufficient, but for chemical processes, it is usually followed by grinding to produce a fine-sized powder. Though many articles have been published on comminution, and [Marshall \(1974\)](#) mentions over 4000, the subject remains essentially empirical. The designer must rely on experience and the advice of the equipment manufacturers when selecting and sizing crushing and grinding equipment. The power requirements are best estimated based on grinding tests carried out on existing plant or in consultation with an equipment vendor. For a fuller treatment of the subject, the reader should refer to the books by [Lowrison \(1974\)](#), [Prasher \(1987\)](#), and [Fuerstenau and Han \(2003\)](#).